

GEN 1.7 DIFFERENCES FROM ICAO STANDARDS, RECOMMENDED PRACTICES AND PROCEDURES

1. Annex 1 - Personnel Licensing, 14th Edition, July 2022, Amendment 178.

Chapter 1

1.2.2.1 PCAR Part 2, 2.2.4.1 (c) (d) (e) (f) (g) and Part 2, 2.2.4.5. 2.2.4.1 VALIDATION OF FLIGHT CREW LICENSES (c) A validation certificate with PPL privileges, based upon at least a PPL, will be issued provided: (1) The applicant for the validation certificate shall present to the Authority the foreign license. (2) The applicant for the validation certificate shall hold a current medical certificate issued under Part 2 or a current medical certificate issued by the Contracting State that issued the applicant's pilot license provided that the foreign medical certificate meets the requirements of Part 2, relevant to the license held. (3) The validation certificate will be valid provided the foreign license or in the case of a continuing license, the rating/medical certificate remains valid. (d) A validation certificate with PPL/IR, CPL, CPL/IR, ATPL or FE privileges, based upon the relevant license, will be issued provided the following requirements are met. (1) The applicant for the validation certificate shall present to the Authority the foreign license and evidence of the experience required by presenting the record (e.g. logbook). (2) The validation certificate will be valid for one year, provided the foreign license or in the case of a continuing license the rating/medical certificate remains valid. (3) Ratings will only be validated together with the validation of a license. (4) The applicant for the validation certificate shall: (i) hold a current medical certificate issued under Part 2 or a current medical certificate issued by the Contracting State that issued the applicant's pilot license provided that the foreign medical certificate meets the requirements of Part 2, relevant to the license held; (ii) complete a skill test for the relevant ratings in the license that he or she wants to be validated relevant to the privileges of the license held; (iii) demonstrate to the satisfaction of the Authority the knowledge relevant to the license to be validated of: (A) Air Law; (B) Aeronautical Weather codes; (C) Flight Performance Planning and; (D) Human Performance (e) Demonstrate a knowledge and comprehension of the English language as required by subpart 2.3.7 (21 March 2011). (f) While exercising the privileges of his license, a valid medical certificate the license on which the validation is based and the certificate of the validation shall be carried, and; (g) Comply with the experience requirements set out in the table below: Others...,

2.2.4.5 VALIDATION OF FOREIGN MECHANICS LICENSES (a) The Authority may render a valid license issued by a foreign country on a reciprocal basis as an alternative to the issuance of its own license. It shall establish validity by suitable authorization to be carried with the former license accepting it as the equivalent of the latter. The validity of the authorization shall not extend beyond the period of validity of his foreign license.

1.2.2.2 PCAR Part 2.2.4.1 (b) and (d). (b) The Republic of the Philippines will verify the authenticity of the license, ratings authorizations and the medical certificate with the state of license issue. (d) A validation certificate with PPL/IR, CPL, CPL/IR, ATPL or FE privileges, based upon the relevant license, will be issued provided the following requirements are met. (1) The applicant for the validation certificate shall present to the Authority the foreign license and evidence of the experience required by presenting the record (e.g. logbook). (2) The validation certificate will be valid for one year, provided the foreign license or in the case of a continuing license the rating/medical certificate remains valid. (3) Ratings will only be validated together with the validation of a license. (4) The applicant for the validation certificate shall: (i) hold a current medical certificate issued under Part 2 or a current medical certificate issued by the Contracting State that issued the applicant's pilot license provided that the foreign medical certificate meets

the requirements of Part 2, relevant to the license held; (ii) complete a skill test for the relevant ratings in the license that he or she wants to be validated relevant to the privileges of the license held; (iii) demonstrate to the satisfaction of the Authority the knowledge relevant to the license to be validated of: (A) Air Law; (B) Aeronautical Weather codes; (C) Flight Performance and Planning; and (D) Human Performance. (E) Demonstrate a knowledge and comprehension of the English language as required by subpart 2.3.7 (F) While exercising the privileges of his license, a valid medical certificate, the license on which the validation is based and the certificate of validation shall be carried; and (G) Comply with the experience requirements.

- 1.2.3 PCAR Part 2, 2.2.1 (b). 2.2.1 GENERAL (b) Privileges. A license and/or certificate holder is not permitted to exercise privileges other than those granted by the license and/or certificate.
- 1.2.4.1 PCAR Part 2, 2.2.1 (c). GENERAL (c) Medical fitness. An applicant for a flight crew or air traffic controller license shall hold a medical certificate issued in accordance with the provisions of this Part.
- 1.2.4.2 PCAR Part 2.2.2.5 (a) (1) and (2). (a) The Authority shall apply, as part of its State safety programme, basic safety management principles to the medical assessment process of license holders, that as a minimum include: (1) routine analysis of in-flight incapacitation events and medical findings during medical assessments to identify areas of increased medical risk; and (2) continuous re-evaluation of the medical assessment process to concentrate on identified areas of increased medical risk.
- 1.2.4.3 PCAR Part 2.2.2.5 (b). (b) The Licensing Authority shall implement appropriate aviation-related health promotion for licence holders subject to a Medical Assessment to reduce future medical risks to flight safety. Note 1. - Standard 2.2.2.5 indicates how appropriate topics for health promotion activities may be determined. Note 2.- Guidance on the subject is contained in the Manual of Civil Aviation Medicine (Doc 8984). Note 3. - Guidance on the relationship between the Licensing Authority and the implementation of Medical Assessment for license holders is contained in the Manual of Procedures for Establishment and Management of a State's Personnel Licensing System (Doc 9379).
- 1.2.4.4 PCAR Part 2, 2.10.1.8 (c) (1) (2). 2.10.1.8 MEDICAL CERTIFICATES (c) Validity: (1) The validity period of the medical certificate is: (i) 6 months for the Class 1 for ATPL. (ii) 12 months for CPL, MPL, flight engineer license, flight navigator license and FOO license. (iii) 24 months for the Class 2 for the PPL, SPL, glider pilot license and free balloon pilot license; and Class 3 for air traffic controller license. (iv) When the holders have passed their 40th birthday: (A) the 12-month interval specified for CPL, MPL, flight engineer license and flight navigator license will be reduced to 6 months; and (B) the 24-month interval specified for the PPL, SPL, glider pilot license, free balloon pilot license and air traffic controller license will be reduced to 12 months. (2) For initial issuance of the medical certificate, the period of validity shall begin on the date the medical examination is performed. For any renewal or re-issuance of a medical certificate, based on a medical examination that takes place during the period of validity of the current medical certificate, but no more than 28 days before its expiry date, the new period of validity shall begin on that date. For any renewal or re-issuance, based on a medical examination taking place after the expiry date or earlier than 28 days before the expiry date, the new period of validity shall begin on the date of the examination.
- 1.2.4.6 PCAR Part 2, 2.10.1.3 (a). 2.10.1.3 AVIATION MEDICAL EXAMINERS (AME) (a) Subject to compliance with the requirements specified in this Part, the Authority will designate and authorize qualified and licensed physicians in the practice of medicine, to be authorized as an AME and conduct medical examinations of fitness of applicants

- for the issue, renewal or reissue of the licenses or ratings specified in this Part. AMEs may be designated outside of Republic of the Philippines.
- 1.2.4.6.1 PCAR Part 2, 2.10.1.3 (b) (1) (d). 2.10.1.3 AVIATION MEDICAL EXAMINERS (AME) (b) AMEs shall have had, or shall receive: (1) Basic training in aviation medicine for Class 2 and 3 medical examinations on the subjects listed in IS: 2.10.1.3 Appendix A, supported by a diploma or a certificate of completion; (d) Prior to a designation, medical examiners shall demonstrate adequate competency in aviation medicine.
- 1.2.4.8.2 PCAR Part 2, 2.10.1.3 (h). 2.10.1.3 AVIATION MEDICAL EXAMINERS (AME) (h) If the medical examination is carried out by a constituted group of AMEs, the head of the group will be appointed by the Authority, who will be responsible for coordinating the results of the examination, evaluating the findings with regard to the medical fitness, and signing the report.
- 1.2.4.9 PCAR Part 2, 2.10.1.3 (i). 2.10.1.3 AVIATION MEDICAL EXAMINERS (AME) (i) The Authority will use the services of physicians experienced in the practice of aviation medicine, when it is necessary to evaluate reports submitted to the Authority by medical examiners.
- 1.2.4.9.1 PCAR Part 2, 2.10.1.3 (g). 2.10.1.3 AVIATION MEDICAL EXAMINERS (AME) (g) The medical examiner shall submit sufficient medical information to the Authority to enable the Authority to audit medical assessment, and to evaluate the medical reports submitted by the examiners, using the services of the medical assessors appointed by the Authority.
- 1.2.5.2 PCAR Part 2, 2.10.1.8 (c) and (c) (1) (v). 2.10.1.8 MEDICAL CERTIFICATES (c) Validity: (1) The validity period of the medical certificate is: (i) 6 months for the Class 1 for ATPL. (ii) 12 months for CPL, MPL, flight engineer license, flight navigator license and FOO license. (iii) 24 months for the Class 2 for the PPL, SPL, glider pilot license and free balloon pilot license; and Class 3 for air traffic controller license. (iv) When the holders have passed their 40th birthday: (A) the 12-month interval specified for CPL, MPL, flight engineer license and flight navigator license will be reduced to 6 months; and (B) the 24-month interval specified for the PPL, SPL, glider pilot license, free balloon pilot license and air traffic controller license will be reduced to 12 months. (v) The period of validity of a medical certificate may be extended, at the discretion of the Licensing Authority, up to 45 days.
- 1.2.5.2.2 PCAR Part 2, 2.10.1.8 (c) (1) (iv) (A) (B). 2.10.1.8 MEDICAL CERTIFICATES (c) Validity: (1) The validity period of the medical certificate is: (iv) When the holders have passed their 40th birthday: (A) the 12-month interval specified for CPL, MPL, flight engineer license and flight navigator license will be reduced to 6 months; and (B) the 24-month interval specified for the PPL, SPL, glider pilot license, free balloon pilot license and air traffic controller license will be reduced to 12 months.
- 1.2.5.2.6 PCAR Part 2, 2.10.1.5 - Special Circumstances. (a) If the medical requirements prescribed in Part 2 for a particular license are not met, the appropriate medical certificate will not be issued, renewed or re-issued unless the following conditions are fulfilled: (1) Accredited medical conclusion indicates that in special circumstances the applicant's failure to meet any requirement, whether numerical or otherwise. Is such that exercise of the privileges of the license applied for is not likely to jeopardize flight safety; (2) Relevant ability, skill and experience of the applicant and operational conditions have been given due consideration; and (3) The license is endorsed by the Authority with any special limitation or limitations when the safe performance of the license holder's duties is dependent on compliance with such limitation or limitations. (b) The AME shall report to the Authority any individual case where, in the AME's judgment, an applicant's failure to meet any requirement, whether numerical or

otherwise, is such that exercise of the privileges of the license being applied for, or held, is not likely to jeopardize flight safety.

1.2.8.3 Not implemented.

Chapter 2

2.1.1.1 PCAR Part 2, 2.2.2.2 (a) (1) - Ratings and PCAR Part 2.3.1.2 (b). 2.2.2.2 RATINGS (a) The following ratings are placed on a pilot license when an applicant satisfactorily accomplishes the requirements in this Part for the rating sought: (1) Category ratings in the following aircraft: (i) Airplane (ii) Helicopter (iii) Glider (iv) Free Balloon.

2.3.1.2 GENERAL RULE CONCERNING PILOT LICENSES. RATINGS AND AUTHORIZATIONS (b) A person shall not act either as pilot-in-command or as co-pilot of an aircraft in any of the categories unless that person is the holder of a pilot license issued in accordance with the provisions of Part 2.

2.1.1.2 PCAR Part 2, 2.3.2.2 (a). 2.3.2.2 CATEGORY RATING (a) The category of aircraft shall be endorsed on the license as a rating.

2.1.1.2.1 PCAR Part 2, 2.3.2.2 (b) (c). 2.3.2.2 CATEGORY RATING (b) Any additional category rating endorsed on a pilot license shall indicate the level of licensing privileges at which the category rating is granted. (c) The holder of a pilot license seeking additional category ratings shall meet the requirements of this Part appropriate to the privileges for which the category rating is sought.

2.1.1.3.1 PCAR Part 2, 2.3.2.1 (c). 2.3.2 CATEGORY, CLASS and TYPE RATINGS AND CATEGORY II/III AUTHORIZATIONS.

2.3.2.1 General (c) When an applicant demonstrates skill and knowledge for the initial issue or re-issue of a pilot license, the category and ratings appropriate to the class or type of aircraft used in the demonstration will be entered on the license.

2.1.1.4 PCAR Part 2.3.2.4 TYPE RATINGS - AIRPLANE AND HELICOPTER AND POWERED-LIFTS (a) Knowledge. The applicant for a type rating shall have completed the theoretical knowledge instruction and demonstrated in a test the relevant knowledge subjects as listed in IS.

Appendix A. (b) Experience. (1) An applicant for a type rating shall: (i) have at least 100 hours as pilot-in-command applicable to the category of aircraft; (ii) where applicable, have an instrument rating applicable to the category of aircraft; (iii) have completed a CRM course as listed in IS.

Appendix B; and (iv) have demonstrated in a test, the ATPL knowledge on the basis of the requirements listed in Subpart 2.3.3.4 (b) (for airplane) or Subpart 2.3.3.8 (b) (for helicopter), as applicable. (c) Flight instruction. (1) The applicant for a type rating shall have completed the flight instruction for the type rating: (i) for single-pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (ii) for multi-pilot aircraft: on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the flight instruction shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure and upset prevention and recovery training. (d) Skill. (1) The applicant for a type rating shall: (i) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; (ii) pass the required skill test: (A) for single pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (B) for multi-pilot aircraft on the subjects listed in IS 2.3.3.4 (for airplane) or IS 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the skill test shall

include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure. (e) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a type rating are to act as a pilot on the type of aircraft specified in the rating. (1) When the skill test for a type rating has been performed under VFR the type rating will be issued limiting the privileges to VFR flight and such limitation will be endorsed on the rating. (f) Validity. Subject to compliance with the requirements in this Part, the validity period of a type rating is 1 year. (g) Renewal. For the renewal of a type rating the pilot shall: (1) within the preceding 12 calendar months, complete a proficiency check: (i) for single pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable: and (ii) for multi-pilot aircraft on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or IS 2.3.3.8 Appendix B (for helicopter), as applicable. (2) have completed 10 route sectors within the 3 months preceding the expiry date. (3) Where applicable the proficiency check shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure. (4) If a pilot takes the proficiency check required in this section in the calendar month before or the calendar month after the month in which it is due, the pilot is considered to have taken it in the month in which it was due for the purpose of computing when the next proficiency check is due. (h) Re-issue. If the type rating has been expired the applicant shall: (1) have received refresher training from an authorized instructor with an endorsement that the person is prepared for the required skill test: and (2) pass the required skill test: (i) for single pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B (for helicopter), as applicable: and (ii) for multi-pilot aircraft on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or IS 2.3.3.8 Appendix B (for helicopter), as applicable. (3) Where applicable the skill test shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure.

2.1.2.1 PCAR Part 2, 2.2.2.2 (a)(1). 2.2.2.2 RATINGS (a) The following ratings are placed on a pilot license when an applicant satisfactorily accomplishes the requirements in this Part for the rating sought: (1) Category ratings in the following aircraft: (i) Airplane (ii) Helicopter (iii) Glider (iv) Free Balloon.

2.1.2.2 PCAR Part 2, 2.3.2.2 CATEGORY RATING (a) The category of aircraft shall be endorsed on the license as a rating.

2.1.3.1 PCAR Part 2, 2.2.2.2 (a)(2). 2.2.2.2 RATINGS (a) The following ratings are placed on a pilot license when an applicant satisfactorily accomplishes the requirements in this Part for the rating sought: (2) Class ratings in the following aircraft: (i) Single-engine land - airplane (ii) Single-engine sea - airplane (iii) Multi-engine land - airplane (iv) Multi-engine sea- airplane.

2.1.3.2 PCAR Part 2, 2.2.2.2 (a)(3) (i) (ii) (iii). (3) Type ratings in the following aircraft: (i) Each type of aircraft certificated for operation with a minimum crew of at least two pilots; (ii) Each type of helicopter certificated for single-pilot except where a class rating has been established under (a)(2)(v); and (iii) Any aircraft considered necessary by the Authority.

2.1.4.1 PCAR Part 2, 2.3.2.1 (a)(1)(2). 2.3.2 CATEGORY, CLASS and TYPE RATINGS AND CATEGORY II / III AUTHORIZATIONS

2.3.2.1 GENERAL (a) The holder of a pilot license shall not be permitted to act as pilot-in-command or as copilot of an airplane or helicopter unless the holder has received Authorization as follows: (1) the appropriate class rating specified in this Part, or (2) a type rating when required in accordance with this Part.

2.1.5.1 PCAR Part 2, 2.3.2.3 (b)(1)(ii). 2.3.2.3 CLASS RATINGS - AIRPLANE AND HELICOPTER AND POWERED-LIFTS (b) Skill. (1) The applicant for a class rating shall: (ii) pass the required skill test on the subjects listed in IS 2.3.3.2 or 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B (for helicopter), as applicable.

2.1.5.2 PCAR Part 2, 2.3.2.4, IS 2.3.2.4 APPENDIX A. 2.3.2.4 TYPE RATINGS - AIRPLANE AND HELICOPTER AND POWERED-LIFTS (a) Knowledge. The applicant for a type rating shall have completed the theoretical knowledge instruction and demonstrated in a test the relevant knowledge subjects as listed in IS 2.3.2.4 Appendix A. (b) Experience. (1) An applicant for a type rating shall: (i) have at least 100 hours as pilot-in-command applicable to the category of aircraft; (ii) where applicable, have an instrument rating applicable to the category of aircraft; (iii) have completed a CRM course as listed in IS 2.3.2.4 Appendix B; and (iv) have demonstrated in a test, the ATPL knowledge on the basis of the requirements listed in Subpart 2.3.3.4 (b) (for airplane) or Subpart 2.3.3.8 (b) (for helicopter), as applicable. (c) Flight instruction. (1) The applicant for a type rating shall have completed the flight instruction for the type rating: (i) for single-pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (ii) for multi-pilot aircraft: on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the flight instruction shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure. (d) Skill. (1) The applicant for a type rating shall: (i) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; (ii) pass the required skill test: (A) for single pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (B) for multi-pilot aircraft on the subjects listed in IS 2.3.3.4 (for airplane) or IS 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the skill test shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure. (e) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a type rating are to act as a pilot on the type of aircraft specified in the rating. (1) When the skill test for a type rating has been performed under VFR the type rating will be issued limiting the privileges to VFR flight and such limitation will be endorsed on the rating.

IS 2.3.2.4 APPENDIX A: CLASS/TYPE RATING [SP (A) and MP (A)] - KNOWLEDGE (a) The knowledge instruction and test for the type rating for multi-pilot - airplane shall include the following subjects: (1) Airplane structure and equipment, normal operation of systems and malfunctions (i) Dimensions (ii) Engine including auxiliary power unit (iii) Fuel system (iv) Pressurization and air-conditioning (v) Ice protection, windshield wipers and rain repellent (vi) Hydraulic systems (vii) Landing gear (viii) Flight controls, lift devices (ix) Electrical power supply (x) Flight instruments, communication, radar and navigation equipment (xi) Cockpit, cabin and cargo compartment (xii) Emergency equipment (2) Limitations: (i) General limitations (ii) Engine limitations (iii) System limitations (iv) Minimum equipment list (3) Performance, flight planning and monitoring (4) Load, balance and servicing (i) Load and balance (ii) Servicing on the ground (5) Emergency procedures (6) Special requirements for extension of a type rating for instrument approaches down to a decision height of less than 200 FT (60 M) (i) Airborne and ground equipment: technical requirements, operational requirements, operational reliability, fail operational, fail-passive, equipment reliability, operating procedures, preparatory measures, operational downgrading, communications (ii) Procedures and limitations: operational procedures, crew co-ordination (7) Special

- requirements for “glass cockpit” airplane with electronic flight instrument systems (e.g. EFTS; EICAS) (8) Flight Management systems (FMS) (i) for single-pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (ii) for multi-pilot aircraft: on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the flight instruction shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure.
- 2.1.5.3 PCAR Part 2, 2.3.2.4 (a)(b)(c)(d). 2.3.2.4 TYPE RATINGS - AIRPLANE AND HELICOPTER AND POWERED-LIFTS (a) Knowledge. The applicant for a type rating shall have completed the theoretical knowledge instruction and demonstrated in a test the relevant knowledge subjects as listed in IS 2.3.2.4 Appendix A. (b) Experience. (1) An applicant for a type rating shall: (i) have at least 100 hours as pilot-in-command applicable to the category of aircraft; (ii) where applicable, have an instrument rating applicable to the category of aircraft; (iii) have completed a CRM course as listed in IS 2.3.2.4 Appendix B; and (iv) have demonstrated in a test, the ATPL knowledge on the basis of the requirements listed in Subpart 2.3.3.4 (b) (for airplane) or Subpart 2.3.3.8 (b) (for helicopter), as applicable. (c) Flight instruction. (1) The applicant for a type rating shall have completed the flight instruction for the type rating: (i) for single-pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (ii) for multi-pilot aircraft: on the subjects listed in IS 2.3.3.4 Appendix B (for airplane) or 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the flight instruction shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure. (d) Skill. (1) The applicant for a type rating shall: (i) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; (ii) pass the required skill test: (A) for single pilot aircraft: on the subjects listed in IS 2.3.3.2 or IS 2.3.3.3 Appendix B (for airplane) or IS 2.3.3.6 or IS 2.3.3.7 Appendix B for helicopter, as applicable; and (B) for multi-pilot aircraft on the subjects listed in IS 2.3.3.4 (for airplane) or IS 2.3.3.8 Appendix B (for helicopter), as applicable. (2) Where applicable the skill test shall include instrument procedures, including instrument approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure.
- 2.1.7 PCAR Part 2, 2.3.3.6 INSTRUMENT RATING - AIRPLANE (a) General. The holder of a pilot license shall not act either as pilot-in-command or as copilot of an aircraft under instrument flight rules (IFR) unless such holder has received proper Authorization from the Authority. Proper Authorization shall comprise an instrument rating appropriate to the aircraft category.
- 2.1.8.1 PCAR Part 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (a) General. The applicant for an instructor rating or Authorization - airplane or helicopter, shall (1) hold at least the license and rating for which instruction is being given, in the appropriate category; and (2) hold the license and rating necessary to act as the pilot-in-command of the aircraft on which the instruction is given; or (3) hold a specific Authorization granted by the Authority.
- 2.1.8.2 PCAR Part 2, 2.2.6.2 USE OF SYNTHETIC FLIGHT TRAINERS The use of a synthetic flight trainer for performing training, testing and checking for which a flight crew member is to receive credit, shall be approved by the Authority, which shall ensure that the synthetic flight trainer is appropriate to the task. Note: See the Manual of Criteria for the Qualification of Flight Simulators (ICAO Doc 9625).

- 2.1.10 PCAR Part 2, 2.3.1.5(a)(b) and Part 8, 8.10.1.1. 2.3.1.5 LIMITATION OF PRIVILEGES OF PILOTS WHO HAVE ATTAINED THEIR 60TH BIRTHDAY (a) No person who holds a pilot license under this Part shall serve as pilot-in-command on a Philippine-registered civil aircraft engaged in international commercial air transport in a single pilot operation if that person has reached his or her 60th birthday. (21 March 2011) (b) No person who holds a pilot license issued under this Part shall serve as a pilot-in-command on a Philippine-registered civil aircraft engaged in international commercial air transport in a multi-crew operation if that person has reached his or her 60th birthday unless the other pilot is younger than 60 years of age. (21 March 2011). 8.10.1.1 LIMITATION OF PRIVILEGES OF PILOTS WHO HAVE ATTAINED THEIR 60TH BIRTHDAY AND CURTAILMENT OF PRIVILEGES OF PILOTS WHO HAVE ATTAINED THEIR 65TH BIRTHDAY (a) No person may serve nor may any AOC holder use a person as a required PIC in single pilot operations on aircraft engaged in commercial air transport operations if that person has reached his or her 60th birthday. (b) For aircraft engaged in commercial air transport operations requiring more than one pilot as flight crew members, the AOC holder may use one pilot up to age 65 provided that the other pilot is less than age 60. (c) Check airmen who have reached their 65th birthday or who do not hold an appropriate medical certificate may continue their check airman functions, but may not serve as or occupy the position of a required pilot flight crew member on an airplane engaged in commercial air transport operations. (21 March 2011)
- 2.2.1 PCAR Part 2, 2.2.2.3 (a) (1) Part 2, 2.3.3.1 (c). 2.2.2.3 AUTHORIZATIONS (a) The following Authorizations are issued when an applicant satisfactorily accomplishes the requirements in this Part for the Authorization sought: (1) Student pilot Authorization Note: if the State prefers, a license or certificate can be issued.
- 2.3.3.1 STUDENT PILOTS (c) Pre-solo flight instruction. Prior to conducting a solo flight, a student pilot shall have: (1) received and logged flight training for the maneuvers and procedures as listed in IS 2.3.3.1. (2) demonstrated satisfactory proficiency and safety, as judged by an authorized instructor, on the maneuvers and procedures as listed in IS. 2.3.3.1.
- 2.2.2 PCAR Part 2, 2.3.3.1.(d) and (e). STUDENT PILOTS (d) Pre-solo flight instruction. Prior to conducting a solo flight, a student pilot shall have: (1) received and logged flight training for the maneuvers and procedures as listed in IS 2.3.3.1. (2) demonstrated satisfactory proficiency and safety, as judged by an authorized instructor, on the maneuvers and procedures as listed in IS 2.3.3.1. (e) Solo flight requirement (1) Holding at least a Class 2 Medical Certificate (2) An Authorized student pilot had received and logged ground training from an authorized instructor on the following subject: (i) Applicable sections of Part 2 and Part 8; (ii) Airspace rules and procedures for the airport where the student will perform solo flight; and (iii) Flight characteristics and operation for the make and model of aircraft to be flown. (3) Must be authorized by a Flight Instructor.
- 2.2.2.1 IS 1.2.1.8 (d). SANCTION TABLES TABLE - 2 SANCTIONS III. Individuals and General Aviation – Owners, Pilots, Aviation Maintenance Organizations, Maintenance personnel 4. Student Operation c. Operation on international flight 60 to 90 day suspension.
- 2.3.1.2 PCAR Part 2, 2.3.3.2 (b) (1) and (b) (3) 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (b) Knowledge. The applicant for a PPL(A) shall: (1) Receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to the holder of a PPL(A); rules of the air; appropriate air traffic services practices and procedures (ii) Aircraft general knowledge: (A) Principles of operation of airplane power-plants, systems and instruments; (B) Operating limitations of airplanes

and power-plants: relevant operational information from the flight manual or other appropriate document; (iii) Flight performance and planning: (A) effects of loading and weight distribution on flight characteristics; weight and balance calculations; (B) use and practical application of take-off, landing and other performance data; (C) pre-flight and en-route flight planning appropriate to private operations under VFR; preparation and filing of air traffic services flight plans; appropriate air traffic services procedures; position reporting procedures; altimeter setting procedures; operations in areas of high-density traffic; (iv) Human performance: human performance relevant to the PPL(A) (v) Meteorology: application of elementary aeronautical meteorology; use of, and procedures for obtaining, meteorological information; altimetry (vi) Navigation: practical aspects of air navigation and dead-reckoning techniques; use of aeronautical charts: (vii) Operational procedures: (A) use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations; (B) appropriate precautionary and emergency procedures, including action to be taken to avoid hazardous weather, wake turbulence and other operating hazards; (viii) Principles of flight: principles of flight relating to airplanes; (ix) Radiotelephony: (A) radiotelephony procedures and phraseology as applied to VFR operations; action to be taken in case of communication failure; (B) as listed in IS 2.3.3.2 Appendix A; (3) pass the required knowledge test on the knowledge areas listed in IS 2.3.3.2 Appendix A.

- 2.3.1.3 PCAR Part 2, 2.3.3.2 (e). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (e) Skill. The applicant for a PPL(A) shall: (2) have demonstrated by passing a skill test the ability to perform as pilot in-command of an airplane, the areas of operation described in IS 2.3.3.2 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of a PPL(A), and to (i) operate the airplane within its limitations; (ii) complete all maneuvers with smoothness and accuracy; (iii) exercise good judgment and airmanship; (iv) apply aeronautical knowledge; and (v) maintain control of the airplane at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.
- 2.3.2.2 PCAR Part 2, 2.3.3.2 (d)(3). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (d) Flight Instruction (3) If the privileges of the PPL(A) are to be exercised at night, the applicant shall have received 4 hours dual instruction in airplanes in night flying, including take-offs, landings and 1 hour of navigation and that information shall be endorsed on the license. Note 1: Training can be performed by an individually authorized flight instructor, by an authorized flight instructor in a flying club, or in an Aviation Training Organization.
- 2.3.3.1.1 PCAR Part 2, 2.3.3.2(c)(1). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (c) Experience (1) The applicant for a PPL(A) shall have completed not less than 40 hours of flight time as pilot of airplanes, a total of 5 hours may have been completed in a flight simulator or flight procedures trainer.
- 2.3.3.1.1.1 PCAR Part 2, 2.3.3.2(c)(3). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (c) Experience (3) The holder of pilot licenses in other categories may be credited with 10 hours of the total flight time as pilot-in-command towards a PPL(A).
- 2.3.3.1.2 PCAR Part 2, 2.3.3.2(c)(2). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (c) Experience (2) The applicant shall have completed in airplanes not less than 10 hours of solo flight time under the supervision of an authorized flight instructor, including 5 hours of solo cross-country flight time with at least one cross-country flight totaling not less than 270 KM (150 MM) in the course of which full-stop landings at two difference aerodromes shall be made.
- 2.3.3.2 PCAR Part 2, 2.3.3.2(d)(1)(2). 2.3.3.2 PRIVATE PILOT LICENSE - AIRPLANE (d) Flight Instruction. (1) The applicant for a PPL(A) shall receive and log not less than 20 hours of dual instruction from an authorized instructor on the subjects listed in IS

2.3.3.2 Appendix B. These 20 hours may include 5 hours completed in a flight simulator or flight procedures trainer. The 20 hours of dual instruction shall include at least 5 hours of solo cross-country flight time with at least one cross-country flight totaling not less than 270 KM (150 NM) in the course of which full-stop landings at two different aerodromes shall be made. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the private pilot: (i) pre-flight operations, including mass and balance determination, airplane inspection and servicing; (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures; (iii) control of the airplane by external visual reference; (iv) flight at critically slow airspeeds; recognition of, and recovery from, incipient and full stalls; (v) flight at critically high airspeeds; recognition of, and recovery from, spiral dives, (vi) normal and cross-wind take-offs and landings; (vii) maximum performance (short field and obstacle clearance take-offs, short field landings; (viii) flight by reference solely to instruments, including the completion of a level 180 degrees turn: (ix) cross-country flying using visual reference, dead reckoning and, where available, radio navigation aids; (x) emergency operations, including simulated airplane equipment malfunctions; and (xi) operations to, from and transmitting controlled aerodromes, compliance with air traffic services procedures, radiotelephony procedures and phraseology as further specified in IS 2.3.3.2 Appendix B.

2.3.4.1.1 PCAR Part 2, 2.3.3.7(c)(1). 2.3.3.7 PRIVATE PILOT LICENSE - HELICOPTER (c) Experience. (1) The applicant for a PPL(H) shall have completed not less than 40 hours of flight time as pilot of airplanes, a total of 5 hours may have been completed in a flight simulator or flight procedures trainer.

2.3.4.1.1.1 PCAR Part 2, 2.3.3.7(c)(3). 2.3.3.7 PRIVATE PILOT LICENSE - HELICOPTER (c) Experience (3) The holder of pilot licenses in other categories may be credited with 10 hours of the total flight time as pilot-in-command towards a PPL(H).

2.3.4.2.1 PCAR Part 2, 2.3.3.7(d)(1)(2) 2.3.3.7 PRIVATE PILOT LICENSE - HELICOPTER (d) Flight Instruction. (1) The applicant for a PPL(H) shall receive and log not less than 20 hours of dual instruction from an authorized instructor on the subjects listed in IS 2.3.3.6 Appendix B. These 20 hours may include 5 hours completed in a flight simulator or flight procedures trainer. The 20 hours of dual instruction shall include at least 5 hours of solo cross-country flight time with at least one cross-country flight totaling not less than 180 KM (100 NM) in the course of which landings at two different points shall be made. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the private pilot: (i) pre-flight operations. including mass and balance determination, helicopter inspection and servicing; (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures; (iii) control of the helicopter by external visual reference; (iv) recovery at the incipient stage from settling with power; recovery techniques from low rotor rpm within the normal range of engine rpm; ground maneuvering and run-ups; hovering; take-offs and landings - normal, out of wind and sloping ground; (v) take-offs and landings with minimum necessary power; maximum performance take-off and landing techniques; restricted site operations; quick stops; (vi) cross-country flying using visual reference, dead reckoning and, where available, radio navigation aids including a flight of at least one hour; (vii) emergency operations, including simulated helicopter equipment malfunctions; autorotative approach and landing; and (viii) operations to, from and transmitting controlled aerodromes, compliance with air traffic services procedures, radiotelephony procedures and phraseology (ix) as further specified in IS 2.3.3.6 Appendix B.

2.3.6.1 Not implemented.

2.3.6.2 Not implemented.

2.4.1.2 PCAR Part 2, 2.3.3.3(ii)(1)(2)(3) and 2.3.3.8 (b)(1)(2)(3). EFOD Page 175 PCAR 2 (2-35) 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (ii) Knowledge. The applicant for a CPL (A) shall: (1) receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to the holder of a CPL (A); rules of the air; appropriate air traffic services practices and procedures (ii) Aircraft general knowledge: (A) principles of operation and functioning of airplane power-plants, systems and instruments: (B) operating limitations of appropriate airplanes and power-plants; relevant operational information from the flight manual or other appropriate document; (C) use and serviceability checks of equipment and systems of appropriate airplanes; (D) maintenance procedures for airframes, systems and power-plants of appropriate airplanes; (iii) Flight performance and planning: (A) effects of loading and mass distribution on airplane handling, flight characteristics and performance; mass and balance calculations; (B) use and practical application of take-off, landing and other performance data; (C) pre-flight and en-route flight planning appropriate to operations under VFR; preparation and filing of air traffic services flight plans; appropriate air traffic services procedures; (iv) Human performance: human performance relevant to the CPL (A); (v) Meteorology: (A) interpretation and application of aeronautical meteorological reports, charts and forecasts; use of, and procedures for obtaining, meteorological information, pre-flight and in-flight; altimetry; (B) aeronautical meteorology; climatology of relevant areas in respect of the elements having an effect upon aviation; the moment of pressure systems, the structure of fronts; and the origin and characteristics of significant weather phenomena which affect take-off, en-route and landing conditions; hazardous weather avoidance (vi) Navigation: air navigation, including the use of aeronautical charts, instruments and navigation aids; understanding of the principles and characteristics of appropriate navigation systems; operation of airborne equipment (vii) Operation procedures: (A) use of aeronautical documentation such as AIR; NOTAM, aeronautical codes and abbreviations: (B) appropriate precautionary and emergency procedures; (C) operational procedures for carriage of freight; potential hazards associated with dangerous goods; (D) requirements and practices for safety briefing to passengers, including precautions to be observed when embarking and disembarking from airplanes; (viii) Principles of flight: principles of flight relating to airplanes; (ix) Radiotelephony: (A) radiotelephony procedures and phraseology as applied to VFR operations; action to be taken in case of communication failure: (B) as further specified in IS 2.3.3.3 Appendix A.(2) Have received an endorsement for the knowledge test from an authorized instructor who: (i) conducted the training on the knowledge subjects: (ii) certifies that the person is prepared for the required knowledge test: and(3) Pass the required knowledge test on the knowledge subjects listed in IS 2.3.3.3 Appendix B.

2.3.3.8 COMMERCIAL PILOT LICENSE - HELICOPTER (b) Knowledge. The applicant for an CPL(H) shall: (1) receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to the holder of a CPL (H); rules of the air; appropriate air traffic services practices and procedures (ii) Aircraft general knowledge (A) principles of operation and functioning of helicopter power-plants, transmission (power-trains) systems and instruments; (B) operating limitations of appropriate helicopters and power-plants; relevant operational information from the flight manual; (C) use and serviceability checks of equipment and systems of appropriate helicopters; (D) maintenance procedures for airframes, systems and power-plants of appropriate helicopters; (iii) Flight performance and planning: (A) effects of loading and mass distribution, including external loads, on helicopter handling, flight characteristics and performance; mass and balance calculations; (B) use and practical application of take-off, landing and other

performance data; (C) pre-flight and en-route flight planning appropriate to operations under VFR; preparation and filing of air traffic services flight plans; appropriate air traffic services procedures; (iv) Human performance: human performance relevant to the CPL (H); (v) Meteorology: (A) interpretation and application of aeronautical meteorological reports, charts and forecasts; use of, and procedures for obtaining, meteorological information, pre-flight and in-flight; altimetry; (B) aeronautical meteorology; climatology of relevant areas in respect of the elements having an effect upon aviation; the moment of pressure systems, the structure of fronts, and the origin and characteristics of significant weather phenomena which affect take-off, en-route and landing conditions, hazardous weather avoidance; (vi) Navigation: air navigation, including the use of aeronautical charts, instruments and navigation aids; understanding of the principles and characteristics of appropriate navigation systems; operation of air borne equipment. (vii) Operation procedures (A) use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations; (B) appropriate precautionary and emergency procedures; settling with power, ground resonance, roll-over and other operating hazards; (C) operational procedures for carriage of freight, including external loads; potential hazards associated with dangerous goods; (D) requirements and practices for safety briefing to passengers, including precautions to be observed when embarking and disembarking from helicopters; (viii) Principles of flight: principles of flight relating to helicopters; (ix) Radiotelephony: (A) radiotelephony procedures and phraseology as applied to VFR operations; action to be taken in case of communication failure; and (B) as further specified in IS 2.3.3.8 Appendix A. (2) have received an endorsement for the knowledge test from an authorized instructor who: (i) conducted the training on the knowledge subjects; (ii) certifies that the person is prepared for the required knowledge test; and (3) pass the required knowledge test on the knowledge subjects listed in IS 2.3.3.8 Appendix A.

- 2.4.1.3 PCAR Part 2, 2.3.3.3 (v) and 2.3.3.8 (e). 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (v) Skill. The applicant for a CPL(A) shall (1) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; and (2) shall have demonstrated by passing a skill test the ability to perform as pilot-in command of an airplane, the areas of operation described in IS 2.3.3.3 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of a CPL(A), and to (i) operate the airplane within its limitations; (ii) complete all maneuvers with smoothness and accuracy; (iii) exercise good judgment and airmanship; (iv) apply aeronautical knowledge; and (v) maintain control of the airplane at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.

2.3.3.8 COMMERCIAL PILOT LICENSE-HELICOPTER (e) Skill. The applicant for a CPL(H) shall: (1) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; and (2) have demonstrated by passing a skill test the ability to perform as pilot-in command of an helicopter, the areas of operation described in IS 2.3.3.7 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of a CPL(H), and to (i) operate the helicopter within its limitations; (ii) complete all maneuvers with smoothness and accuracy; (iii) exercise good judgment and airmanship; (iv) apply aeronautical knowledge; and (v) maintain control of the helicopter at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.

- 2.4.2.1 PCAR Part 2, 2.3.3.3 (vii) and 2.3.3.8 (g). 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (vii) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a CPL(A) shall be: (1) to exercise all the

privileges of the holder of a PPL(A); (2) to act as pilot-in-command in any airplane engaged in operations other than commercial air transportation; (3) to act as pilot-in-command in commercial air transportation in any airplane certificated for single-pilot operation; and (4) to act as co-pilot in commercial air transportation in airplanes required to be operated with a copilot.

2.3.3.8 COMMERCIAL PILOT LICENSE-HELICOPTER (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a CPL(H) shall be: (1) to exercise all the privileges of the holder of a PPL(H); (2) to act as pilot-in-command in any helicopter engaged in operations other than commercial air transportation; (3) to act as pilot-in-command in commercial air transportation in any helicopter certificated for single-pilot operation; and (4) to act as co-pilot in commercial air transportation in helicopters required to be operated with a co-pilot.

2.4.2.2 PCAR Part 2, 2.3.3.3 (iv) (3) and 2.3.3.8 (d)(3) 2.3.3.3. COMMERCIAL PILOT LICENSE - AIRPLANE (iv) Flight Instruction. (3) If the privileges of the CPL(A) are to be exercised at night, the applicant shall have received 4 hours dual instruction in airplanes in night flying, including take-offs, landings and 1 hour of navigation.

2.3.3.8 COMMERCIAL PILOT LICENSE-HELICOPTER (d) Flight Instruction (3) If the privileges of the license are to be exercised at night, the applicant shall have received dual instruction in helicopters in night flying, including take-offs, landings and navigation.

2.4.3.1.1 PCAR Part 2, 2.3.3.3 (iii) (1). 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (iii) Experience. (1) The applicant for a CPL(A) shall have completed not less than 200 hours of flight time, or 150 hours if completed during an Authority-approved training course provided for in an Approved Training Organization under Part 3, as a pilot of airplanes, of which 10 hours may have been completed in a flight simulator or flight procedures trainer.

2.4.3.1.2 PCAR Part 2, 2.3.3.3 (iii) (3). 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (iii) Experience. (3) The holder of a pilot license in another category may be credited towards the 200 hours of flight time as follows: (i) 10 hours as PIC in a category other than helicopters; or (ii) 30 hours as pilot-in-command holding a PPL(H) on helicopters; or (iii) 100 hours as pilot-in-command holding a CPL(H) on helicopters.

2.4.3.2.1 PCAR Part 2, 2.3.3.3 (iv) (1)(2). 2.3.3.3 COMMERCIAL PILOT LICENSE - AIRPLANE (iv) Flight Instruction. (1) The applicant for a CPL(A) shall receive and log not less than 25 hours of dual instruction from an authorized instructor. These 25 hours may include 5 hours completed in a flight simulator or flight procedures trainer. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the commercial pilot: (i) pre-flight operations, including mass and balance determination, airplane inspection and servicing; aerodrome and traffic pattern operations, collision avoidance precautions and procedures; (ii) control of the airplane by external visual reference; (iii) flight at critically slow airspeeds; recognition of, and recovery from, incipient and full stalls; (iv) flight at critically high airspeeds; recognition of, and recovery from, spiral dives; (v) normal and cross-wind take-offs and landings; (vi) maximum performance (short field and obstacle clearance take-offs, short field landings; (vii) basic flight maneuvers and recovery from unusual attitudes by reference solely to basic flight instruments; (viii) cross-country flying using visual reference, dead reckoning and radio navigation aids; diversion procedures (ix) abnormal and emergency procedures and maneuvers; and (x) operations to, from and transmitting controlled aerodromes, compliance with air traffic services procedures, radiotelephony procedures and phraseology as further specified in IS 2.3.3.3 Appendix B.

- 2.4.4.1.1 PCAR Part 2, 2.3.3.8 (c) (1). 2.3.3.8 COMMERCIAL PILOT LICENSE – HELICOPTER (c) Experience. (1) The applicant for a CPL(H) license shall have completed not less than 150 hours of flight time, or 100 hours if completed during an integrated course of approved training provided for in an Aviation Training Organization Part 3, as a pilot of helicopters, of which 10 hours may have been completed in a flight simulator or flight procedures trainer.
- 2.4.4.1.2 PCAR Part 2, 2.3.3.8 (c)(3). 2.3.3.8 COMMERCIAL PILOT LICENSE - HELICOPTER (c) Experience. (3) The holder of a pilot license in the helicopter category may be credited towards the 150 hours of flight time as follows: (i) 20 hours as pilot-in-command holding a PPL(A) in airplanes; or (ii) 50 hours as pilot-in-command holding a CPL(A) in airplanes.
- 2.4.4.2 PCAR Part 2, 2.3.3.8 (d)(1)(2). 2.3.3.8 COMMERCIAL PILOT LICENSE - HELICOPTER (d) Flight Instruction (1) The applicant for a CPL(H) shall have received and log not less than 30 hours of dual instruction in helicopters from an authorized flight instructor under Part 3 on the subjects listed in IS 2.3.3.8 Appendix B. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the commercial pilot: (i) Pre-flight operations, including mass and balance determination, helicopter inspection and servicing; (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures; (iii) control of the helicopter by external visual reference; (iv) recovery at the incipient stage from settling with power; recovery techniques from low rotor rpm within the normal range of engine rpm; (v) ground maneuvering and run-ups; hovering; take-offs and landings - normal, out of wind and sloping ground; steep approaches; (vi) take-offs and landings with minimum necessary power; maximum performance take-off and landing techniques; restricted site operations; quick stops; (vii) hovering out of ground effect; operations with external load, if applicable; flight at high altitude; (viii) basic flight maneuvers and recovery from unusual attitudes by reference solely to basic flight instruments; (ix) cross-country flying using visual reference, dead reckoning and radio navigation aids; diversion procedures (x) abnormal and emergency procedures including simulated helicopter equipment malfunctions; autorotative approach and landing; and (xi) operations to, from and transmitting controlled aerodromes, compliance with air traffic services procedures, radiotelephony procedures and phraseology (xii) as further specified in IS 2.3.3.8 Appendix B.
- 2.4.6.1.1 Not implemented.
- 2.4.6.1.1.1 Not implemented.
- 2.4.6.2 Not implemented.
- 2.5.1.2.1.1 PCAR Part 2, 2.3.3.5.2 (c) 2.3.36(b). 2.3.3.5.2 REQUIREMENTS (c) Knowledge. The applicant shall have demonstrated a level of knowledge appropriate to the privileges granted to the holder of an ATPL (A) and appropriate to the category of aircraft intended to be included in the license, in at least the following subjects (see IS: 2.3.3.5 Appendix A). The applicant for an MPL shall: (1) Receive and log ground training from an authorized instructor on the following subjects: (i) Air Law: Rules and regulations relevant to the holder of an airline transport pilot license; rules of the air; appropriate air traffic services practices and procedures. (ii) Aircraft General Knowledge: (A) General characteristics and limitations of electrical, hydraulic, pressurization and other aircraft systems; flight control systems, including autopilot and stability augmentation. (B) Principles of operation, handling procedures and operating limitations of aircraft power-plants; effects of atmospheric conditions on engine performance; relevant operational information from the flight manual or other appropriate document. (C) Operating procedures and limitations of the relevant category of aircraft; effects of atmospheric conditions on aircraft performance in accordance to the relevant

operational information from the flight manual. (D) Use and serviceability checks of equipment and systems of appropriate aircraft. (E) Flight instruments, compasses, turning and acceleration errors; gyroscopic instruments, operational limits and precession effects; practices and procedures in the event of malfunctions of various flight instruments and electronic display units. (F) Maintenance procedures for airframes, systems and power-plants of appropriate aircraft. (G) For helicopter and powered-lift, transmission (power-trains) where applicable; (iii) Flight Performance and Planning: (A) Effects of loading and mass distribution on aircraft handling. (iv) Human Performance: (A) Human performance including principles of threat and error management. (v) Meteorology: (A) Interpretation and application of aeronautical meteorological reports, charts and forecasts; codes and abbreviations; use of, and procedures for obtaining, meteorological information, pre-flight and in-flight altimetry (B) Aeronautical meteorology; climatology of relevant areas in respect of the elements having an effect upon aviation; the movement of pressure systems; the structure of fronts, and the origin and characteristics of significant weather phenomena which affect take-off, en-route and landing conditions. (C) Causes, recognition and effects of icing; frontal zone penetration procedures; hazardous weather avoidance. (D) In the case of airplane and powered-lift, practical high altitude meteorology, including interpretation and use of weather reports, charts and forecasts, and jet-streams. (vi) Navigation: (A) Air navigation, including the use of aeronautical charts, radio navigation aids and area navigation systems; specific navigation requirements for long-range flights. (B) Use, limitation and serviceability of avionics and instruments necessary for the control and navigation of aircraft. (C) Use, accuracy and reliability of navigation systems used in departure, enroute, approach and landing phases of flight; identification of radio navigation aids. (D) Principles and characteristics of self-contained and external-referenced navigation systems; operation of airborne equipment. (vii) Operational Procedures: (A) Application of threat and error management to operational performance; (B) Interpretation and use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations; (C) Precautionary and emergency procedures; safety practices; (D) Operational procedures for carriage of freight and dangerous goods; (E) Requirements and practices for safety briefing to passengers, including precautions to be observed when embarking and disembarking from aircraft; (F) In the case of the helicopter, and if applicable, powered-lift, settling with power; ground resonance; retreating blade stall; dynamic roll-over and other operation hazards; safety procedures, associated with flight under VFR; (viii) Principles of flight (ix) Radiotelephony (A) Procedures and phraseology; action to be taken in case of communication failure.

- 2.5.2.2 PCAR Part 2, 2.3.3.5.2(g)(2). 2.3.3.5.2 REQUIREMENTS (g) (2) Before exercising the privileges of the instrument rating in a single pilot operation, the license holder shall have demonstrated an ability to act as pilot-in-command in a single pilot operation exercised solely by reference to instruments.
- 2.5.2.3 PCAR Part 2, 2.3.3.5.2(g)(3)(i)(ii). 2.3.3.5.2 REQUIREMENTS (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an MPL shall be: (3) Before exercising the privileges of a commercial pilot license in a single pilot operation, the license holder shall have: (i) completed 70 hours, either as pilot-in-command, or made up by not less than 10 hours as pilot-in-command and the necessary additional flight time as pilot in-command under supervision; (ii) meet the requirements for the commercial pilot license.
- 2.5.3.1 PCAR Part 2, 2.3.3.5.2(f)(1)(i). 2.3.3.5.2 REQUIREMENTS (f) Ratings. Comply with the sections of this regulation that apply to the rating he seeks. (1) Airplane rating: Aeronautical experience. (i) An applicant for a Multi-crew pilot license with an airplane

rating shall have completed an approved training course and shall have not less than 240 hours as pilot flying and pilot not flying of actual and simulated flight.

- 2.5.3.2 PCAR Part 2,2.3.3.5.2(f)(1)(ii). 2.3.3.5.2 REQUIREMENTS (f) Ratings. Comply with the sections of this regulation that apply to the rating he seeks. (1) Airplane rating: Aeronautical experience. (ii) Flight experience in actual flight shall include not less than 40 hours of flight time, or 35 hours if completed during a course of approved training, as a pilot of airplanes appropriate to the class rating sought and shall include upset recovery training, night flying, cross country and flight by reference solely to instruments.
- 2.5.4.1 PCAR Part 2, 2.3.3.5.1(b)(1). 2.3.3.5.1 GENERAL (b) Eligibility requirement: (1) To be eligible for an MPL in the airplane category, the applicant shall have completed an approved training course. The training shall be competency-based and conducted in a multi-crew operational environment.
- 2.5.4.2 PCAR Part 2, 2.3.3.5.1 (b) (2). 2.3.3.5.1 GENERAL (b) Eligibility requirement: (2) During the training, the applicant shall have acquired the knowledge, skills and attitudes required as the underpinning attributes for performing as a co-pilot of a turbine-powered air transport airplane certificated for operation with a minimum crew of at least two pilots.
- 2.6.1.2.1 PCAR Part 2,2.3.3.4 (b) for airplane and 2.3.3.9 (b) for helicopter. 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (b) Knowledge. The applicant for an ATPL(A) shall: (1) receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to the holder of an ATPL(A); rules of the air, appropriate air traffic services practices and procedures (ii) Aircraft general knowledge: (A) general characteristics and limitations of electrical, hydraulic, pressurization and other airplane systems; flight control systems, including auto pilot and stability augmentation; (B) principles of operation, handling procedures and operating limitations of airplane power-plants; effects of atmospheric conditions on engine performance; relevant operational information from the flight manual or other appropriate document, (C) operating procedures and limitations of appropriate airplanes; effects of atmospheric conditions on airplane performance, (D) use and serviceability checks of equipment and systems of appropriate airplanes; (E) flight instruments; compasses, turning and acceleration errors; gyroscopic instruments, operational limits and precession effects; practices and procedures in the event of malfunctions of various flight instruments; (F) maintenance procedures for airframes, systems and power-plants of appropriate airplanes; (iii) Flight performance and planning: (A) effects of loading and mass distribution on airplane handling, flight characteristics and performance; mass and balance calculations; (B) use and practical application of take-off, landing and other performance data, including procedures for cruise control; (C) pre-flight and en-route operational flight planning; preparation and filing of air traffic services flight plans; appropriate air traffic services procedures; altimeter setting procedures (iv) Human performance: human performance relevant to the ATPL(A) (v) Meteorology: (A) interpretation and application of aeronautical meteorological reports, charts and forecasts; codes and abbreviations; use of, and procedures for obtaining, meteorological information; pre-flight and in-flight; altimetry; (B) aeronautical meteorology; climatology of relevant areas in respect of the elements having an effect upon aviation; the moment of pressure systems; the structure of fronts, and the origin and characteristics of significant weather phenomena which affect take-off, en-route and landing conditions; (C) causes, recognition and effects of engine and airframe icing; frontal zone penetration procedures; hazardous weather avoidance; (vi) Navigation: (A) air navigation, including the use of aeronautical charts, radio navigation aids and area navigation systems; specific navigation requirements for long-range flights; use, limitation and serviceability of avionics and instruments

necessary for the control and navigation of airplanes; (B) use, accuracy and reliability of navigation systems used in departure, enroute, approach and landing phases of flight; identification of radio navigation aids; (C) principles and characteristics of self-contained and external-referenced navigation systems; operation of airborne equipment; (vii) Operation procedures: (A) interpretation and use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations, and instrument procedure charts for departure, en-route, descent and approach; (B) precautionary and emergency procedures; safety practices associated with flight under IFR (C) operational procedures for carriage of freight and dangerous goods; (D) requirements and practices for safety briefing to passengers, including precautions to be observed when embarking and disembarking from airplanes; (viii) Principles of flight: principles of flight relating to airplanes; subsonic aerodynamics; compressibility effects, maneuver boundary limits, wing design characteristics, effects of supplementary lift and drag devices; relationships between lift, drag and thrust at various airspeeds and in different flight configuration; (ix) Radiotelephony: radiotelephony procedures and phraseology; action to be taken in case of communication failure; as further specified in IS 2.3.3.4 Appendix A.

2.6.1.2.2 PCAR Part 2,2.3.3.4 (C)(4). 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (c) Experience. (4) The applicant for an ATPL(A) shall be the holder of a CPL(A) with instrument and multi-engine rating issued under this Part.

2.6.1.3.1 PCAR Part 2,2.3.3.4(e)(2)(i) to (v). 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (e) Skill. The applicant for an ATPL(A) shall: (2) have demonstrated by passing a skill test the ability to perform, as pilot-in-command of a multi-engine airplane required to be operated with a co-pilot, the following procedures and maneuvers: (i) pre-flight procedures, including the preparation of the operational flight plan and filing of the air traffic services flight plan; (ii) normal flight procedures and maneuvers during all phases of flight; (iii) procedures and maneuvers for IFR operations under normal, abnormal and emergency conditions, including simulated engine failure, and covering at least the following: (A) transition to instrument flight on take-off; (B) standard instrument departures and arrivals; (C) en-route IFR procedures and navigation; (D) holding procedures; (E) instrument approaches to specified minima; (F) missed approach procedures; (G) landings from instrument approaches; (iv) abnormal and emergency procedures and maneuvers related to failures and malfunctions of equipment, such as power-plant, systems and airframe; and (v) procedures for crew in capacitation and crew coordination, including allocation of pilot tasks, crew cooperation and use of checklists.

2.6.1.3.1.1 PCAR Part 2, 2.3.3.4(e)(3). 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (e) Skill. The applicant for an ATPL(A) shall: (3) have demonstrated by passing a skill test the ability to perform the areas of operation described in IS 2.3.3.4 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of an ATPL(A).

2.6.1.3.1.2 PCAR Part 2,2.3.3.4(e)(2)(iv)(3)(ii)(iii)(vii). 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (e) Skill. The applicant for an ATPL(A) shall: (2) have demonstrated by passing a skill test the ability to perform, as pilot-in command of a multi-engine airplane required to be operated with a co-pilot, the following procedures and maneuvers: (iv) abnormal and emergency procedures and maneuvers related to failures and malfunctions of equipment, such as power-plant, systems and airframe (3) have demonstrated by passing a skill test the ability to perform the areas of operation described in IS 2.3.3.4 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of an ATPL(A). (ii) complete all maneuvers with smoothness and accuracy; (iii) exercise good judgment and airmanship; (vii) communicate effectively with the other flight crew members.

- 2.6.2.1 PCAR Part 2, 2.3.3.4 (g) for airplane and 2.3.3.9 for helicopter. 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an ATPL(A) shall be: (1) to exercise all the privileges of the holder of a PPL(A) and CPL(A) and of an IR(A); and (2) to act as pilot-in-command and co-pilot in airplanes in air transportation service in aircraft or approved simulator training device of the category, class and type for which he is rated. To do so, he must be familiar with the operating procedures of the company for which he is providing pilot instruction. However, he may not instruct for more than 8 hours in one day, and not more than 36 hours in any 7-day period. He may instruct under this Section only in aircraft with functioning dual controls. Unless he has a valid flight instructor license, an airline transport pilot may instruct only as provided in this section.
- 2.3.3.9 AIRLINE TRANSPORT PILOT LICENSE – HELICOPTER (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an ATPL(H) shall be: (1) to exercise all the privileges of the holder of a PPL(H) and CPL(H); and (2) to act as pilot-in-command and co-pilot in helicopters in air transportation.
- 2.6.2.2 PCAR Part 2, 2.3.3.5.2(g)(5). 2.3.3.5.2 REQUIREMENTS (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an MPL shall be: (5) When the holder of an airline transport pilot license in the airplane category has only previously held a MPL, the privileges of the license shall be limited to multicrew operations unless the holder has met the requirements established. Any limitations of privileges shall be endorsed on the license.
- 2.6.3.1.1 PCAR Part 2, 2.3.3.4(c) (1) Part 2, 2.2.6.2. 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE – AIRPLANE (c) Experience. (1) The applicant for an ATPL(A) shall have completed not less than 1500 hours of flight time as a pilot of airplanes of which a maximum of 100 hours may have been completed in a flight simulator.
- 2.2.6.2 USE OF SYNTHETIC FLIGHT TRAINERS - The use of a synthetic flight trainer for performing training, testing and checking for which a flight crew member is to receive credit, shall be approved by the Authority, which shall ensure that the synthetic flight trainer is appropriate to the task. Note: See the Manual of Criteria for the Qualification of Flight Simulators (ICAO Doc 9625).
- 2.6.3.1.1.1 PCAR Part 2, 2.3.3.4 (c)(1) (i) to (iv). 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (c) Experience. (1) The applicant for an ATPL(A) shall have completed not less than 1500 hours of flight time as a pilot of airplanes of which a maximum of 100 hours may have been completed in a flight simulator. The applicant shall have completed in airplanes not less than: (i) 250 hours, either as pilot-in-command, or made up by not less than 100 hours as pilot-in-command and the necessary additional flight time as co-pilot performing, under the supervision of the pilot-in-command, the duties and functions of a pilot-in-command; provided that the method of supervision employed is acceptable to the Authority; (ii) 200 hours of cross-country flight time, of which not less than 100 hours shall be as pilot-in-command or as co-pilot performing, under the supervision of the pilot-in-command, the duties and functions of a pilot-in-command, provided that the method of supervision employed is acceptable to the Authority; (iii) 75 hours of instrument time, of which not more than 30 hours may be instrument ground time; and (iv) 100 hours of night flight as pilot-in-command or as co-pilot.
- 2.6.3.1.2 Not implemented.

- 2.6.3.2 PCAR Part 2, 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (d) Flight Instruction. The applicant for an ATPL(A) shall have received the dual flight instruction required for the issue of the CPL(A) and the IR.
- 2.6.4.1.1 PCAR Part 2, 2.3.3.9(c)(1) and Part 2, 2.2.6.2. 2.3.3.9 AIRLINE TRANSPORT PILOT LICENSE - HELICOPTER (c) Experience. (1) The applicant for an ATPL(H) shall have completed not less than 1000 hours of flight time as a pilot of helicopters of which a maximum of 100 hours may have been completed in a flight simulator.
- 2.2.6.2 USE OF SYNTHETIC FLIGHT TRAINERS. The use of a synthetic flight trainer for performing training, testing and checking for which a flight crew member is to receive credit, shall be approved by the Authority, which shall ensure that the synthetic flight trainer is appropriate to the task. Note: See the Manual of Criteria for the Qualification of Flight Simulators (ICAO Doc 9625).
- 2.6.4.1.1.1 PCAR Part 2, 2.3.3.9 (c)(1)(i) to (iv). 2.3.3.9 AIRLINE TRANSPORT PILOT LICENSE - HELICOPTER (c) Experience. (1) The applicant for an ATPL(H) shall have completed not less than 1000 hours of flight time as a pilot of helicopters of which a maximum of 100 hours may have been completed in a flight simulator. The applicant shall have completed in helicopters not less than: (i) 250 hours, either as pilot-in-command, or made up by not less than 100 hours as pilot-in-command and the necessary additional flight time as co-pilot performing, under the supervision of the pilot-in-command, the duties and functions of a pilot-in-command; provided that the method of supervision employed is acceptable to the Authority; (ii) 200 hours of cross-country flight time, of which not less than 100 hours shall be as pilot-in-command or as co-pilot performing, under the supervision of the pilot-in-command, the duties and functions of a pilot-in-command, provided that the method of supervision employed is acceptable to the Authority; (iii) 30 hours of instrument time, of which not more than 10 hours may be instrument ground time; and (iv) 50 hours of night flight as pilot-in-command or as co-pilot.
- 2.6.4.1.2 PCAR Part 2, 2.3.3.9 (c)(2). 2.3.3.9 AIRLINE TRANSPORT PILOT LICENSE - HELICOPTER (c) Experience. (2) Holders of a CPL(A) will be credited with 50% of their airplane flight time as pilot-in-command towards the flight time required in (1).
- 2.6.4.2 PCAR Part 2, 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE - AIRPLANE (d) Flight Instruction. The applicant for an ATPL(A) shall have received the dual flight instruction required for the issue of the CPL(A) and the IR.
- 2.7.1.1 PCAR Part 2, 2.3.3.6(b) for Airplane and 2.3.3.10 (b) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (b) Knowledge. The applicant for an IR (A) shall: (1) receive and log ground training from an authorized instructor on the following subjects (i) Air law: rules and regulations relevant to flight under IFR: related air traffic services practices and procedures; (ii) Aircraft general knowledge: (A) use, limitation and serviceability of avionics and instruments necessary for the control and navigation of airplanes under IFR and in instrument meteorological conditions; use and limitations of autopilot; (B) compasses, turning and acceleration errors; gyroscopic instruments, operational limits and precession effects; practices and procedures in the event of malfunctions of various flight instruments; (iii) Flight performance and planning (A) pre-flight preparations and checks appropriate to flight under IFR; (B) operational flight planning; preparation and filing of air traffic services flight plans under IFR; altimeter setting procedures; (iv) Human performance: human performance relevant to instrument flight in airplanes; (v) Meteorology: (A) application of aeronautical meteorology; interpretation and use of reports, charts and forecasts; codes and abbreviations; use of, and procedures for obtaining, meteorological information; altimetry; (B) causes, recognition and effects of engine and airframe icing; frontal zone penetration procedures; hazardous weather avoidance; (vi) Navigation: (A) practical

air navigation using radio navigation aids; (B) use, accuracy and reliability of navigation systems used in departure, enroute, approach and landing phases of flight; identification of radio navigation aids; (vii) Operation procedures (A) interpretation and use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations, and instrument procedure charts for departure, en-route; descent and approach; (B) precautionary and emergency procedures; safety practices associated with flight under IFR (viii) Radiotelephony: (A) radiotelephony procedures and phraseology as applied to aircraft operations under IFR, action to be taken in case of communication failure; (B) as listed in IS 2.3.3.6 Appendix A.

2.3.3.10 INSTRUMENT RATING – HELICOPTER (b) Knowledge. The applicant for an IR (H) shall: (1) receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to flight under IFR; related air traffic services practices and procedures; (ii) Aircraft general knowledge: (A) use, limitation and serviceability of avionics and instruments necessary for the control and navigation of helicopters under IFR and in instrument meteorological conditions; use and limitations of autopilot; (B) compasses, turning and acceleration errors; gyroscopic instruments, operational limits and precession effects; practices and procedures in the event of malfunctions of various flight instruments; (iii) Flight performance and planning: (A) pre-flight preparations and checks appropriate to flight under IFR; (B) operational flight planning; preparation and filing of air traffic services flight plans under IFR; altimeter setting procedures; (iv) Human performance: human performance relevant to instrument flight in helicopters; (v) Meteorology: (A) application of aeronautical meteorology; interpretation and use of reports, charts and forecasts; codes and abbreviations: use of, and procedures for obtaining, meteorological information; altimetry; (B) causes, recognition and effects of engine and airframe icing; frontal zone penetration procedures; hazardous weather avoidance; (vi) Navigation: (A) practical air navigation using radio navigation aids, (B) use; accuracy and reliability of navigation systems used in departure, enroute, approach and landing phases of flight; identification of radio navigation aids; (vii) Operation procedures: (A) interpretation and use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations, and instrument procedure charts for departure, en-route, descent and approach; (B) precautionary and emergency procedures; safety practices associated with flight under IFR: (viii) Radiotelephony: (A) radiotelephony procedures and phraseology as applied to aircraft operations under IFR; action to be taken in case of communication failure; (B) as further specified in IS 2.3.3.5 Appendix A.

2.7.1.2.1 PCAR Part 2,2.3.3.6(e) for Airplane and 2.3.3.10 (e) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (e) Skill. The applicant for an IR (A) shall: (1) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; and (2) have demonstrated by passing a skill test the ability to perform the areas of operation described in IS 2.3.3.6 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of an IR (A), and to: (i) operate the airplane within its limitations; (ii) complete all maneuvers with smoothness and accuracy; (iii) exercise good judgment and airmanship; (iv) apply aeronautical knowledge; and (v) maintain control of the airplane at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt, (vi) understand and apply crew coordination and incapacitation procedures, and (vii) communicate effectively with the other flight crew members.

2.3.3.10 INSTRUMENT RATING – HELICOPTER (e) Skill. The applicant for an IR(H) shall: (1) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; and (2) have demonstrated by

passing a skill test the ability to perform the areas of operation described in IS: 2.3.3.5 Appendix B, with a degree of competency appropriate to the privileges granted to the holder of an IR (H), and to: (i) operate the helicopter within its limitations: (ii) complete all maneuvers with smoothness and accuracy: (iii) exercise good judgment and airmanship; (iv) apply aeronautical knowledge; and (v) maintain control of the helicopter at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.

2.7.1.2.1.1 PCAR Part 2, 2.3.3.6(e)(3). 2.3.3.6 INSTRUMENT RATING - AIRPLANE (e) Skill. The applicant for an IR (A) shall: (3) have demonstrated by passing a skill test the ability to operate multi-engine airplanes solely by reference to instruments with one engine in operative, or simulated in operative, described in IS 2.3.3.6 Appendix B, if the privileges of the instrument rating are to be exercised on such airplanes.

2.7.1.3.1 (f) MEDICAL FITNESS

2.7.2.1 PCAR Part 2, 2.3.3.6(g) for Airplane and 2.3.3.10 (g) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an IR (A) shall be to pilot airplanes under IFR.

2.3.3.10 INSTRUMENT RATING – HELICOPTER (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of an IR (H) shall be to pilot helicopters under IFR.

2.7.2.2 PCAR Part 2, 2.3.3.6 (g) (1) Part 2, 2.3.3.10 (d)(3). 2.3.3.6 INSTRUMENT RATING - AIRPLANE (g) (1) Before exercising the privileges on multi-engine airplanes the holder of the rating shall have complied with the requirements of (e) (3).

2.3.3.10 INSTRUMENT RATING – HELICOPTER (d) Flight Instruction. (3) If the privileges of the Instrument rating are to be exercised on multi-engine helicopters, the applicant shall have received dual instrument flight instruction in such an helicopter from an authorized flight instructor. The instructor shall ensure that the applicant has operational experience in the operation of the helicopter solely by reference to instruments with, one engine inoperative or simulated in operative.

2.7.3.1 PCAR Part 2, 2.3.3.6 (c)(1) for Airplane and 2.3.3.10(c)(1) for Helicopter. INSTRUMENT RATING - AIRPLANE (c) Experience. (1) The applicant for an IR (A) shall hold at least a PPL (A).

2.3.3.10 INSTRUMENT RATING – HELICOPTER (c) Experience. (1) The applicant for an IR (H) shall hold a PPL (H), a CPL (H) or an ATPL (H).

2.7.3.2 PCAR Part 2, 2.3.3.6 (c)(2) for Airplane and 2.3.3.10(c)(2) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (c) Experience. (2) The applicant shall have completed not less than: (i) 50 hours of cross-country flight time as pilot-in-command of aircraft in categories acceptable to the Authority, of which not less than 10 hours shall be in airplanes; and (ii) 40 hours of instrument time in airplanes or helicopters of which not more than 20 hours, or 30 hours where a flight simulator is used, may be instrument round time. The ground time shall be under the supervision of an authorized instructor.

2.3.3.10 INSTRUMENT RATING – HELICOPTER (c) Experience. (2) The applicant shall have completed not less than: (i) 50 hours of cross-country flight time as pilot-in-command of aircraft in categories acceptable to the Authority, of which not less than 10 hours shall be in helicopters; and (ii) 40 hours of instrument time in helicopters or airplanes of which not more than 20 hours, or 30 hours where a flight simulator is used, may be instrument ground time. The ground time shall be under the supervision of an authorized instructor.

- 2.7.4.1 PCAR Part 2,2.3.3.6(d)(1)(2) for Airplane and 2.3.3.10(d)(1)(2) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (d) Flight Instruction. (1) The applicant for an IR (A) shall have not less than 10 hours of the instrument flight time required in (c) (2) (ii) while receiving and logging dual instruction in airplanes from an authorized flight instructor, on the subjects listed in IS 2.3.3.6 Appendix B. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the holder of an instrument rating: (i) pre-flight procedures, including the use of the flight manual or equivalent document, and appropriate air traffic services documents in the preparation of an IFR flight plan; (ii) pre-flight inspection, use of checklists, taxiing and pre-take-off checks; (iii) procedures and maneuvers for IFR operation under normal, abnormal and emergency conditions covering at least: (A) transition to instrument flight on take-off; (B) standard instrument departures and arrivals; (C) en-route IFR procedures and navigation; (D) holding procedures; (E) instrument approaches to specified minima; (F) missed approach procedures; (G) landings from instrument approaches; (iv) in flight maneuvers and particular flight characteristics.
- 2.3.3.10 INSTRUMENT RATING – HELICOPTER (d) Flight Instruction. (1) The applicant for an IR (H) shall have gained not less than 10 hours of the instrument flight time required in (c) (2) (ii) while receiving and logging dual instruction in helicopters from an authorized flight instructor, on the subjects listed in IS 2.3.3.5 Appendix B. (2) The instructor shall ensure that the applicant has operational experience in at least the following areas to the level of performance required for the holder of an instrument rating: (i) pre-flight procedures, including the use of the flight manual or equivalent document, and appropriate air traffic services documents in the preparation of an IFR flight plan; (ii) pre-flight inspection, use of checklists, taxiing and pre-take-off checks; (iii) procedures and maneuvers for IFR operation under normal, abnormal and emergency conditions covering at least: (A) transition to instrument flight on take-off; (B) standard instrument departures and arrivals; (C) en-route IFR procedures and navigation; (D) holding procedures; (E) instrument approaches to specified minima; (F) missed approach procedures; (G) landings from instrument approaches; (iv) in flight maneuvers and particular flight characteristics.
- 2.7.4.2 PCAR Part 2,2.3.3.6(d)(3) for Airplane and 2.3.3.10(d)(3) for Helicopter. 2.3.3.6 INSTRUMENT RATING - AIRPLANE (d) Flight Instruction (3) If the privileges of the instrument rating are to be exercised on multi-engine airplanes, the applicant shall have received dual instrument flight instruction in such an airplane from an authorized flight instructor. The instructor shall ensure that the applicant has operational experience in the operation of the airplane solely by reference to instruments with one engine inoperative or simulated inoperative.
- 2.3.3.10 INSTRUMENT RATING – HELICOPTER (d) Flight Instruction. (3) If the privileges of the instrument rating are to be exercised on multi-engine helicopters, the applicant shall have received dual instrument flight instruction in such an helicopter from an authorized flight instructor. The instructor shall ensure that the applicant has operational experience in the operation of the helicopter solely by reference to instruments with one engine inoperative or simulated inoperative.
- 2.8.1.1 PCAR Part 2,2.3.3.11(b)(2) (i)(ii)(A) to (L). 2.3.3.6 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (b) Flight Instructors (2) Knowledge: (i) The applicant for an FI rating shall have met the knowledge requirements for the issue of a CPL as specified in Subparts 2.3.3.3 and 2.3.3.7, as applicable. (ii) In addition, the applicant for an FI rating shall receive and log ground training from an authorized instructor on the following subjects: (A) techniques of applied instruction; (B) assessment of student performance in those subjects in which ground instruction is given; (C) the learning process; (D) elements of effective teaching; (E) student evaluation and testing, training

- philosophies; (F) training program development; (G) lesson planning; (H) classroom instructional techniques; (I) use of training aids; (J) analysis and correction of student errors; (K) human performance relevant to flight instruction; and (L) hazards involved in simulating system failures and malfunctions in the aircraft.
- 2.8.1.2 PCAR Part 2,2.3.3.11(b)(5). 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (b) Flight Instructors (5) Skill. The applicant for an FI rating shall: (i) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required skill test; and (ii) pass the required skill test on the subjects listed in IS 2.3.3.11 Appendix A or B, as applicable.
- 2.8.1.3 PCAR Part 2,2.3.3.11(b)(3). 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (b) Flight Instructors (3) Experience. The applicant for an FI rating shall have completed not less than 200 hours of flight time on single-pilot aircraft of the appropriate category.
- 2.8.1.4 PCAR Part 2,2.3.3.11(b)(4). 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (b) Flight Instructors (4) Flight Instruction. The applicant for an FI rating shall, under the supervision of an instructor accepted by the Authority for that purpose (i) have received instruction in flight instructional techniques including demonstration, student practices, recognition and correction of common student errors; and (ii) have practiced instructional techniques in those flight maneuvers and procedures in which it is intended to provide flight instruction listed in IS 2.3.3.11 Appendix A or B, as applicable.
- 2.8.2.1 PCAR Part 2,2.3.3.11(c)(1)(i)(ii) and (a)(1)(2)(3). 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (c) Privileges. Subject to compliance with the requirements specified in this Part, (1) the privileges for the holder of an FI rating shall be: (i) to supervise solo flights by student pilots; (ii) to carry out flight instruction for the issue of a PPL;
- 2.3.3.11 INSTRUCTOR RATINGS - AIRPLANE AND HELICOPTER (a) General. The applicant for an instructor rating or Authorization - airplane or helicopter, shall (1) hold at least the license and rating for which instruction is being given, in the appropriate category; and (2) hold the license and rating necessary to act as the pilot-in-command of the aircraft on which the instruction is given; or (3) hold a specific Authorization granted by the Authority.
- 2.8.2.2 Not implemented.
- 2.9.1.2.1 PCAR Part 2,2.3.3.13(b)(1). 2.3.3.13 GLIDER PILOT LICENSE (b) Knowledge. The applicant for a glider pilot license shall: (1) receive training on the knowledge subjects: (i) Air law: rules and regulations relevant to the holder of a glider pilot license; rules of the air; appropriate air traffic services practices and procedures; (ii) Aircraft general knowledge: (A) principles of operation of glider systems and instruments; (B) operating limitations of gliders; relevant operational information from the flight manual or other appropriate document; (iii) Flight performance and planning: (A) effects of loading and mass distribution on flight characteristics; mass and balance considerations; (B) use and practical application of launching, landing and other performance data; (C) pre flight and en-route flight planning appropriate to operations under VFR; appropriate air traffic services procedures; altimeter setting procedures; operations in areas of high density traffic; (iv) Human performance: human performance relevant to the glider pilot; (v) Meteorology: application of elementary aeronautical meteorology; use of, and procedures for obtaining, meteorological information; altimetry; (vi) Navigation: practical aspects of air navigation and dead-reckoning techniques, use of aeronautical charts; (vii) Operation procedures: (A) use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations; (B) different launch methods and

associated procedures; (C) appropriate precautionary and emergency procedures, including action to be taken to avoid hazardous weather and wake turbulence and other operating hazards; (viii) Principles of flight: principles of flight relating to gliders.

2.9.1.3.1.1 Not implemented.

2.9.1.4 PCAR Part 2, 2.3.3.13 GLIDER PILOT LICENSE (d) Skill. The applicant for a glider pilot license shall have demonstrated by passing the required skill test the ability to perform as pilot-in-command of a glider, the procedures and maneuvers described in (c) with a degree of competency appropriate to the privileges granted to the holder of a glider pilot license, and to: (1) operate the glider within its limitations; (2) complete all maneuvers with smoothness and accuracy; (3) exercise good judgment and airmanship; (4) apply aeronautical knowledge; and (5) Maintain control of the glider at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.

2.10.1.2.1 PCAR Part 2, 2.3.3.14 FREE BALLOON PILOT LICENSE (b) Knowledge. The applicant for a free balloon pilot license shall: (1) receive training on the following knowledge subjects: (i) Air law: rules and regulations relevant to the holder of a free balloon pilot license; rules of the air; appropriate air traffic services practices and procedures; (ii) Aircraft general knowledge: (A) principles of operation of free balloon systems and instruments; (B) operating limitations of free balloons; relevant operational information from the flight manual or other appropriate document; (C) physical properties and practical application of gases used in free balloons; (iii) Flight performance and planning: (A) effects of loading and mass distribution on flight characteristics; mass calculations; (B) use and practical application of launching, landing and other performance data, including the effect of temperature; (C) pre-flight and en-route flight planning appropriate to operations under VFR; appropriate air traffic services procedures; altimeter setting procedures; operations in areas of high-density traffic; (iv) Human performance: human performance relevant to the free balloon pilot; (v) Meteorology: application of elementary aeronautical meteorology; use of, and procedures for obtaining, meteorological information; altimetry; (vi) Navigation: practical aspects of air navigation and dead-reckoning techniques; use of aeronautical charts; (vii) Operation procedures: (A) use of aeronautical documentation such as AIP, NOTAM, aeronautical codes and abbreviations; (B) appropriate precautionary and emergency procedures, including action to be taken to avoid hazardous weather and wake turbulence and other operating hazards; (viii) Principles of flight: principles of flight relating to gliders (ix) Radiotelephony: the applicant should have demonstrated a level of knowledge appropriate to the privileges to be granted to the holder of a free balloon pilot license, in radiotelephony procedures and phraseology as appropriate to VFR operations and on action to be taken in case of communication failure; (2) pass the required knowledge test.

2.10.1.3.2 PCAR Part 2, 2.3.3.14(c)(2). 2.3.3.14 FREE BALLOON PILOT LICENSE (c) Experience. (2) The applicant for a free balloon pilot license shall have gained in free balloons under appropriate supervision operational experience.

2.10.1.4 PCAR Part 2, 2.3.3.14 FREE BALLOON PILOT LICENSE (d) Skill. The applicant for a free balloon pilot license shall have demonstrated by passing the required skill test the ability to perform as pilot-in-command of a free balloon, the procedures and maneuvers described in (c) with a degree of competency appropriate to the privileges granted to the holder of a free balloon pilot license, and to: (1) operate the free balloon within its limitations; (2) complete all maneuvers with smoothness and accuracy; (3) exercise good judgment and airmanship; (4) apply aeronautical knowledge; and (5) Maintain control of the free balloon at all times in a manner such that the successful outcome of a procedure or maneuver is never seriously in doubt.

Chapter 3

- 3.1.1 PCAR Part 2,2.5.2(a) for flight navigator and 2.4.2 (c) for flight engineer. 2.5.2 GENERAL (a) An applicant shall, before being issued with a flight navigator license, meet such requirements in respect of age, knowledge; experience, skill, medical fitness and language proficiency as are specified for that license.
- 2.4.2 GENERAL (c) An applicant shall, before being issued with a flight engineer license, meet such requirements in respect of age, knowledge, experience, skill, medical fitness and language proficiency as are specified for that license or rating.
- 3.1.1.1 PCAR Part 2,2.5.3 (b) (3) and (d) for Flight Navigator and 2.4.4 (a)(3) and (e) for Flight Engineer. 2.5.3 FLIGHT NAVIGATOR LICENSE (b) Knowledge. (3) pass the required knowledge test on the subjects listed in paragraph (b) (1). (d) Skill. The applicant shall have demonstrated by passing the required skill test the ability to perform as flight navigator of an aircraft with a degree of competency appropriate to the privileges granted to the holder of a flight navigator license, and to: (1) exercise good judgment and airmanship; (2) apply aeronautical knowledge; (3) perform all duties as part of an integrated crew; and (4) communicate effectively with the other flight crew members. 2.4.4 FLIGHT ENGINEER LICENSE (a) Knowledge. The applicant for a flight engineer license shall: (3) pass the required knowledge test. (e) Skill. The applicant for a flight engineer license shall: (1) have received an endorsement from an authorized instructor who certifies that the person is prepared for the required. skill test; and (2) have demonstrated by passing the required skill test, the ability to perform as flight engineer of an aircraft, the duties and procedures described in paragraph (c) (3) above with a degree of competency appropriate to the privileges granted to the holder of a flight engineer license, and to (i) use aircraft systems within the aircraft's capabilities and limitations; (ii) exercise good judgment and airmanship; (iii) apply aeronautical knowledge; (iv) perform all the duties as part of an integrated crew with the successful outcome never in doubt; and (v) communicate effectively with the other flight crew members.
- 3.2.1.2 PCAR Part 2,2.5.3(b)(1)(3). 2.5.3 FLIGHT NAVIGATOR LICENSE (b) Knowledge. (1) The applicant for a flight navigator license shall receive and log ground training from an authorized instructor on the following subjects: (i) Air Law: rules and regulations relevant to the holder of a flight navigator license; appropriate air traffic services practices and procedures; (ii) Flight performance and planning (A) effects of loading and mass distribution on aircraft performance; (B) use of take-off, landing and other performance data including procedures for cruise control; (C) pre-flight and en route operational flight planning; preparation and filing of air traffic services flight plans; appropriate air traffic services procedures; altimeter setting procedures; (iii) Human performance: human performance relevant to the flight navigator; (iv) Meteorology (A) Interpretation and practical application of aeronautical meteorological reports, charts and forecasts; codes and abbreviations; use of, and procedures for obtaining, meteorological information; pre-flight and inflight; altimetry; (B) aeronautical meteorology; climatology of relevant areas in respect of the elements having an effect upon aviation; the movement of pressure systems; the structure of fronts, and the origin and characteristics of significant weather phenomena which affect take-off, en route and landing conditions; (v) Navigation (A) dead-reckoning; pressure-pattern and celestial navigation procedures, the use of aeronautical charts, radio navigation aids and area navigation systems; specific navigation requirements for long-range flights; (B) use; limitation and serviceability of avionics and instruments necessary for the navigation of the aircraft, (C) use, accuracy and reliability of navigation systems used in departure, enroute and approach phases of flight; identification of radio navigation aids; (D) principles, characteristics and use of self-contained and external referenced navigation systems; operation of airborne equipment; (E) the celestial sphere including

the movement of heavenly bodies and their selection and identification for the purpose of observation and reduction of sights; calibration of sextants; the completion of navigation documentation; (F) definitions, units and formulae used in air navigation; (vi) Operational procedures: interpretation and use of aeronautical documentation such as AIP, NOTAM, aeronautical codes, abbreviations, and instrument procedure charts for departure, en-route, descent and approach; (vii) Principles of flight: principles of flight; (viii) Radiotelephony: radiotelephony procedures and phraseology. (3) pass the required knowledge test on the subjects listed in paragraph (b) (1).

- 3.2.1.3.1.1 PCAR Part 2, 2.5.3 FLIGHT NAVIGATOR LICENSE (c) Experience. (2) The holder of a pilot license may be credited with 30 hours towards the 200 hours of flight time.
- 3.2.1.4 PCAR Part 2, 2.5.3 FLIGHT NAVIGATOR LICENSE (d) Skill. The applicant shall have demonstrated by passing the required skill test the ability to perform as flight navigator of an aircraft with a degree of competency appropriate to the privileges granted to the holder of a flight navigator license, and to: (1) exercise good judgment and airmanship; (2) apply aeronautical knowledge; (3) perform all duties as part of an integrated crew; and (4) communicate effectively with the other flight crew members.
- 3.2.1.5 PCAR Part 2, 2.5.3 FLIGHT NAVIGATOR LICENSE (e) Medical fitness. The applicant shall hold a current Class 1 Medical Certificate.
- 3.2.2 PCAR Part 2, 2.5.3 FLIGHT NAVIGATOR LICENSE (f) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a flight navigator license shall be to act as flight navigator of any aircraft.
- 3.3.1.2.1 PCAR Part 2, 2.4.4 FLIGHT ENGINEER LICENSE (b) Knowledge. The applicant for a flight engineer license shall: (1) receive and log ground training from an authorized instructor on the following subjects: (i) Air law: rules and regulations relevant to the holder of a flight engineer license; rules and regulations governing the operation of civil aircraft pertinent to the duties of a flight engineer; (ii) Aircraft general knowledge: (A) basic principles of power-plants; gas turbines and/or piston engines; characteristics of fuels; fuel systems including fuel control; lubricants and lubrication systems; after burners and injection systems, function and operation of engine ignition and starter systems; (B) principles of operation; handling procedures and operating limitations of aircraft power-plants; effects of atmospheric conditions on engine performance; (C) airframes; flight controls; structures; wheel assemblies; brakes and anti-skid units; corrosion and fatigue life; identification of structural damage and defects; (D) ice and rain protection systems; (E) pressurization and air-conditioning systems; oxygen systems; (F) hydraulic and pneumatic systems; (G) basic electrical theory, electric systems (AC and DC); aircraft wiring systems; bonding and screening; (H) principles of operation of instruments, compasses, autopilots, radio communication equipment, radio and radar navigation aids, flight management systems, displays and avionics; (I) limitations of appropriate aircraft; (J) fire protection, detection, suppression and extinguishing systems; (K) use and serviceability checks of equipment and systems of appropriate aircraft; (iii) Flight performance and planning: (A) effects of loading and mass distribution on aircraft handling, flight characteristics and performance; mass and balance calculations; (B) use and practical application of performance data including procedures for cruise control; (iv) Human performance: human performance relevant to the flight engineer; (v) Operational procedures: (A) principles of maintenance; procedures for the maintenance of airworthiness; defect reporting; pre-flight inspections; precautionary procedures for fuelling and use of external power; installed equipment and cabin systems; (B) normal, abnormal and emergency procedures; (C) operational procedures for carriage of freight and dangerous goods; (vi) Principles of flight: fundamentals of aerodynamics; (vii) Radiotelephony: radiotelephony procedures and phraseology; (3) pass the required knowledge test.

- 3.3.1.3.1 PCAR Part 2,2.2.6.2 and Part 2,2.4.4(c)(1)(2). 2.2.6.2 USE OF SYNTHETIC FLIGHT TRAINERS The use of a synthetic flight trainer for performing training, testing and checking for which a flight crew member is to receive credit, shall be approved by the Authority, which shall ensure that the synthetic flight trainer is appropriate to the task. Note: See the Manual of Criteria for the Qualification of Flight Simulators (ICAO Doc 9625).
- 2.4.4 FLIGHT ENGINEER LICENSE (c) Experience. (1) The applicant for a flight engineer license shall have completed under the supervision of a person accepted by the Authority for that purpose, not less than 100 hours of flight time in the performance of the duties of a flight engineer, of which 50 hours may have been completed in a flight simulator. (2) The holder of a pilot license may be credited with 30 hours towards the 100 hours of night time.
- 3.3.1.3.1.1 Not implemented.
- 3.3.1.4.1 PCAR Part 2,2.4.4(e)(2). 2.4.4 FLIGHT ENGINEER LICENSE (e) Skill. The applicant for a flight engineer license shall: (2) have demonstrated by passing the required skill test, the ability to perform as flight engineer of an aircraft, the duties and procedures described in paragraph (c) (3) above with a degree of competency appropriate to the privileges granted to the holder of a flight engineer license, and to (i) use aircraft systems within the aircraft's capabilities and limitations; (ii) exercise good judgment and airmanship; (iii) apply aeronautical knowledge; (iv) perform all the duties as part of an integrated crew with the successful outcome never in doubt; and (v) communicate effectively with the other flight crew members.
- 3.3.1.4.2 PCAR Part 2, 2.2.6.2 USE OF SYNTHETIC FLIGHT TRAINERS. The use of a synthetic flight trainer for performing training, testing and checking for which a flight crew member is to receive credit, shall be approved by the Authority, which shall ensure that the synthetic flight trainer is appropriate to the task. Note: See the Manual of Criteria for the Qualification of Flight Simulators (ICAO Doc 9625).
- 3.3.1.5 PCAR Part 2, 2.4.4 FLIGHT ENGINEER LICENSE (f) Medical fitness. The applicant for a flight engineer license shall hold a current Class 1 Medical Certificate.
- 3.3.2.1 PCAR Part 2, 2.4.4 FLIGHT ENGINEER LICENSE (g) Privileges. Subject to compliance with the requirements specified in this Part, the privileges of the holder of a flight engineer license shall be to act as flight engineer of any type of aircraft on which the holder has demonstrated a level of knowledge and skill, on the basis of those requirements specified in Subpart 2.4.4 (b) and (d) which are applicable to the safe operation of that type of aircraft.
- 3.3.2.2 PCAR Part 2, 2.2.2.2(b)(1). 2.2.2.2 RATINGS (b) The following ratings are placed on a flight engineer's license when an applicant satisfactorily accomplishes the requirements in this Part for the rating sought: (1) Type rating.

Chapter 4

- 4.1.1 Not implemented.
- 4.1.2 Not implemented.
- 4.2.1.1 PCAR Part 2, 2.6.2.2 (a) (1). 2.6.2.2 ELIGIBILITY REQUIREMENTS: GENERAL (a) An applicant for an AMT license and any associated rating shall (1) Be at least 18 years of age.
- 4.2.1.2 PCAR Part 2, 2.6.2.4 KNOWLEDGE REQUIREMENTS (a) The applicant for an Aviation Maintenance Technician (AMT) license shall have pass a knowledge test covering at least the following areas: (1) Air law and airworthiness requirements: (i) rules and regulations relevant to an Aviation Maintenance Technician (AMT) license

holder including applicable airworthiness requirements governing certification and continuing airworthiness of aircraft and approved aircraft maintenance organization procedures; (2) Natural science and aircraft general knowledge (i) basic mathematics; units of measurement; fundamental principles and theory of physics and chemistry applicable to aircraft maintenance; (3) Aircraft engineering (i) characteristics and applications of the materials of aircraft construction including principles of construction and functioning of aircraft structures, fastening techniques; power-plants and their associated systems, mechanical, fluid, electrical and electronic power sources, aircraft instrument and display systems, aircraft control systems, and airborne navigation and communication systems; (4) Aircraft maintenance: (i) tasks required to ensure the continuing airworthiness of an aircraft including methods and procedures for the overhaul, repair, inspection, replacement, modification or defect rectification of aircraft structures, components and systems in accordance with the methods prescribed in the relevant Maintenance Manuals and the applicable requirements of airworthiness; (5) Human performance: (i) human performance and limitations relevant to the duties of an aviation maintenance license holder.

- 4.2.1.3 PCAR Part 2, 2.6.2.7 EXPERIENCE REQUIREMENTS (a) An applicant for an AMT license and associated ratings may qualify by either practical experience or through training in an ATO; (b) Practical experience only. Each applicant for an AMT license and rating(s) relying solely on practical experience shall provide documentary evidence, acceptable to the Authority, of the following experience in the inspection, servicing and maintenance of aircraft or its components: (1) Airframe rating - 30 months, (2) Power-plant rating - 30 months; (3) Airframe and Power-plant ratings - 60 months; (c) Approved Training. Each applicant for an AMT license relying on completion of training in an Approved Training Organization (ATO) shall provide documentary evidence, acceptable to the Authority, of the following training: (1) Airframe rating - 24 months (2) Power-plant rating - 24 months (3) Airframe and Power plant ratings - 30 months.
- 4.2.1.5 PCAR Part 2, 2.6.2.6 SKILL REQUIREMENTS (a) Each applicant for an AMT license or rating must pass an oral and a skill test on the license or rating he seeks. The tests cover the applicant's basic skill in performing practical projects on the subjects covered by the written test for the license or rating, and shall contain at least the subjects in the IS 2.6.2.6, appropriate to the license or rating sought.
- 4.2.2.1 PCAR Part 2, 2.6.2.8 (a) thru (f). 2.6.2.8 PRIVILEGES AND LIMITATIONS (a) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with A&P may perform or supervise the maintenance, preventive maintenance, or modification of, or after inspection, sign a maintenance release and a return to service for any aircraft, airframe aircraft engine, propeller, appliance, component, or part thereof for which he or she is rated, provided the licensed AMT has: (1) Satisfactorily performed the work at an earlier date; (2) Demonstrated the ability to perform the work to the satisfaction of the Authority; (3) Received training acceptable to the Authority on the tasks to be performed; or (4) Performed the work while working under the direct supervision of a licensed AMT who is appropriately rated and has: (i) Had previous experience in the specific operation concerned; or (ii) Received training acceptable to the Authority on the task to be performed. (b) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with an Airframe rating may after he/she has performed the 100-hour inspection required by Part 8 of this chapter on an airframe, or any related part or appliance, and sign a maintenance release. (c) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with a Powerplant rating may perform the 100-hour inspection required by Part 8 of this chapter on a powerplant or propeller or any related part or appliance, and sign a maintenance release. (d) An AMT with an Airframe or Power-plant rating may not (1)

Supervise the maintenance, repair, or modification of, or approve and provide a maintenance release for any aircraft, airframe, aircraft engine, propeller, appliance, component or part thereof, for which he/she is rated unless he/she has satisfactorily performed the work concerned at an earlier date. (2) Exercise the privileges of the license unless the licensed AMT understands the current instructions for continued airworthiness and the maintenance instructions for the specific operation concerned. (e) An AMT with an Airframe or Power-plant rating may (1) Perform or supervise under the direct supervision and control of an AMO, any repair or alteration of instruments. (2) Sign a Maintenance Release for any aircraft, airframe, engine, propeller, appliance, component, or part thereof after completion of a major alteration or major repair. (f) A licensed mechanic shall not exercise the privileges of his license and ratings unless he is familiar with the current CAR, manufacturer's instructions and the maintenance manuals pertinent to the particular aircraft operation to be performed.

4.2.2.2 PCAR Part 2, 2.6.2.8 (a) thru (f) Part 2, 2.6.2.9 and Part 2, 2.6.2.10. 2.6.2.8 PRIVILEGES AND LIMITATIONS (a) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with A&P may perform or supervise the maintenance, preventive maintenance, or modification of, or after inspection, sign a maintenance release and a return to service for any aircraft, airframe aircraft engine, propeller, appliance, component, or part thereof for which he or she is rated, provided the licensed AMT has: (1) Satisfactorily performed the work at an earlier date; (2) Demonstrated the ability to perform the work to the satisfaction of the Authority; (3) Received training acceptable to the Authority on the tasks to be performed; or (4) Performed the work while working under the direct supervision of a licensed AMT who is appropriately rated and has: (i) Had previous experience in the specific operation concerned; or (ii) Received training acceptable to the Authority on the task to be performed. (b) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with an Airframe rating may after he/she has performed the 100-hour inspection required by Part 8 of this chapter on an airframe, or any related part or appliance, and sign a maintenance release. (c) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with a Powerplant rating may perform the 100-hour inspection required by Part 8 of this chapter on a powerplant or propeller or any related part or appliance, and sign a maintenance release. (d) An AMT with an Airframe or Power-plant rating may not (1) Supervise the maintenance, repair, or modification of, or approve and provide a maintenance release for any aircraft, airframe, aircraft engine, propeller, appliance, component or part thereof, for which he/she is rated unless he/she has satisfactorily performed the work concerned at an earlier date. (2) Exercise the privileges of the license unless the licensed AMT understands the current instructions for continued airworthiness and the maintenance instructions for the specific operation concerned. (e) An AMT with an Airframe or Power-plant rating may (1) Perform or supervise under the direct supervision and control of an AMO, any repair or alteration of instruments. (2) Sign a Maintenance Release for any aircraft, airframe, engine, propeller, appliance, component, or part thereof after completion of a major alteration or major repair. (f) A licensed mechanic shall not exercise the privileges of his license and ratings unless he is familiar with the current CAR, manufacturer's instructions and the maintenance manuals pertinent to the particular aircraft operation to be performed.

2.6.2.9 DURATION OF THE LICENSE (a) The duration of the AMT license is two years. (b) The holder of a license with an expiration date may not, after that date, exercise the privileges of that license. (c) The license shall remain valid as long as the holder thereof maintains his/her competency.

2.6.2.10 RECENT EXPERIENCE REQUIREMENTS (a) A licensed AMT may not exercise the privileges of his/her license or rating unless, within the preceding 24

months: (1) The Authority has found that he/she is able to do that work; or (2) For at least 6 months within the preceding 24 months - (i) Served as an AMT under his/her license and rating; (ii) Technically supervised other AMT's; (iii) Provided aviation maintenance instruction or served as the direct supervisor of persons providing aviation maintenance instruction for an AMT course or program acceptable to the Authority; (iv) Supervised the maintenance, preventive maintenance, or alteration of any aircraft, airframe, aircraft engine, propeller, appliance, component, or part thereof; or been engaged in any combination of paragraphs (a)(2)(i) through (a)(2)(iii) of this subsection; and (b) Has received appropriate recurrent training from an AMO within the preceding 24 months and is familiar both with current manufacturers and approved maintenance organization's instructions and the maintenance manuals pertinent to the particular aircraft operation to be performed. (1) The requirements of recurrent training applies to all AME's when certifying aircraft or components used in commercial operations.

4.2.2.3 PCAR Part 2, 2.6.2.8 (a) to (f). 2.6.2.8 PRIVILEGES AND LIMITATIONS (a) Except as specified in paragraphs (d), (e) and (f) of this subsection, a licensed AMT with A&P may perform or supervise the maintenance, preventive maintenance, or modification of, or after inspection, sign a maintenance release and a return to service for any aircraft, airframe aircraft engine, propeller, appliance, component, or part thereof for which he or she is rated, provided the licensed AMT has: (1) Satisfactorily performed the work at an earlier date; (2) Demonstrated the ability to perform the work to the satisfaction of the Authority; (3) Received training acceptable to the Authority on the tasks to be performed; or (4) Performed the work while working under the direct supervision of a licensed AMT who is appropriately rated and has: (i) Had previous experience in the specific operation concerned; or (ii) Received training acceptable to the Authority on the task to be performed. (f) A licensed mechanic shall not exercise the privileges of his license and ratings unless he is familiar with the current CAR, manufacturer's instructions and the maintenance manuals pertinent to the particular aircraft operation to be performed.

4.2.2.4 Not implemented.

4.3.1 Not implemented.

4.3.2 Not implemented.

4.4.1 PCAR Part 2, 2.7.2 GENERAL (a) An applicant shall, before being issued with an air traffic controllers license, meet such requirements in respect of age, knowledge, experience, skill, medical fitness and language proficiency as are specified for that license or rating.

4.5.1 PCAR Part 2, 2.7.4 (1) aerodrome control rating; (2) approach control rating; (3) approach radar control rating; (4) approach precision radar control rating; (5) area control rating; and (6) area radar control rating.

4.5.2.1 PCAR Part 2, 2.7.5(a)(1)(3). 2.7.5 AIR TRAFFIC CONTROLLER RATING REQUIREMENTS (a) Knowledge. The applicant for an air traffic controller rating shall (1) receive an approved training course from an authorized instructor on the following subjects: (i) aerodrome control rating: (A) aerodrome layout; physical characteristics and visual aids; (B) airspace structure; (C) applicable rules, procedures and source of information; (D) air navigation facilities; (E) air traffic control equipment and its use; (F) terrain and prominent landmarks; (G) characteristics of air traffic; (H) weather phenomena; and emergency and search and rescue plans; (ii) approach control and area control ratings: (A) airspace structure; (B) applicable rules, procedures and source of information; (C) air navigation facilities; (D) air traffic control equipment and its use; (E) terrain and prominent landmarks; (F) characteristics of air traffic and traffic

flow; (G) weather phenomena; and (H) emergency and search and rescue plans; and (iii) approach radar, approach precision radar and area radar control ratings. The applicant shall meet the requirements specified in (ii) in so far as they affect the area of responsibility. and shall have demonstrated a level of knowledge appropriate to the privileges granted. in at least the following additional subjects: (A) principles, use and limitations of radar, other surveillance systems and associated equipment; and (B) procedures for the provision of approach, precision approach or area radar control services, as appropriate, including procedures to ensure appropriate terrain clearance; (3) pass the required knowledge test.

- 4.5.2.2.1 PCAR Part 2, 2.7.5 AIR TRAFFIC CONTROLLER RATING REQUIREMENTS (b) Experience. (1) The applicant for an air traffic controller license shall have: (i) satisfactorily completed an approved training course; (ii) provided; satisfactorily, under the supervision of an appropriately rated air traffic controller: (A) aerodrome control rating: an aerodrome control service, for a period of not less than 90 hours or one month, whichever is greater, at the unit for which the rating is sought; (B) approach, approach radar, area or area radar control rating: the control service for which the rating is sought. for a period of not less than 180 hours or three months, whichever is greater, at the unit for which the rating is sought; and (C) approach precision radar control rating: not less than 200 precision approaches of which not more than 100 shall have been carried out on a radar simulator approved for that purpose by the Authority. Not less than 50 of those precision approaches shall have been carried out at the unit and on the equipment for which the rating is sought; and (iii) if the privileges of the approach radar control rating include surveillance radar approach duties, the experience shall include not less than 25 plan position indicator (PPI) approaches on the surveillance equipment of the type in use at the unit for which the rating is sought and under the supervision of an appropriately rated approach radar controller.
- 4.5.2.2.3 Not implemented.
- 4.5.2.3 PCAR Part 2, 2.7.5 AIR TRAFFIC CONTROLLER RATING REQUIREMENTS (c) Skill. The applicant shall have demonstrated by passing the required skill test, at a level appropriate to the privileges being granted, the skill, judgment and performance required to provide a safe orderly and expeditious control service.
- 4.5.2.4 Not implemented.
- 4.5.3.1 PCAR Part 2, 2.7.5 (d) Privileges. (1) Subject to compliance with the requirements specified in this Part, the privileges of the holder of an air traffic controller license (i) aerodrome control rating: to provide or to supervise the provision of aerodrome control service for the aerodrome for which the license holder is rated; (ii) approach control rating: to provide or to supervise the provision of approach control service for the aerodrome or aerodromes for which the license holder is rated, within the airspace or portion thereof, under the jurisdiction of the unit providing approach control service; (iii) approach control rating: to provide or to supervise the provision of approach control service for the aerodrome or aerodromes for which the license holder is rated, within the airspace or portion thereof, under the jurisdiction of the unit providing approach control service; (iv) approach radar control rating: to provide and/or supervise the provision of approach control service with the use of radar or other surveillance systems for the aerodrome or aerodromes for which the license holder is rated, within the airspace or portion thereof, under the jurisdiction of the unit providing approach control service: (A) subject to compliance with the provisions of paragraph (b) (1) (iii), the privileges shall include the provision of surveillance radar approaches; (v) approach precision radar control rating: to provide and/or supervise the provision of precision approach radar service at the aerodrome for which the license holder is rated; (vi) area control rating: to provide and/or supervise the provision of area control

service within the control area or portion thereof, for which the license holder is rated; and (vii) area radar control rating: to provide and/or supervise the provision of area control service with the use of radar, within the control area or portion thereof, for which the license holder is rated.

4.5.3.3 Not implemented.

4.6.1.2 PCAR Part 2, 2.8.3 FLIGHT OPERATIONS OFFICER LICENSE (b) Knowledge. The applicant for a flight operations officer license shall (1) receive an approved training course from an authorized instructor on the knowledge areas: (i) Air Law: rules and regulations relevant to the holder of a flight operations officer license; appropriate air traffic services practices and procedures; (ii) Aircraft general knowledge: (A) principles of operation of airplane power-plants, systems and instruments (B) operating limitations of airplanes and power-plants; (C) minimum equipment list; (iii) Flight performance calculation and planning procedures: (A) effects of loading and mass distribution on aircraft performance and flight characteristics; mass and balance calculations; (B) operational flight planning; fuel consumption and endurance calculations; alternate airport selection procedures; en-route cruise control; extended range operation; (C) preparation and filing of air traffic services flight plans; (D) basic principles of computer-assisted planning systems; (iv) Human performance: human performance relevant to dispatch duties; (v) Meteorology (A) aeronautical meteorology; the moment of pressure systems; the structure of fronts, and the origin and characteristics of significant weather phenomena which affect take-off, en-route and landing conditions; (B) interpretation and application of aeronautical meteorological reports, charts and forecasts, codes and abbreviations; use of, and procedures for obtaining, meteorological information; (vi) Navigation: principles of air navigation with particular reference to instrument flight; (vii) Operational procedures: (A) use of aeronautical documentation; (B) operational procedures for the carriage of freight and dangerous goods; (C) procedures relating to aircraft accidents and incidents; emergency flight procedures; (D) procedures relating to unlawful interference and sabotage of aircraft; (viii) Principles of flight: principles of flight relating to the appropriate category of aircraft; (ix) Radio communication: procedures for communicating with aircraft and relevant ground stations; (3) pass the required knowledge test.

4.6.1.4 PCAR Part 2, 2.8.3 FLIGHT OPERATIONS OFFICER LICENSE (d) Skill. The applicant shall have demonstrated the ability to: (1) make an accurate and operationally acceptable weather analysis from a series of daily weather maps and weather reports; provide an operationally valid briefing on weather conditions prevailing in the general neighborhood of a specific air route; forecast weather trends pertinent to air transportation with particular reference to destination and alternates; (2) determine the optimum flight path for a given segment, and create accurate manual and/or computer generated flight plans; and (3) provide operating supervision and all other assistance to a flight in actual or simulated adverse weather conditions as appropriate to the duties of the holder of a flight operations officer license.

4.7.1.1 PCAR Part 2, 2.9.1 and Part 2, 2.9.2(a). 2.9.1 APPLICABILITY. This section prescribes the requirements for the issue, renewal or re-issue of an aeronautical station operator license.

2.9.2 GENERAL (a) An applicant shall, before being issued with an aeronautical station operator license, meet such requirements in respect of age, knowledge, experience, skill, medical fitness and language proficiency as are specified for that license.

4.7.1.3 PCAR Part 2, 2.9.3(b)(1)(3). 2.9.3 AERONAUTICAL STATION OPERATOR LICENSE (b) Knowledge. The applicant for an aeronautical station operator license shall: (1)

receive and log ground training from an authorized instructor on the following subjects: (i) General Knowledge: air traffic services provided within Republic of the Philippines; (ii) Language: the language or languages nationally designated for use in air-ground communications and ability to speak such language or languages without accent or impediment which would adversely affect radio communication; (iii) Operational Procedures: radiotelephony procedures; phraseology; telecommunication network: (iv) Rules and regulations: rules and regulations applicable to the aeronautical station operator; (v) Telecommunication equipment: principles, use and limitations of telecommunication equipment in an aeronautical station (3) pass the required knowledge test.

Chapter 6

- 6.1.1 PCAR Part 2, 2.10.1.8(1)(i)(ii); 2.10.2.2.1; 2.3.3.3(vi); 2.3.3.5.1(e); 2.3.3.4(f); 2.3.3.9(f); 2.5.3(e); 2.4.4(f); 2.3.3.2(f); 2.3.3.7(f); 2.3.3.13(e); 2.3.3.14(e); and 2.7.3(d). 2.10.1.8 MEDICAL CERTIFICATES (b) Issue of medical certificates (1) A medical certificate will be issued to any person who meets the medical requirements prescribed in this Subpart, based on medical examination and evaluation of the applicants history and condition. (i) The Authority will issue the Class 1 medical certificate (ii) The issue of Class 2 and 3 medical certificates may be delegated to the authorized Aviation Medical Examiner.
- 2.10.2.2.1 CERTIFICATE ISSUE AND RENEWAL (a) An applicant for a CPL, MPL, ATPL, Flight Engineer or Flight Navigator license shall undergo an initial medical examination for the issue of a Class 1 Medical Certificate. (b) Except where otherwise stated in this subpart, holders of CPL, MPL, ATPL, Flight Engineer or Flight Navigator license shall have their Class 1 Medical Certificate renewed at intervals not exceeding those specified in Subpart 2.10.1.8 (c). (c) A Class 1 Medical Certificate will be issued when the applicant complies with the requirements of this Part.
- 2.3.3.3 COMMERCIAL PILOT LICENSE – AIRPLANE (vi) Medical fitness. The applicant for a CPL(A) shall hold a current Class 1 Medical Certificate.
- 2.3.3.5 MULTI-CREW PILOT LICENSE (MPL) 2.3.3.5.1 GENERAL (e) Nationality: Be a citizen of the Philippines or a citizen of a foreign country granting similar rights and privileges to citizens of the Philippines subject, however, to existing treaty or treaties and agreements entered into by the Philippine Government with foreign countries and subject further to security measures adopted by the Philippine Government.
- 2.3.3.4 AIRLINE TRANSPORT PILOT LICENSE – AIRPLANE (f) Medical fitness. The applicant for an ATPL(A) shall hold a current Class 1 Medical Certificate, except for: (i) Check Airman Qualifications for Flight Simulation Training Device as provided for under 8.10.1.39 (b)(1); and (ii) Check Airmen who have reached their 65th birthday or who do not have an appropriate medical certificate as provided for under 8.10.1.39 (d) and 8.10.1.1 (c). (21 March 2011).
- 2.3.3.9 AIRLINE TRANSPORT PILOT LICENSE – HELICOPTER (f) Medical fitness. The applicant for an ATPL (H) shall hold a current Class 1 Medical Certificate, except for: (i) Check Airman Qualifications for Flight Simulation Training Device as provided for under 8.10.1.39 (b) (1); and (ii) Check Airmen who have reached their 65th birthday or who do not have an appropriate medical certificate as provided for under 8.10.1.39 (d) and 8.10.1.1 (c). (21 March 2011).
- 2.5.3 FLIGHT NAVIGATOR LICENSE (e) Medical fitness. The applicant shall hold a current Class 1 Medical Certificate, except for: (i) Check Airman Qualifications for Flight Simulation Training Device as provided for under 8.10.1.39 (b)(1); and (ii) Check Airmen who have reached their 65th birthday or who do not have an appropriate

medical certificate as provided for under 8.10.1.39 (d) and 8.10.1.1 (c). (21 March 2011)

2.4.4 FLIGHT ENGINEER LICENSE (f) Medical fitness, The applicant for a flight engineer license shall hold a current Class 1 Medical Certificate, except for: (i) Check Airman Qualifications for Flight Simulation Training Device as provided for under 8.10.1.39 (b)(1);and (ii) Check Airmen who have reached their 65th birthday or who do not have an appropriate medical certificate as provided for under 8.10.1.39 (d) and 8.10.1.1 (c). (21 March 2011)

2.3.3.2 PRIVATE PILOT LICENSE – AIRPLANE (f) Medical fitness. The applicant for a PPL(A) shall hold a current Class 2 Medical Certificate.

2.3.3.7 PRIVATE PILOT LICENSE – HELICOPTER (f) Medical fitness. The applicant for a PPL(H) shall hold a current Class 2 Medical Certificate.

2.3.3.13 GLIDER PILOT LICENSE (e) Medical Fitness. The applicant of a glider pilot license shall holder a current Class 2 Medical Certificate.

2.3.3.14 FREE BALLOON PILOT LICENSE (e) Medical fitness. The applicant for a free balloon pilot license shall hold a current Class 2 Medical Certificate.

2.7.3 AIR TRAFFIC CONTROLLER LICENSE (d) Medical fitness. The applicant for an air traffic controller license shall hold a current Class 3 Medical Certificate.

6.1.2 PCAR Part 2, 2.10.1.4 (a) and (c). 2.10.1.4 AVIATION MEDICAL EXAMINATIONS (a) Each applicant for a medical certificate shall provide the AME with a personally certified statement of medical facts concerning personal, familial and hereditary history. (c) Any false declaration to an AME made by an applicant for a license or rating shall be reported to the Authority for such action as may be considered appropriate.

6.1.4 PCAR, Part 2, 2.10.1.8 (d) (1). 2.10.1.8 MEDICAL CERTIFICATES (d) Renewal or re-issue of a medical certificate (1) The requirements to be met for the renewal or re-issue of a medical certificate are the same as those for the initial certificate except where otherwise specifically stated.

6.2.3.1 PCAR Part 2, 2.10.2.1.3 VISUAL ACUITY TEST REQUIREMENTS (a) Visual acuity tests must be conducted in an environment with a level of illumination that corresponds to ordinary office illumination (30-60cd/m²). (b) Visual acuity must be measured by means of a series of Landolt rings or similar opto types, placed at a distance from the applicant appropriate to the method of testing adopted.

6.2.4.1 PCAR Part 2, 2.10.2.1.4 COLOR PERCEPTION REQUIREMENTS (a) The applicant shall be required to demonstrate the ability to perceive readily those colors the perception of which is necessary for the safe performance of duties. (b) The applicant shall be tested for the ability to correctly identify a series of pseudoisochromatic plates in daylight or in artificial light of the same color temperature such as that provided by CIE standard illuminants C or 02 as specified by the International Commission of Illumination (CIE). (c) An applicant obtaining a satisfactory result as prescribed by the Authority shall be assessed as fit. An applicant failing to obtain a satisfactory result in such a test shall be assessed as unfit unless able to readily distinguish the colors used in air navigation and correctly identify aviation colored lights. Applicants who fail to meet these criteria shall be assessed as unfit except for Class 2 assessment with the following restriction: valid daytime only.

6.2.5.1 PCAR Part 2.10.2.1.5 HEARING REQUIREMENTS Hearing requirements are established in addition to the ear examination conducted during the medical examination for the physical and mental requirements.

- 6.2.5.2 PCAR Part 2, 2.10.2.1.5 HEARING REQUIREMENTS (a) The applicant shall be required to be free from any hearing defect that would interfere with the safe performance of duties in exercising the privileges of the license. Note 1: The reference zero for calibration of pure-tone audiometers used for applying Subparts 2.10.2.2.4 (a) and 2.10.2.4.4 (a) is that of the International Organization for Standardization (ISO) Recommendation R389, 1964. Note 2: The frequency composition of the background noise referred to in Subparts 2.10.2.4 (a)(1) and 2.10.2.4.4.(a)(1) is defined only to the extent that the frequency range 600 to 4800 HZ is adequately represented. Note 3: In the choice of speech material, aviation-type material is not to be used exclusively for the above tests. Lists of phonetically balanced words in use by a number of Contracting States have given satisfactory results. Note 4: A quiet room for the purposes of testing the hearing requirements is a room in which the intensity of the background noise is less than 50 dB when measured on 'slow response of an 'A"- weighted sound level meter. Note 5: For the purposes of hearing requirements, the sound level of an average conversational voice at point of output ranges from 85 to 95 dB.
- 6.2.5.3 PCAR Part 2, 2.10.2.2.4 HEARING REQUIREMENTS The medical examination shall be based on the following requirements. (a) The applicant, tested on a pure-tone audiometer at first issue of license, not less than once every five years up to the age of 40 years, and thereafter not less than once every three years,
- 6.2.5.4 PCAR Part 2, 2.10.2.4.4 HEARING REQUIREMENTS The medical examination shall be based on the following requirements. (a) The applicant, tested on a pure-tone audiometer at first issue of license, not less than once every five years up to the age of 40 years, and thereafter not less than once every three years,
- 6.2.5.6 PCAR Part 2, 2.10.2.1.5(a) (Note 4). 2.10.2.1.5 HEARING REQUIREMENTS (a) The applicant shall be required to be free from any hearing defect that would interfere with the safe performance of duties in exercising the privileges of the license. Note 4: A quiet room for the purposes of testing the hearing requirements is a room in which the intensity of the background noise is less than 50 dB when measured on slow response of an 'A" - weighted sound level meter
- 6.3.1.1 PCAR Part 2, 2.10.2.2.1 CERTIFICATE ISSUE AND RENEWAL (a) An applicant for a CPL, MPL, ATPL, Flight Engineer or Flight Navigator license shall undergo an initial medical examination for the issue of a Class 1 Medical Certificate.
- 6.3.1.2 PCAR Part 2, 2.10.2.2.1 CERTIFICATE ISSUE AND RENEWAL (b) Except where otherwise stated in this subpart, holders of CPL, MPL, ATPL, Flight Engineer or Flight Navigator license shall have their Class 1 Medical Certificate renewed at intervals not exceeding those specified in Subpart 2.10.1.8 (c).
- 6.3.1.3 PCAR Part 2, 2.10.2.2.1 CERTIFICATE ISSUE AND RENEWAL (c) A Class 1 Medical Certificate will be issued when the applicant complies with the requirements of this Part.
- 6.3.2.2 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (b) The applicant shall have no established medical history or clinical diagnosis of: (1) a psychosis; (2) alcoholism; (3) drug dependence; (4) any personality disorder, particularly if severe enough to have repeatedly resulted in overt acts; (5) a mental abnormality, or neurosis of a significant degree; such as might render the applicant unable to safely exercise the privileges of the license applied for or held, unless accredited medical conclusion indicates that in special circumstances, the applicant's failure to meet the requirement is such that exercise of the privileges of the license applied for is not likely to jeopardize flight safety.
- 6.3.2.4 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (e) Cases of head injury, the effects of which, according to accredited medical conclusion, are likely

to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.

- 6.3.2.5 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (f) The applicant shall not possess any abnormality of the heart, congenital or acquired, which is likely to interfere with the safe exercise of the applicant's license and rating privileges. A history of proven myocardial infarction shall be disqualifying. Note: Such commonly occurring conditions as respiratory arrhythmia, occasional extra systoles which disappear on exercise, increase of pulse rate from excitement or exercise, or a slow pulse not associated with auriculoventricular dissociation may be regarded as being "within-normal" limits.
- 6.3.2.5.1 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (f) (2) An applicant who has undergone coronary bypass, grafting or angioplasty (with or without stenting) or other cardiac intervention or who suffers from any other potentially incapacitating cardiac condition shall be assessed unfit, unless the applicant's cardiac condition has been investigated and evaluated in accordance with best medical practices and is assessed not likely to interfere with the safe exercise of the applicant's license or rating privileges.
- 6.3.2.5.2 Not implemented.
- 6.3.2.6 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (f) (1) Electrocardiography shall form part of the heart examination for the first issue of a license and shall be included in re-examination of applicants between the ages of 30 and 40 no less frequently than every two years, and thereafter no less frequently than annually. Note 1: The purpose of routine electrocardiography is case finding. It does not provide sufficient evidence to justify disqualification without further thorough cardiovascular investigation. Note 2: Guidance on resting and exercise electrocardiography is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.3.2.6.1 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (f) (1) Electrocardiography shall form part of the heart examination for the first issue of a license and shall be included in re-examination of applicants between the ages of 30 and 40 no less frequently than every two years, and thereafter no less frequently than annually. Note 1: The purpose of routine electrocardiography is case finding. It does not provide sufficient evidence to justify disqualification without further thorough cardiovascular investigation. Note 2: Guidance on resting and exercise electrocardiography is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.3.2.7.1 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS g) The systolic and diastolic blood pressures shall be within normal limits. Note 1: The use of drugs for control of high blood pressure is disqualifying except for those drugs. the use of which, according to accredited medical conclusion is compatible with the safe exercise of the applicant's license and rating privileges. Note 2: Extensive guidance on the subject is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.3.2.9 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (i) There shall be no acute disability of the lungs, nor any active disease of the structures of the lungs, mediastinum or pleura.
- 6.3.2.10 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (k) Cases of chronic obstructive pulmonary disease should be assessed as unfit if the condition is causing symptoms.
- 6.3.2.11 Not implemented.
- 6.3.2.11.1 Not implemented.

- 6.3.2.12 PCAR Part 2,2.10.2.2.2 (l) first sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (l) Cases of active pulmonary tuberculosis, duly diagnosed, shall be assessed as unfit.
- 6.3.2.12.1 PCAR Part 2,2.10.2.2.2 (l) second sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (l) xxx. Cases of quiescent or healed lesions which are known to be tuberculous, or are presumably tuberculous in origin, may be assessed as fit. Note 1: Guidance material on assessment of respiratory diseases is published in the Manual of Civil Aviation Medicine (Doc 8984). Note 2: Guidance material on hazards of the medications is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.3.2.13 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (m) Cases of disabling disease with important impairment of function of the gastrointestinal tract or its adnexae shall be assessed as unfit.
- 6.3.2.13.1 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (n) The applicant shall be required to be completely free from those hernias that might give rise to incapacitating symptoms.
- 6.3.2.16.1 PCAR Part 2,2.10.2.2.2(q) second sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (q) Cases of insulin-treated diabetes mellitus shall be assessed as unfit. Proven cases of non-insulin treated diabetes mellitus shall be assessed as unfit unless the condition is shown to be satisfactorily controlled by diet alone or by diet combined with oral anti diabetic medication, the use of which, according to the accredited medical conclusion are compatible with the safe exercise of the applicant's license and rating privileges.
- 6.3.2.17 PCAR Part 2,2.10.2.2.2(s)(1). (2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (s) Cases of significant localized and generalized enlargement of the lymphatic glands and of diseases of the blood shall be assessed as unfit, except in cases where accredited medical conclusion indicates that the condition is not likely to affect the safe exercise of the applicant's license and rating privileges. (1) Possession of the sickle cell trait should not be a reason for disqualification unless there is positive medical evidence to the contrary.
- 6.3.2.18 PCAR Part 2,2.10.2.2.2(t) first sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (t) Cases presenting any signs of organic disease of the kidney shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.
- 6.3.2.18.1 PCAR Part 2,2.10.2.2.2(t) second sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (t) Cases presenting any signs of organic disease of the kidney shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit. Urine examination shall form part of the medical examination and shall contain no abnormal element considered by the medical examiner to be of pathological significance. Cases of affections of the urinary passages and of the genital organs shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.
- 6.3.2.19 PCAR, Part 2, 2.10.2.2.2 (u) first sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (u) Any sequelae of disease or surgical procedures on the kidneys and the urinary tract likely to cause incapacity, in particular any obstructions due to stricture or compression, shall be assessed as unfit. Compensated nephrectomy without hypertension or uremia may be assessed as fit.
- 6.3.2.19.1 PCAR Part 2,2.10.2.2.2(u) second sentence and (u)(1). 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (u) xxx. Compensated nephrectomy without hypertension or uremia may be assessed as fit. (1) An applicant who has undergone a major surgical operation on the urinary system which has involved a total or partial excision or a

diversion of any of its organs should be assessed as unfit until such time as the medical Authority designated for the purpose by Republic of the Philippines and having access to the details of the operation concerned considers that the effects of the operation are not likely to cause in capacity in the air.

- 6.3.2.20 CAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (aa) Applicants with acquired immunodeficiency syndrome (AIDS) shall be assessed as unfit. Those who are seropositive for human immunodeficiency virus (HIV) shall be assessed as unfit unless full investigation provides no evidence of clinical disease.
- 6.3.2.21 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (w) Applicants who have a history of severe menstrual disturbances that have proved unamenable to treatment and that are likely to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.
- 6.3.2.22 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (y) Following confinement or termination of pregnancy, the applicant shall not be permitted to exercise the privileges of her license until she has undergone reexamination and has been assessed as fit.
- 6.3.2.23 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (z) Any active disease of the bones, joints, muscles or tendons and all serious functional sequelae of congenital or acquired disease shall be assessed as unfit. Functions after-effects of lesion affecting the bones, joints, muscles or tendons and certain anatomical defects compatible with the safe exercise of the applicants license and rating privileges may be assessed as fit.
- 6.3.2.24 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (bb) There shall be: (1) no active pathological process, acute or chronic, of the internal ear or of the middle ear.
- 6.3.2.25 PCAR Part 2, 2.10.2.2.2 (bb)(2)(3)(4) and Part 2, 2.10.2.4.2. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (bb) There shall be: (2) no unhealed (unclosed) perforation of the tympanic membranes. A single dry perforation need not render the applicant unfit. Licenses shall not be issued or renewed in these circumstances unless the appropriate hearing requirements in Subpart 2.10.2.2.4 are complied with; (3) no permanent obstruction of the Eustachian tubes; (4) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in Subpart 2.10.2.4.
- 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (y) There shall be: (1) no active pathological process, acute or chronic, of the internal ear or of the middle ear; (2) no unhealed (unclosed) perforations of the tympanic membrane. A single dry perforation need not render the applicant unfit. License shall not be issued or renewed in these circumstances unless the appropriate hearing requirements in Subpart 2.10.2.2.4 are complied with; (3) no permanent obstruction of the Eustachian tubes; and (4) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in Subpart 2.10.2.4.4.
- 6.3.2.25.1 PCAR Part 2, 2.10.2.2.2 (bb) (2)second sentence and (bb)(4). 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (bb) There shall be: (2) no unhealed (unclosed) perforation of the tympanic membranes. A single dry perforation need not render the applicant unfit. Licenses shall not be issued or renewed in these circumstances unless the appropriate hearing requirements in Subpart 2.10.2.2.4 are complied with; (4) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in Subpart 2.10.2.4.

- 6.3.2.26 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (cc) There shall be free nasal air entry on both sides. There shall be no serious malformation, nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract. Cases of speech defects and stuttering shall be assessed as unfit.
- 6.3.2.27 CAR Part 2, 2.10.2.2.2(cc) last sentence. 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (cc) There shall be free nasal air entry on both sides. There shall be no serious malformation, nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract. Cases of speech defects and stuttering shall be assessed as unfit.
- 6.3.3.6.1 PCAR Part 2, 2.10.2.2.3 VISUAL REQUIREMENTS (h) The applicant shall be required to have normal binocular function. Note: Defective stereopsis, abnormal convergence not interfering with near vision, and ocular misalignment where the fusional reserves are sufficient to prevent asthenopia and diplopia may not be disqualifying.
- 6.3.4.1 PCAR Part 2, 2.10.2.2.4 HEARING REQUIREMENTS The medical examination shall be based on the following requirements. a) The applicant, tested on a pure-tone audiometer at first issue of license, not less than once every five years up to the age of 40 years, and thereafter not less than once every three years, shall not have a hearing loss in either ear separately, of more than 35 dB at any of the frequencies 500, 1000 or 2000 HZ, or more than 50 dB at 3000 HZ. However, an applicant with a hearing loss greater than the above may be declared fit provided that:
- 6.3.4.1.1 PCAR Part 2, 2.10.2.2.4 (a) the last sentence and (1). 2.10.2.2.4 HEARING REQUIREMENTS (a) xxx. However, an applicant with a hearing loss greater than the above may be declared fit provided that: (1) the applicant has a hearing performance in each ear separately equivalent to that of a normal person, against a background noise that will simulate the masking properties of flight deck noise upon speech and beacon signals;
- 6.3.4.1.2 PCAR Part 2, 2.10.2.2.4 HEARING REQUIREMENTS (b) Alternatively, other methods providing equivalent results to those specified in paragraph (a) shall be used.
- 6.4.1.1 PCAR Part 2, 2.10.2.3.1 CERTIFICATE ISSUE AND RENEWAL (a) An applicant for a PPL, SPL, FOO and a Glider Pilot license or a Free Balloon Pilot license shall undergo an initial medical examination for the issue of a Class 2 Medical Certificate.
- 6.4.1.2 PCAR Part 2, 2.10.2.3.1 CERTIFICATE ISSUE AND RENEWAL (b) Except where otherwise stated in this subpart, holders of a PPL, SPL, FOO and a Glider Pilot license or a Free Balloon Pilot license shall have their Class 2 Medical Certificate renewed at intervals not exceeding those specified in Subpart 2.10.1.8 (c).
- 6.4.1.3 PCAR Part 2, 2.10.2.3.1 CERTIFICATE ISSUE AND RENEWAL (c) A Class 2 Medical Certificate will be issued when the applicant complies with the requirements of this Part.
- 6.4.2.2 PCAR Part 2, 2.10.2.3.2 (b) The applicant shall have no established medical history or clinical diagnosis of: (1) a psychosis; (2) alcoholism; (3) drug dependence; (4) any personality disorder, particularly if severe enough to have repeatedly resulted in overt acts; (5) a mental abnormality, or neurosis of a significant degree; (6) such as might render the applicant unable to safely exercise the privileges of the license applied for or held, unless accredited medical conclusion indicates that in special circumstances, the applicant's failure to meet the requirement is such that exercise of the privileges of the license applied for is not likely to jeopardize flight safety.
- 6.4.2.4 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (e) Cases of head injury, the effects of which, according to accredited medical conclusion, are likely

to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.

- 6.4.2.5 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (f) The applicant shall not possess any abnormality of the heart, congenital or acquired, which is likely to interfere with the safe exercise of the applicant's license and rating privileges. A history of proven myocardial infarction shall be disqualifying. Note: Such commonly occurring conditions as respiratory arrhythmia, occasional extrasystoles which disappear on exercise, increase of pulse rate from excitement or exercise, or a slow pulse not associated with auriculoventricular dissociation may be regarded as being "within normal" limits.
- 6.4.2.5.2 Not implemented.
- 6.4.2.6 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (f)(1) Electrocardiography should form part of the heart examination for the first issue of a license, at the first re-examination after the ages 30 and 40, no less frequently than every two years, and thereafter no less frequently than annually. Note 1: The purpose of routine electrocardiography is case finding. It does not provide sufficient evidence to justify disqualification without further thorough cardiovascular investigation. Note 2: Guidance on resting and exercise electrocardiography is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.4.2.6.1 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (f)(1) Electrocardiography should form part of the heart examination for the first issue of a license, at the first re-examination after the ages 30 and 40, no less frequently than every two years, and thereafter no less frequently than annually. Note 1: The purpose of routine electrocardiography is case finding. It does not provide sufficient evidence to justify disqualification without further thorough cardiovascular investigation. Note 2: Guidance on resting and exercise electrocardiography is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.4.2.7.1 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (g) The systolic and diastolic blood pressures shall be within normal limits. Note 1: The use of drugs for control of high blood pressure is disqualifying except for those drugs, the use of which, according to accredited medical conclusion is compatible with the safe exercise of the applicant's license and rating privileges. Note 2: Extensive guidance on the subject is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.4.2.8 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (h) There shall be no significant functional nor structural abnormality of the circulatory tree. The presence of varicosities does not necessarily entail unfitness.
- 6.4.2.9 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (i) There shall be no acute disability of the lungs nor any active disease of the structures of the lungs, mediastinum or pleura. Radiography shall form a part of the medical examination in all doubtful clinical cases.
- 6.4.2.10 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (k) Cases of chronic obstructive pulmonary disease should be assessed as unfit if the condition is causing symptoms.
- 6.4.2.11 Not implemented.
- 6.4.2.11.1 Not implemented.
- 6.4.2.12 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (l) Cases of active pulmonary tuberculosis, duly diagnosed, shall be assessed as unfit. Cases of quiescent or healed lesions which are known to be tuberculous, or are presumably tuberculous in origin, may be assessed as fit. Note 1: Guidance material on

- assessment of respiratory diseases is published in the Manual of Civil Aviation Medicine (Doc 8984). Note 2: Guidance material on hazards of the medications is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.4.2.12.1 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (l) Cases of active pulmonary tuberculosis, duly diagnosed, shall be assessed as unfit. Cases of quiescent or healed lesions which are known to be tuberculous, or are presumably tuberculous in origin, may be assessed as fit. Note 1: Guidance material on assessment of respiratory diseases is published in the Manual of Civil Aviation Medicine (Doc 8984). Note 2: Guidance material on hazards of the medications is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.4.2.13.1 PCAR Part 2, 2.10.2.3.2(m) Cases of disabling disease with important impairment of function of the gastrointestinal tract or its adnexae shall be assessed as unfit.
- 6.4.2.15 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (o) Cases of metabolic, nutritional or endocrine disorders likely to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.
- 6.4.2.16.1 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (p) Cases of insulin-treated diabetes mellitus shall be assessed as unfit. Proven cases of diabetes mellitus shown to be satisfactorily controlled without the use of any anti-diabetic drug may be assessed as fit. The use of anti-diabetic drugs for the control of diabetes mellitus is disqualifying except for those oral drugs administered under conditions permitting appropriate medical supervision and control and which according to accredited medical conclusion, are compatible with the safe exercise of the applicant's license and rating privileges.
- 6.4.2.17 PCAR Part 2, 2.10.2.3.2(r)(1). 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (r) Cases of significant localized and generalized enlargement of the lymphatic glands and of diseases of the blood shall be assessed as unfit, except in cases where accredited medical conclusion indicates that the condition is not likely to affect the safe exercise of the applicant's license and rating privileges. (1) Possession of the sickle cell trait should not be a reason for disqualification unless there is positive medical evidence to the contrary.
- 6.4.2.18 PCAR Part 2, 2.10.2.3.2 (s) first sentence. 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (s) Cases presenting any signs of organic disease of the kidney shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.
- 6.4.2.18.1 PCAR Part 2, 2.10.2.3.2 (s) from second sentence. 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (s) xxx. Urine examination shall form part of the medical examination and shall contain no abnormal element considered by the medical examiner to be of pathological significance. Cases of affections of the urinary passages and of the genital organs shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.
- 6.4.2.19 PCAR Part 2, 2.10.2.3.2 (t) first sentence. 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (t) Any sequelae of disease or surgical procedures on the kidneys and the urinary tract likely to cause incapacity, in particular any obstructions due to stricture or compression, shall be assessed as unfit. Compensated nephrectomy without hypertension or uremia may be assessed as fit.
- 6.4.2.19.1 PCAR Part 2, 2.10.2.3.2 (t) 2nd sentence and (1). 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (t) Any sequelae of disease or surgical procedures on the kidneys and the urinary tract likely to cause incapacity, in particular any obstructions due to stricture or compression, shall be assessed as unfit. Compensated nephrectomy without hypertension or uremia may be assessed as fit. (1) An applicant who has

- undergone a major surgical operation on the urinary system which has involved a total or partial excision or a diversion of any of its organs should be assessed as unfit until such time as the medical Authority designated for the purpose by the Republic of the Philippines and having access to the details of the operation concerned considers that the effects of the operation are not likely to cause incapacity in the air.
- 6.4.2.20 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (z) Applicants with acquired immunodeficiency syndrome (AIDS) shall be assessed as unfit. Those who are seropositive for human immunodeficiency virus (HIV) shall be assessed as unfit unless full investigation provides no evidence of clinical disease.
- 6.4.2.21 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (x) Following confinement or termination of pregnancy, the applicant shall not be permitted to exercise the privileges of her license until she has undergone re-examination and has been assessed as fit.
- 6.4.2.22 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (y) Any active disease of the bones, joints, muscles or tendons and all serious functional sequelae of congenital or acquired disease shall be assessed as unfit. Certain qualifying functional after-effects of lesion affecting the bones, joints, muscles or tendons and certain anatomical defects compatible with the safe exercise of the applicants license and rating privileges may be assessed as fit.
- 6.4.2.23 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (z) Applicants with acquired immunodeficiency syndrome (AIDS) shall be assessed as unfit. Those who are seropositive for human immunodeficiency virus (HIV) shall be assessed as unfit unless full investigation provides no evidence of clinical disease.
- 6.4.2.24 PCAR 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (aa) There shall be: (1) no active pathological process, acute or chronic, of the internal ear or of the middle ear;
- 6.4.2.25 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (aa) There shall be: (2) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit.
- 6.4.2.25.1 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (aa) There shall be: (2) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in 2.10.2.3.4. (cc) There shall be no serious malformation nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract.
- 6.4.2.26 PCAR Part 2, 2.10.2.3.2 PHYSICAL AND MENTAL REQUIREMENTS (cc) There shall be no serious malformation nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract.
- 6.4.2.27 PCAR Part 2, 2.10.2.2.2 PHYSICAL AND MENTAL REQUIREMENTS (cc) There shall be free nasal air entry on both sides. There shall be no serious malformation, nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract. Cases of speech defects and stuttering shall be assessed as unfit.
- 6.4.4.1 PCAR Part 2, 2.10.2.3.4(a) (ii). 2.10.2.3.4 HEARING REQUIREMENTS (a) The medical examination shall be based on the following requirements: (ii) the applicant has the ability to hear an average conversational voice in a quiet room, using both ears, at a distance of 2M from the examiner, with the back turned to the examiner.
- 6.4.4.2 PCAR Part 2, 2.10.2.3.4(a) (1). 2.10.2.3.4 HEARING REQUIREMENTS (a) The medical examination shall be based on the following requirements: (1) The applicant tested on a pure-tone audiometer at first issue of license, and every three (3) years after the age of 50 years, shall not have a hearing loss in either ear separately, of more

- than 35 dB at any of the frequencies 500, 1000 or 2000 HZ, or more than 50 dB at 3000 HZ.
- 6.5.1.3 PCAR Part 2, 2.10.2.4.1 CERTIFICATE ISSUE AND RENEWAL (c) A Class 3 Medical Certificate will be issued when the applicant complies with the requirements of this Part.
- 6.5.2.2 PCAR Part 2, 2.10.2.4.2 (b) The applicant shall have no established medical history or clinical diagnosis of: (1) a psychosis; (2) alcoholism; (3) drug dependence, (4) any personality disorder, particularly if severe enough to have repeatedly resulted in overt acts; (5) a mental abnormality, or neurosis of a significant degree, (i) such as might render the applicant unable to safely exercise the privileges of the license applied for or held, unless accredited medical conclusion indicates that in special circumstances, the applicant's failure to meet the requirement is such that exercise of the privileges of the license applied for is not likely to jeopardize flight safety.
- 6.5.2.4 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (e) Cases of head injury, the effects of which, according to accredited medical conclusion, are likely to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.
- 6.5.2.5.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (f) (2) An applicant who has undergone coronary bypass, grafting or angioplasty (with or without stenting) or other cardiac intervention or who suffers from any other potentially incapacitating cardiac condition shall be assessed unfit, unless the applicant's cardiac condition has been investigated and evaluated in accordance with best medical practices and is assessed not likely to interfere with the safe exercise of the applicant's license or rating privileges.
- 6.5.2.5.2 Not implemented.
- 6.5.2.6.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (f) (1) Electrocardiography shall form part of the heart examination for the first issue of a license, and shall be included in re-examination of the applicant between the ages 30 and 40 no less frequently than every two years, and thereafter no less frequently than annually. Note 1: The purpose of routine electrocardiography is case finding. It does not provide sufficient evidence to justify disqualification without further thorough cardiovascular investigation. Note 2: Guidance on resting and exercise electrocardiography is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.5.2.7.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (g) The systolic and diastolic blood pressures shall be within normal limits. Note 1: The use of drugs for control of high blood pressure is disqualifying except for those drugs, the use of which, according to accredited medical conclusion is compatible with the safe exercise of the applicant's license and rating privileges. Note 2: Extensive guidance on the subject is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.5.2.8 PCAR Part 2, 2.10.2.4.2 (h) first sentence. 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (h) There shall be no significant functional nor structural abnormality of the circulatory tree.
- 6.5.2.9 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (i) There shall be no acute disability of the lungs nor any active disease of the structures of the lungs, mediastinum or pleura. Radiography shall form a part of the medical examination in all doubtful clinical cases.
- 6.5.2.10 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (j) Cases of chronic obstructive pulmonary disease should be assessed as unfit if the condition is causing symptoms.

- 6.5.2.11 Not implemented.
- 6.5.2.11.1 Not implemented.
- 6.5.2.12.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (k) Cases of active pulmonary tuberculosis, duly diagnosed, shall be assessed as unfit. Cases of quiescent or healed lesions which are known to be tuberculous, or are presumably tuberculous in origin, may be assessed as fit. Note 1: Guidance material on assessment of respiratory diseases is published in the Manual of Civil Aviation Medicine (Doc 8984). Note 2: Guidance material on hazards of the medications is published in the Manual of Civil Aviation Medicine (Doc 8984).
- 6.5.2.13 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (l) Cases of disabling disease with important impairment of function of the gastrointestinal tract or its adnexae shall be assessed as unfit.
- 6.5.2.14 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (n) Any sequelae of disease or surgical intervention on any part of the digestive tract or its adnexae, likely to cause incapacity in flight, in particular any obstructions due to structure or compression shall be assessed as unfit.
- 6.5.2.16 PCAR Part 2, 2.10.2.4.2 (p) first sentence. 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (p) Cases of insulin treated diabetes mellitus shall be assessed as unfit. Proven cases of diabetes mellitus shown to be satisfactorily controlled without the use of any anti-diabetic drug may be assessed as fit. The use of anti-diabetic drugs for the control of diabetes mellitus is disqualifying except for those oral drugs administered under conditions permitting appropriate medical supervision and control and which according to accredited medical conclusion, are compatible with the safe exercise of the applicant's license and rating privileges.
- 6.5.2.16.1 PCAR Part 2, 2.10.2.4.2 (p) from second sentence. 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (p) Proven cases of diabetes mellitus shown to be satisfactorily controlled without the use of any anti-diabetic drug, may be assessed as fit. The use of anti-diabetic drugs for the control of diabetes mellitus is disqualifying except for those oral drugs administered under conditions permitting appropriate medical supervision and control and which according to accredited medical conclusion, are compatible with the safe exercise of the applicant's license and rating privileges.
- 6.5.2.17 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (q) Cases of significant localized and generalized enlargement of the lymphatic glands and of diseases of the blood shall be assessed as unfit, except in cases where accredited medical conclusion indicates that the condition is not likely to affect the safe exercise of the applicant's license and rating privileges.
- 6.5.2.18 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (r) Cases presenting any signs of organic disease of the kidney shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit. Urine examination shall form part of the medical examination and shall contain no abnormal element considered by the medical examiner to be of pathological significance. Cases of affections of the urinary passages and of the genital organs shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.
- 6.5.2.18.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (r) Cases presenting any signs of organic disease of the kidney shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit. Urine examination shall form part of the medical examination and shall contain no abnormal element considered by the medical examiner to be of pathological significance. Cases of affections of the urinary passages and of the genital organs shall be assessed as unfit; those due to a transient condition may be assessed as temporarily unfit.

- 6.5.2.19 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (s) Any sequelae of disease or surgical procedures on the kidneys and the urinary tract likely to cause incapacity, in particular any obstructions due to stricture or compression, shall be assessed as unfit. Compensated nephrectomy without hypertension or uremia may be assessed as fit.
- 6.5.2.19.1 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (s) Any sequelae of disease or surgical procedures on the kidneys and the urinary tract likely to cause incapacity, in particular any obstructions due to stricture or compression, shall be assessed as unfit. Compensated nephrectomy without hypertension or uremia may be assessed as fit.
- 6.5.2.20 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (x) Applicants with acquired immunodeficiency syndrome (AIDS) shall be assessed as unfit. Those who are seropositive for human immunodeficiency virus (HIV) shall be assessed as unfit unless full investigation provides no evidence of clinical disease.
- 6.5.2.21 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (u) Applicants who have a history of severe menstrual disturbances that have proved unamenable to treatment and that are likely to interfere with the safe exercise of the applicant's license and rating privileges shall be assessed as unfit.
- 6.5.2.22 PCAR Part 2, (2) 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (v) (2) Following confinement or termination of pregnancy, the applicant shall not be permitted to exercise the privileges of her license until she has undergone re-examination and has been assessed as fit.
- 6.5.2.23 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (w) Any active disease of the bones, joints, muscles or tendons and all serious functional sequelae of congenital or acquired disease shall be assessed as unfit. Functional after-effects of lesion affecting the bones, joints, muscles or tendons and certain anatomical defects compatible with the safe exercise of the applicants license and rating privileges may be assessed as fit.
- 6.5.2.24 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (y) There shall be: (1) no active pathological process, acute or chronic, of the internal ear or of the middle ear; (2) no unhealed (unclosed) perforations of the tympanic membrane. A single dry perforation need not render the applicant unfit. License shall not be issued or renewed in these circumstances unless the appropriate hearing requirements in Subpart 2.10.2.2.4 are complied with; (3) no permanent obstruction of the Eustachian tubes; and (4) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in Subpart 2.10.2.4.4.
- 6.5.2.25 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (y) There shall be: (1) no active pathological process, acute or chronic, of the internal ear or of the middle ear; (2) no unhealed (unclosed) perforations of the tympanic membrane. A single dry perforation need not render the applicant unfit. License shall not be issued or renewed in these circumstances unless the appropriate hearing requirements in Subpart 2.10.2.2.4 are complied with; (3) no permanent obstruction of the Eustachian tubes; and (4) no permanent disturbances of the vestibular apparatus. Transient conditions may be assessed as temporarily unfit. Note: The details of the hearing requirements are set out in Subpart 2.10.2.4.4.
- 6.5.2.26 PCAR Part 2, 2.10.2.4.2 PHYSICAL AND MENTAL REQUIREMENTS (z) There shall be no serious malformation, nor serious, acute or chronic affection of the buccal cavity or upper respiratory tract. Cases of speech defects and stuttering shall be assessed as unfit.

- 6.5.3.6.1 PCAR Part 2, 2.10.2.4.3 VISUAL REQUIREMENTS (i) The applicant shall be required to have normal binocular function. Note: Defective stereopsis: abnormal convergence not interfering with near vision, and ocular misalignment where the fusional reserves are sufficient to prevent asthenopia and diplopia may not be disqualifying.
- 6.5.4.1 PCAR Part 2, 2.10.2.4.4 HEARING REQUIREMENTS. The medical examination shall be based on the following requirements. (a) The applicant, tested on a pure-tone audiometer at first issue of license, not less than once every five years up to the age of 40 years, and thereafter not less than once every three years, shall not have a hearing loss in either ear separately, of more than 35 dB at any of the frequencies 500, 1000 or 2000 HZ, or more than 50 dB at 3000 HZ.
- 6.5.4.1.2 PCAR Part 2, 2.10.2.4.4 HEARING REQUIREMENTS (b) Alternatively, other methods providing equivalent results to those specified in (a) shall be used.

2. Annex 2 - Rules of the Air, 10th Edition, July 2005, Amendment 47.

DOC 4444 - PANS-ATM/501 - PROCEDURES FOR AIR NAVIGATION SERVICES - AIR TRAFFIC MANAGEMENT, 16th Edition, 2016, Amendment 10.

3. Annex 3 - Meteorological Service for International Air Navigation, 20th Edition, July 2018, Amendment 80.

Chapter 2

- 2.1.4 PAGASA (Philippine Atmospheric, Geophysical and Astronomical Services) as the Philippines' designated Meteorological Authority, in coordination with the Civil Aviation Authority of the Philippines (CAAP), hereby assumed the mandate to provide or to arrange for the provision of meteorological service for air navigation on its behalf. Details of the meteorological authority so designated shall be included in the Aeronautical Information Publication (AIP).
- 2.1.5 Civil Aviation Authority of the Philippines (CAAP) ensures that the designated MET Authority complies with the requirements of the WMO, in respect of qualifications, competencies, education and training of meteorological personnel providing service for air navigation.
- 2.2.3 To standardize the quality management and assurance systems, the ISO 9000 series shall be followed in conjunction with ICAO Doc 9873 - Manual on the Quality System for the Provision of Meteorological Service for International Air Navigation.
- 2.2.4 The quality system is mandatory to the MET authority to supply correct and timely information to users.
- 2.2.5 Verification and validation of procedures are needed for the exchange of MET information for operational purposes.

Chapter 3

- 3.1 For global use in the MET information. This is one of the responsibilities of PAGASA is supplying users for international air navigation.
- 3.2.1 There is no WAFC in the Philippines. PAGASA coordinates with WAFC in terms of global MET information.
- 3.2.2 In case of interruption of the operations of a WAFC, its function shall be carried out by the other WAFC.

Note: Back-up procedures to be used in case of interruption of the operation of a WAFC are updated by the World Area Forecast System Operations Group

(WAFSOPSG) as necessary; the latest revision can be found on the ICAO WAFSOPSG website.

- 3.3.3 A function of PAGASA in all international airports.
- 3.4.1 Close coordination between CAAP and PAGASA in terms of MET information in most airports including international airports.
- 3.4.3 The MET Authority (PAGASA) covers coincidentally the boundaries of the Manila FIR. The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.5.1 This function is under PHIVOLCS. Tri-partite agreement between CAAP, PAGASA and PHIVOLCS for coordination and contingency procedures.
- 3.6 PHILVOCS monitor these volcanoes, information are disseminated to all stakeholders. A tri-partite agreement between CAAP, PAGASA and PHIVOLCS for coordination and information sharing.
- 3.7 The meteorological authority of the Philippines disseminates tropical cyclone advisory from associated TCAC. Advisories are disseminated to concerned agencies.

Chapter 4

- 4.1.1 Part of the service provision of PAGASA in support of the MET information to air navigation. PAGASA stations are located in international and some domestic airports.
- 4.1.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.2. An amended/updated Letter-of-Agreement (LOA) between CAA of the Philippines (ATS) and PAGASA (MET) has been signed in December 15, 2016. This LOA forms part of the additional obligations between agencies to enhance coordination procedures to provide MET information to air navigation and vice versa.
- 4.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.5.3 There is no information regarding Optional elements under supplementary information in the METAR/SPECI issued by MET Authority (PAGASA).
- 4.6.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations. Wind sensors are installed within the TDZ - only in RPLL, RPLB and RPLC.
- 4.6.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.2.3 The recommendation was adopted by mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.
- 4.6.3.1 A requirement for flight operations during landing and take-off phases of the aircraft adopted as mandatory requirement to Philippine regulations.
- 4.6.3.2 Important safety requirements for all 4.6.3.2 categories. Adopted as mandatory requirement to Philippine regulations.
- 4.6.3.3 Safety requirements for flight operations. The recommendation was adopted to mandatory requirements in the Philippine regulations. RPLL and RPKV RVR are operational, all other airports no RVR. Will be included in the audit and inspection checklist.

- 4.6.3.4 Safety requirements for flight operations. The recommendation was adopted to mandatory requirements in the Philippine regulations. RPLL and RPVK RVR are operational, all other airports no RVR. Will be included in the audit and inspection checklist.
- 4.6.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.5.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.6.8 The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.
- 4.7.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.7.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.8.0.3 Safety requirement for flight operations. Provision in the Letter of Agreement between CAAP (ATS) and PAGASA (MET) in terms of responsibilities and coordination. The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.

Chapter 5

- 5.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations. Will be included in the audit and inspection checklist.
- 5.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 6

- 6.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.4.2 Safety requirement for flight operations. Provision in the Letter of Agreement between CAAP (ATS) and PAGASA (MET) service providers in terms of responsibilities and procedures for flight operations. The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.
- 6.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.
- 6.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations. Forecasting is provided by NAIA MET Office and disseminated to various international and domestic airports.
- 6.5.1 There is no GAMET forecast issued by MET 6.5.1 Authority (PAGASA). GAMET is still on developing stages.

Chapter 7

- 7.1.4 SIGMET messages issued concerning volcanic ash cloud and tropical cyclones shall be based on advisory information provided by VAACs and TCACs respectively. SIGMET TC is primarily based from PAGASA forecasting center.
- 7.2.1 AIRMET information shall be issued by a meteorological watch office in accordance with regional air navigation agreement, taking into account the density of air traffic operating below flight level 100. AIRMET information shall give a concise description in abbreviated plain language concerning the occurrence and/or expected occurrence of specified en-route weather phenomena and may affect safety of flights, and of the development of those phenomena in time and space.
- 7.3.1 Safety requirement for flight operations. Included in the audit checklist. Provisions of information is included in the Letter of Agreement between CAAP and PAGASA signed in 15 December 2016.
- 7.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.4.1 Significant information for low-level flights. Currently, PAGASA has no wind shear detection equipment. It will be developed soon as part of PAGASA's Modernization Program.
- 7.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations. A proposal in the PAGASA Modernization Program.
- 7.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit and inspection checklist.

Chapter 8

- 8.1.1 No information regarding the issuance of Aeronautical climatological information by the MET Authority (PAGASA) however, this provision is internal within the MET Authority (PAGASA). Aeronautical climatological information will be included in the audit checklist.
- 8.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations. Will be included in the audit and inspection checklist.
- 8.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations. Will be included in the audit and inspection checklist.
- 8.2.0.1 The recommendation was adopted to mandatory requirements in the Philippine regulations. Will be included in the audit and inspection checklist.
- 8.3.0.2 The recommendation was adopted to mandatory requirements in the Philippine regulations. Will be included in the audit and inspection checklist.
- 8.4 There is no information regarding the application of meteorology to international air navigation, meteorological observation data required for research, investigation or operational analysis between MET Authority operators and others concerned.

Chapter 9

- 9.4.1 There is no information regarding automated pre-flight information systems to supply and display of meteorological information to operators and flight and flight crew members for self-briefing, flight planning and flight documentation purposes. Proper consultation with the operators shall be established. Included in the audit checklist.
- 9.4.2 There is no information regarding automated pre-flight information systems providing for a harmonized, common point of access to meteorological information and aeronautical information services by operators, flight crew members and other

aeronautical personnel concerned. Proper consultation with the operators shall be established. Included in the audit checklist.

- 9.5.3 There is no information regarding meteorological information supplied through D-VOLMET or VOLMET broadcasts.

Chapter 10

- 10.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations. Provision in the Letter of Agreement between CAAP and PAGASA signed in 15 December 2016. Included in the audit checklist. Covered by PAGASA internal Memorandum. Included in the audit checklist.

- 10.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations. Internal to PAGASA memorandum. Included in the audit checklist.

Chapter 11

- 11.1.1 Limited number of aerodrome meteorological offices and aeronautical meteorological stations are connected with CAAP AFTN facility. This provision will be included in the audit checklist.

- 11.1.2 Limited number of aerodrome meteorological offices and aeronautical meteorological stations are connected with CAAP AFTN facility. This provision will be included in the audit checklist.

- 11.1.5 The recommendation was adopted to mandatory requirements in the Philippine regulations. Provision in the Letter of Agreement between CAAP and PAGASA signed in 15 December 2016. Included in the audit checklist.

- 11.1.6 The recommendation was adopted to mandatory requirements in the Philippine regulations. To be included in the audit checklist.

- 11.1.7 The recommendation was adopted to mandatory requirements in the Philippine regulations. Included in the audit checklist.

- 11.1.9 The recommendation was adopted to mandatory requirements in the Philippine regulations. PAGASA is connected to CAAP AFTN network. Included in the audit checklist.

- 11.3.0.1 Essential information used for air navigation. CAAP is a recipient of PAGASA's direct coordination with WAFC. Reception of WAFS products is in place. Connectivity with the AFTN network provides the essential information in relation to flight operations.

4. Annex 4 - Aeronautical Charts, 11th Edition, July 2009, Amendment 61.

DOC 8697 - AN/669 - Aeronautical Chart Manual

Chapter 1

- 1.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 2

Some charts will deviate slightly from specifications embodied in the provisions.

- 2.1.8 The basic sheet charts shall be 210 x 297 MM (8.27 2.1.8 x 11.69 inches) (A4). A5 size cannot clearly accommodate the aeronautical information provided in the charts, thus the use of A4 is preferred. Furthermore, ICAO Annex 4 recommendations adopted to mandatory requirements in Philippine regulations.

- 2.4.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.11.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.12.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.14.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.15.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.15.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.15.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.5.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.8.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.8.4.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.8.4.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.9.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.9.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.9.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.9.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.9.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.10.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 5

- 5.5.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.5.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.5.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.5.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.7.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 6

6.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.5.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 7

7.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

7.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

7.6.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

7.7.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

7.8.1 Bearings, tracks and radials shall be magnetic. Where bearings and tracks are additionally provided as true values for RNAV segments, they shall be shown in parentheses to the nearest tenth of a degree, e.g. 290° (294.9°T).

Chapter 8

8.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

8.8.1 Bearings, tracks and radials shall be magnetic. Where bearings and tracks are additionally provided as true values for RNAV segments, they shall be shown in parentheses to the nearest tenth of a degree, e.g. 290° (294.9°T).

Chapter 9

9.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

9.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

9.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

9.8.1 Bearings, tracks and radials shall be magnetic. Where bearings and tracks are additionally provided as true values for RNAV segments, they shall be shown in parentheses to the nearest tenth of a degree, e.g. 290° (294.9°T).

9.9.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 10

- 10.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 10.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 10.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 10.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 10.8.1 Bearings, tracks and radials shall be magnetic. Where bearings and tracks are additionally provided as true values for RNAV segments, they shall be shown in parentheses to the nearest tenth of a degree, e.g. 290° (294.9°T).
- 10.9.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 11

- 11.3.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.4.1 The sheet size may be 210 x 148 MM (8.27 x 5.82 inches).
- 11.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.7.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.8.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.9.1 Bearings, tracks and radials shall be magnetic. Where bearings and tracks are additionally provided as true values for RNAV segments, they shall be shown in parentheses to the nearest tenth of a degree, e.g. 290° (294.9°T).
- 11.10.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.6.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.8.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.10.8.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 12

- 12.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 12.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 12.4.1 The sheet size shall be 210 x 297 MM (8.27 x 11.69 inches). It would be advantageous to print the charts in several colors, selected to provide maximum legibility in varying degrees and kinds of light.
- 12.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 12.7.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 12.7.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 12.9.1 Bearings, tracks and radials shall be magnetic. Annex 4 para 12.9.2 not applied since the Philippines is located in lower latitude.
- 12.10.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 13

- 13.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 14

- 14.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 14.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 14.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 15

- 15.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 15.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 15.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 16

- 16.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.3.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 16.4.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.5.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.6.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.9.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 16.7.11.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 17

- 17.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.3.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.4.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.5.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.5.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.6.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.7.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.7.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.7.6.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 17.7.9.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

17.7.11.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

17.7.12.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 18

18.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.3.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.6.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.9.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.11 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.12 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.14 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.6.14.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.8.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

18.8.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 19

19.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

19.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

19.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 19.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 19.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 19.5.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 19.7.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 19.9.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 21

- 21.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 21.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 21.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 21.8.1 Bearings, tracks and radials shall be magnetic. ICAO Annex 4 para 21.8.2 not applicable since the Philippines is located in lower latitude.

5. Annex 5 - Units of Measurement to be used in Air and Ground Operations, 5th Edition, July 2010, Amendment 17.

6. Annex 6 - Operation of Aircraft.

Part I - International Commercial Air Transport - Aeroplanes, 12th Edition, July 2022, Amendment 48

Chapter 2

- 2 PCAR Part 8, 8.1.1.1 Applicability (a) (1) The Standards and Recommended Practices contained in Annex 6 Part I shall be applicable to the operation of airplanes by operators authorized to conduct commercial air transport operations international and domestic. (2) The Standards and Recommended Practices contained in Annex 6, Part II (Airplanes), and Annex 6 Part III Section III (Helicopters) shall be applicable to general aviation operations (International and Domestic). (3) The Standards and Recommended Practices contained in Annex 6, Part III Sections I and II, shall be applicable to all helicopters engaged in commercial air transport operations (International and Domestic), except for these Standards and Recommended Practices are not applicable to helicopters in aerial work.

Chapter 3

- 3.1.7 PCAR Part 8, 8.5.1.10 (a) (b). 8.5.1.10 SEARCH AND RESCUE INFORMATION (a) An operator shall ensure that PIC has available on board the aircraft all the essential information concerning the search and rescue services in the area over which the aircraft will be flown; (b) For all flights, the PIC shall have available on board the aircraft essential information concerning the search and rescue services in the areas over which it is intended the aircraft will be flown.

Chapter 4

- 4.2.1.8 PCAR Part 9, 9.1.1.1 APPLICABILITY (b) This Part of the regulations prescribes requirements for the original certification and continued validity of air operator certificates (AOC) issued by Republic of the Philippines.
- 4.2.2.3 PCAR Part 10, 10.1.1.6 OPERATIONAL REQUIREMENTS (a) Subject to the applicable laws and regulations governing ownership and operations of radio equipment, each foreign air operator shall equip its aircraft with such radio equipment, as is necessary to properly use the air navigation facilities, and to maintain communications with ground stations, along and adjacent to their routes in the Philippines. (b) Each flight crew member shall be familiar with the applicable rules, the navigational and communication facilities, and the air traffic control and other procedures, of the areas to be traversed by him/her within Republic of the Philippines. (c) Each foreign air carrier shall establish procedures to ensure that each of the flight crew members has the knowledge required by paragraph (a) above, and shall check the ability of each of its flight crew member to operate safely in accordance with the applicable rules and procedures. (d) Each foreign air carrier shall conform to the practices, procedures and other requirements prescribed by the Authority for the Philippines air operators for the areas to be operate in. (e) Subject to applicable immigration laws and regulations, each foreign air operator shall furnish the ground personnel necessary to provide for two way voice communication between its aircraft and ground stations, at places where the Authority finds that voice communication is necessary and that communications can not be maintained in a language familiar with which ground station operators are familiar. (f) Each person furnished by a foreign air operator under paragraph (e) above, must be able to speak English and the language necessary to maintain communications with the aircraft concerned, and shall assist ground personnel in directing air traffic.
- 4.2.3.1 PCAR Part 9, 9.3.1.2 (c) (d). 9.3.1.2 OPERATIONS MANUAL (c) Each operator shall prepare and keep current an Operations Manual that contains the operator's procedures and policies for the use and guidance of its personnel. (d) Each operator shall issue the Operations Manual, or pertinent portions, together with all amendments and revisions to all personnel that are required to use it. The Operations Manual shall be amended or revised as is necessary to ensure that the information contained therein is kept up to date. All such amendments or revisions shall be issued to all personnel that are required to use this manual.
- 4.2.3.2 PCAR Part 9, 9.3.1.2 OPERATIONS MANUAL (e) No person may provide for use of its personnel in commercial air transport any Operations Manual, or portion of this manual, together with all amendments or revisions, which has not been reviewed and found acceptable or approved for the operator by the Authority. The operator shall incorporate in the Operations Manual such mandatory material as the Authority may require.
- 4.2.5 PCAR Part 8, 8.8.1.17 IN-FLIGHT SIMULATION OF ABNORMAL SITUATIONS (a) An operator shall ensure that when passengers or cargo are being carried, no emergency or abnormal situations shall be simulated.
- 4.2.8.1 PCAR Part 8, 8.8.1.7 INSTRUMENT APPROACH OPERATING MINIMA (b) An operator shall establish airport/heliport-operating minima for each airport/heliport to be used in operation, and shall approve the method of determination of such minima. That minima shall not be lower than any that may be established for such airports by the State in which the airport is located, except when specifically approved by that State.
- 4.2.8.2 PCAR Part 8, IS: 8.8.1.7 INSTRUMENT APPROACH OPERATING MINIMA (a) Each Operator establishing airport/heliport-operating minima shall have its method for

determining such minima approved by the Authority. (b) Each Operator's method for determining airport/heliport-operating minima which will apply to any particular operation, shall accurately account for: (1) the type, performance and handling characteristics of the aircraft; (2) the composition of the flight crew, their competence and experience; (3) the dimensions and characteristics of the runways selected for use; (4) the adequacy and performance of the available visual and non-visual ground aids; (5) the equipment available on the aircraft for the purpose of navigation and/or control of flight path during the approach to landing and the missed approach; (6) the obstacles in the approach and missed approach areas and the obstacle clearance altitude/height for the intended instrument approach procedures; (7) the means used to determine and report meteorological conditions; and (8) the obstacles in the climb-out areas and the necessary clearance margins; (9) the declared distances, for helicopters.

- 4.2.8.4 PCAR Part 8, 8.8.1.8 (a) (1) to (4). 8.8.1.8 CATEGORY II AND III OPERATIONS: GENERAL OPERATING RULES (a) No person may operate a civil aircraft in a Category II or III operation unless: (1) The PIC and CP of the aircraft hold the appropriate authorizations and ratings prescribed in Subpart 2.2.2. (2) Each flight crew member has adequate knowledge of, and familiarity with, the aircraft and the procedures to be used; (3) The instrument panel in front of the pilot who is controlling the aircraft has appropriate instrumentation for the type of flight control guidance system that is being used; and (4) RVR information is provided.
- 4.2.8.7 Not implemented.
- 4.2.10.1 PCAR Part 8, 8.6.2.13.1 (f) first sentence. 8.6.2.13.1 FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (f) Each operator shall maintain fuel and oil records to enable the Authority to ascertain that, for each flight, the requirements of Subparts 8.6.2.14 and 8.6.2.15 have been complied with. Fuel and oil records shall be retained by the operator for a period of three (3) months.
- 4.2.12.2 PCAR Part 8, 8.9.1.4 PASSENGER BRIEFING (b) The PIC shall ensure that all persons on board are aware of the locations and general manner of use of the principal emergency equipment carried for collective use. Note: For commercial air transport operations, the briefing shall contain all subjects approved /by the Authority for the specific operations conducted as included in the pertinent Operations Manual. Note: When cabin crew members are required in a commercial air transport operation, the PIC may delegate this responsibility, but shall ascertain that the proper briefing has been conducted prior to take-off.
- 4.2.12.3 PCAR Part 8, 8.9.1.5 IN-FLIGHT EMERGENCY INSTRUCTION (a) In an emergency during flight, the PIC shall ensure that all persons on board are instructed in such emergency action as may be appropriate to the circumstances. Note: When cabin crew members are required in a commercial air transport operation, the PIC may delegate this responsibility, but shall ascertain that the proper briefing has been conducted.
- 4.3.1 PCAR Part 8, 8.6.2.1 (a) (1) to (7). 8.6.2.1 AIRCRAFT AIRWORTHINESS AND SAFETY PRECAUTIONS (a) A flight, or series of flights, shall not be commenced until flight preparation forms have been completed certifying that the PIC is satisfied that: (1) the aircraft is airworthy, duly registered and that appropriate certificates with respect thereto are aboard the aircraft; (2) the instruments and equipment prescribed in Part 7 for the particular type of operation to be undertaken, are appropriate, taking in account the expected flight conditions, installed and sufficient for the flight; (3) any necessary maintenance has been performed and a maintenance release, if applicable, has been issued in respect to the aircraft; (4) the mass of the aircraft and center of gravity location are such that the flight can be conducted safely, taking into account the flight conditions expected; (5) any load carried is properly distributed and safely

secured; (6) a check has been completed indicating that the operating limitations, contained in the flight manual or its equivalent, can be complied with for the flight to be undertaken; and (7) the Standards of Subpart 8.6.2.16 relating to the operational flight planning have been complied with.

- 4.3.3.2 PCAR Part 9 - 9.3.1.2 OPERATIONS MANUAL (g) Unless otherwise acceptable to the Authority, each operator shall provide an Operations Manual containing information on operations administration and supervision, safety management programs, personnel training, flight crew and cabin crew flight duty period and rest period requirements, flight operations, operational flight planning including contents and use of operational flight plan, aircraft performance, routes, guides and charts, minimum flight altitudes, airport operating minima, search and rescue, dangerous goods, navigation, communications, security, and human factors. The operations manual shall encompass the matters set forth above. The operations manual may be published in parts, as a single document, or as a series of volumes. Specific subjects are listed below. Subjects presented with reference to a specific section shall be addressed in accordance with the requirements of the referenced section. (1) Aircraft Operating Manual. (9.3.1.4) (2) Minimum Equipment List and Configuration Deviation List. (9.3.1.12) (3) Training Program. (9.3.1.3) (4) Aircraft Performance Planning Manual. (9.3.1.13) (5) Route Guide. (9.3.1.20) (6) Dangerous Goods Procedures. (7) Accident Reporting Procedures. (8) Security Procedures. (9) Aircraft Loading and Handling Manual. (9.3.1.15) (10) Cabin Crew Safety Manual. (9.3.1.17) (h) The Operations Manual shall conform to the outline contained in IS: 9.3.1.2.
- 4.3.4.1.1 PCAR Part 8, 8.6.2.9 (a) No person may release or take-off an aircraft without a suitable take-off alternate specified in the flight release if either the meteorological conditions at the airport/heliport of departure are below the operator's established airport/heliport operating landing minima for that operation or if it would not be possible to return to the airport/heliport of departure for other reasons.
- 4.3.4.2 PCAR Part 8, 8.6.2.12 EN ROUTE ALTERNATE AIRPORTS: ETOPS OPERATIONS (a) The PIC shall ensure that the required en route alternates for EDTO are selected and specified in ATC flight plans in accordance with the EDTO diversion time approved by the Authority.
- 4.3.4.4 PCAR Part 8, 8.6.2.15.1 (e) (1) to (3). 8.6.2.15.1 COMMERCIAL AIR TRANSPORT: PISTON-ENGINED AIRPLANES (e) Notwithstanding the provisions in 8.6.2.15.1(c) (1) - (7) above, the Authority may approve a variation to these requirements provided the operator can demonstrate an equivalent level of safety will be maintained through a safety risk assessment that includes at least the following: (1) Flight fuel calculations; (2) Capabilities of the operator to include: (i) A data-driven method that includes a fuel consumption monitoring program; and/or (ii) The advanced use of alternate aerodromes; and (3) Specific mitigation measures. Note: ICAO Doc 9976, Flight Planning and Fuel Management Manual, contains guidance on the specific safety risk assessment, fuel consumption monitoring program and the advanced use of alternate aerodromes.
- 4.3.5.2 PCAR Part 8, 8.6.2.6.1 IFR DESTINATION ALTERNATE AIRPORT/HELIPORT For a flight to be conducted in accordance with the IFR, at least one destination alternate airport/heliport shall be selected and specified in the operational and ATS flight plans, unless: (a) the duration of the flight from the departure aerodrome, or from the point of in-flight re-planning to the destination aerodrome is such that, taking into account all meteorological conditions and operational information relevant to the flight, at the estimated time of use, a reasonable certainty exists that: (1) the approach and landing may be made under VMC, (for helicopters: the weather conditions in Subpart 8.6.2.6.2 prevails); and (2) separate runways are usable at the estimated time of use of the

destination aerodrome with at least one runway having an operational instrument approach procedure, in case of helicopter operations, a point of no return (PNR) shall be determined; or (b) the aerodrome is isolated. Operations into isolated aerodromes do not require the selection of a destination alternate aerodrome(s) and shall be planned in accordance with 8.6.2.15.1(c) (4) (D); (1) for each flight into an isolated aerodrome a point of no return shall be determined; and (2) a flight to be conducted to an isolated aerodrome shall not be continued past the point of no return unless a current assessment of meteorological conditions, traffic, and other operational conditions indicate that a safe landing can be made at the estimated time of use. Note 1: Separate runways are two or more runways at the same aerodrome configured such that if one runway is closed, operations to the other runway(s) can be conducted. Note 2: Guidance on planning operations to isolated aerodromes is contained in the Flight Planning and Fuel Management Manual (ICAO Doc 9976). (c) Two destination alternate aerodromes shall be selected and specified in the operational and ATS flight plans when, for the destination aerodrome: (1) meteorological conditions at the estimated time of use will be below the operator's established aerodrome operating minima for that operation; or meteorological information is not available. (d) For a heliport to be selected as a destination alternate, the available information shall indicate that, at the estimated time of use, the conditions will be at or above the heliport operating minima for that operation. (e) For helicopters: For a flight departing to a destination which is forecast to be below the heliport operating minima, two destination alternates should be selected. The first destination alternate should be at or above the heliport operating minima for destination and the second at or above the heliport operating minima for alternate.

- 4.3.6.1 PCAR Part 8, 8.6.2.13.1 FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (a) A flight shall not be commenced unless, taking into account both the meteorological conditions and any delays that are expected in flight, the aircraft carries sufficient fuel and oil to ensure that it can safely complete the flight, and, applicable special provisions are complied with, as stated in Subpart 8.6.2.14 or 8.6.2.15. In addition, a reserve shall be carried to provide for contingencies.
- 4.3.6.3 PCAR Part 8, 8.6.2.15.1 COMMERCIAL AIR TRANSPORT: PISTON-ENGINED AIRPLANES. The fuel and oil carried in order to comply with Subpart 8.6.2.13.1 shall, in the case of piston-engined airplanes, be at least the amount sufficient to allow the airplane: (a) When a destination alternate airport is required, either: (1) to fly to the airport to which the flight is planned thence to the most critical (in terms of fuel consumption) alternate airport specified in the operational and ATS flight plans and thereafter for a period of 45 minutes; or (2) to fly to the airport to which the flight is planned, then 15 minutes of holding at 1500 FT, thence to the most critical (in terms of fuel consumption) alternate airport specified in the operational and ATS flight plans and thereafter for a period of 30 minutes holding at 1500 FT; or (3) to fly to the alternate airport via any predetermined point and thereafter for 45 minutes, provided that this shall not be less than the amount required to fly to the airport to which the flight is planned and thereafter for: (i) 45 minutes plus 15 per cent of the flight time planned to be spent at the cruising level(s), or (ii) two hours, whichever is less. (b) When a destination alternate airport is not required: (1) in terms of Subpart 8.6.2.6.1, to fly to the airport to which the flight is planned and thereafter for a period of 45 minutes; or (2) in terms of Subpart 8.6.2.6.1, to fly to the airport to which the flight is planned and thereafter for: (i) 45 minutes plus 15 per cent of the flight time planned to be spent at the cruising level(s), or (ii) two hours, whichever is less. (c) Airplanes. No person may commence a flight under IFR, or continue past the point of in-flight re-planning, unless there is enough fuel supply, considering meteorological conditions and any delays that are expected in flight, to include the following: (1) Taxi fuel – which shall be the amount of fuel expected to be consumed before take-off; (2) Trip fuel – which shall be the

amount of fuel required to enable the airplane to fly from take-off, or the point of in-flight re-planning, until landing at the destination aerodrome taking into account the operating conditions in the data provided by the manufacturer; (3) Contingency fuel – which shall be the amount of fuel required to compensate for unforeseen factors. It shall be five percent of the planned trip fuel or of the fuel required from the point of in-flight re-planning based on the consumption rate used to plan the trip fuel, but in any case, shall not be lower than the amount required to fly for five minutes at holding speed at 450 M (1500 FT) above the destination aerodrome in standard conditions; (4) Destination alternate fuel – which shall be (i) Where a destination alternate aerodrome is required, the amount of fuel required to enable the airplane to: (A) Perform a missed approach at the destination aerodrome; (B) Climb to the expected cruising altitude; (C) Fly the expected routing; (D) Descend to the point where the expected approach is initiated; and (E) Conduct the approach and landing at the destination alternate aerodrome; or (ii) Where two destination alternate aerodromes are required, the amount of fuel, as calculated in (4)(i) above, required to enable the airplane to proceed to the destination alternate aerodrome which requires the greater amount of alternate fuel; or (iii) Where a flight is operated without a destination alternate aerodrome, the amount of fuel required to enable the airplane to fly for 15 minutes at holding speed at 450 M (1500 FT) above destination aerodrome elevation in standard conditions; or (iv) Where the aerodrome of intended landing is an isolated aerodrome: (A) For helicopters, the amount of fuel required to fly for 45 minutes plus 15 percent of the flight time planned to be spent at cruising level, including final reserve fuel, or two hours, whichever is less; or (B) For a turbine-engined airplane, the amount of fuel required to fly for two hours at normal cruise consumption above the destination aerodrome, including final reserve fuel; (3) Final reserve fuel – which shall be the amount of fuel calculated using the estimated mass on arrival at the destination alternate aerodrome, or the destination aerodrome when no destination alternate aerodrome is required, or a pre-calculated value for each airplane type and variant in the fleet rounded up to an easily recalled figure: (i) For a helicopters, the amount of fuel required to fly for 45 minutes, under speed and altitude conditions specified by the Authority; or (ii) For a turbine-engined airplane, the amount of fuel required to fly for 30 minutes at holding speed at 450 M (1500 FT) above aerodrome elevation in standard conditions; (4) Additional fuel – which shall be the supplementary amount of fuel required if the minimum fuel calculated in accordance with trip fuel, contingency fuel, destination alternate fuel and final reserve fuel above is not sufficient to: (i) Allow the airplane to descend as necessary and proceed to an alternate aerodrome in the event of engine failure or loss of pressurization, whichever requires the greater amount of fuel based on the assumption that such a failure occurs at the most critical point along the route; (A) To fly for 15 minutes at holding speed at 450 M (1500 FT) above the aerodrome elevation in standard conditions; and (B) Make an approach and landing; (C) Allow an airplane engaged in EDTO to comply with the EDTO critical fuel scenario as established by the Authority; (D) Meet additional requirements not covered above. Note: Fuel planning for a failure that occurs at the most critical point along a route may place the airplane in a fuel emergency situation. (5) Discretionary fuel – shall be the extra amount of fuel to be carried at the discretion of the PIC, or (d) An airplane shall not take off or continue from the point of in-flight re-planning unless the usable fuel on board meets the requirements in 8.6.2.15.1(c) (2), (4), (5) and (6) if required. (e) Notwithstanding the provisions in 8.6.2.15.1 (c) (1) - (7) above, the Authority may approve a variation to these requirements provided the operator can demonstrate an equivalent level of safety will be maintained through a safety risk assessment that includes at least the following: (1) Flight fuel calculations; (2) Capabilities of the operator to include: (i) A data-driven method that includes a fuel consumption monitoring program; and/or (ii) The advanced use of alternate aerodromes; and (3) Specific mitigation measures. Note: ICAO Doc 9976, Flight Planning and Fuel

- Management Manual, contains guidance on the specific safety risk assessment, fuel consumption monitoring program and the advanced use of alternate aerodromes.
- 4.3.6.5 PCAR Part 8, 8.6.2.15.1 COMMERCIAL AIR TRANSPORT: PISTON-ENGINED AIRPLANES (d) An airplane shall not take off or continue from the point of in-flight re-planning unless the usable fuel on board meets the requirements in 8.6.2.15.1(c) (2), (4), (5) and (6) if required.
- 4.3.6.7 Not implemented.
- 4.3.8.2 PCAR Part 8, 8.9.1.2 (a) (2). 8.9.1.2 REFUELING WITH PASSENGERS ON BOARD (a) Airplanes: No PIC may allow an airplane to be refueled when passengers are embarking, on board or disembarking unless: 2) two-way communication is maintained by the aircraft's intercommunication system or other suitable means between the ground crew supervising the refueling and the qualified personnel on board the aircraft.
- 4.3.9.1 PCAR Part 7, IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 2. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. Flight deck seat occupants, not supplied by the flight crew source, are to be considered as passengers for the purpose of oxygen supply. (4) Cabin crew members, additional crew members, and passengers (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. Notes: 1. The supply provided shall take account of the cabin pressure altitude and descent profile for the routes concerned. 2. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 20 minutes at 10000 FT 3. The required

minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 110 minutes at 10000 FT. The oxygen required to meet the Crew Protective Breathing Equipment provisions of this Part may be included in determining the supply required. 4. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 15000 FT (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not operate a non-pressurized helicopter at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

4.3.9.2

PCAR Part 7, IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2.) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with oxygen in supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the

aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. Notes: 1. The supply provided shall take account of the cabin pressure altitude the descent profile for the routes concerned. 2. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certified operating altitude to 10000 FT in 10 minutes and followed by 20 minutes at 10000 FT. 3. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 10 minutes at 10000 FT. The oxygen requires to meet the crew protective breathing equipment provisions of this part maybe included in determining the supply required. 4. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated maximum altitude to 15000 FT. (c) The supplemental oxygen supply requirements of non-pressurized helicopters are as follows: 1. An operator shall not operate of non-pressurized helicopters at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. 2. The amount of supplemental oxygen for substance required for a particular operations shall be determined on the basis of flight altitudes and flight durations, consistent with the operating procedures, and established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. 3. A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. 3. Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on the flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

- 4.4.1.2 PCAR Part 8, 8.8.4.12 (a)(b). 8.8.4.12 INSTRUMENT APPROACH PROCEDURES AND IFR LANDING MINIMUMS (a) No person may make an instrument approach at an airport/heliport except in accordance with IFR weather minimums and instrument approach procedures established for that airport/heliport set forth by the Authority. (b) No AOC holder may make an instrument approach at an airport/heliport except as set forth in the AOC holder's operations specifications.
- 4.4.1.3 PCAR Part 8, IS: 8.8.4.13 CONTINUATION OF AN INSTRUMENT APPROACH (a) In commercial air transport operations, no pilot may continue an approach past the final approach fix (FAF), or where a FAF is not used, begin the final approach segment of an instrument approach procedure, at any airport/heliport unless: (1) A source approved by the Authority issues a weather report for that airport/heliport; and (2) The latest weather report for that airport/heliport reports the visibility or controlling RVR to be equal to or more that the minimums prescribed for that procedure. (b) If a pilot begins the final approach segment of an instrument approach procedure and subsequently receives a weather report indicating below-minimum conditions, the pilot may continue the approach to DH or MDA. Note: For the purpose of this subsection, the final approach segment begins at the final approach fix or facility prescribed in the instrument approach procedure. When a final approach fix is not prescribed for a procedure that includes a procedure turn, the final approach segment begins at the point where the procedure turn is completed and the aircraft is established inbound toward the airport on the final approach course within the distance prescribed in the procedure. (c) An instrument approach shall not be continued below 300 M (1000 FT) above the aerodrome elevation or into the final approach segment unless the reported visibility or controlling RVR is at or above the aerodrome operating minima. Note: Criteria for the Final Approach Segment is contained in PANS-OPS (Doc 8168),

Volume II. (d) If, after entering the final approach segment or after descending below 300 M (1000 FT) above the aerodrome elevation the reported visibility or controlling RVR falls below the specified minimum, the approach may be continued to DA/H or MDA/H. In any case, an airplane shall not continue approach-to-land any aerodrome beyond a point at which the limits of the operating minima specified for that aerodrome would be infringed. Note: Controlling RVR means the reported values of one or more RVR reporting locations (touchdown, mid-point and stop-end) used to determine determined whether minima are or are not met. Where RVR is used, the controlling RVR, unless otherwise specified by State criteria. (per MC No. 04-14 series of 2014 dated 30 January 2014).

- 4.4.3 PCAR Part 8, 8.5.1.21 (a)(1). 8.5.1.21 REPORTING OF HAZARDOUS CONDITIONS (a) The PIC shall report to the appropriate ATC facility, without delay and with enough details to be pertinent to the safety of other aircraft, any hazardous flight conditions encountered en-route, including those associated with meteorological conditions. (1) Those associated with meteorological observations during en-route and climb-out phases of flight.
- 4.4.4.1 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (a) Each required flight crew member shall remain at the assigned duty station during take-off and landing and critical phases of flight. (b) Each flight crew member shall remain at his or her station during all phases of flight unless: (1) Absence is necessary for the performance of his or her duties in connection with the operation; (2) Absence is necessary for physiological needs, provided one qualified pilot remains at the controls at all times; or (3) The crew member is taking a rest period and a qualified relief crew member replaces him or her at the duty station. (i) For the assigned PIC during the en route cruise portion of the flight by a pilot who holds an airline transport pilot license and an appropriate type rating, and who is currently qualified as PIC or CP, and is qualified as PIC of that aircraft during the en-route cruise portion of the flight; and (ii) In the case of the assigned CP, by a pilot qualified to act as PIC or Co-pilot of that aircraft during en route operations.
- 4.4.4.2 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (a) Each required flight crew member shall remain at the assigned duty station during take-off and landing and critical phases of flight. (b) Each flight crew member shall remain at his or her station during all phases of flight unless: (1) Absence is necessary for the performance of his or her duties in connection with the operation; (2) Absence is necessary for physiological needs, provided one qualified pilot remains at the controls at all times; or (3) The crew member is taking a rest period and a qualified relief crew member replaces him or her at the duty station. (i) For the assigned PIC during the en route cruise portion of the flight by a pilot who holds an airline transport pilot license and an appropriate type rating, and who is currently qualified as PIC or CP, and is qualified as PIC of that aircraft during the en route cruise portion of the flight; and (ii) In the case of the assigned CP, by a pilot qualified to act as PIC or Co-pilot of that aircraft during en route operations.
- 4.4.4.3 PCAR Part 8, 8.5.1.6 FLIGHT CREW MEMBER USE OF SEAT BELTS AND SAFETY HARNESES (a) Each flight crew member shall have his or her seat belts fastened during take-off and landing and all other times when seated at his or her station.
- 4.5.4 PCAR Part 8, 8.5.1.18 COMPLETION OF THE TECHNICAL LOG: COMMERCIAL AIR TRANSPORT (a) The PIC shall ensure that all portions of the technical log are completed with known or suspected defects in the aircraft before, during and after flight operations, including: (1) The journey logbook; and (2) The aircraft maintenance records section.
- 4.5.5 PCAR Part 8, 8.5.1.18 COMPLETION OF THE TECHNICAL LOG: COMMERCIAL AIR TRANSPORT (a) The PIC shall ensure that all portions of the technical log are completed with known or suspected defects in the aircraft before, during and after flight operations, including: (1) The journey logbook; and (2) The aircraft maintenance records section.

- 4.7.2.1 PCAR Part 8, 8.6.2.11 REQUIREMENTS FOR EXTENDED DIVERSION TIME OPERATIONS - AIRPLANES (a) An AOC holder shall not conduct operations beyond the threshold distance determined in accordance with Subpart 8.6.2.10 unless approved to do so by the Authority; (b) In requesting EDTO approval, each AOC holder shall show to the satisfaction of the Authority that: (1) For airplanes: (i) For all airplanes, (A) the most limiting EDTO significant system time limitation, if any indicated in the Airplane Flight Manual (directly or by reference) and relevant to that particular operation is not exceeded; and (B) the additional fuel required by Subpart 8.6.2.15 shall include the fuel necessary to comply with the EDTO critical fuel scenario as established by the Authority. (ii) For airplanes with two turbine engines, the airplanes EDTO certified and has verified the - (A) Reliability of the propulsion system; (B) Airworthiness certification for EDTO of the airplane type; and (C) EDTO maintenance program. (D) Crew training programs; for two power-unit airplanes are consistent with the level of safety required for current extended range of operations with the three and four unit turbine-powered airplanes. (2) It has conducted a safety risk assessment which demonstrates how an equivalent level of safety will be maintained, taking into account the following: (i) Capabilities of the operator; (ii) Overall reliability of the airplane; (iii) Reliability of each time limited system; (iv) Relevant information from the airplane manufacturer; and (v) Specific mitigation measures. (c) Before conducting an EDTO flight, an AOC holder shall ensure that a suitable EDTO en-route alternate is available, within either the approved diversion time or a diversion time based on MEL generated serviceability status of the airplane whichever is shorter. (d) No AOC holder shall commence a flight unless, during the possible period of arrival, the required en-route alternate aerodrome will be available and the available information indicates that conditions at the aerodrome will be at or above the aerodrome operating minima approved for the operation. (e) No AOC holder shall conduct operations beyond 60 minutes, from a point on a route to an en-route alternate aerodrome unless it ensures that: (1) For all airplanes; (i) En-route alternate aerodromes are identified; and (ii) The most up-to-date information is provided to the flight crew on identified enroute alternate aerodromes, including operational status and meteorological conditions; (2) For airplanes with two turbine engines, the most up-to-date information provided to the flight crew indicates that conditions at identified en-route alternate aerodromes will be at or above the operator's established aerodrome operating minima for the operation at the estimated time of use. (3) These requirements are incorporated into the operators: (i) operational control and flight dispatch procedures; (ii) operating procedures; and (iii) training programs. (f) No AOC Holder shall proceed beyond the threshold time approved by the Authority unless: (1) the identified en-route alternate aerodromes have been re-evaluated for availability; and (2) the most up to date information indicates that, during the estimated time of use, conditions at those aerodromes will be at or above the operator's established aerodrome operating minima for that operation; or (3) conditions are identified that would preclude a safe approach and landing at that aerodrome during the estimated time of use and an alternative course of action has been determined. Note 1: ICAO Annex 6, Part I, Attachment D contains guidance on the requirements of this provision. Note 2: FAA AC 120-42B (as amended), Extended Operations (EDTO and Polar Operations), provides additional guidance. (g) Commercial air transport where the Authority has approved alternate minima as an equivalent level of safety based on the results of a specific safety risk assessment demonstrated by the operator, which contains the following: (1) Capabilities of the operator; (2) Overall capability of the airplane and its systems; (3) Available aerodrome technologies, capabilities and infrastructure; (4) Quality and reliability of meteorological information; (5) Identified hazards and safety risks associated with each alternate aerodrome variation; (6) Specific mitigation measures. Note: ICAO Doc 9859, Safety Management Manual, and ICAO Doc 9976, Flight Planning and Fuel Management Manual, contain guidance on performing a safety risk assessment and on determining variations, including examples of variations.
- 4.7.2.2 PCAR Part 8, 8.6.2.9 (b) (3). 8.6.2.9 TAKE-OFF ALTERNATE AIRPORTS/ HELIPORTS: COMMERCIAL AIR TRANSPORT OPERATIONS (b)(3) For aircraft

- engaged in Extended Diversion Time Operations (EDTO) where an alternate aerodrome meeting the distance criteria of a) or b) is not available, the first available alternate aerodrome located within the distance of the operator's approved maximum diversion time considering the actual take-off mass.
- 4.7.2.3 PCAR Part 8, 8.6.2.11 (b)(i) (A)(B). 8.6.2.11 REQUIREMENTS FOR EXTENDED DIVERSION TIME OPERATIONS - AIRPLANES (b) In requesting EDTO approval, each AOC holder shall show to the satisfaction of the Authority that: (1) For airplanes: (i) For all airplanes, (A) the most limiting EDTO significant system time limitation, if any indicated in the Airplane Flight Manual (directly or by reference) and relevant to that particular operation is not exceeded; and (B) the additional fuel required by Subpart 8.6.2.15 shall include the fuel necessary to comply with the EDTO critical fuel scenario as established by the Authority.
- 4.7.2.3.1 PCAR Part 8, 8.6.2.11 (b)(2)(i) to (v). 8.6.2.11 REQUIREMENTS FOR EXTENDED DIVERSION TIME OPERATIONS - AIRPLANES [AOC] (b) (2) It has conducted a safety risk assessment which demonstrates how an equivalent level of safety will be maintained, taking into account the following: (i) Capabilities of the operator; (ii) Overall reliability of the airplane; (iii) Reliability of each time limited system; (iv) Relevant information from the airplane manufacturer; and (v) Specific mitigation measures.
- 4.7.2.6 PCAR Part 8, 8.6.2.11 (b)(1)(ii)(A) to (D). 8.6.2.11 REQUIREMENTS FOR EXTENDED DIVERSION TIME OPERATIONS - AIRPLANES [AOC] (b) (1) (ii) For airplanes with two turbine engines, the airplanes EDTO certified and has verified the - (A) Reliability of the propulsion system; (B) Airworthiness certification for EDTO of the airplane type; and (C) EDTO maintenance program. (D) Crew training programs; for two power-unit airplanes are consistent with the level of safety required for current extended range of operations with the three and four unit turbine-powered airplanes.
- 4.8 PCAR Part 8, 8.9.2.14 CARRY-ON BAGGAGE (a) No person may allow the boarding of carry-on baggage in to the passenger cabin unless it is adequately and securely stowed in accordance with the operator's approved Operations Manual procedures. (b) No person may allow aircraft passenger entry doors to be closed in preparation for taxi or pushback unless at least one required crew member has verified that each article of baggage has been properly stowed in overhead racks with approved restraining devices or doors, or in approved locations aft of the bulkhead. (c) No person may allow carry-on baggage to be stowed in a location that would cause that location to be loaded beyond its maximum placard mass limitation. Note: The stowage locations shall be capable of restraining the articles in crash impacts severe enough to induce the ultimate inertia forces specified in the emergency landing conditions under which the aircraft was type-certified.
- 4.10.1 PCAR Part 8, 8.11, 8.11.1.1 (d). 8.11 FATIGUE MANAGEMENT FOR FLIGHT TIME, FLIGHT DUTY PERIODS, DUTY PERIODS AND REST PERIODS: COMMERCIAL AIR TRANSPORT 8.11.1.1 APPLICABILITY (d) For the purpose of managing fatigue, the Operator shall establish regulations specifying the limitations applicable to flight time, flight duty periods, duty periods and rest periods for flight and cabin crew members. These regulations shall be based upon scientific principles and knowledge, where available, with the aim of ensuring that the flight and cabin crew members are performing at an adequate level of alertness. Note: Guidance for the development of prescriptive fatigue management regulations is given in ICAO Annex 6 Part 1, Attachment A.
- 4.10.5 PCAR Part 8, 8.11.1.2 (d) (1) to (3). 8.11.1.2. MANAGING FATIGUE-RELATED SAFETY RISKS (d) Operators using an FRMS must adhere to the following provisions of the FRMS approval process that allows the Authority to ensure that the approved FRMS meets the requirements of Subpart 8.11.1.2(c). (1) Establish maximum values for flight times and/or flight duty period(s) and duty period(s), and minimum values for rest periods that shall be based upon scientific principles and

knowledge, subject to safety assurance processes. Note: ICAO Doc 9966, Fatigue Risk Management Systems Manual, provides a definition for Safety Assurance Processes. (2) Adhere to Authority mandates to decrease maximum values and increase in minimum values in the event that the operator's data indicates these values are too high to too low, respectively; and (3) Provide justification to the Authority for any increase in maximum values or decrease in minimum values based on accumulated FRMS experience and fatigue-related data before such changes will be approved by the Authority.

- 4.10.8 PCAR Part 8, 8.11.1.7 FLIGHT TIME, FLIGHT DUTY, DUTY AND REST PERIODS RECORDS (a) Each AOC holder shall maintain records for each crew member and flight operations officer/flight dispatcher of flight time, flight duty periods, duty periods, and rest periods.

Chapter 5

- 5.1.1 PCAR Part 8, 8.7.1.1 APPLICABILITY (a) Each civil aircraft shall be operated in accordance with the comprehensive code of performance established by the Authority and in compliance with the acceptable standards, as prescribed in CAR Part 5, Subpart 5.4.1.2 and this Subpart.
- 5.1.2 PCAR Part 8, 8.7.2.2 GENERAL (d) Except as provided in Subpart 8.8.4.21, no person may operate a single-engine aircraft used for revenue passenger carrying operations unless that aircraft is continually operated in daylight, VFR, excluding over the top, and over routes and diversions there-from, that do not permit a safe forced landing to be executed in the event of a power-unit failure.
- 5.2.1 PCAR Part 8, 8.7.2.3.3 AIRCRAFT PERFORMANCE: HELICOPTERS CERTIFICATED IN ACCORDANCE WITH PART IV OF ANNEX 8 (a) The Standards contained in paragraphs (b) to (f) inclusive below and Subpart 8.7.2.4.2 (a) are applicable to the helicopters to which Part IV of Annex 8 is applicable.
- 5.2.2 PCAR Part 8, 8.7.2.3.3(b) (for helicopter). 8.7.2.3.3 AIRCRAFT PERFORMANCE: HELICOPTERS CERTIFICATED IN ACCORDANCE WITH PART IV OF ANNEX 8 (b) The level of performance defined by the appropriate parts of the code of performance referred to in Subpart 8.7.1.1 for the helicopters designated in paragraph (a) above shall be consistent with to the overall level embodied in the Standards of this chapter.
- 5.2.3 PCAR Part 8, 8.7.2.3.3 AIRCRAFT PERFORMANCE: HELICOPTERS CERTIFICATED IN ACCORDANCE WITH PART IV OF ANNEX 8 (c) A helicopter shall be operated in compliance with the terms of its certificate of airworthiness and within the approved operating limitations contained in its flight manual.
- 5.2.5 PCAR Part 8, 8.7.1.3 (a) (last sentence) 8.7.1.3 AIRCRAFT PERFORMANCE CALCULATIONS (a) Each operator shall ensure that the performance data contained in the AFM/RFM, or other authorized source is used to determine compliance with the appropriate requirements of Subpart 8.7. A flight shall not be commenced unless the performance information provided in the flight manual indicates that the Standards of Subpart 8.7 can be complied with for the flight to be undertaken.
- 5.2.7 PCAR Part 8, 8.7.2.4.1(a)(1) Part 8, 8.6.2.18(a)(2) Part 8, 8.7.1.2(a)(4). 8.7.2.4.1 TAKE-OFF LIMITATIONS: AIRPLANES (a) The mass of the airplane at the start of take-off shall not exceed the mass at which paragraph (5) below is complied with, nor the mass at which Subparts 8.7.2.6, 8.7.2.7 and 8.7.2.8 are complied with, allowing for expected reductions in mass as the flight proceeds, and for such fuel jettisoning as is envisaged in applying Subparts 8.7.2.6 and 8.7.2.7 and, in respect of alternate airports, Subparts 8.7.1.4 (d) and 8.7.2.8. No person may take-off an airplane used in commercial air transport, unless the following requirements are met when determining the maximum permitted take-off mass: (1) In no case shall the mass at the

start of take-off exceed the maximum take-off mass specified in the flight manual for the pressure-altitude appropriate to the elevation of the airport, and, if used as a parameter to determine the maximum take-off mass, any other local atmospheric condition. 8.6.2.18 MAXIMUM ALLOWABLE MASS TO BE CONSIDERED ON ALL LOAD MANIFESTS (a) The PIC shall ensure that the maximum allowable mass for a flight does not exceed the maximum allowable take-off mass: (2) Considering anticipated fuel and oil consumption that allows compliance with applicable en route performance, landing mass, and landing distance limitations for destination and alternate airports. 8.7.1.2 GENERAL (a) No person may operate an aircraft that: (4) Exceeds the mass limitations, if applicable, imposed by the terms of its noise certification standards, as contained in the applicable part of ICAO Annex 16, Volume I, unless otherwise approved by the Authority.

5.2.8 PCAR Part 8, 8.7.2.4.1(a) (5). 8.7.2.4.1 TAKE-OFF LIMITATIONS: AIRPLANES (a) The mass of the airplane at the start of take-off shall not exceed the mass at which paragraph (5) below is complied with, nor the mass at which Subparts 8.7.2.6, 8.7.2.7 and 8.7.2.8 are complied with, allowing for expected reductions in mass as the flight proceeds, and for such fuel jettisoning as is envisaged in applying Subparts 8.7.2.6 and 8.7.2.7 and, in respect of alternate airports, Subparts 8.7.1.4 (d) and 8.7.2.8. No person may take-off an airplane used in commercial air transport, unless the following requirements are met when determining the maximum permitted take-off mass: (5) The airplane shall be able, in the event of a critical power-unit failing at any point in the takeoff, either to discontinue the take-off and stop within the accelerate-stop distance available, or to continue the take-off and clear all obstacles along the flight path by an adequate margin, described in paragraph (6), until the airplane is in a position to comply with Subpart 8.7.2.6.

5.2.9 PCAR Part 8, 8.7.2.6 EN ROUTE LIMITATIONS: ONE POWER UNIT INOPERATIVE (a) Airplane. No person may take off an airplane used in commercial air transport having two power units unless that airplane can, in the event of the critical power unit becoming inoperative at any point along the route, or planned diversion therefrom, to continue the flight to a suitable airport or planned diversions therefrom, to continue the flight to an airport, at which the Standard of Subpart 8.7.2.7 can be met, without flying below the minimum flight altitude at any point, where a landing can be made while allowing: (1) For reciprocating engine powered airplanes - (i) At least a rate of climb of 0.079 - (0.106/number of engines installed) VSO2 (when VSO is expressed in knots) at an altitude of 300 M (1000 FT) above all terrain and obstructions within 9.3 KM (5SM), on each side of the intended track; and (ii) A positive slope at an altitude of at least 450 M (1500 FT) above the airport where the airplane is assumed to land. (2) For turbine engine powered transport category airplanes - (i) A positive slope at an altitude of at least 300 M (1000 FT) above all terrain and obstructions within 9.3 KM (5 SM), on each side of the intended track; (ii) A net flight path from cruising altitude to the intended landing airport that allows at least 600 M (2000 FT) clearance above all terrain and obstructions within 9.3 KM (5 SM), on each side of the intended track: and (iii) A positive slope at an altitude of at least 450 M (1500 FT) above the airport where the airplane is assumed to land; Note: The climb rate specified in paragraph (a) (1)(i) may be amended to 0.026 Vso2 for large transport category aircraft issued a type certificate prior to 1953. Note: The 9.3 KM (5 NM) clearance margin stated in paragraph (a) shall be increased to 18.5 KM (10 NM) if navigational accuracy does not meet the 95% containment level.

5.2.10 PCAR Part 8, 8.7.2.7 EN ROUTE LIMITATIONS: TWO POWER UNITS INOPERATIVE (a) Airplane. No person may take-off an airplane used in commercial air transport having three or more power units, on any part of a route where the location of en-route alternate airports and the total duration of the flight are such that the

probability of a second power-unit becoming inoperative must be allowed for if the general level of safety implied by the Standards of this chapter is to be maintained, the airplane shall be able, in the event of any two power-units becoming inoperative, to continue the flight to an en-route alternate airport and land while allowing: (1) For turbine engine powered airplanes - (i) A net flight path (considering the ambient temperatures anticipated along the track) clearing vertically by at least 600 M (2000 FT) all terrain and obstructions within 9.3 KM (5 NM) on each side of the intended track; (ii) A positive slope at 450 M (1500 FT) above the airport of intended landing; and (iii) Enough fuel to continue to the airport of intended landing, to arrive at an altitude of at least 450 M (1500 FT) directly over the airport, and thereafter to fly for 15 minutes at cruise power. Note: The consumption of fuel and oil after the engine failure is the same as the consumption that is allowed for in the net flight path data in the AFM. (2) For reciprocating engine powered airplanes - (i) A rate of climb at 0.013 V_{SO2} feet per minute (that is, the number of feet per minute is obtained by multiplying the number of knots squared by 0.013) at an altitude of 300 M (1000 FT) above the highest ground or obstruction within 18.6 KM (10 NM) on each side of the intended track, or at an altitude of 1500 M (5000 FT), which ever is higher; and (ii) Enough fuel to continue to the airport of intended landing and to arrive at an altitude of at least 300 M (1000 FT) directly over that airport. Note: When the two engines of the reciprocating airplane are predicted to fail at an altitude above the prescribed minimum altitude, compliance with the prescribed rate of climb need not be shown during the descent from the cruising altitude to the prescribed minimum altitude, if those requirements can be met once the prescribed minimum altitude is reached and assuming descent to be along a net flight path and the rate of descent to be 0.013 V_{SO2} greater than the rate in the approved performance data. Note: If fuel jettisoning is authorized (or planned), the airplane's mass at the point where the two engines fail is considered to be not less than that which would include enough fuel to proceed to an airport and to arrive at an altitude of at least 300 M (1000 FT) directly over that airport.

- 5.2.11 PCAR Part 8, 8.7.2.8 LANDING LIMITATIONS (a) Airplane: No person may take off an airplane used in commercial operations unless its mass is such on arrival at the airport of intended landing and at any alternate airport, that after clearing all obstacles in the approach path by a safe margin, be able to land, with assurance that it can come to a stop or; for a seaplane, to a satisfactorily low speed, within the landing distance available (Allowance shall be made for expected variations in the approach and landing techniques, if such allowance has not been made in the scheduling of performance data.), and within: (1) For turbine engine powered airplanes, 60 percent of the effective length of each runway.(2) For reciprocating engine powered airplanes, 70 percent of the effective length of each runway.

- 5.3.1 PCAR Part 9, 9.3.1.14 PERFORMANCE DATA CONTROL SYSTEM (b) The system approved by the Authority shall provide current obstacle data for departure and arrival performance calculations to comply with Subpart 8.7.2.4.

Chapter 6

- 6.1.1 PCAR Part 7, 7.1.4(a)(b). 7.1.4 GENERAL INSTRUMENT AND EQUIPMENT REQUIREMENTS (a) In addition to the minimum equipment necessary for the issuance of a certificate of airworthiness, the instruments, equipment and flight documents prescribed in Part 7 shall be installed or carried, as appropriate, in aircraft according to the aircraft used and to the circumstances under which the flight is to be conducted; (b) All required instruments and equipment shall be approved and installed in accordance with applicable airworthiness requirements.

- 6.1.4 PCAR Part 9, 9.3.1.2(g)(1) and Part 9, 9.3.1.4(a)(b). 9.3.1.2 OPERATIONS MANUAL (g) Unless otherwise acceptable to the Authority, each operator shall provide

an Operations Manual containing information on operations administration and supervision, safety management programs, personnel training, flight crew and cabin crew flight duty period and rest period requirements, flight operations, operational flight planning including contents and use of operational flight plan, aircraft performance, routes, guides and charts, minimum flight altitudes, airport operating minima, search and rescue, dangerous goods, navigation, communications, security, and human factors. The operations manual shall encompass the matters set forth above. The operations manual may be published in parts, as a single document, or as a series of volumes. Specific subjects are listed below. Subjects presented with reference to a specific section shall be addressed in accordance with the requirements of the referenced section. (1) Aircraft Operating Manual. 9.3.1.4 AIRCRAFT OPERATING MANUAL (a) Each operator or applicant shall submit proposed aircraft operating manuals for each type and variant of aircraft operated, containing the normal, abnormal and emergency procedures relating to the operation of the aircraft for approval by the Authority; (b) Each Aircraft Operating Manual shall be based upon the aircraft manufacturer's data for the specific aircraft type and variant operated by the operator and shall include specific operating parameters, details of the aircraft systems, and of the check lists to be used applicable to the operations of the operator that are approved by the Authority. The design of the manual shall observe Human Factors principles.

- 6.2.2 PCAR Part 7,7.8.11(a)(b), Part 7,7.8.6(a)(1)to(5), Part 7,7.9.1(a)(1)(2)(i)(ii), Part 8,8.9.1.4(a)(1)to(4) and Part 7,7.9.8 7.8.11 FIRST-AID AND EMERGENCY MEDICAL KIT (a) No person may operate and aircraft unless it is equipped with first-aid kits readily accessible to cabin crew located near an exit and, on passenger flights, an approved emergency medical kit for treatment of injuries or medical emergencies that might occur during flight time or in minor accidents. (b) The number of first-aid kits to be carried shall be to the following scale: Number of Passenger seats installed, number of first-aid kits required 0 to 50 1 51 to 150 2 151 to 250 3 More than 250 4
- 7.8.6 PORTABLE FIRE EXTINGUISHERS (a) [AAC] No person may operate an aircraft unless it is equipped with portable fire extinguishers accessible for use in crew, passenger, and cargo compartments as follows: (1) The type and quantity of extinguishing agent shall be suitable for the kinds of fires likely to occur in the compartment where the extinguisher is intended to be used. Note: For passenger compartments, the extinguisher shall be designed to minimize the hazard of toxic gas concentrations. (2) At least one portable fire extinguisher shall be provided and conveniently located for use in each Class E cargo compartment which is accessible to crew members during flight, and at least one shall be located in each upper and lower lobe galley. (3) At least one portable fire extinguisher shall be conveniently located on the flight deck for use by the flight crew. (4) At least one portable fire extinguisher shall be conveniently located in the passenger compartment of airplanes having a passenger seating capacity of 30 or less. (5) For each airplane having a passenger seating capacity of more than 30, there shall be at least the following number of portable fire extinguishers conveniently located and uniformly distributed throughout the compartment. 7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (1) A seat with safety belt for each person on board over the age of two (2) years old; and a restraining belt for each berth on board the aircraft (2) A safety harness for each flight crew member seat. (i) The safety harness for each pilot seat shall incorporate a device, which will automatically restrain the occupant's torso in the event of rapid deceleration. (ii) The safety harness for each pilot seat, which includes shoulder straps and a seat belt, should incorporate a device to prevent a suddenly incapacitated pilot from interfering

- with the flight controls. 8.9.1.4 PASSENGER BRIEFING (a) The PIC shall ensure that crew members and passengers are made familiar, by means of an oral briefing or by other means, with the location and use of the following items: (1) Seat belts or harnesses, as appropriate; (2) Emergency exits; (3) Life jackets, if carriage of life jackets is prescribed; (4) Oxygen dispensing equipment, if provision of oxygen for the passengers is prescribed 7.9.8 PROTECTIVE CIRCUIT FUSES No person may operate an aircraft in which protective fuses are installed unless there are spare fuses available for use in flight equal to at least 10% of the number of fuses of each rating or three of each rating whichever is the greater.
- 6.2.2.1 PCAR Part 7,7.8.7(a)(b). 7.8.7 LAVATORY FIRE EXTINGUISHER (a) No person may operate a passenger-carrying transport category airplane unless each lavatory in the airplane is equipped with a built-in fire extinguisher for each disposal receptacle for towels, paper, or waste located within the lavatory. (b) Built-in lavatory fire extinguishers shall be designed to discharge automatically into each disposal receptacle upon occurrence of a fire in the receptacle.
- 6.2.3 PCAR Part 8,8.2.1.8(a)(10) (11)(20). 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (a) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in commercial air transport operations unless it has within it the following current and approved documents: (10) AFM (for airplanes) or RFM (for helicopters). (11) Part(s) of the Operations Manual relevant to operation(s) conducted. (20) Maps and charts for routes of proposed flight or possibly diverted flights.
- 6.3.1.3 PCAR Part 7,7.7.2.4(a) (1)to(3). 7.7.2.4 DURATION (a) FDRs shall be capable of retaining the information recorded during the last - (1) Type I and II - 25 hours of operation. (2) Type IIA – 30 minutes of operation. (3) Type IV, IVA and V - 10 hours of operation.
- 6.3.2.4.1 PCAR Part 7,7.7.3.5(a) (1-3)&(b)(1-2). 7.7.3.5 COCKPIT VOICE RECORDER ALTERNATE POWER (a) No person may operate an airplane required to be equipped with a CVR unless it is equipped with CVR alternate power that: (1) automatically engages and provides ten minutes, plus or minus one minute, of operation whenever airplane power to the recorder ceases, either by normal shutdown or by any other loss of power; (2) powers the CVR and its associated cockpit area microphone components, and (3) is located as close as practicable to the alternate power source. (b) No person may operate an airplane of a maximum certificated take-off mass of over 27000 KG for which the individual certificate of airworthiness is first issued on or after 1 January 2018 unless is it equipped with an alternate power source, as described in (a) above, that powers - (1) the forward CVR in the case of combination recorders, or (2) at least one CVR. Note 1: "Alternate" means separate from the power source that normally provides power to the CVR. The use of airplane batteries or other power sources is acceptable provided that the requirements are above are met and electrical power to essential and critical loads is not compromised. Note 2: When the CVR function is combined with other recording functions within the same unit, powering the other functions is allowed.
- 6.3.2.4.2 PCAR Part 7,7.7.3.5(b)(1). 7.7.3.5 COCKPIT VOICE RECORDER ALTERNATE POWER (b) [AOC] No person may operate an airplane of a maximum certificated take-off mass of over 27000 KG for which the individual certificate of airworthiness is first issued on or after 1 January 2018 unless is it equipped with an alternate power source, as described in (a) above, that powers– (1) the forward CVR in the case of combination recorders.
- 6.3.3.1.1 PCAR Part 7,7.7.4 (Note) and PCAR Part 7,7.7.4.1 (a). 7.7.4 DATA LINK RECORDERS (DLR) AND DATA LINK RECORDING SYSTEMS (DLRS) Note: Data link recorders performance requirements are as contained in the EUROCAE ED-112,

- Minimum Operational Performance Specifications (MOPS) for Crash Protected Airborne Recorder Systems, or equivalent documents. 7.7.4.1 APPLICABILITY (a) No person may operate an airplane or helicopter for which the individual certificate of airworthiness is first issued on or after 1 January 2016, which utilize any of the data link communications applications listed in IS 7.7.4.1 and are required to carry a CVR, unless the aircraft records on a flight recorder the data link communications messages.
- 6.3.5.3 PCAR Part 7,7.7.1.3 (a)(b). (a) The operator shall conduct operational checks and evaluations of recordings from the flight recorder systems to ensure the continued serviceability of the recorders. (b) The procedures for the inspections of the flight recorder systems are given in IS 7.7.1.3.
- 6.3.5.5.2 PCAR Part 7,7.7.1.5(b)(1) (2). 7.7.1.5 COMBINATION RECORDERS (b) No person may operate an airplane of a maximum certificated take-off mass of over 5700 KG and which is required to be equipped with both a FDR and CVR unless - (1) The airplane is equipped with an FDR and a CVR or alternatively equipped with two combination recorders (FDR/CVR). (2) The airplane is equipped with two combination recorders (FDR/CVR) for airplanes type certificated on or after 1 January 2016.
- 6.4.1 PCAR Part 7, 7.2.2(a)(1)(2) (5)(7)(b)(1)(2). 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (5) A magnetic compass; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional (2) gyroscope).
- 6.4.2 Not implemented .
- 6.5.1 PCAR Part 7, 7.8, Part 7, 7.9, Part 7, 7.8.16(a)(2), Part 7, 7.9.15 and Part 7, 7.9.16. 7.8 EMERGENCY, RESCUE, AND SURVIVAL EQUIPMENT 7.8.1 EMERGENCY EQUIPMENT: ALL AIRCRAFT (a) Each item of emergency and flotation equipment shall be - (1) Readily accessible to the crew and, with regard to equipment located in the passenger compartment, to passengers without appreciable time for preparatory procedures; (2) Clearly identified and clearly marked to indicate its method of operation; (3) Marked as to date of last inspection; and (4) Marked as to contents when carried in a compartment or container. PCAR Part 7, 7.9 MISCELLANEOUS SYSTEMS AND EQUIPMENT 7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (1) A seat with safety belt for each person on board over the age of two (2) years old; and a restraining belt for each berth on board the aircraft (2) A safety harness for each flight crew member seat. (i) The safety harness for each pilot seat shall incorporate a device, which will automatically restrain the occupant's torso in the event of rapid deceleration. (ii) The safety harness for each pilot seat, which includes shoulder straps and a seat belt, should incorporate a device to prevent a suddenly incapacitated pilot from interfering with the flight controls. (3) A forward or rearward facing (within 15 degrees of the longitudinal axis of the airplane) seat equipped with a safety harness for each cabin crew member station in the passenger compartment. (4) The cabin crew member's seats shall be located near floor level and other emergency exits for emergency evacuation. 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (2) One life-jacket or equivalent flotation device equipped with a means of

- electric illumination shall be carried for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.
- 7.9.15 MARITIME SOUND SIGNALLING DEVICE All seaplanes for all flights shall be equipped with equipment for making the sound signals prescribed in the International Regulations for Preventing Collisions at Sea, where applicable.
- 7.9.16 ANCHORS All seaplanes for all flights shall be equipped with one anchor, and one sea anchor (drogue), when necessary to assist in manoeuvring (approval for the anchors not required). Note: "Seaplanes" includes amphibians operated as seaplanes.
- 6.5.2.1 PCAR Part 7, 7.8.16(a)(1)(i)(ii)(iii). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (1) Shall carry the equipment prescribed in paragraph (2): (i) When flying en-route over water beyond gliding distance from the shore; (ii) When flying over water at a distance of more than 93 KM (50 NM) away from the shore for aircraft capable of maintaining safe altitude after the failure of one engine for two engine aircraft and the failure of two engines for three or four-engine aircraft.; or (iii) When taking off or landing at an aerodrome where the (Authority) has determined the take off or approach path is so disposed over water that in the event of a mishap there would be the likelihood of a ditching.
- 6.5.3.1 PCAR Part 7, 7.8.17(a) (1) to (3) (b)(c)(d)(3). 7.8.17 LIFE RAFT (a) In addition to the equipment prescribed in 7.8.16 and 7.8.18 of this Part, life saving rafts in sufficient numbers to carry all persons on board shall be installed in: (1) All airplanes when operating in "Extended over-water operations" as defined in Subpart 8.1.1.2 (b). (2) Class 1 and 2 helicopters when they are operated over water at a distance from land corresponding to more than 10 minutes at normal cruise speed. (3) Class 3 helicopters when they are operated over water beyond autorotational or safe forced landing distance from land. (b) An aircraft shall have life saving rafts with a sufficient capacity to carry all persons on board in the event of the loss of one raft of the largest capacity. (c) All life saving rafts shall be stowed so as to facilitate their ready use in an emergency. (d) Life rafts shall be equipped with the following life sustaining equipment - (3) A pyrotechnical signalling device.
- 6.5.3.2 PCAR Part 7, 7.8.16(a)(2). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be carried for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.
- 6.6 PCAR Part 7, 7.8.3 VISUAL SIGNALLING DEVICES (a) No person may operate an aircraft over water or across land areas which have been designated by the Republic of the Philippines as areas in which search and rescue would be especially difficult, unless equipped with such signaling devices as may be appropriate to the area overflown.
- 6.7.1 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated; (3) Sufficient

spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure; (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin. (e) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft.

6.7.2 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated; (3) Sufficient spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure; (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin. (e) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft.

6.7.3 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated; (3) Sufficient spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure; (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed

- throughout the cabin. (e) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft.
- 6.7.5 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated; (3) Sufficient spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure; (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin. (e) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft.
- 6.8 PCAR Part 7, 7.9.9 ICING PROTECTION EQUIPMENT (a) No person may operate an aircraft in icing conditions unless it is certified and equipped for the prevention or removal of ice on windshields, wings, empennage, propellers, and other parts of the aircraft where ice formation will adversely affect the safety of the aircraft. (b) A flight to be planned or expected to operate in suspected or known ground icing conditions shall not be commenced unless the aircraft has been inspected for icing and, if necessary, has been given appropriate de-icing/anti-icing treatment. Accumulation of ice or other naturally occurring contaminants shall be removed so that the aircraft is kept in an airworthy condition prior to take-off. Note: Guidance material can be found in ICAO Manual of Aircraft Ground De-icing/Anti-icing Operations (Doc 9640). (c) No AOC holder may operate an aircraft in expected or actual icing conditions at night unless it is equipped with a means to illuminate or detect the formation of ice. Note: Any illumination that is used shall be of a type that will not cause glare or reflection that would handicap crew members in the performance of their duties.
- 6.9.1 PCAR Part 7, 7.2.2 (a) (1) to (7) and Part 7, 7.2.3(a)(3). 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator; (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An

accurate timepiece indicating the time in hours minutes and seconds (approval not required). 7.2.3 IFR INSTRUMENTS (a) All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments, shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate.

- 6.9.2.1 PCAR Part 7, 7.2.4(a)(3)(b)(c). 7.2.4 STANDBY ATTITUDE INDICATOR (a) No person may operate an airplane with a maximum certified take-off mass exceeding 5700 KG, aircraft having a maximum approved passenger seating configuration of more than 9 seats and all helicopters when operating in accordance with IFR, unless it is equipped with a single standby attitude indicator (artificial horizon) that - (3) After a total failure of the normal electrical generating system, is automatically powered for the purpose of operating and illuminating for a minimum of 30 minutes from a source independent of the normal electrical generating system. (b) When the standby attitude indicator is being operated by emergency power, it shall be clearly evident to the flight crew. (c) Where the standby attitude indicator has its own dedicated power supply there shall be an associated indication, either on the instrument or on the instrument pane when this supply is in use.
- 6.10 PCAR Part 7, 7.2.2 and Part 7, 7.4.1(a)(5)(6)(7)(b)(1). 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator; (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope). PCAR Part 7 7.4.1 AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (a) All aircraft operated at night shall be equipped with: (5) Illumination for all flight instruments and equipment that are essential for the safe operation of the aircraft; (6) Lights in all passenger compartments; (7) A flashlight for each crew member station (approval not required) and; (b) No AOC holder may operate an aircraft by night unless, in addition to the equipment specified in paragraph (a) above, it is equipped with; (1) Two landing lights or a single light having two separately energized filaments.
- 6.12 PCAR Part 7, 7.9.14 COSMIC RADIATION DETECTION EQUIPMENT (a) No person shall operate an airplane in commercial air transport operations in an airplane intended to be operated above 15000 M (49000 FT) unless it is equipped with - (1) an instrument to measure and indicate continuously the dose rate of total cosmic radiation being received (i.e., the total of ionising and neutron radiation of galactic and solar origin) and the cumulative dose on each flight or (2) A system of on-board quarterly radiation sampling acceptable to the Authority as described in IS: 7.9.14.
- 6.15.1 PCAR Part 7, 7.6.5 GROUND PROXIMITY WARNING SYSTEM (a) No person may operate an airplane with a maximum certified take-off mass in excess of 5700 KG, unless it is equipped with a ground proximity warning system which has a forward looking terrain avoidance function.
- 6.15.5 PCAR Part 7, 7.6.5 GROUND PROXIMITY WARNING SYSTEM (a) No person may operate an airplane with a maximum certified take-off mass in excess of 5700 KG,

- unless it is equipped with a ground proximity warning system which has a forward looking terrain avoidance function.
- 6.15.6 PCAR Part 7, 7.6.5 GROUND PROXIMITY WARNING SYSTEM (b) Each ground proximity warning system shall automatically provide, by means of aural signals which may be supplemented by visual signals, timely and distinctive warning to the flight crew of sink rate, ground proximity, altitude loss after take-off or go around, incorrect landing configuration and downward glideslope deviation.
- 6.15.7 PCAR Part 7, 7.6.5 (c) (1) to (5). GROUND PROXIMITY WARNING SYSTEM (c) A ground proximity warning system shall provide, as a minimum, warnings of the following circumstances - (1) Excessive descent rate. (2) Excessive terrain closure rate. (3) Excessive altitude loss after take-off or go-around. (4) Unsafe terrain clearance while not in landing configuration; and (5) Excessive descent below the instrument glide path.
- 6.16.1 PCAR Part 7, 7.9.1(a)(3). 7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (3) A forward or rearward facing (within 15 degrees of the longitudinal axis of the airplane) seat equipped with a safety harness for each cabin crew member station in the passenger compartment.
- 6.16.3 PCAR Part 7, 7.9.1(a)(4). 7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) (4) The cabin crew member's seats shall be located near floor level and other emergency exits for emergency evacuation.
- 6.17.2 PCAR Part 7, 7.8.5 and Part 7, IS:7.8.5. 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on all flights shall be equipped with a 406 MHZ ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use. (b) All aircraft should carry an ELT however an aircraft may be operated without an ELT on board where the aircraft is (1) a glider, balloon, airship, ultra-light airplane, gyroplane, restricted or non-Type Certificated aircraft; (2) operated by the holder of a flight training unit operating certificate, engaged in flight training and operated within 25 NM of the aerodrome of departure; (3) engaged in a flight test. (c) Batteries used in ELTs shall be replaced (or recharged if the battery is rechargeable) when - (1) The transmitter has been in use for more than one cumulative hour; or (2) 50 percent of their useful life (or for rechargeable batteries, 50 percent of their useful life of charge) has expired. (d) The expiration date for a replacement or recharged ELT battery shall be legibly marked on the outside of the transmitter. Note: The battery useful life (or useful life of charge) requirements do not apply to batteries (such as water-activated batteries) that are essentially unaffected during probable storage intervals. IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) The following requirements shall be used to determine the quantity and type of 406 MHZ ELT which must be carried. (a) Except as provided for in (b), all airplanes authorized to carry more than 19 passengers shall be equipped with at least one automatic ELT or two ELTs of any type. (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic. (c) Except as provided for in (d), all airplanes authorized to carry 19 passengers or less shall be equipped with at least one ELT of any type. (d) All airplanes authorized to carry 19 passengers or less for which the individual certificate of airworthiness is first issued

after 1 July 2008 shall be equipped with at least one automatic ELT. (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.

6.17.3 PCAR Part 7,7.8.5 and Part 7,IS:7.8.5. 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on all flights shall be equipped with a 406 MHZ ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use. (b) All aircraft should carry an ELT however an aircraft may be operated without an ELT on board where the aircraft is (1) a glider, balloon, airship, ultra-light airplane, gyroplane, restricted or non-Type Certificated aircraft; (2) operated by the holder of a flight training unit operating certificate, engaged in flight training and operated within 25 NM of the aerodrome of departure; (3) engaged in a flight test. (c) Batteries used in ELTs shall be replaced (or recharged if the battery is rechargeable) when - (1) The transmitter has been in use for more than one cumulative hour; or (2) 50 percent of their useful life (or for rechargeable batteries, 50 percent of their useful life of charge) has expired. (d) The expiration date for a replacement or recharged ELT battery shall be legibly marked on the outside of the transmitter. Note: The battery useful life (or useful life of charge) requirements do not apply to batteries (such as water-activated batteries) that are essentially unaffected during probable storage intervals. IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) The following requirements shall be used to determine the quantity and type of 406 MHZ ELT which must be carried. (a) Except as provided for in (b), all airplanes authorized to carry more than 19 passengers shall be equipped with at least one automatic ELT or two ELTs of any type. (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic. (c) Except as provided for in (d), all airplanes authorized to carry 19 passengers or less shall be equipped with at least one ELT of any type. (d) All airplanes authorized to carry 19 passengers or less for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least one automatic ELT. (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.

6.17.4 PCAR Part 7, 7.8.5 and Part 7, IS:7.8.5. 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on all flights shall be equipped with a 406 MHZ ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use. (b) All aircraft should carry an ELT however an aircraft may be operated without an ELT on board where the aircraft is (1) a glider, balloon, airship, ultra-light airplane, gyroplane, restricted or non-Type Certificated aircraft; (2) operated by the holder of a flight training unit operating certificate, engaged in flight training and operated within 25 NM of the aerodrome of departure; (3) engaged in a flight test. (c) Batteries used in ELTs shall be replaced (or recharged if the battery is rechargeable) when - (1) The transmitter has been in use for more than one cumulative hour; or (2) 50 percent of their useful life (or for rechargeable batteries, 50 percent of their useful life of charge) has expired. (d) The expiration date

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6.17.5 PCAR Part 7, 7.8.5 and Part 7, IS: 7.8.5. 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on all flights shall be equipped with a 406 MHZ ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use. (b) All aircraft should carry an ELT however an aircraft may be operated without an ELT on board where the aircraft is (1) a glider, balloon, airship, ultra-light airplane, gyroplane, restricted or non-Type Certificated aircraft; (2) operated by the holder of a flight training unit operating certificate, engaged in flight training and operated within 25 nautical miles of the aerodrome of departure; (3) engaged in a flight test. (c) Batteries used in ELTs shall be replaced (or recharged if the battery is rechargeable) when - (1) The transmitter has been in use for more than one cumulative hour; or (2) 50 percent of their useful life (or for rechargeable batteries, 50 percent of their useful life of charge) has expired. (d) The expiration date for a replacement or recharged ELT battery shall be legibly marked on the outside of the transmitter. Note: The battery useful life (or useful life of charge) requirements do not apply to batteries (such as water-activated batteries) that are essentially unaffected during probable storage intervals. IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) The following requirements shall be used to determine the quantity and type of 406 MHZ ELT which must be carried. (a) Except as provided for in (b), all airplanes authorized to carry more than 19 passengers shall be equipped with at least one automatic ELT or two ELTs of any type. (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic. (c) Except as provided for in (d), all airplanes authorized to carry 19 passengers or less shall be equipped with at least one ELT of any type. (d) All airplanes authorized to carry 19 passengers or less for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least one automatic ELT. (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.

6.17.6 PCAR Part 7, 7.8.5 and Part 7, IS: 7.8.5. 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) [AAC] Except for 7.8.5 (b) all aircraft on all flights shall be

equipped with a 406 MHZ ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use. (b) All aircraft should carry an ELT however an aircraft may be operated without an ELT on board where the aircraft is (1) a glider, balloon, airship, ultra-light airplane, gyroplane, restricted or non-Type Certificated aircraft; (2) operated by the holder of a flight training unit operating certificate, engaged in flight training and operated within 25 nautical miles of the aerodrome of departure; (3) engaged in a flight test. (c) Batteries used in ELTs shall be replaced (or recharged if the battery is rechargeable) when - (1) The transmitter has been in use for more than one cumulative hour; or (2) 50 percent of their useful life (or for rechargeable batteries, 50 percent of their useful life of charge) has expired. (d) The expiration date for a replacement or recharged ELT battery shall be legibly marked on the outside of the transmitter. Note: The battery useful life (or useful life of charge) requirements do not apply to batteries (such as water-activated batteries) that are essentially unaffected during probable storage intervals. IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) The following requirements shall be used to determine the quantity and type of 406 MHZ ELT which must be carried. (a) Except as provided for in (b), all airplanes authorized to carry more than 19 passengers shall be equipped with at least one automatic ELT or two ELTs of any type. (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic. (c) Except as provided for in (d), all airplanes authorized to carry 19 passengers or less shall be equipped with at least one ELT of any type. (d) All airplanes authorized to carry 19 passengers or less for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least one automatic ELT. (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.

- 6.19.1 PCAR Part 7, 7.6.7 AIRBORNE COLLISION AVOIDANCE SYSTEM II (ACAS II) (a) No person may operate a turbine powered airplane with a maximum certificated takeoff mass in excess of 5700 KG, unless it is equipped with an Airborne Collision Avoidance System II (ACAS II) and operate in accordance with the relevant portions of ICAO Annex 10, Volume IV.
- 6.19.3 PCAR Part 7, 7.6.7 AIRBORNE COLLISION AVOIDANCE SYSTEM II (ACAS II) (a) No person may operate a turbine powered airplane with a maximum certificated takeoff mass in excess of 5700 KG, unless it is equipped with an Airborne Collision Avoidance System II (ACAS II) and operate in accordance with the relevant portions of ICAO Annex 10, Volume IV.
- 6.20.1 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV.
- 6.20.2 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV. (1) The aircraft will be assigned a distinct serialized 24-bit address identity code supplied by the Authority for the transponder, this must be programmed prior to operation of the aircraft. (b) No person may operate an aircraft at altitudes above FL 290 unless it is equipped with a system that is automatically

reporting pressure altitudes. (c) No person may operate an aircraft in commercial air transportation unless it is equipped with a pressure-altitude reporting transponder that operates in accordance with the requirements of the Republic of the Philippines.

6.20.3 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV. (1) The aircraft will be assigned a distinct serialized 24-bit address identity code supplied by the Authority for the transponder, this must be programmed prior to operation of the aircraft. (b) No person may operate an aircraft at altitudes above FL 290 unless it is equipped with a system that is automatically reporting pressure altitudes. (c) No person may operate an aircraft in commercial air transportation unless it is equipped with a pressure-altitude reporting transponder that operates in accordance with the requirements of the Republic of the Philippines ATS.

6.25.1 Not implemented.

6.25.2.1 Not implemented.

6.25.2.2 Not implemented.

6.25.3 Not implemented.

Chapter 7

7.1.1 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.

7.1.2 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.

7.2.1 PCAR Part 7, 7.2.9(a)(1)(3). 7.2.9 NAVIGATION EQUIPMENT - GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (1) It's operational flight plan, (3) The requirements of air traffic services

7.2.2 PCAR Part 7, 7.2.9(a)(2)(b). 7.2.9 NAVIGATION EQUIPMENT - GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (2) The prescribed Performance-Based Navigation Specification, and (b) No person may operate an aircraft in Performance-Based Navigation specification airspace unless authorized by CAAP for such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in RNP airspace.

7.2.5 PCAR Part 7, 7.2.7(a)(1)(2). 7.2.7 NAVIGATION EQUIPMENT FOR OPERATIONS IN MNPS AIRSPACE (a) No person may operate an aircraft in MNPS airspace unless it is equipped with navigation equipment that - (1) Continuously provides indications to the flight crew of adherence to or departure from track to the required degree of accuracy at any point along that track; and (2) Be authorized by the State of Registry for operations in such airspace. Note: Equipment shall comply with Minimum

Navigation Performance Specifications (MNPS) prescribed in ICAO Doc 7030 in the form of Regional Supplementary Procedures.

- 7.2.6 PCAR Part 8, 8.8.1.22(d) (1) thru (2) and Part 7, 7.2.8(a) thru (b)(3). 8.8.1.22 OPERATIONS IN MNPS OR RVSM AIRSPACE (d) For flights in designated RVSM airspace: For flights in defined portions of airspace where, based on Regional Air Navigation Agreement, a reduced vertical separation minimum (RVSM) of 300 M (1000 FT) is applied between FL 290 and FL 410 inclusive, an aircraft: (1) shall be provided with equipment which is capable of: (i) indicating to the flight crew the flight level being flown; (iii) automatically maintaining a selected flight level; (iv) providing an alert to the flight crew when a deviation occurs from the selected flight level. The threshold for the alert shall not exceed ± 90 M (300 FT); and (v) automatically reporting pressure-altitude; and (2) shall be authorized by the Authority for operation in the airspace concerned. 7.2.8 NAVIGATION EQUIPMENT FOR OPERATIONS IN RVSM AIRSPACE (a) For flights in defined portions of airspace where, based on Regional Air Navigation Agreement, a reduced vertical separation minimum (RVSM) of 300 M (1000 FT) is applied between FL 290 and FL 410 inclusive, an aircraft: (1) Shall be provided with equipment that is capable of: (i) Indicating to the flight crew the flight level being flown; (ii) Automatically maintaining a selected flight level; (iii) Providing an alert to the flight crew when a deviation occurs from the selected flight level. The threshold for the alert shall not exceed + or - 90 M (300 FT); and (iv) Automatically reporting pressure-altitude (b) Prior to granting an RVSM approval required by paragraph (a)(2), the Republic of the Philippines shall be satisfied that: (1) The vertical navigation performance capability of the aircraft satisfies the requirements specified in ICAO Annex 6, Part I, Appendix 4 (AOC) and ICAO Annex 6, Part II, Appendix 2 (General Aviation). (2) The operator has instituted appropriate procedures in respect of continued airworthiness (maintenance and repair) practices and programmes; and (3) The operator has instituted appropriate flight crew procedures for operations in RVSM airspace. Note: An RVSM approval is valid globally on the understanding that any operating procedures specific to a given region will be stated in the operations manual or appropriate crew guidance.
- 7.2.10 PCAR Part 7, 7.2.8 NAVIGATION EQUIPMENT FOR OPERATIONS IN RVSM AIRSPACE (d) The Authority will take appropriate action in respect of aircraft and operators found to be operating in RVSM airspace in the Republic of the Philippines without a valid RVSM approval. Note 1: These provisions and procedures need to address both the situation where the aircraft in question is operating without approval in the airspace of the State, and the situation where an operator for which the State has regulatory oversight responsibility is found to be operating without the required approval in the airspace of another State. Note 2: See ICAO Doc 9574, Manual on Implementation of a 300 M (1000 FT) Vertical Separation Minimum Between FL 290 and FL 410 Inclusive, for guidance relating to the approval for operations in RVSM airspace.
- 7.4 PCAR Part 7, 7.2.1(a)(4). 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.

Chapter 8

- 8.1.1 PCAR Part 9, 9.4.1.2 MAINTENANCE RESPONSIBILITY (a) Each operator shall ensure the airworthiness of the aircraft and the serviceability of both operational and emergency equipment by (1) Assuring the accomplishment of preflight inspections; (2) Assuring the correction of any defect and/or damage affecting safe operation of an

- aircraft to an approved standard, taking into account the MEL and CDL if available for the aircraft type; (3) Assuring the accomplishment of all maintenance in accordance with the approved operator's aircraft maintenance program; (4) The analysis of the effectiveness of the operator's approved aircraft maintenance program; (5) Assuring the accomplishment of any operational directive, airworthiness directive and any other continued, airworthiness requirement made mandatory by the Authority; and (6) Assuring the accomplishment of modifications in accordance with an approved standard and for non-mandatory modifications, the establishment of an embodiment policy. (b) Each operator shall ensure that the Certificate of Airworthiness for each aircraft operated remains valid in respect to (1) The requirements in paragraph (a); (2) The expiration date of the Certificate; and (3) Any other maintenance condition specified in the Certificate.
- 8.1.3 PCAR Part 9, 9.4.1.5 MAINTENANCE MANAGEMENT (c) Each operator shall employ a person or group of persons, acceptable to the Authority; to ensure that all maintenance is carried out to an approved standard such that the maintenance requirements of Subpart 9.4.1.2 and requirements of the AOC holder's maintenance control manual are satisfied; and to ensure the functioning of the quality system.
- 8.1.4 PCAR Part 9, 9.4.1.2(a)(3). 9.4.1.2 MAINTENANCE RESPONSIBILITY (a) Each operator shall ensure the airworthiness of the aircraft and the serviceability of both operational and emergency equipment by (3) Assuring the accomplishment of all maintenance in accordance with the approved operator's aircraft maintenance program.
- 8.3.1 PCAR Part 9, 9.4.1.11 OPERATOR'S MAINTENANCE PROGRAM (a) Each operator's aircraft maintenance program and any subsequent amendment shall be submitted to the State of Registry for approval; acceptance by the Authority will be conditioned upon prior approval by the State of Registry, or where appropriate, upon the operator complying with recommendations provided by the State of Registry. The design and application of the operator's maintenance program shall observe Human Factors principles.
- 8.4.3 PCAR Part 9, 9.4.1.7(c)(d). 9.4.1.7 MAINTENANCE RECORDS (c) Each operator shall ensure that in the event of temporary change of operator, the records specified in paragraph (a) shall be made available to the new operator. (d) Each operator shall ensure that when an aircraft is permanently transferred from one operator to another operator, the records specified in paragraph (a) are also transferred.
- 8.5.1 PCAR Part 8, 8.3.1.1 APPLICABILITY (f) The owner of an airplane over 5700 KG maximum certificated take-off mass, or in the case where it is leased, the lessee, shall, as prescribed by the State of Registry, ensure that the information resulting from maintenance and operational experience with respect to continuing airworthiness, is transmitted as required by Subpart 8.3.16.
- 8.5.2 PCAR Part 9, 9.4.1.4(b)(8) (9). 9.4.1.4 MAINTENANCE CONTROL MANUAL (b) The AOC holder's maintenance control manual shall contain the following information which may be issued in separate parts (8) A description of the procedures for obtaining and assessing continuing airworthiness information and implementing any resulting actions for all aircraft over 5700 KG maximum certificated take-off mass, from the organization responsible for the type design, and shall implement such actions considered necessary by the State of Registry; (9) A description of the procedures for implementing mandatory continuing airworthiness information as required in Subpart 9.4.1.2(a)(5) .
- 8.6 PCAR Part 9, 9.4.1.10 MODIFICATION AND REPAIRS (a) All modifications and repairs shall comply with airworthiness requirements of this CAR and acceptable to the

authority and the State of Registry. Procedures shall be established to ensure that the substantiating data supporting compliance with the airworthiness requirements are retained. However, in the case of a major repair or major modification, the work must have been done in accordance with technical data approved by the Authority.

Chapter 9

- 9.4.1.1 PCAR Part 2, 2.3.1.6(a) and Part 8, 8.4.1.9(a)(1). 2.3.1.6 RECENT EXPERIENCE REQUIREMENTS (a) A pilot shall not operate an aircraft carrying passengers as pilot-in-command or copilot to operate at the flight controls of a type or a variant of a type of aircraft during take-off and landing unless that pilot has operated the flight controls during at least three take-offs and landings within the preceding 90 days on the same type of aircraft or in the flight simulator approved for the purpose. 8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (a) No person may act as PIC or a co-pilot of an aircraft unless, within the preceding 90 days that pilot has: (1) Made three take-offs and landings as the sole manipulator of the flight controls in an aircraft of the same category and class and if a type rating is required, of the same type or in a flight simulation training device approved for the purpose. PCAR Part 2, 9.4.2.2 PCAR Part 2, 2.3.1.6(c)(1) (2) Part 8, 8.4.1.9(d)(1)(2). 2.3.1.6 RECENT EXPERIENCE REQUIREMENTS (c) A pilot shall not act in the capacity of a cruise relief pilot in a type or variant of a type of aircraft unless, within the preceding 90 days that pilot has either (1) Operated as a PIC, C P or cruise relief pilot on the same type of aircraft; or (2) Carried out flying skill refresher training including normal, abnormal and emergency procedures specific to cruise flight on the same type of aircraft or in a flight simulator approved for the purpose, and has practiced approach and landing procedures, where the approach and landing procedure practice may be performed as the pilot who is not flying the aircraft. 8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (d) No person may act as a cruise relief pilot in a commercial air transport unless within the preceding 90 days, that pilot has either: (1) Operated as PIC, C P or cruise pilot on the same type of aircraft; or (2) Carried out flying skill refresher training including normal, abnormal and emergency procedures specific to cruise flight on the same type of aircraft or in a flight simulator approved for the purpose, and has practiced approach and landing procedures, where the approach and landing procedure practice may be performed as the pilot who is not flying the aircraft.
- 9.4.2.2 PCAR Part 2, 9.4.2.2 and PCAR Part 2, 2.3.1.6(c)(1) (2) and Part 8, 8.4.1.9(d)(1)(2). 2.3.1.6 RECENT EXPERIENCE REQUIREMENTS (c) A pilot shall not act in the capacity of a cruise relief pilot in a type or variant of a type of aircraft unless, within the preceding 90 days that pilot has either (1) Operated as a PIC, C P or cruise relief pilot on the same type of aircraft; or (2) Carried out flying skill refresher training including normal, abnormal and emergency procedures specific to cruise flight on the same type of aircraft or in a flight simulator approved for the purpose, and has practiced approach and landing procedures, where the approach and landing procedure practice may be performed as the pilot who is not flying the aircraft. 8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (d) No person may act as a cruise relief pilot in a commercial air transport unless within the preceding 90 days, that pilot has either: (1) Operated as PIC, CP or cruise pilot on the same type of aircraft; or (2) Carried out flying skill refresher training including normal, abnormal and emergency procedures specific to cruise flight on the same type of aircraft or in a flight simulator approved for the purpose, and has practiced approach and landing procedures, where the approach and landing procedure practice may be performed as the pilot who is not flying the aircraft.
- 9.4.4.1 PCAR Part 9, 9.3.1.9(b) and Part 8, 8.10.1.20(b)(c). 9.3.1.9 CREW MEMBER CHECKING AND STANDARDIZATION PROGRAM (b) An operator shall check pilots'

proficiency on those maneuvers and procedures that are prescribed by the Authority for pilot proficiency checks, which shall include emergency procedures and, where applicable, instrument flight rules. Note: A standardized process is defined to address the operator unique fleet differences and compliance methods. Note: See Part 8 for specific checking requirements 8.10.1.20 AIRCRAFT AND INSTRUMENT PROFICIENCY CHECKS (b) An operator shall ensure that piloting technique and the ability to execute emergency procedures is checked in such a way as to demonstrate the pilot's competence on each type or variant of type of aircraft. Where the operation may be conducted under IFR, an operator shall ensure that the pilot's competence to comply with such rules is demonstrated to either a check pilot of the operator or to a representative of the Authority. A pilot may complete both these requirements simultaneously in a specific aircraft type. (c) Such checks shall be performed twice within any period of one (1) year. Any two (2) such checks which are similar and which occur within a period of four (4) consecutive months shall not alone satisfy this requirement.

Chapter 10

- 10.1 PCAR Part 8, 8.10.1.7(a) thru (b). 8.10.1.7 PERSONS QUALIFIED TO FLIGHT RELEASE (a) No person may act as a flight operations officer in releasing a Scheduled passenger carrying commercial air transport operation unless that person: (1) Holds a flight operations officer license or an ATP license; and (2) Is currently qualified with the Operator for the operation and type of aircraft used. (b) When the Authority requires that a flight operations officer/flight dispatcher, employed in conjunction with an approved method of control and supervision of flight operations, be licensed, that flight operations officer/flight dispatcher shall be licensed in accordance with the provisions of Part 2 of these Regulations.the Philippines, for more than 10 consecutive hours of duty in a 24-hour period if that flight operations officer/flight dispatcher is relieved of all duty with the operator for at least eight hours during each 24-hour period.

Chapter 12

- 12.3 PCAR Part 8, 8.9.2.4 CABIN CREW MEMBERS AT DUTY STATIONS (b) Each cabin crew member assigned to emergency evacuation duties shall occupy a seat provided in accordance with Subpart 7.9.1 during take-off and landing and whenever the PIC so directs, with seat belt or, when provided, safety harness fastened.
- 12.4 PCAR Part 8, 8.10.1.34(c)(1) to (6). 8.10.1.34 RECURRENT TRAINING: CABIN CREW (c) The recurrent training program shall ensure that each cabin crew member is: (1) Competent to execute those safety duties and functions which the cabin crew member is assigned to perform in the event of an emergency or in a situation requiring emergency evacuation; (2) Drilled and capable in the use of emergency and life-saving equipment required to be carried, such as life jackets, life rafts, evacuation slides, emergency exits, portable fire extinguishers, oxygen equipment and first-aid kits; (3) When serving on airplanes operated above 3000 M (10000 FT), knowledgeable as regards the effect of lack of oxygen and, in the case of pressurized airplanes, as regards physiological phenomena accompanying a loss of pressurization; (4) Aware of other crew members' assignments and functions in the event of an emergency so far as is necessary for the fulfillment of the cabin crew member's own duties; (5) Aware of the types of dangerous goods which may, and may not, be carried in a passenger cabin and has completed the dangerous goods training programme required by Annex 18; and (6) Knowledgeable about human performance as related to passenger cabin safety duties including flight crew-cabin crew coordination.

Chapter 14

14.2 Not implemented.

14.3 Not implemented.

14.4 Not implemented.

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Chapter 1

1.2 PCAR Part 8, 8.1.1.1(a) to (c) 8.1.1.1. APPLICABILITY (a) (1) The Standards and Recommended Practices contained in Annex 6 Part I shall be applicable to the operation of airplanes by operators authorized to conduct commercial air transport operations international and domestic). (2) The Standards and Recommended Practices contained in Annex 6, Part II (airplanes), and Annex 6 Part III Section III (helicopters) shall be applicable to general aviation operations (international and domestic). (3) The Standards and Recommended Practices contained in Annex 6, Part III Sections I and II, shall be applicable to all helicopters engaged in commercial air transport operations (international and domestic), except for these Standards and Recommended Practices are not applicable to helicopters in aerial work. (b) CAR Part 8 prescribes the requirements for: (1) Operations conducted by airman certified in Republic of the Philippines while operating aircraft registered in Republic of the Philippines. (2) Operations of foreign registered aircraft by Republic of the Philippines Operators. (3) Operations of aircraft within Republic of the Philippines by airman or Operators of a foreign State. (c) For operations outside of Republic of the Philippines, all Republic of the Philippines pilots and operators shall comply with these requirements unless compliance would result in a violation of the laws of the foreign State in which the operation is conducted. Note: Where a particular requirement is applicable only to a particular segment of aviation operations, it will be identified by a reference to those particular operations, such as "commercial air transport" or "small non-turbojet or turbofan airplanes". Note: Those specific subsections not applicable to foreign operators will include the phrase "This requirement is not applicable to foreign operators".

Chapter 2

2.1.1.1 PCAR Part 8, 8.5.1.2(a). 8.5.1.2 COMPLIANCE WITH LOCAL REGULATIONS (a) The PIC shall comply with the relevant laws, regulations and procedures of the States in which the aircraft is operated.

2.1.1.3 PCAR Part 8, 8.12.1.2(d)(1). 8.12.1.2 QUALIFIED PERSONS REQUIRED FOR OPERATIONAL CONTROL FUNCTIONS (d) For all flights, the PIC shares in the responsibility for operational control of the aircraft and has the situational authority to make decisions regarding operational control issues in-flight. (1) Where a decision of the PIC differs from that recommended, the person making the recommendation shall make a record of the associated facts.

2.2.1.1 PCAR Part 8, 8.6.2.2(c). 8.6.2.2 ADEQUACY OF OPERATING FACILITIES (c) The PIC shall not commence a flight unless it has been ascertained by every reasonable means available that the ground and/or water areas and facilities available and directly required for such flight and for the safe operation of the aircraft are adequate, including communication facilities and navigation aids. Note: Reasonable means "denotes use, at the point of departure, of information available to the PIC either through official information published by the aeronautical information services or readily obtainable in other sources.

2.2.2.2.1 PCAR Part 8, 8.8.1.7(a). PCAR Part 8 8.8.1.7 INSTRUMENT APPROACH OPERATING MINIMA (a) No person may operate to or from an airport/heliport using

- operating minima lower than those which may be established for that airport/heliport by the State in which it is located, unless that State specifically approves that operation, in accordance with the provisions of Implementing Standard: IS: 8.8.1.7.
- 2.2.3.1 PCAR Part 8, 8.6.2.1(a) (1)to(6). 8.6.2.1 AIRCRAFT AIRWORTHINESS AND SAFETY PRECAUTIONS (a) A flight, or series of flights, shall not be commenced until flight preparation forms have been completed certifying that the PIC is satisfied that: (1) the aircraft is airworthy, duly registered and that appropriate certificates with respect thereto are aboard the aircraft; (2) the instruments and equipment prescribed in Part 7 for the particular type of operation to be undertaken, are appropriate, taking in account the expected flight conditions, installed and sufficient for the flight; (3) any necessary maintenance has been performed and a maintenance release, if applicable, has been issued in respect to the aircraft; (4) the mass of the aircraft and center of gravity location are such that the flight can be conducted safely, taking into account the flight conditions expected; (5) any load carried is properly distributed and safely secured; (6) a check has been completed indicating that the operating limitations, contained in the flight manual or its equivalent, can be complied with for the flight to be undertaken.
- 2.2.3.4.1 PCAR Part 8, 8.6.2.4 WEATHER LIMITATIONS FOR VFR FLIGHTS (a) No person will commence a flight to be conducted in accordance with the VFR unless available current meteorological reports, or a combination of current reports and forecasts, indicate that the meteorological conditions along the route, or that part of the route to be flown under the VFR, will, at the appropriate time, allow VFR operations.
- 2.2.3.6.1 PCAR Part 8, 8.6.2.13.1(a), PCAR Part 8.6.2.15.1(a) (b)(1)(2)(3)(i)(ii) and PCAR Part 8, 8.6.2.14(a)(1)(2). 8.6.2.13.1 FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (a) A flight shall not be commenced unless, taking into account both the meteorological conditions and any delays that are expected in flight, the aircraft carries sufficient fuel and oil to ensure that it can safely complete the flight, and, applicable special provisions are complied with, as stated in Subpart 8.6.2.14 or 8.6.2.15. In addition, a reserve shall be carried to provide for contingencies. 8.6.2.15.1 COMMERCIAL AIR TRANSPORT: PROPELLER DRIVEN AIRPLANES The fuel and oil earned in order to comply with Subpart 8.6.2.13.1 shall, in the case of propeller driven airplanes, be at least the amount sufficient to allow the airplane: (a) When a destination alternate airport is required, either: (1) to fly to the airport to which the flight is planned thence to the most critical (in terms of fuel consumption) alternate airport specified in the operational and ATS flight plans and thereafter for a period of 45 minutes; or (2) to fly to the airport to which the flight is planned, then 15 minutes of holding at 1500 FT, thence to the most critical (in terms of fuel consumption) alternate airport specified in the operational and ATS flight plans and thereafter for a period of 30 minutes holding at 1500 FT; or (3) to fly to the alternate airport via any predetermined point and thereafter for 45 minutes, provided that this shall not be less than the amount required to fly to the airport to which the flight is planned and thereafter for: (i) 45 minutes plus 15 per cent of the flight time planned to be spent at the cruising level(s), or (ii) two hours, whichever is less. 8.6.2.14 MINIMUM FUEL AND OIL SUPPLY FOR VFR FLIGHTS (a) For airplanes: No person may commence a flight in an airplane under VFR unless, considering the wind and forecast weather conditions, there is enough fuel to fly to the first point of intended landing and, assuming normal cruising speed (1) For turbo-jet aircraft: flights during the day, for at least 30 minutes thereafter; or (2) For Propeller-driven aircraft: flights at night, for at least 45 minutes thereafter.
- 2.2.3.6.2 PCAR Part 8, 8.6.2.15.1 (c) Airplanes. No person may commence a flight under IFR, or continue past the point of in-flight re-planning, unless there is enough fuel supply, considering meteorological conditions and any delays that are expected in flight, to include the following: (1) Taxi fuel – which shall be the amount of fuel expected to be

consumed before take-off; (2) Trip fuel – which shall be the amount of fuel required to enable the airplane to fly from take-off, or the point of in-flight re-planning, until landing at the destination aerodrome taking into account the operating conditions in the data provided by the manufacturer; (3) Contingency fuel – which shall be the amount of fuel required to compensate for unforeseen factors. It shall be five percent of the planned trip fuel or of the fuel required from the point of in-flight re-planning based on the consumption rate used to plan the trip fuel, but in any case, shall not be lower than the amount required to fly for five minutes at holding speed at 450 M (1500 FT) above the destination aerodrome in standard conditions; (4) Destination alternate fuel – which shall be (i) Where a destination alternate aerodrome is required, the amount of fuel required to enable the airplane to: (A) Perform a missed approach at the destination aerodrome; (B) Climb to the expected cruising altitude; (C) Fly the expecting routing; (D) Descend to the point where the expected approach is initiated; and (E) Conduct the approach and landing at the destination alternate aerodrome; or (ii) Where two destination alternate aerodromes are required, the amount of fuel, as calculated in (4)(i) above, required to enable the airplane to proceed to the destination alternate aerodrome which requires the greater amount of alternate fuel; or (iii) Where a flight is operated without a destination alternate aerodrome, the amount of fuel required to enable the airplane to fly for 15 minutes at holding speed at 450 M (1500 FT) above destination aerodrome elevation in standard conditions; or (iv) Where the aerodrome of intended landing is an isolated aerodrome: (A) For helicopters, the amount of fuel required to fly for 45 minutes plus 15 percent of the flight time planned to be spend at cruising level, including final reserve fuel, or two hours, whichever is less; or (B) For a turbine-engined airplane, the amount of fuel required to fly for two hours at normal cruise consumption above the destination aerodrome, including final reserve fuel; (3) Final reserve fuel – which shall be the amount of fuel calculated using the estimated mass on arrival at the destination alternate aerodrome, or the destination aerodrome when no destination alternate aerodrome is required, or a pre-calculated value for each airplane type and variant in the fleet rounded up to an easily recalled figure: (i) For a helicopters, the amount of fuel required to fly for 45 minutes, under speed and altitude conditions specified by the Authority; or (ii) For a turbine-engined airplane, the amount of fuel required to fly for 30 minutes at holding speed at 450 M (1500 FT) above aerodrome elevation in standard conditions; (4) Additional fuel – which shall be the supplementary amount of fuel required if the minimum fuel calculated in accordance with trip fuel, contingency fuel, destination alternate fuel and final reserve fuel above is not sufficient to: (i) Allow the airplane to descend as necessary and proceed to an alternate aerodrome in the event of engine failure or loss or pressurization, whichever requires the greater amount of fuel based on the assumption that such a failure occurs at the most critical point along the route; (A) To fly for 15 minutes at holding speed at 450 M (1500 FT) above the aerodrome elevation in standard conditions; and (B) Make an approach and landing; (C) Allow an airplane engaged in EDTO to comply with the EDTO critical fuel scenario as established by the Authority; (D) Meet additional requirements not covered above. Note: Fuel planning for a failure that occurs at the most critical point along a route may place the airplane in a fuel emergency situation. (5) Discretionary fuel – shall be the extra amount of fuel to be carried at the discretion of the PIC.

- 2.2.4.1.3 PCAR Part 8, 8.8.4.13 CONTINUATION OF AN INSTRUMENT APPROACH (d) If, after entering the final approach segment or after descending below 300 M (1000 FT) above the aerodrome elevation the reported visibility or controlling RVR falls below the specified minimum, the approach may be continued to DA/H or MDA/H. In any case, an airplane shall not continue approach-to-land any aerodrome beyond a point at which the limits of the operating minima specified for that aerodrome would be infringed. Note: Controlling RVR means the reported values of one or more RVR

- reporting locations (touchdown, mid-point and stop-end) used to determine determined whether minima are or are not met. Where RVR is used, the controlling RVR, unless otherwise specified by State criteria. (per MC No. 04-14 series of 2014 dated 30 January 2014)
- 2.2.4.5.1 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (a) Each required flight crew member shall remain at the assigned duty station during take-off and landing and critical phases of flight.
 - 2.2.4.5.2 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (b) Each flight crew member shall remain at his or her station during all phases of flight unless: (1) Absence is necessary for the performance of his or her duties in connection with the operation; (2) Absence is necessary for physiological needs, provided one qualified pilot remains at the controls at all times; or (3) The crew member is taking a rest period and a qualified relief crew member replaces him or her at the duty station. (i) For the assigned PIC during the en route cruise portion of the flight by a pilot who holds an airline transport pilot license and an appropriate type rating, and who is currently qualified as PIC or CP, and is qualified as PIC of that aircraft during the en route cruise portion of the flight; and (ii) In the case of the assigned CP, by a pilot qualified to act as PIC or Co-pilot of that aircraft during en route operations.
 - 2.2.4.5.3 PCAR Part 8, 8.5.1.6 FLIGHT CREW MEMBER USE OF SEAT BELTS AND SAFETY HARNESES (a) Each flight crew member shall have his or her seat belts fastened during take-off and landing and all other times when seated at his or her station.
 - 2.2.5.1 PCAR Part 8, 8.5.1.1 AUTHORITY AND RESPONSIBILITY OF THE PIC (b) The PIC shall be responsible for the operations and safety of the aircraft and for the safety of all crew members, passengers and cargo on board, when the doors are closed. The PIC shall also be responsible for the operation and safety of the aircraft from the moment the aircraft is ready to move for the purpose of taking off until the moment it finally comes to rest at the end of the flight and the engine(s) used as primary propulsion units are shut down (and in the case of helicopters, rotor blades stopped).
 - 2.2.6 PCAR Part 8, 8.9.2.14 CARRY-ON BAGGAGE (a) No person may allow the boarding of carry-on baggage in to the passenger cabin unless it is adequately and securely stowed in accordance with the operator's approved Operations Manual procedures.
 - 2.3.1.1 PCAR Part 8, 8.7.1.2 GENERAL (a) No person may operate an aircraft that: (1) Exceeds its designed performance limitations for any operation, as established by the State of Registry; (2) Exceeds the operating limitations contained in the aircraft flight manual, or its equivalent; or (3) Exceeds the terms of its certificate of airworthiness. (4) Exceeds the mass limitations, if applicable, imposed by the terms of its noise certification standards, as contained in the applicable part of ICAO Annex 16, Volume I, unless otherwise approved by the Authority. 8.2.1.6 CIVIL AIRCRAFT FLIGHT MANUAL, MARKING AND PLACARD REQUIREMENTS (b) No person may operate a civil aircraft within or over Republic of the Philippines without complying with the operating limitations specified in the approved AFM or RFM, markings and placards, or as otherwise prescribed by the certifying authority for the aircraft's State of Registry.
 - 2.3.1.2 PCAR Part 8, 8.2.1.6 CIVIL AIRCRAFT FLIGHT MANUAL, MARKING AND PLACARD REQUIREMENTS (d) Each operator shall display in the aircraft all placards, listings, instrument markings or combination thereof, containing those operating limitations prescribed by the certifying authority for the aircraft's State of Registry for visual presentation.
 - 2.3.1.3 PCAR Part 8, 8.7.1.4 No person may take-off an aircraft without ensuring that the maximum allowable mass for a flight does not exceed the maximum allowable take-off

or landing mass, or any applicable en route performance or landing distance limitations considering the (1) Condition of the take-off and landing areas to be used; (2) Gradient of runway to be used (landplanes only); (3) Pressure altitude; (4) Ambient temperature, (5) Current and forecast winds; and (6) Any known conditions (e.g., atmospheric and aircraft configuration) which may adversely affect aircraft performance, or compliance with noise certification standards. (b) No person may take-off an aircraft at a mass that, assuming normal engine operation, cannot safely clear all obstacles during all phases of flight, including all points along the intended en route path or any planned diversions. (c) In no case shall the mass at the start of take-off, or at the expected time of landing at the airport of intended landing and at any destination alternate airport, exceed the relevant maximum masses at which compliance has been demonstrated with the applicable noise certification Standards in Annex 16, Volume I, unless otherwise authorized in exceptional circumstances for a certain airport or a runway where there is no noise disturbance problem, by the competent authority of the State in which the airport is situated. (d) In no case shall the estimated mass for the expected time of landing at the airport of intended landing and at any destination alternate airport, exceed the maximum landing mass specified in the flight manual for the pressure-altitude appropriate to the elevation of those airports, and if used as a parameter to determine the maximum landing mass, any other local atmospheric condition.

- 2.4.1 PCAR Part 7, 7.1.4 GENERAL INSTRUMENT AND EQUIPMENT REQUIREMENTS (a) In addition to the minimum equipment necessary for the issuance of a certificate of airworthiness, the instruments, equipment and flight documents prescribed in Part 7 shall be installed or carried, as appropriate, in aircraft according to the aircraft used and to the circumstances under which the flight is to be conducted. (b) All required instruments and equipment shall be approved and installed in accordance with applicable airworthiness requirements.
- 2.4.2.1 PCAR Part 7, 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (1) Control the flight path of the aircraft; (2) Carry out any required procedural manoeuvres; (3) Observe the operating limitations of the aircraft in the expected operating conditions; and (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.
- 2.4.2.2 PCAR Part 7, 7.8.11 FIRST-AID AND EMERGENCY MEDICAL KIT (a) No person may operate an aircraft unless it is equipped with first-aid kits readily accessible to cabin crew located near an exit and, on passenger flights, an approved emergency medical kit for treatment of injuries or medical emergencies that might occur during flight time or in minor accidents. 7.8.6 PORTABLE FIRE EXTINGUISHERS (a) No person may operate an aircraft unless it is equipped with portable fire extinguishers accessible for use in crew, passenger, and cargo compartments as follows: (1) The type and quantity of extinguishing agent shall be suitable for the kinds of fires likely to occur in the compartment where the extinguisher is intended to be used. Note: For passenger compartments, the extinguisher shall be designed to minimize the hazard of toxic gas concentrations. (3) At least one portable fire extinguisher shall be conveniently located on the flight deck for use by the flight crew. (4) At least one portable fire extinguisher shall be conveniently located in the passenger compartment of airplanes having a passenger seating capacity of 30 or less. 7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (1) A seat with safety belt for each person on board over the age of two (2) years old; and a restraining belt for each berth on board the aircraft PCAR Part 8 8.2.1.8 DOCUMENTS TO BE

- CARRIED ON AIRCRAFT: ALL OPERATIONS (10) AFM (for airplanes) or RFM (for helicopters). (14) Operational Flight Plan. 8.8.1.28 INTERCEPTION (a) When intercepted by a military or government aircraft, each PIC shall comply with the international standards when interpreting and responding to visual signals and communication as specified in IS: 8.8.1.28. (b) No pilot may conduct an international flight unless the procedures and signals relating to interception of aircraft, as specified in IS: 8.8.1.28, are readily available on the flight deck. 7.8.3 VISUAL SIGNALLING DEVICES (a) [AAC] No person may operate an aircraft over water or across land areas which have been designated by the Republic of the Philippines as areas in which search and rescue would be especially difficult, unless equipped with such signaling devices as may be appropriate to the area overflown, to include (1) Visual signals for use by intercepting and intercepted aircraft 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (b) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in general aviation operations unless it has within it the following current and approved documents: (3) Aircraft Journey Log. 7.9.8 PROTECTIVE CIRCUIT FUSES [AAC] No person may operate an aircraft in which protective fuses are installed unless there are spare fuses available for use in flight equal to at least 10% of the number of fuses of each rating or three of each rating whichever is the greater.
- 2.4.3.1 PCAR Part 7, 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) [AAC] No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (5) A magnetic compass; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required).
- 2.4.4.1 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided. (b) Seaplanes. (1) For all flights, seaplanes shall be equipped with the equipment prescribed in paragraph (2). 7.9.15 MARITIME SOUND SIGNALLING DEVICE All seaplanes for all flights shall be equipped with equipment for making the sound signals prescribed in the International Regulations for Preventing Collisions at Sea, where applicable. 7.9.16 ANCHORS All seaplanes for all flights shall be equipped with one anchor, and one sea anchor (drogue), when necessary to assist in maneuvering (approval for the anchors not required). Note. - "Seaplanes" includes amphibians operated as seaplanes.
- 2.4.4.3.1 PCAR Part 7, IS: 7.8.4 SURVIVAL EQUIPMENT FLIGHTS OVER WATER (b) Where life rafts are required to be carried in accordance with subpart 7.8.17 (d), they shall be equipped with an attached survival kit containing at least the following: 1) a pyrotechnic signaling device; 2) a radar reflector; 3) a life raft repair kit; 4) a bailing bucket and sponge; 5) a signaling mirror; 6) a whistle; (7) a raft knife; 8) an inflation pump; 9) dye marker; 10) a waterproof flashlight; 11) a two day supply of water, calculated using the overload capacity of the raft, consisting of one pint of water per day for each person or a means of desalting or distilling salt water sufficient to provide an equivalent amount; 12) a fishing kit; 13) a book on sea survival; and 14) a first aid kit containing antiseptic swabs, burn dressing compresses, bandages and anti-motion sickness pills.
- 2.4.4.3.2 PCAR Part 7, 7.8.17 LIFE RAFT (b) An aircraft shall have lifesaving rafts with a sufficient capacity to carry all persons on board in the event of the loss of one raft of the largest capacity. (d) Life rafts shall be equipped with the following life sustaining equipment - (1) A electric survivor locator light; (2) A survival kit equipped in

accordance with the requirements of subsection 7.8.4; (3) A pyrotechnical signaling device.

- 2.4.5 PCAR Part 7, 7.8.4 SURVIVAL EQUIPMENT (a) Subject to subsection (b), no person shall operate an aircraft over land unless there is carried on board survival equipment, sufficient for the survival on the ground of each person on board, given the geographical area, the season of the year and anticipated seasonal climatic variations, that provides the means for (1) starting a fire; (2) providing shelter; (3) providing or purifying water; and (4) visually signaling distress.
- 2.4.6.1 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority.
- 2.4.6.2 PCAR Part 7, 7.6.2 LOSS OF PRESSURIZATION INDICATOR All pressurized aircraft intended to be operated at altitudes at or above 25,000 above sea level shall be equipped with a device to provide positive warning to the flight crew of any dangerous loss of pressurization.
- 2.4.7 PCAR Part 7, 7.2.3 IFR INSTRUMENTS (a) All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments, shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate. 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator; (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight Instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope).
- 2.4.8 PCART Part 7, 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator; (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight Instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope). 7.2.3 IFR INSTRUMENTS (a) All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments, shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - (1) An airspeed indicating system with a means of preventing malfunctioning due to either condensation or icing; (2) A sensitive pressure

- altimeter with counter drum-pointer or equivalent presentation, calibrated in feet with a sub-scale setting calibrated in hectopascals/millibars, adjustable for any barometric pressure likely to be set during flight; (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate. 7.4.1 AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (a) All aircraft operated at night shall be equipped with: (1) A landing light; (2) An anti-collision light system; (3) Navigation/position lights; (4) Lights to conform to the International regulations for preventing collisions at sea if the aircraft is a seaplane or an amphibian aircraft. (5) Illumination for all flight instruments and equipment that are essential for the safe operation of the aircraft; (6) Lights in all passenger compartments; (7) A flashlight for each crew member station (approval not required).
- 2.4.9 PCAR Part 8,8.2.1.8(a)(9). 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (a) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in commercial air transport operations unless it has within it the following current and approved documents: (9) Noise Certificate, or a document attesting noise certification. When the document, or a suitable statement attesting noise certification as contained in another document approved by the State of Registry, is issued in a language other than English, it shall include an English translation.
- 2.4.11.1 PCAR Part 7, 7.6.5 GROUND PROXIMITY WARNING SYSTEM (a) No person may operate an airplane with a maximum certified take-off mass in excess of 5700 KG, unless it is equipped with a ground proximity warning system which has a forward looking terrain avoidance function.
- 2.4.11.4 PCAR Part 7, 7.6.5 GROUND PROXIMITY WARNING SYSTEM (b) Each ground proximity warning system shall automatically provide, by means of aural signals which may be supplemented by visual signals, timely and distinctive warning to the flight crew of sink rate, ground proximity, altitude loss after take-off or go around, incorrect landing configuration and downward glideslope deviation.
- 2.4.11.7 PCAR Part 7, 7.6.5(a)(c) (1) to (5). 7.6.5 GROUND PROXIMITY WARNING SYSTEM (a) [AAC] No person may operate an airplane with a maximum certified take-off mass in excess of 5700 KG, unless it is equipped with a ground proximity warning system which has a forward looking terrain avoidance function. (c) A ground proximity warning system shall provide, as a minimum, warnings of the following circumstances - (1) Excessive descent rate. (2) Excessive terrain closure rate. (3) Excessive altitude loss after take-off or go-around. (4) Unsafe terrain clearance while not in landing configuration; and (5) Excessive descent below the instrument glide path.
- 2.4.12.2 PCAR Part 7, 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on fall flights shall be equipped with a 406 MHz ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3.
- 2.4.12.3 PCAR Part 7, IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic.
- 2.4.12.4 PCAR Part 7, 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (a) Except for 7.8.5 (b) all aircraft on fall flights shall be equipped with a 406 MHz ELT of a type and quantity as specified in IS: 7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3.
- 2.4.13.1 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless

- that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV.
- 2.4.13.2 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV.
- 2.4.16.1.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (b) Flight data recorder media not acceptable for use in aircraft registered in the Republic of the Philippines, are - (1) Engraving metal foil.
- 2.4.16.2.2 PCAR Part 7, 7.7.3.3 (a) CVS media not acceptable for use in aircraft registered in the Philippines, or operated in commercial air transport operations in Philippines, are - (1) Magnetic tape and wire.
- 2.4.16.2.3.1 PCAR Part 7, 7.7.2 COCKPIT VOICE RECORDERS (c) A CVR shall be capable of retaining the information recorded during at least the last (2) 2 hours, for airplanes of a maximum certificated take-off mass of over 5700 KG or a helicopter that receives type certification after 1 January 2003.
- 2.4.16.3.1.1 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio. (b) Sufficient information to derive the content of the data link communications message and, whenever practical, the time the message was displayed to or generated by the crew shall be recorded. Note: Data link communications include, but are not limited to, automatic dependent surveillance — contract (ADS-C), controller-pilot data link communications (CPDLC), data link-flight information services (D-FIS) and aeronautical operational control messages.
- 2.4.16.3.1.2 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio. (b) Sufficient information to derive the content of the data link communications message and, whenever practical, the time the message was displayed to or generated by the crew shall be recorded. Note: Data link communications include, but are not limited to, automatic dependent surveillance - contract (ADS-C), controller-pilot data link communications (CPDLC), data link-flight information services (D-FIS) and aeronautical operational control messages.
- 2.4.16.3.2 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
- 2.4.16.3.3 PCAR Part 7, 7.7.3(a) last sentence 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.

- 2.4.16.4.2.2 PCAR Part 7,7.7.1(a)(3) and Part 8,8.5.1.24(c). 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall (3) Be deactivated upon completion of a flight following an accident or incident. 8.5.1.24 OPERATION OF FLIGHT DECK VOICE AND FUGHT DATA RECORDERS (c) In event of an accident or incident, the PIC shall act to preserve the recorded data for subsequent investigation.
- 2.4.16.4.3 PCAR Part 8,8.5.1.24(c) 8.5.1.24 OPERATION OF FLIGHT DECK VOICE AND FUGHT DATA RECORDERS (c) In event of an accident or incident, the PIC shall act to preserve the recorded data for subsequent investigation.
- 2.4.16.4.4 PCAR Part 7,7.7.1(a)(4) Part 8,8.5.1.24(a)(1)(2) 7.7.1 FLIGHT RECORDERS (a) [AAC] Flight recorders shall— (4) The FDR system shall be recalibrated at least every five years to determine any discrepancies in the engineering conversion routines for the mandatory parameters, in accordance with the requirements of the aircraft manufacturer to ensure that the parameters are being recorded within the calibration tolerances; and when the parameters of altitude and airspeed are provided by sensors that are dedicated to the FDR system, there shall be a recalibration performed as recommended by the sensor manufacturer, or at least every two years. 8.5.1.24 OPERATION OF FLIGHT DECK VOICE AND FUGHT DATA RECORDERS (a) The PIC shall ensure that whenever an aircraft has flight recorders installed, those recorders are operationally checked and operated continuously from the instant: (1) For a flight data recorder, the aircraft begins its take-off roll until it has completed the landing roll, and (2) For a flight deck voice recorder, the initiation of the pre-start checklist until the end of the securing aircraft checklist.
- 2.4.17.1 Not implemented.
- 2.4.17.2.1 Not implemented.
- 2.4.17.2.2 Not implemented.
- 2.5.1.1 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 2.5.1.3 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 2.5.1.4 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) [AAC] All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 2.5.1.5 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency

frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.

- 2.5.1.6 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (i) For flights in defined portions of airspace or on routes where a Required Communication Performance (RCP) type has been prescribed, an aircraft shall: (1) have installed communication equipment which will enable it to operate in accordance with the prescribed RCP types; and (2) be authorized by the State of the Operator for operations in such airspace.
- 2.5.2.1 PCAR Part 7, 7.2.9 NAVIGATION EQUIPMENT — GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (1) Its operational flight plan, (2) The prescribed Performance-Based Navigation Specification, and (3) The requirements of air traffic services.
- 2.5.2.2 PCAR Part 8, 8.8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (a) No operator shall permit, and no pilot shall conduct operations in airspace designated as Performance-based Navigation (PBN) airspace, unless so approved in the operator's Operations Specifications. (b) For flights in defined portions of airspace or on routes where a PBN type has been prescribed, an aircraft shall be provided with navigation equipment which will enable it to operate in accordance with the prescribed RNP type(s) and the ATS requirements as specified in Subparts 7.2.7, 7.2.8, 7.2.9 and 7.2.10. (c) An operator shall equip the aircraft as prescribed in subparagraph (b) above by incorporating the necessary airworthiness requirements, and submit to the Authority for approval the company manuals and amendments thereof, including the pre-flight and en-route procedures to be followed for such flights, the training and qualifications required of maintenance personnel, flight operations officers/flight dispatchers, and flight crew members; and such other information necessary in the conduct of operations in PBN designated airspace. On successful demonstration of competency of operations in PBN designated airspace by an operator, the Authority shall authorize such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in PBN airspace.
- 2.5.2.3 PCAR Part 8, 8.8.1.31 8.8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (a) No operator shall permit, and no pilot shall conduct operations in airspace designated as Performance-based Navigation (PBN) airspace, unless so approved in the operator's Operations Specifications. (b) For flights in defined portions of airspace or on routes where a PBN type has been prescribed, an aircraft shall be provided with navigation equipment which will enable it to operate in accordance with the prescribed RNP type(s) and the ATS requirements as specified in Subparts 7.2.7, 7.2.8, 7.2.9 and 7.2.10. (c) An operator shall equip the aircraft as prescribed in subparagraph (b) above by incorporating the necessary airworthiness requirements, and submit to the Authority for approval the company manuals and amendments thereof, including the pre-flight and en-route procedures to be followed for such flights, the training and qualifications required of maintenance personnel, flight operations officers/flight dispatchers, and flight crew members; and such other information necessary in the conduct of operations in PBN designated airspace. On successful demonstration of competency of operations in PBN designated airspace by an operator, the Authority shall authorize such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in PBN airspace.
- 2.5.2.4 PCAR Part 8, 8.8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (a) No operator shall permit, and no pilot shall conduct operations in airspace designated as Performance-based Navigation (PBN) airspace, unless so

approved in the operator's Operations Specifications. (b) For flights in defined portions of airspace or on routes where a PBN type has been prescribed, an aircraft shall be provided with navigation equipment which will enable it to operate in accordance with the prescribed RNP type(s) and the ATS requirements as specified in Subparts 7.2.7, 7.2.8, 7.2.9 and 7.2.10. (c) An operator shall equip the aircraft as prescribed in subparagraph (b) above by incorporating the necessary airworthiness requirements, and submit to the Authority for approval the company manuals and amendments thereof, including the pre-flight and en-route procedures to be followed for such flights, the training and qualifications required of maintenance personnel, flight operations officers/flight dispatchers, and flight crew members; and such other information necessary in the conduct of operations in PBN designated airspace. On successful demonstration of competency of operations in PBN designated airspace by an operator, the Authority shall authorize such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in PBN airspace.

2.5.2.10 PCAR Part 7, 7.2.8 NAVIGATION EQUIPMENT FOR OPERATIONS IN RVSM AIRSPACE (d) The Authority will take appropriate action in respect of aircraft and operators found to be operating in RVSM airspace in the Republic of the Philippines without a valid RVSM approval. Note 1: These provisions and procedures need to address both the situation where the aircraft in question is operating without approval in the airspace of the State, and the situation where an operator for which the State has regulatory oversight responsibility is found to be operating without the required approval in the airspace of another State. Note 2: See ICAO Doc 9574, Manual on Implementation of a 300 M (1000 FT) Vertical Separation Minimum Between FL 290 and FL 410 Inclusive, for guidance relating to the approval for operations in RVSM airspace.

2.5.2.11 PCAR Part 7, 7.2.8 NAVIGATION EQUIPMENT FOR OPERATIONS IN RVSM AIRSPACE (d) The Authority will take appropriate action in respect of aircraft and operators found to be operating in RVSM airspace in the Republic of the Philippines without a valid RVSM approval. Note 1: These provisions and procedures need to address both the situation where the aircraft in question is operating without approval in the airspace of the State, and the situation where an operator for which the State has regulatory oversight responsibility is found to be operating without the required approval in the airspace of another State. Note 2: See ICAO Doc 9574, Manual on Implementation of a 300 M (1000 FT) Vertical Separation Minimum Between FL 290 and FL 410 Inclusive, for guidance relating to the approval for operations in RVSM airspace.

2.6.1.1 PCAR Part 8, 8.3.1.2 GENERAL (a) The registered owner or operator of an aircraft is responsible for maintaining that aircraft in an airworthy condition, including compliance with all airworthiness directives.

2.6.1.3 PCAR Part 8, 8.3.1.2 (d) The owner or operator shall use one of the following inspection programs as appropriate for aircraft and the type operation: (1) Annual inspection; (2) Annual/100 hours inspection; (3) Progressive; or (4) Continuous airworthiness program.

2.6.2.1 PCAR Part 8, 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (a) (1) Records of the maintenance, preventive maintenance, minor modifications, and records of the 100-hour; annual, and other required or approved inspections, as appropriate, for each aircraft (including the airframe) and each engine, propeller, rotor, and appliance of an aircraft (2) Records containing the following information - (i) The total time-in-service of the airframe, each engine, each propeller, and each rotor. (ii) The current status of all life-limited aeronautical products; (iii) The time since last overhaul of all items installed on the aircraft which are required to be overhauled on a

- specified time basis; (iv) The current inspection status of the aircraft, including the time since the last inspection required by the inspection program under which the aircraft and its appliances are maintained. (vi) Copies of the forms prescribed by this chapter for each major modification to the airframe and currently installed engines, rotors, propellers, and appliances.
- 2.6.2.2 PCAR Part 8, 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (a) Except for records maintained by an operator, each registered owner or operator shall retain the following records for a minimum period of 90 days, or until the work is repeated or superseded by other work of equivalent scope and detail, or the unit to which they refer has been permanently withdrawn from service, and the records in paragraph (1) below for a minimum period of one year after the signing of the maintenance release.
- 2.7.1 PCAR Part 8, 8.4.1.1 COMPOSITION OF THE FLIGHT CREW (a) The number and composition of the flight crew shall not be less than that specified in the flight manual (for helicopters: operations manual). The flight crew members shall include flight crew members in addition to the minimum numbers specified in the flight manual or other documents associated with the certificate of airworthiness, when necessitated by considerations related to the type of aircraft used, the type of operation involved and the duration of flight between points where flight crew members are changed.
- 2.7.2.1 PCAR Part 8, 8.4.1.2 FLIGHT CREW QUALIFICATIONS (a) The PIC shall ensure that the licenses of each flight crew member have been issued or rendered valid by the State of Registry, contain the proper ratings, and that all flight crew members have maintained recent experience and competence.
- 2.8.2.1 PCAR Part 9, 9.3.1.5 AIRCRAFT TECHNICAL LOG ENTRIES - JOURNEY RECORDS SECTION (a) Each aircraft journey log book (technical log) shall contain the following items and corresponding Roman numerals for each flight: (See 9.4.1.9 for maintenance section of the aircraft technical log).
- 2.9.1 PCAR Part 8, 8.5.1.1 AUTHORITY AND RESPONSIBILITY OF THE PIC (a) Pilot-in-command (PIC): Each operator shall designate one pilot to act as PIC for each flight. (b) The PIC shall be responsible for the operations and safety of the aircraft and for the safety of all crew members, passengers and cargo on board, when the doors are closed. The PIC shall also be responsible for the operation and safety of the aircraft from the moment the aircraft is ready to move for the purpose of taking off until the moment it finally comes to rest at the end of the flight and the engine(s) used as primary propulsion units are shutdown (and in the case of helicopters, rotor blades stopped). (c) The PIC of an aircraft shall have final authority as to the operation of the aircraft while he or she is in command. (d) The PIC of an aircraft shall, whether manipulating the controls or not, be responsible for the operation of the aircraft in accordance with the rules of the air, except that the PIC may depart from these rules in emergency circumstances that render such departure absolutely necessary in the interests of safety. (e) In an emergency during flight, the PIC shall ensure that all persons on board are instructed in such emergency action as may be appropriate to the circumstances.
- 3.1.1 PCAR Part 8.1.1.1 (2) The Standards and Recommended Practices contained in Annex 6, Part II (airplanes), and Annex 6 Part III Section III (helicopters) shall be applicable to general aviation operations (international and domestic).
- 3.3.1.3 Not implemented.
- 3.3.2.1 PCAR Part 13 SUBPART I: MISCELLANEOUS 13.220 NON-DISCLOSURE OF RECORDS (a) Subject to paragraph (b), no person shall disclose or make available to any other person any of the following records for any purpose other than an investigation carried out under this Part unless the High Court determines that the

benefits resulting from disclosure outweighs the adverse domestic and international impact the disclosure may have on that or any future investigation: (1) any statement taken from persons in the course of an investigation; (2) any communication between persons having been involved in the operation of the aircraft; (3) any medical or private information regarding persons involved in the accident or incident; (4) any cockpit voice recording and transcript from such recording; (5) any ATS recording and transcript from such recording; (6) any opinion expressed in the analysis of information, including flight recorder information. (b) A record referred to in paragraph (a) shall be included in a Final Report or its appendices or in any other report only when it is relevant to the analysis of the accident or incident. (c) Such parts of the record which are not relevant to the analysis of the accident or incident shall not be included in the Final Report.

- 3.4.1.1 PCAR Part 8, 8.6.2.2 ADEQUACY OF OPERATING FACILITIES (a) An operator shall ensure that a flight will not be commenced unless it has been ascertained by every reasonable means available that the ground and/or water facilities available and directly required on such flight, for the safe operation of the airplane or helicopter and the protection of the passengers, are adequate for the type of operation under which the flight is to be conducted and are adequately operated for this purpose.
- 3.4.2.1.1 Not implemented.
- 3.4.2.1.2 Not implemented.
- 3.4.2.2 Not implemented.
- 3.4.2.3.1 Not implemented.
- 3.4.2.5 PCAR Part 8, 8.5.1.9 COMPLIANCE WITH CHECKLISTS The PIC shall ensure that the checklists specified in Subpart 9.3.1.11 are complied with in detail.
- 3.4.2.6 PCAR Part 8, 8.8.1.30 (c) The method for establishing the minimum flight altitudes shall be approved by the Authority.
- 3.4.2.7 PCAR Part 8.8.1.7 INSTRUMENT APPROACH OPERATING MINIMA (a) No person may operate to or from an airport/heliport using operating minima lower than those which may be established for that airport/heliport by the State in which it is located, unless that State specifically approves that operation, in accordance with the provisions of Implementing Standard: IS: 8.8.1.7.
- 3.4.2.8 Not implemented.
- 3.4.3.1 PCAR Part 8, 8.6.2.1 AIRCRAFT AIRWORTHINESS AND SAFETY PRECAUTIONS (a) A flight, or series of flights, shall not be commenced until flight preparation forms have been completed certifying that the PIC is satisfied that: (1) the aircraft is airworthy, duly registered and that appropriate certificates with respect thereto are aboard the aircraft; (2) the instruments and equipment prescribed in Part 7 for the particular type of operation to be undertaken, are appropriate, taking in account the expected flight conditions, installed and sufficient for the flight; (3) any necessary maintenance has been performed and a maintenance release, if applicable, has been issued in respect to the aircraft; (4) the mass of the aircraft and center of gravity location are such that the flight can be conducted safely, taking into account the flight conditions expected; (5) any load carried is properly distributed and safely secured; (6) a check has been completed indicating that the operating limitations, contained in the flight manual or its equivalent, can be complied with for the flight to be undertaken.
- 3.4.3.3 Not implemented.
- 3.4.3.4.1.1 PCAR Part 8, 8.6.2.9 TAKE-OFF ALTERNATE AIRPORTS/HELIPORTS: COMMERCIAL AIR TRANSPORT OPERATIONS (a) No person may release or take-off an aircraft without a suitable take-off alternate specified in the flight release if the

weather conditions at the airport/heliport of departure are at or below the applicable airport/heliport operating minima or it would not be possible to return to the airport/heliport of departure for other reasons.

- 3.4.3.4.1.2 PCAR Part 8, 8.6.2.9 TAKE-OFF ALTERNATE AIRPORTS/HELIPORTS: COMMERCIAL AIR TRANSPORT OPERATIONS (b) Each operator shall ensure that each take-off alternate specified shall be located within the following distance from the airport of departure: (1) For two power-unit aircraft, not more than a distance equivalent to a flight time of one hour at single-engine cruise speed unless the aircraft and crew members are authorized for ETOPS; or (2) For three or four power-unit aircraft, not more than a distance equivalent to a flight time of two hours at single-engine cruise speed. Note: All calculations are based on the one-engine-inoperative cruising speed according to the AFM in still air conditions based on the actual take-off mass.
- 3.4.3.4.3 Not implemented.
- 3.4.3.5.1 PCAR Part 8, 8.6.2.13.1 (a) A flight shall not be commenced unless, taking into account both the meteorological conditions and any delays that are expected in flight, the aircraft carries sufficient fuel and oil to ensure that it can safely complete the flight, and, applicable special provisions are complied with, as stated in Subpart 8.6.2.14 or 8.6.2.15. In addition, a reserve shall be carried to provide for contingencies.
- 3.4.5.2 PCAR Part 8, 8.5.1.22 REPORTING OF INCIDENTS (b) Air traffic incident report. The PIC shall submit, without delay, an air traffic incident report whenever an aircraft in flight has been endangered by: (1) A near collision with another aircraft or object; (2) Faulty air traffic procedures or lack of compliance with applicable procedures by ATC or by the flight crew; or (3) A failure of ATC facilities. (c) Birds. In the event a bird constitutes an in-flight hazard or an actual bird strike occurs, the PIC shall, without delay: (1) Inform the appropriate ground station whenever a potential bird hazard is observed; and (2) Submit a written bird strike report after landing. (d) Dangerous Goods. The PIC shall inform the appropriate ATC facility, if the situation permits, when an in-flight emergency occurs, involving dangerous goods on board. (e) Unlawful Interference. The PIC shall submit a report to the local authorities and to the Authority, without delay, following an act of unlawful interference with the crew members on board an aircraft. 8.5.1.23 ACCIDENT NOTIFICATION (a) The PIC shall notify the nearest appropriate authority, by the quickest available means, of any accident involving his or her aircraft that results in serious injury or death of any person, or substantial damage to the aircraft or property.
- 3.4.5.3 PCAR Part 8, 8.5.1.19 REPORTING MECHANICAL IRREGULARITIES (a) The PIC shall ensure that all mechanical irregularities occurring during flight time are: (1) For general aviation operations, entered in the aircraft logbook and disposed of in accordance with the MEL or other approved or prescribed procedure.
- 3.4.5.4 PCAR Part 8, 8.2.1.8 (b) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in general aviation operations unless it has within it the following current and approved documents: (3) Aircraft Journey Log.
- 3.4.6 PCAR Part 8, 8.9.2.14 CARRY-ON BAGGAGE (a) No person may allow the boarding of carry-on baggage in to the passenger cabin unless it is adequately and securely stowed in accordance with the operator's approved Operations Manual procedures. (b) No person may allow aircraft passenger entry doors to be closed in preparation for taxi or pushback unless at least one required crew member has verified that each article of baggage has been properly stowed in overhead racks with approved restraining devices or doors, or in approved locations aft of the bulkhead. (c) No person may allow carry-on baggage to be stowed in a location that would cause that location to be loaded beyond its maximum placard mass limitation. Note: The stowage locations

- shall be capable of restraining the articles in crash impacts severe enough to induce the ultimate inertia forces specified in the emergency landing conditions under which the aircraft was type-certified.
- 3.5.2.2 Not implemented.
 - 3.5.2.3 Not implemented.
 - 3.5.2.4 Not implemented.
 - 3.5.2.6 Not implemented.
 - 3.5.2.7.1 Not implemented.
 - 3.5.2.9 Not implemented.
 - 3.6.1.1 PCAR Part 8, 8.2.1.5 INOPERATIVE INSTRUMENTS AND EQUIPMENT (a) No person may take-off an aircraft with inoperative instruments or equipment installed, except as authorized by the Authority. (b) An Operator shall not operate a multi-engine aircraft with inoperative instalments and equipment installed unless the following conditions are met: (1) An approved Minimum Equipment List exists for that aircraft. (2) The Authority has issued the Operator specific operating provisions authorizing operations in accordance with an approved Minimum Equipment List. The flight crew shall have direct access at all times prior to flight to all of the information contained in the approved Minimum Equipment List through printed or other means approved by the Authority in the Operators specific operating provisions. An approved Minimum Equipment List, as authorized by the specific operating provisions, constitutes an approved change to the type design without requiring recertification. (3) The approved Minimum Equipment List must: (i) Be prepared in accordance with the limitations specified in paragraph (c) of this section. (ii) Provide for the operation of the aircraft with certain instruments and equipment in an inoperative condition. (4) Records identifying the inoperative instruments and equipment and the information required by subparagraph (b)(3)(ii) of this section must be available to the pilot. (5) The aircraft is operated under all applicable conditions and limitations contained in the Minimum Equipment List and the specific operating provisions authorizing use of the Minimum Equipment List. (c) The following instruments and equipment may not be included in the Minimum Equipment List: (1) Instruments and. equipment that are either specifically or otherwise required by the airworthiness requirements under which the aircraft is type certificated and which are essential for safe operations under all operating conditions. (2) Instruments and equipment required by an airworthiness directive to be in operable condition unless the airworthiness directive provides otherwise. (3) Instruments and equipment required for specific operations under Part 7, Part 8 and/or Part 9 of these regulations. (d) Notwithstanding paragraphs (c)(1) and (c)(3) of this section, an aircraft with inoperative instruments or equipment may be operated under a special flight permit under Subpart 5.4.1.11 of these regulations
Implementing Standard: See IS: 8.2.1.5 for specific limitation on inoperative instruments and equipment.
 - 3.6.2.1 PCAR Part 7, 7.8.11 FIRST- AIDE AND EMERGENCY MEDICAL KIT (a) No person may operate and aircraft unless it is equipped with first-aide kits readily accessible to cabin crew located near an exit and, on passenger flights, and approved emergency medical kit for treatment or injuries or medical emergencies that might occur during flight time or in minor accidents. (b) The number of the first-aide kits to be carried shall be the following scale: Number of Passenger Number of first-aide kits Seats installed required 0-50 1 51-150 2 151-250 3 More than 250 4 PCAR Part 77.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of

that aircraft:(2) A safety harness for each flight crew member seat.(i) The safety harness for each pilot seat shall incorporate a device, which will automatically restrain the occupant's torso in the event of rapid deceleration. PCAR Part 77.9.3 PASSENGER INFORMATION SIGNS (a) [AOC] No AOC holder may operate a passenger carrying aircraft unless it is equipped with - (1) At least one passenger information sign (using either letters or symbols) notifying when smoking is prohibited and one sign (using either letters or symbols) notifying when safety belts should be fastened shall, when illuminated, be legible to each person seated in the passenger cabin under all probable conditions of cabin illumination; (2) Signs which notify when safety belts should be fastened and when smoking is prohibited shall be so constructed that the crew can turn them on and off; (3) At least one legible sign or placard that reads "Fasten Seat Belt While Seated" is visible from each passenger seats;(4) A sign or placard in each lavatory that reads: "Philippine law provides for a penalty for tampering with the smoke detector installed in this lavatory"; (5) Each seat that is equipped with passenger floatation devices or life vests will be placarded, with instructions on where they are located. PCAR Part 8 8.9.1.4 PASSENGER BRIEFING (a) The PIC shall ensure that crew members and passengers are made familiar, by means of an oral briefing or by other means, with the location and use of the following items: (1) Seat belts or harnesses, as appropriate; (2) Emergency exits; (3) Life jackets, if carriage of life jackets is prescribed; (4) Oxygen dispensing equipment, if provision of oxygen for the passengers is prescribed; and(5) Other emergency equipment provided for individual use, including passenger emergency briefing cards. PCAR Part 88.9.1.8 RESTRICTION ON SMOKING ON BOARD AIRCRAFT (a) No operator may permit smoking nor may any person smoke on board any part of an aircraft involved in scheduled and non-scheduled international or domestic flights.

- 3.6.2.2 PCAR Part 8, 8.2.1.6 CIVIL AIRCRAFT FLIGHT MANUAL, MARKING AND PLACARD REQUIREMENTS (a) No person may operate a Republic of the Philippines-registered civil aircraft unless there is available in the aircraft (1) A current, approved AFM or RFM; or (2) An AOM approved by the Authority for the Operator, (3) If no AFM or RFM exists, approved manual material, markings and placards, or any combination thereof, which provide the PIC with the necessary limitations for safe operation.
- 3.6.3.1.1.1 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (h) No person shall operate an airplane of a maximum certificated take-off mass of over 5700 KG, for which the individual certificate of airworthiness is first issued after 1 January 2005, unless it is equipped with a Type IA FDR.
- 3.6.3.1.1.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (c) No person shall operate an airplane of a maximum certificated take-off mass of over 27000 KG for which the individual certificate of airworthiness is first issued after 1 January 1989, unless it is equipped with a Type I FDR
- 3.6.3.2.1.2 PCAR Part 7, 7.7.2 COCKPIT VOICE RECORDERS (a) No person may operate the following aircraft unless it is equipped with a cockpit voice recorder capable of recording the aural environment of the flight deck during flight time. (2) All turbine-engined airplanes with a maximum certificated take-off mass of more than 27000 KG that are of types of which the prototype was certificated by the appropriate national authority after 30 September 1969 and for which the individual certificate of airworthiness was first issued before 1 January 1987.
- 3.6.3.4.1 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (1) Shall carry the equipment prescribed in paragraph (2): (i) When flying en-route over water beyond gliding distance from the shore; (ii) When flying over water at a distance of more than 93 KM (50 NM) away from the shore for aircraft capable of maintaining safe altitude after the failure of one engine for two-engine aircraft and the

failure of two engines for three or four-engine aircraft.; or (iii) When taking off or landing at an aerodrome where the (Authority) has determined the take-off or approach path is so disposed over water that in the event of a mishap there would be the likelihood of a ditching. (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided. (b) Seaplanes. (1) For all flights, seaplanes shall be equipped with the equipment prescribed in paragraph (2). (c) Helicopters. (1) For offshore operations the life jacket shall be worn unless the occupant is wearing an integrated survival suit.

PCAR Part 8 8.5.1.10 SEARCH AND RESCUE INFORMATION (a) An operator shall ensure that PIC has available on board the aircraft all the essential information concerning the search and rescue services in the area over which the aircraft will be flown. (b) For all flights, the PIC shall have available on board the aircraft essential information concerning the search and rescue services in the areas over which it is intended the aircraft will be flown.

PCAR Part 7 7.8.17 LIFE RAFT (a) In addition to the equipment prescribed in 7.8.16 and 7.8.18 of this Part, lifesaving rafts in sufficient numbers to carry all persons on board shall be installed in: (1) All airplanes when operating in "Extended over-water operations" as defined in Subpart 8.1.1.2(b). (2) Class 1 and 2 helicopters when they are operated over water at a distance from land corresponding to more than 10 minutes at normal cruise speed. (3) Class 3 helicopters when they are operated over water beyond auto-rotational or safe forced landing distance from land. (b) An aircraft shall have life saving rafts with a sufficient capacity to carry all persons on board in the event of the loss of one raft of the largest capacity. (c) All life saving rafts shall be stowed so as to facilitate their ready use in an emergency. (d) Life rafts shall be equipped with the following life sustaining equipment - (1) A electric survivor locator light; (2) A survival kit equipped in accordance with the requirements of subsection 7.8.4; (3) A pyrotechnical signaling device. (e) In helicopters, life rafts which are not deployable by remote control and which have a mass of more than 40 KG shall be equipped with a means of mechanically assisted deployment.

PCAR Part 7 7.8.4 SURVIVAL EQUIPMENT (a) Subject to subsection (b), no person shall operate an aircraft over land unless there is carried on board survival equipment, sufficient for the survival on the ground of each person on board, given the geographical area, the season of the year and anticipated seasonal climatic variations, that provides the means for (1) starting a fire; (2) providing shelter; (3) providing or purifying water; and (4) visually signaling distress. (b) Subsection (a) does not apply in respect of (1) a balloon, a glider, a hang glider, a gyroplane or an ultra-light airplane; (2) an aircraft that is operated-within 25 NM of the aerodrome of departure and that has the capability of radio communication with a surface-based radio station for the duration of the flight. (3) a multi-engined aircraft that is operated south of 66 30' north latitude or north of 66 30'south latitude (i) in IFR flight Within-controlled airspace, or (ii) along designated air routes; (4) an aircraft that is operated by an air operator, where the aircraft is equipped with equipment" specified in the air operator's company operations manual, but not with the equipment required by subsection (a); or (5) an aircraft that is operated in a geographical area what a time of year when the survival of the persons on board is not jeopardised.

PCAR Part 7 7.8.3 VISUAL SIGNALLING DEVICES (a) No person may operate an aircraft over water or across land areas which have been designated by the. Republic of the Philippines as areas in which search and rescue would be especially difficult, unless equipped with such signaling devices as may be appropriate to the area overflown, to include - (1) Visual signals for use by intercepting and intercepted aircraft;

and (2) At least one pyrotechnic signaling device for each life raft required for overwater, operations.

3.6.3.4.2 PCAR Part 7, 7.8.17 LIFE RAFT (d) Life rafts shall be equipped with the following life sustaining equipment - (1) A electric survivor locator light; (2) A survival kit equipped in accordance with the requirements of subsection 7.8.4; (3) A pyrotechnical signaling device.

3.6.3.5.1 PCAR Part 7, 7.6.2 LOSS OF PRESSURIZATION INDICATOR All pressurized aircraft intended to be operated at altitudes at or above 25,000 above sea level shall be equipped with a device to provide positive warning to the flight crew of any dangerous loss of pressurization.

3.6.3.5.2 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated. (3) Sufficient spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure. (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin. (e) [AAC] The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft.

PCAR Part 7 IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin

pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not operate a non-pressurized helicopter at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

PCAR Part 7 7.8.13 PROTECTIVE BREATHING EQUIPMENT (PBE) (a) No person may operate an airplane with a maximum certified take-off mass exceeding 5700 KG or having a maximum approved seating configuration of more than 19 seats unless - (1) It has PBE to protect the eyes, nose and mouth of each flight crew member while on flight deck duty and to provide oxygen for a period of not less than 15 minutes; and (2) It has sufficient portable PBE to protect the eyes, nose and mouth of all required cabin crew members and to provide breathing gas for a period of not less than 15 minutes. (b) The oxygen supply for PBE may be provided by the required supplemental oxygen system. (c) The PBE intended for flight crew use shall be conveniently located on the flight deck and be easily accessible for immediate use by each required flight crew member at their assigned duty station. (d) The PBE intended for cabin crew use shall be installed adjacent to each required cabin crew member duty station. (e) Easily accessible portable PBE shall be provided and located at or adjacent to the required hand Are extinguishers except that, where the fire extinguisher is located inside a cargo compartment, the PBE shall be stowed outside but adjacent to the entrance to that compartment. (f) The PBE while in use shall not prevent required communication.

3.6.3.5.3 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) All aircraft intended to be operated at altitudes requiring the use of supplemental oxygen shall be equipped with adequate oxygen storage and dispensing apparatus. (b) The

oxygen apparatus, the minimum rate of oxygen flow, and the supply of oxygen shall meet applicable airworthiness standards for type certification in the transport category as specified by the Authority. (c) No person may operate an aircraft at altitudes above 10000 FT unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station. (d) No person may operate a pressurized aircraft at altitudes above 25000 FT unless: (1) Flight crew member oxygen masks are of a quick donning type; (2) An automatically deployable oxygen-dispensing unit is connected to oxygen supply terminals installed so as to be immediately available to each occupant, wherever seated. (3) Sufficient spare outlets and masks and/or sufficient portable oxygen units with masks are distributed evenly throughout the cabin to ensure immediate availability of oxygen to each required cabin crew member regardless of his location at the time of cabin pressurization failure. (4) The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin. (e) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual Implementing Standard: See IS: 7.8.12 to determine the amount of supplemental oxygen needed for non-pressurized and pressurized aircraft. PCAR Part 7 IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not operate a non-pressurized helicopter at pressure

altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. PCAR Part 7 7.8.13 PROTECTIVE BREATHING EQUIPMENT (PBE) (a) No person may operate an airplane with a maximum certified take-off mass exceeding 5700 KG or having a maximum approved seating configuration of more than 19 seats unless - (1) It has PBE to protect the eyes, nose and mouth of each flight crew member while on flight deck duty and to provide oxygen for a period of not less than 15 minutes; and (2) It has sufficient portable PBE to protect the eyes, nose and mouth of all required cabin crew members and to provide breathing gas for a period of not less than 15 minutes. (b) The oxygen supply for PBE may be provided by the required supplemental oxygen system. (c) The PBE intended for flight crew use shall be conveniently located on the flight deck and be easily accessible for immediate use by each required flight crew member at their assigned duty station. (d) The PBE intended for cabin crew use shall be installed adjacent to each required cabin crew member duty station. (e) Easily accessible portable PBE shall be provided and located at or adjacent to the required hand fire extinguishers except that, where the fire extinguisher is located inside a cargo compartment, the PBE shall be stowed outside but adjacent to the entrance to that compartment. (f) The PBE while in use shall not prevent required communication.

- 3.6.4 PCAR Part 7, 7.9.9 ICING PROTECTION EQUIPMENT (a) No person may operate an aircraft in icing conditions unless it is certified and equipped for the prevention or removal of ice on windshields, wings, empennage, propellers, and other parts of the aircraft where ice formation will adversely affect the safety of the aircraft.
- 3.6.5.1 PCAR Part 7, 7.2.3 IFR INSTRUMENTS (a) All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - (1) An airspeed indicating system with a means of preventing malfunctioning due to either condensation or icing; (2) A sensitive pressure altimeter with counter drum-pointer or equivalent presentation, calibrated in feet with a sub-scale setting calibrated in hectopascals/millibars, adjustable for any barometric pressure likely to be set during flight; (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate.
- 3.6.5.2.1 PCAR Part 7, 7.2.4 STANDBY ATTITUDE INDICATOR (a) No person may operate an airplane with a maximum certified take-off mass exceeding 5700 KG, aircraft having a maximum approved passenger seating configuration of more than 9 seats and all helicopters when operating in accordance with IFR, unless it is equipped with a single standby attitude indicator (artificial horizon) that - (3) After a total failure of the normal

- electrical generating system, is automatically powered for the purpose of operating and illuminating for a minimum of 30 minutes from a source independent of the normal electrical generating system.
- 3.6.6 PCAR Part 7, 7.6.6 WEATHER RADAR No person may operate a helicopter carrying passengers or a pressurized airplane in commercial air transport in an area where potentially hazardous weather conditions may be expected unless it is equipped with operative weather radar.
- 3.6.8.1 PCAR Part 7, 7.9.1 SEATS. SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (3) A forward or rearward facing (within 15 degrees of the longitudinal axis of the airplane) seat equipped with a safety harness for each cabin crew member station in the passenger compartment.
- 3.6.8.2.2 PCAR Part 7, 7.9.1 SEATS. SAFETY BELTS, AND SHOULDER HARNESSSES (a) (4) The cabin crew member's seats shall be located near floor level and other emergency exits for emergency evacuation.
- 3.6.9.2 Not implemented.
- 3.6.10 PCAR Part 7, 7.6.4 ALTITUDE ALERTING SYSTEM (a) No person may operate a turbine propeller powered airplane with a maximum certified take-off mass in excess of 5700 KG or having a maximum approved passenger seating configuration of more than 9 seats, or a turbojet powered airplane, unless it is equipped with an altitude alerting system capable of - (1) Alerting the flight crew upon approaching pre-selected altitude in either ascent or descent; and (2) Alerting the flight crew by at least an aural signal, when deviating above or below a pre-selected altitude. (b) For operations in defined portions of airspace where, based on Regional Air Navigation Agreement, a RVSM of 300 M (1000 FT) is applied above FL 290, an aircraft shall be provided with equipment which is capable of; (1) indicating to the flight crew the flight level being flown (2) automatically maintaining a selected flight level (3) Providing an alert to the flight crew when a deviation occurs from the selected flight level. The threshold for the alert may not exceed ± 90 M (300 FT). (4) automatically reporting pressure-altitude, and (5) Shall be authorized by the State of the Operator for operation in the airspace concerned.
- 3.7.1 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 3.7.2 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (d) When more than one communications equipment unit is required, each shall be independent of the other or others to the extent that a failure in any one will not result in failure of any other.
- 3.7.3.1 PCAR Part 7, 7.2.10 ELECTRONIC NAVIGATION DATA MANAGEMENT (a) No person shall employ electronic navigation data products that have been processed for application in the air and on the ground unless the Authority has approved: (1) The operators procedures for ensuring that the process applied and the products delivered have acceptable • standards of integrity and that the products are compatible with the intended function of the equipment that will use them; (2) The operator's program for continual monitoring of both process and products; and (3) The operator's procedures to ensure the timely distribution and insertion of current and unaltered

electronic navigation data to all aircraft that require it. Note: Guidance relating to the processes that data suppliers may follow is contained in RTCA DO-200A/EUROCAE ED-76.

3.8.1.1

PCAR Part 8, 8.3.1.1 APPLICABILITY (a) This Subpart prescribes the rules governing the inspection of Republic of the Philippines registered civil aircraft operating within or outside Republic of the Philippines (b) Subparts 8.3.1.3 and 8.3.1.4 do not apply to aircraft subject to an approved continuous maintenance program approved by the Authority for an Operator in Part 9. (c) This Subpart applies to all aircraft, as designated below, operated as commercial air transport in Republic of the Philippines if the operator has not been designated an Operator by Republic of the Philippines. (d) This Subpart applies to all general aviation large, complex aircraft operated in Republic of the Philippines, whether or not the aircraft is registered in Republic of the Philippines: (e) Where any aircraft, not registered in Republic of the Philippines and operating under an inspection program approved or accepted by the State of Registry, does not have the equipment required by Republic of the Philippines for operations within Republic of the Philippines; the owner/operator shall ensure that such equipment is installed and inspected in accordance with the requirements of the State of Registry, acceptable to the Authority prior to operation of that aircraft in Republic of the Philippines. (f) The owner of an airplane over 5700 KG maximum certificated take-off mass, or in the case where it is leased, the lessee, shall, as prescribed by the State of Registry, ensure that the information resulting from maintenance and operational experience with respect to continuing airworthiness, is transmitted as required by Subpart 8.3.16. PCAR Part 8 8.3.1.2 GENERAL (a) The registered owner or operator of an aircraft is responsible for maintaining that aircraft in an airworthy condition, including compliance with all airworthiness directives. (b) No person may perform maintenance, preventive maintenance, or alterations on an aircraft other than as prescribed in this Subpart and other applicable regulations, including Parts. (c) No person may operate an aircraft for which a manufacturer's maintenance manual or instructions for continued airworthiness has been issued that contains an airworthiness limitations section unless the mandatory replacement times, inspection intervals and related procedures set forth in operations specifications approved by the Authority under Part 9 or in accordance with an inspection program approved under this Subsection are complied with: (d) The owner or operator shall use one of the following inspection programs as appropriate for aircraft and the type operation: (1) Annual inspection; (2) Annual/100 hours inspection; (3) Progressive; or (4) Continuous airworthiness program. (e) No aircraft shall have a Maintenance Release signed after inspection unless the replacement times for life-limited parts specified in the aircraft specification-type data sheets are complied with and the airplane, including airframe, power-units, propellers, rotors, appliances, and survival and emergency equipment, is inspected in accordance with the selected inspection program. (f) Each person wishing to establish or change an approved inspection program shall submit the program for approval by the Authority and shall include in writing: (1) Instructions and procedures for conduct of inspection for the particular make and model aircraft, including necessary tests and checks. The instructions shall set forth in detail the parts and areas of the aeronautical products, including survival and emergency equipment required to be inspected; and (2) A schedule for the inspections that shall be performed expressed in terms of time in service, calendar time, number of system operations or any combination of these. (g) When an operator changes from one inspection program to another, the operator shall apply the time in service, calendar times, or cycles of operation accumulated under the previous program, in determining when the inspection is due under the new program. PCAR Part 8 8.3.1.6 CONTINUOUS AIRWORTHINESS MAINTENANCE INSPECTION (a) The registered owner or operator of each large airplane certificated with a maximum take-off mass of over

5700 KG, turbine-powered multi-engine airplane, and turbine powered rotorcraft shall select, identify in the aircraft maintenance records, and use one of the following continuous airworthiness maintenance inspection programs for the inspection of the aircraft: (1) A current inspection program recommended by the manufacturer; (2) A continuous airworthiness maintenance program for that make and model of aircraft currently approved by the Authority for use by an AOC holder, or (3) Any other inspection program established by the registered owner or operator of that aircraft and approved by the Authority. (b) Each owner/operator shall include in the selected inspection program the name and address of the person responsible for the scheduling of the inspections required by the program and provide a copy of the program to the person performing inspection on the aircraft. Note: The aircraft manufacturers inspection program and any other inspection program approved by the Authority will specify who can perform aircraft maintenance, inspections and return of the aircraft to service.

- 3.8.3.1 PCAR Part 8, 8.3.1.6 CONTINUOUS AIRWORTHINESS MAINTENANCE INSPECTION (a) The registered owner or operator of each large airplane certificated with a maximum take-off mass of over 5700 KG, turbine-powered multi-engine airplane, and turbine powered rotorcraft shall select, identify in the aircraft maintenance records, and use one of the following continuous airworthiness maintenance inspection programs for the inspection of the aircraft: (2) A continuous airworthiness maintenance program for that make and model of aircraft currently approved by the Authority for use by an AOC holder.
- 3.8.3.2 Not implemented.
- 3.9.3.1 PCAR Part 8, 8.10.1.8.3 FLIGHT CREW MEMBER TRAINING PROGRAMS An operator shall establish and maintain a ground and flight training programme, approved by the Authority, which ensures that all flight crew members are adequately trained to perform their assigned duties. The training program shall: (a) Include ground and flight training facilities and properly qualified instructors as determined by the Authority; (e) Ensure that all flight crew members know the functions for which they are responsible and the relation of these functions to the functions of other crew members, particularly in regard to abnormal or emergency procedures.
- 3.9.4.1.1 PCAR Part 8, 8.4.1.2 FLIGHT CREW QUALIFICATIONS (a) The PIC shall ensure that the licenses of each flight crew member have been issued or rendered valid by the State of Registry, contain the proper ratings, and that all flight crew members have maintained recent experience and competence.
- 3.9.4.2 PCAR Part 8, 8.4.1.2 FLIGHT CREW QUALIFICATIONS (a) The PIC shall ensure that the licenses of each flight crew member have been issued or rendered valid by the State of Registry, contain the proper ratings, and that all flight crew members have maintained recent experience and competence. 8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (a) No person may act as PIC or a co-pilot of an aircraft unless, within the preceding 90 days that pilot has: (1) Made three take-offs and landings as the sole manipulator of the flight controls in an aircraft of the same category and class and if a type rating is required, of the same type or in a flight simulation training device approved for the purpose.
- 3.9.4.3 PCAR Part 8, 8.4.1.2 FLIGHT CREW QUALIFICATIONS (a) The PIC shall ensure that the licenses of each flight crew member have been issued or rendered valid by the State of Registry, contain the proper ratings, and that all flight crew members have maintained recent experience and competence. 8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (a) No person may act as PIC or a co-pilot of an aircraft unless, within the preceding 90 days that pilot has: (1) Made three take-offs and landings as the sole manipulator of the flight controls in an aircraft of the

same category and class and if a type rating is required, of the same type or in a flight simulation training device approved for the purpose.

- 3.9.4.4 PCAR Part 8, 8.4.1.11 PILOT CURRENCY: GENERAL AVIATION OPERATIONS (a) In addition to the requirements in Subparts 8.4.1.9 and 8.4.1.10: (1) No person may act as PIC of an aircraft type certified for more than one pilot or a turbojet aircraft unless, since the beginning of the past 12 calendar months, he or she has passed a proficiency check in an aircraft with an authorized representative of the Authority. (2) No person may act as PIC of an aircraft type certified for more than one pilot or a turbojet aircraft unless, since the beginning of the past 24 calendar months, he or she has passed a proficiency check in the type aircraft to be operated. (3) No person may act as PIC of an aircraft type certified for a single pilot unless, since the beginning of the 24 calendar months, he or she has passed a proficiency check with an authorized representative of the Authority. (4) No person may act as CP of an aircraft type certified for more than one pilot unless, since the beginning of the 12 calendar months, he or she has logged 3 take-off and landings as the sole manipulator of the controls in the aircraft of the same type.
- 3.11.2.1 Not implemented.
- 3.11.2.2 Not implemented.
- 3.11.3 PCAR Part 8, 8.5.1.24 OPERATION OF FLIGHT DECK VOICE AND FLIGHT DATA RECORDERS (b) The PIC may not permit a flight data recorder or flight deck voice recorder to be disabled, switched off or erased during flight, unless necessary to preserve the data for an accident or incident investigation. (c) In event of an accident or incident, the PIC shall act to preserve the recorded data for subsequent investigation.
- 3.12.2 PCAR Part 8, 8.9.2.4 CABIN CREW MEMBERS AT DUTY STATIONS (b) Each cabin crew member assigned to emergency evacuation duties shall occupy a seat provided in accordance with Subpart 7.9.1 during take-off and landing and whenever the PIC so directs, with seat belt or, when provided, safety harness fastened.
- 3.12.3 PCAR Part 8, 8.9.2.4 CABIN CREW MEMBERS AT DUTY STATIONS (a) During taxi, cabin crew members shall remain at their duty stations with safety belts and shoulder harness fastened except to perform duties related to the safety of the aircraft and its occupants. (b) Each cabin crew member assigned to emergency evacuation duties shall occupy a seat provided in accordance with Subpart 7.9.1 during take-off and landing and whenever the PIC so directs, with seat belt or, when provided, safety harness fastened. (c) During take-off and landing, cabin crew members shall be located as near as practicable to required floor level exits and shall be uniformly distributed throughout the aircraft to provide the most effective egress of passengers in event of an emergency evacuation. (d) When passengers are on board a pariced aircraft, cabin crew members (or another person qualified in emergency evacuation procedures for the aircraft) will be placed in the following manner: (1) If only one qualified person is required, that person shall be located in accordance with the Operator's Operations Manual procedures. (2) If more than one qualified person is required, those persons shall be spaced throughout the cabin to provide the most effective assistance for the evacuation in case of an emergency.
- 3.12.4.1 PCAR Part 8, 8.10.1.8.1 APPLICABILITY AND TERMS USED (c) For the purpose of this Subpart, the following terms and definitions apply - (ii) Cabin crew members, who have not previously qualified on a particular type of aircraft or have not served on a specific type of aircraft for longer than 24 months from the date of the last flight; 8.10.1.34 RECURRENT TRAINING: CABIN CREW (a) No person may serve nor may any operator use a person as a cabin crew unless within the preceding 12 calendar

months that person has completed the recurrent training program approved by the Authority relevant to the type(s) and/or variant(s) of aircraft and operations to which he or she is assigned.

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Chapter 1

- 1.1.3 PCAR Part 8, 8.12.1.2 (a) (first sentence) QUALIFIED PERSONS REQUIRED FOR OPERATIONAL CONTROL FUNCTIONS (a) An operator or a designated representative shall have responsibility for operational control. Responsibility for operational control shall be delegated by an operator only to the PIC and to a flight operations officer/flight dispatcher, if the operator's approved method of control and supervision of flight operations requires the use of flight operations officer/flight dispatcher personnel.
- 1.1.4 PCAR Part 8, 8.12.1.2(a) (last sentence) QUALIFIED PERSONS REQUIRED FOR OPERATIONAL CONTROL FUNCTIONS (a) An operator or a designated representative shall have responsibility for operational control. Responsibility for operational control shall be delegated by an operator only to the PIC and to a flight operations officer/flight dispatcher, if the operator's approved method of control and supervision of flight operations requires the use of flight operations officer/flight dispatcher personnel.
- 1.1.6 PCAR Part 8, 8.5.1.2 COMPLIANCE WITH LOCAL REGULATIONS (b) If an emergency situation which endangers the safety of the aircraft or persons necessitates the taking of action which involves a violation of local regulations or procedures, the PIC shall - (1) Notify the appropriate local authority without delay; (2) Submit a report on any such violation to the appropriate authority of such State, if required by the State in which the incident occurs; and (3) Submit a copy of this report to the State of Registry.
- 1.3.3 PCAR Part 13, SUBPART I: MISCELLANEOUS 13.220 NON-DISCLOSURE OF RECORDS (a) Subject to paragraph (b), no person shall disclose or make available to any other person any of the following records for any purpose other than an investigation carried out under this Part unless the High Court determines that the benefits resulting from disclosure outweighs the adverse domestic and international impact the disclosure may have on that or any future investigation: (1) any statement taken from persons in the course of an investigation; (2) any communication between persons having been involved in the operation of the aircraft; (3) any medical or private information regarding persons involved in the accident or incident; (4) any cockpit voice recording and transcript from such recording; (5) any ATS recording and transcript from such recording; (6) any opinion expressed in the analysis of information, including flight recorder information. (b) A record referred to in paragraph (a) shall be included in a Final Report or its appendices or in any other report only when it is relevant to the analysis of the accident or incident. (c) Such parts of the record which are not relevant to the analysis of the accident or incident shall not be included in the Final Report.

Chapter 2

- 2 PCAR Part 8, 8.1.1.1 APPLICABILITY (a) (1) The Standards and Recommended Practices contained in Annex 6 Part I shall be applicable to the operation of airplanes by operators authorized to conduct commercial air transport operations international and domestic). (2) The Standards and Recommended Practices contained in Annex 6, Part II (airplanes), and Annex 6 Part III Section III (helicopters) shall be applicable to general aviation operations (international and domestic). (3) The Standards and Recommended Practices contained in Annex 6, Part III Sections I and II, shall be applicable to all helicopters engaged in commercial air transport operations

(international and domestic), except for these Standards and Recommended Practices are not applicable to helicopters in aerial work.

- 2.2.1.7 PCAR Part 9, 9.1.1.7 CONTENTS OF AIR OPERATOR CERTIFICATE (d) Air operator certificates and their associated operations specifications first issued from 20 November 2008 shall follow the layouts prescribed in IS: 9.1.1.7. Note: The MEL constitutes an integral part of the Operations Manual. IS: 9.1.1.7 AIR OPERATOR CERTIFICATE (AOC) AND ASSOCIATED OPERATIONS SPECIFICATIONS (a) General: (1) The AOC and its associated aircraft model-specific operations specifications shall contain minimum information required in paragraph (c) below in standardized format. (2) The AOC and its associated operations specifications shall define the operations for which an operator is authorized. (3) Subpart 9.1.1.7 contains additional information that may be listed in the operations specifications associated with the AOC. (b) A certified true copy of the AOC and operations specifications shall be carried aboard. (c) The layout of AOC and operations specifications summary: (1) Name of the State of the Operator. (2) Identification of the issuing Authority of the State of the Operator. (3) For use of the State of the Operator. (4) Unique AOC number, as issued by the State of the Operator. (5) Date after which the AOC ceases being valid (dd-mm-yyyy). (6) The operator registered name. (7) Operator trading name, if different. Insert "DbA" before the trading name (for "Doing business as"). (8) The contact details include the telephone and fax numbers, including, the country code, and the e-mail address (if available) at which operational management can be contacted without undue delay for issues related to flight operations, airworthiness, flight and cabin crew competency, dangerous goods and other matters as appropriate. (9) Operator principle place of business address. (10) Operator principle place of business telephone and fax details, including the country code, and the e-mail to be provided (if available). (11) Insertion of the controlled document, carried on board, on which the Contact details are listed, with the appropriate paragraph or page reference, e.g.: "Contact details are listed in the Operations Manual, Chapter 1,1.1"; or "are listed in the Operations Specifications, page 1"; or are listed in an attachment to this document". (12) Operator registered name. (13) Insertion of reference to the appropriate Civil Aviation Regulations. (14) Issuance date of the AOC (dd-mm-yyyy). (15) Title, name and signature of the Authority representative. In addition, an official stamp may be applied to the AOC.
- 2.2.1.8 PCAR Part 9, 9.1.1.1 APPLICABILITY (b) This Part of the regulations prescribes requirements for the original certification and continued validity of air operator certificates (AOC) issued by Republic of the Philippines.
- 2.2.2.1 PCAR Part 9, PCAR Part 9.1.1.1 APPLICABILITY d) Republic of the Philippines shall recognize as valid an AOC issued by another Contracting State, provided that the requirements under which the certificate was issued are at least equal to the applicable Standards specified in these Civil Aviation Regulations.
- 2.2.2.2 PCAR Part 10, 10.1.1.1 APPLICABILITY (c) Each foreign air operator operating within the territories of Republic of the Philippines is required to be a part of the Philippines operational safety surveillance program and follow the procedures established by this Part to preserve aviation safety.
- 2.2.2.3 PCAR Part 10, 10.1.1.4 COMPLIANCE (a) A foreign air operator may not operate an aircraft in commercial air transportation operations in Republic of the Philippines contrary to the requirements of: (1) Part 10; (2) Applicable paragraphs of Parts 7, 8 and 9; and (3) Applicable standards contained in the Annexes to the Convention on International Civil Aviation (Chicago, 1944).
- 2.2.3.1 PCAR Part 9, 9.3.1.2 OPERATIONS MANUAL (a) Each operator shall issue to the crew members and persons assigned operational control functions, an Operations

Manual acceptable to the Authority. (b) The Operations Manual shall contain the overall (general) company policies and procedures regarding the flight operations it conducts. (c) Each operator shall prepare and keep current an Operations Manual that contains the operator's procedures and policies for the use and guidance of its personnel. (d) Each operator shall issue the Operations Manual, or pertinent portions, together with all amendments and revisions to all personnel that are required to use it. The Operations Manual shall be amended or revised as is necessary to ensure that the information contained therein is kept up to date. All such amendments or revisions shall be issued to all personnel that are required to use this manual. (e) No person may provide for use of its personnel in commercial air transport any Operations Manual, or portion of this manual, together with all amendments or revisions, which has not been reviewed and found acceptable or approved for the operator by the Authority. The operator shall incorporate in the Operations Manual such mandatory material as the Authority may require. (f) Each operator shall ensure that the contents of the Operations Manual includes, at least those subjects designated by the Authority that are applicable to the AOC holder's operations. (g) Unless otherwise acceptable to the Authority, each operator shall provide an Operations Manual containing information on operations administration and supervision, safety management programs, personnel training, flight crew and cabin crew flight duty period and rest period requirements, flight operations, operational flight planning including contents and use of operational flight plan, aircraft performance, routes, guides and charts, minimum flight altitudes, airport operating minima, search and rescue, dangerous goods, navigation, communications, security, and human factors. The operations manual shall encompass the matters set forth above. The operations manual may be published in parts, as a single document, or as a series of volumes. Specific subjects are listed below. Subjects presented with reference to a specific section shall be addressed in accordance with the requirements of the referenced section. (1) Aircraft Operating Manual. (9.3.1.4) (2) Minimum Equipment List and Configuration Deviation List. (9.3.1.12) (3) Training Program. (9.3.1.3) (4) Aircraft Performance Planning Manual. (9.3.1.13) (5) Route Guide. (9.3.1.20) (6) Dangerous Goods Procedures. (7) Accident Reporting Procedures. (8) Security Procedures. (9) Aircraft Loading and Handling Manual. (9.3.1.15) (10) Cabin Crew Safety Manual. (9.3.1.17) (h) The Operations Manual shall conform to the outline contained in IS: 9.3.1.2.

- 2.2.3.2 PCAR Part 9, 9.3.1.2 OPERATIONS MANUAL (a) Each operator shall issue to the crew members and persons assigned operational control functions, an Operations Manual acceptable to the Authority. (e) No person may provide for use of its personnel in commercial air transport any Operations Manual, or portion of this manual, together with all amendments or revisions, which has not been reviewed and found acceptable or approved for the operator by the Authority. The operator shall incorporate in the Operations Manual such mandatory material as the Authority may require. IS: 9.3.1.2 OPERATIONS MANUAL (a) Each AOC holder shall ensure that the contents and structure of the operations manual are in accordance with rules and regulations of the Authority, and is relevant to the area(s) and type(s) of operation. (b) An AOC holder may design a manual to be more restrictive than the Authority's requirements. (c) Each AOC holder shall ensure that the operations manual presents the items of information listed below, to meet the requirements of Subpart 9.3.1.2 (g). The manual may consist of two or more parts containing together all such information in a format and manner based upon the outline presented in paragraph (d) below. Each part of the operations manual must contain all information required by each group of personnel addressed in that part. (1) General policies. (2) Duties and responsibilities of each crew member, appropriate members of the ground organization, and management personnel. (3) Reference to appropriate Civil Aviation Regulations. (4) Flight dispatching, supervision and operational control, including procedures for

coordinated dispatch or flight control or flight monitoring procedures and maintenance control procedures, as applicable. (5) Enroute flight, navigation, and communication procedures, including procedures for the dispatch or release or continuance of flight if any item of equipment required for the particular type of operation becomes inoperative or unserviceable en route. (6) Appropriate information from the en route operations specifications, including for each approved route the types of aircraft authorized the type of operation such as VFR, IFR, day, night, etc., and any other pertinent information. (7) Appropriate information from the airport operations specifications, including for each airport - (i) Its location (domestic and flag operations only); (ii) Its designation (regular, alternate, provisional, etc.) (domestic and flag operations only); (iii) The types of aircraft authorized (domestic and flag operations only); (iv) Instrument approach procedures; (v) Landing and takeoff minimums; and (vi) Any other pertinent information. (8) Procedures for familiarizing passengers with the use of emergency equipment, during flight. (9) Emergency equipment and procedures. (10) The method of designating succession of command of flight crew members. (11) Procedures for determining the usability of landing and takeoff areas, and for disseminating pertinent information thereon to operations personnel. (12) Procedures for operating in periods of ice, hail, thunderstorms, turbulence, or any potentially hazardous meteorological condition. (13) Airman training programs, including appropriate ground, flight, and emergency, phases. (14) Procedures for refueling aircraft, eliminating fuel contamination, protection from fire excluding electrostatic protection), and supervising and protecting passengers during refueling. - (15) Methods and procedures for maintaining the aircraft weight and centre of gravity within approved limits. (16) Where applicable, pilot and dispatcher route and airport qualification procedures. (17) Accident notification procedures. (18) Procedures and information to assist personnel to identify packages marked or labelled as containing hazardous materials and, if these materials are to be earned, stored, or handled, procedures and instructions relating to the carriage, storage, or handling of Dangerous Goods, including the following: (i) Procedures for determining the proper shipper certification and proper packaging, marking, labeling, shipping documents, compatibility of materials, and instructions on the loading, storage, and handling. (ii) Notification procedures for reporting hazardous material incidents. (iii) Instructions and procedures for the notification of the pilot in command when there are Dangerous Goods aboard. (19) Other information or instructions relating to safety. (d) The operations manual may be based upon the following outline.

- 2.2.9.1 PCAR 8, 8.6.2.13.1(f) (first sentence) FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (f) Each operator shall maintain fuel and oil records to enable the Authority to ascertain that, for each flight, the requirements of Subparts 8.6.2.14 and 8.6.2.15 have been complied with. Fuel and oil records shall be retained by the operator for a period of three (3) months. Note: If the Authority requires that fuel, in addition to any other requirement herein, is necessary on a particular route or flight operation in the interest of safety, this additional fuel will be included in the minimum fuel supply for that route.
- 2.2.9.2 PCAR Part 8, 8.6.2.13.1(f) (last sentence) FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (f) Each operator shall maintain fuel and oil records to enable the Authority to ascertain that, for each flight, the requirements of Subparts 8.6.2.14 and 8.6.2.15 have been complied with. Fuel and oil records shall be retained by the operator for a period of three (3) months. Note: If the Authority requires that fuel, in addition to any other requirement herein, is necessary on a particular route or flight operation in the interest of safety, this additional fuel will be included in the minimum fuel supply for that route.

- 2.3.3.1 PCAR Part 8, 8.6.2.16 FLIGHT PLANNING DOCUMENT DISTRIBUTION AND RETENTION: COMMERCIAL AIR TRANSPORT (a) An operational flight plan shall be completed for every intended flight or series of flights. The operational flight plan shall be approved and signed by the PIC and, where applicable, signed by the flight operations officer/flight dispatcher, and a copy shall be lodged with the appropriate authority. The operator shall determine the most efficient means of lodging the operational flight plan. For commercial air transport operations, the PIC shall complete and sign the following flight preparation documents prior to departure: 8.12.1.4 OPERATIONAL CONTROL DUTIES (a) A flight operations officer/flight dispatcher in conjunction with a method of control and supervision of flight operations in accordance with Subpart 9.1.1.6 (c) shall: (1) Assist the PIC in flight preparation and provide the relevant information required; (2) Assist the PIC in preparing the operational and ATS flight plans; (3) Sign the dispatch copy of the flight release and file the ATS flight plan with the appropriate ATS unit; (4) Furnish the PIC while in flight, by appropriate means, with information which may be necessary for the safe conduct of the flight.
- 2.3.3.2 PCAR Part 9, 9.3.1.2 OPERATIONS MANUAL (g) Unless otherwise acceptable to the Authority, each operator shall provide an Operations Manual containing information on operations administration and supervision, safety management programs, personnel training, flight crew and cabin crew flight duty period and rest period requirements, flight operations, operational flight planning including contents and use of operational flight plan, aircraft performance, routes, guides and charts, minimum flight altitudes, airport operating minima, search and rescue, dangerous goods, navigation, communications, security, and human factors. The operations manual shall encompass the matters set forth above. The operations manual may be published in parts, as a single document, or as a series of volumes. Specific subjects are listed below. Subjects presented with reference to a specific section shall be addressed in accordance with the requirements of the referenced section. IS: 9.3.1.2 OPERATIONS MANUAL (a) Each AOC holder shall ensure that the contents and structure of the operations manual are in accordance with rules and regulations of the Authority, and is relevant to the area(s) and type(s) of operation. (b) An AOC holder may design a manual to be more restrictive than the Authority's requirements. (c) Each AOC holder shall ensure that the operations manual presents the items of information listed below, to meet the requirements of Subpart 9.3.1.2 (g). The manual may consist of two or more parts containing together all such information in a format and manner based upon the outline presented in paragraph (d) below. Each part of the operations manual must contain all information required by each group of personnel addressed in that part. (1) General policies. (2) Duties and responsibilities of each crew member, appropriate members of the ground organization, and management personnel. (3) Reference to appropriate Civil Aviation Regulations. (4) Flight dispatching, supervision and operational control, including procedures for coordinated dispatch or flight control or flight monitoring procedures and maintenance control procedures, as applicable. (5) En route flight, navigation, and communication procedures, including procedures for the dispatch or release or continuance of flight if any item of equipment required for the particular type of operation becomes inoperative or unserviceable en route. (6) Appropriate information from the en route operations specifications, including for each approved route the types of aircraft authorized, the type of operation such as VFR, IFR, day, night, etc., and any other pertinent information. (7) Appropriate information from the airport operations specifications, including for each airport - (i) Its location (domestic and flag operations only); (ii) Its designation (regular, alternate, provisional, etc.) (domestic and flag operations only); (iii) The types of aircraft authorized (domestic and flag operations only); (iv) Instrument approach procedures; (v) Landing and takeoff minimums; and (vi) Any other pertinent information.

(8) Procedures for familiarizing passengers with the use of emergency equipment, during flight. (9) Emergency equipment and procedures. (10) The method of designating succession of command of flight crew members. (11) Procedures for determining the usability of landing and takeoff areas, and for disseminating pertinent information thereon to operations personnel. (12) Procedures for operating in periods of ice, hail, thunderstorms, turbulence, or any potentially hazardous meteorological condition. (13) Airman training programs, including appropriate ground, flight, and emergency, phases. (14) Procedures for refueling aircraft, eliminating fuel contamination, protection from fire excluding electrostatic protection), and supervising and protecting passengers during refueling. - (15) Methods and procedures for maintaining the aircraft weight and centre of gravity within approved limits. (16) Where applicable, pilot and dispatcher route and airport qualification procedures. (17) Accident notification procedures. (18) Procedures and information to assist personnel to identify packages marked or labelled as containing hazardous materials and, if these materials are to be earned, stored, or handled, procedures and instructions relating to the carriage, storage, or handling of Dangerous Goods, including the following: (i) Procedures for determining the proper shipper certification and proper packaging, marking, labeling, shipping documents, compatibility of materials, and instructions on the loading, storage, and handling. (ii) Notification procedures for reporting hazardous material incidents. (iii) Instructions and procedures for the notification of the pilot in command when there are Dangerous Goods aboard. (19) Other information or instructions relating to safety. (d) The operations manual may be based upon the following outline. PCAR Part 8 8.6.2.16 FLIGHT PLANNING DOCUMENT DISTRIBUTION AND RETENTION: COMMERCIAL AIR TRANSPORT (a) An operational flight plan shall be completed for every intended flight or series of flights. The operational flight plan shall be approved and signed by the PIC and, where applicable, signed by the flight operations officer/flight dispatcher, and a copy shall be lodged with the appropriate authority. The operator shall determine the most efficient means of lodging the operational flight plan. For commercial air transport operations, the PIC shall complete and sign the following flight preparation documents prior to departure: (1) An operational flight plan, including NOTAMs and weather pertinent to the flight planning decisions regarding minimum fuel supply, en route performance, and destination and alternate airports. (2) A load manifest; showing the distribution of the load, center of gravity, take-off and landing mass and compliance with maximum operating mass limitations, and performance analysis. (3) An applicable technical log page, if mechanical irregularities were entered after a previous flight, maintenance or inspection functions were performed or a maintenance release was issued at the departure airport/heliport. (b) No person may take-off an aircraft in commercial air transport unless all flight release documents, signed by the PIC, are retained and available at the point of departure. (c) The PIC shall carry a copy of the documents specified in paragraph (a) on the aircraft to the destination airport/heliport. (d) Completed flight preparation documents shall be kept by the Operator for a period of three (3) months. Note: These documents are in addition to those specified in Subpart 8.2 for all aircraft operations. Note: The Authority may approve a different retention location where all documents can be available for subsequent review.

2.3.4.1.1 PCAR Part 8, 8.6.2.9 TAKE-OFF ALTERNATE AIRPORTS/HELIPORTS: COMMERCIAL AIR TRANSPORT OPERATIONS (a) No person may release or take-off an aircraft without a suitable take-off alternate specified in the flight release if the weather conditions at the airport/heliport of departure are at or below the applicable airport/heliport operating minima or it would not be possible to return to the airport/heliport of departure for other reasons.

2.3.5.1 PCAR Part 8, 8.6.2.4 WEATHER LIMITATIONS FOR VFR FLIGHTS (a) No person will commence a flight to be conducted in accordance with the VFR unless available

current meteorological reports, or a combination of current reports and forecasts, indicate that the meteorological conditions along the route, or that part of the route to be flown under the VFR, will, at the appropriate time, allow VFR operations.

- 2.3.5.2 PCAR Part 8, 8.6.2.5 IFR DESTINATION AIRPORT/HELIPORT (a) When a destination alternate airport/heliport is required, a flight to be conducted in accordance with IFR shall not be commenced unless the information is available which indicates that will, at the estimated time of arrival, be at or above the airport/heliport operating minima.
- 2.3.6.3.2 PCAR Part 8, 8.6.2.15.4 HELICOPTERS: IFR FLIGHTS (b) When a destination alternate airport is required, to fly to and execute an instrument approach, and a missed approach, at the heliport to which the flight is planned, and thereafter: (1) to fly to the alternate specified in the flight plan; and then (2) to fly 30 minutes at holding speed at 450 M (1500 FT) above the destination heliport under standard temperature conditions, and approach and land; and (3) to have an additional amount of fuel, sufficient to provide for the increased consumption on the occurrence of any of the potential contingencies specified by the operator to the satisfaction of the Authority.
- 2.4.1.2 PCAR Part 8, 8.8.4.13 CONTINUATION OF AN INSTRUMENT APPROACH (a) In commercial air transport operations, no pilot may continue an approach past the final approach fix (FAF), or where a FAF is not used, begin the final approach segment of an instrument approach procedure, at any airport/heliport unless: (1) A source approved by the Authority issues a weather report for that airport/heliport; and (2) The latest weather report for that airport/heliport reports the visibility or controlling RVR to be equal to or more than the minimums prescribed for that procedure. (b) If a pilot begins the final approach segment of an instrument approach procedure and subsequently receives a weather report indicating below minimum conditions, the pilot may continue the approach to DH or MDA. Note: For the purpose of this subsection, the final approach segment begins at the final approach fix or facility prescribed in the instrument approach procedure. When a final approach fix is not prescribed for a procedure that includes a procedure turn, the final approach segment begins at the point where the procedure turn is completed and the aircraft is established inbound toward the airport on the final approach course within the distance prescribed in the procedure.
- 2.4.1.3 PCAR Part 8, 8.8.4.13 CONTINUATION OF AN INSTRUMENT APPROACH (a) In commercial air transport operations, no pilot may continue an approach past the final approach fix (FAF), or where a FAF is not used, begin the final approach segment of an instrument approach procedure, at any airport/heliport unless: (1) A source approved by the Authority issues a weather report for that airport/heliport; and (2) The latest weather report for that airport/heliport reports the visibility or controlling RVR to be equal to or more than the minimums prescribed for that procedure. (b) If a pilot begins the final approach segment of an instrument approach procedure and subsequently receives a weather report indicating below minimum conditions, the pilot may continue the approach to DH or MDA. Note: For the purpose of this subsection, the final approach segment begins at the final approach fix or facility prescribed in the instrument approach procedure. When a final approach fix is not prescribed for a procedure that includes a procedure turn, the final approach segment begins at the point where the procedure turn is completed and the aircraft is established inbound toward the airport on the final approach course within the distance prescribed in the procedure.
- 2.4.3 PCAR Part 8, 8.5.1.21 REPORTING OF HAZARDOUS CONDITIONS (a) The PIC shall report to the appropriate ATC facility, without delay and with enough details to be

- pertinent to the safety of other aircraft, any hazardous flight conditions encountered en-route, including those associated with meteorological conditions.
- 2.4.4.1 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (a) Each required flight crew member shall remain at the assigned duty station during take-off and landing and critical phases of flight.
- 2.4.4.2 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (b) Each flight crew member shall remain at his or her station during all phases of flight unless: (1) Absence is necessary for the performance of his or her duties in connection with the operation; (2) Absence is necessary for physiological needs, provided one qualified pilot remains at the controls at all times; or (3) The crew member is taking a rest period and a qualified relief crew member replaces him or her at the duty station. (i) For the assigned PIC during the en route cruise portion of the flight by a pilot who holds an airline transport pilot license and an appropriate type rating, and who is currently qualified as PIC or CP, and is qualified as PIC of that aircraft during the en route cruise portion of the flight; and (ii) In the case of the assigned CP, by a pilot qualified to act as PIC or Co-pilot of that aircraft during en route operations.
- 2.4.4.3 PCAR Part 8, 8.5.1.6 FLIGHT CREW MEMBER USE OF SEAT BELTS AND SAFETY HARNESES (a) Each flight crew member shall have his or her seat belts fastened during take-off and landing and all other times when seated at his or her station.
- 2.4.7.2 PCAR Part 8, 8.8.4.1 APPLICABILITY (a) All aircraft operated in accordance with instrument flight procedures shall comply with the IFR, and the airport/heliport instrument approach procedures approved by the State in which the operation will take place.
- 2.4.9.2 PCAR Part 8, 8.6.2.15.6 IN-FLIGHT FUEL MANAGEMENT (b) The pilot-in-command shall continually ensure that the amount of usable fuel remaining on board is not less than the fuel required to proceed to an aerodrome where a safe landing can be made with the planned final reserve fuel remaining upon landing.
- 2.5.1 PCAR Part 8, 8.5.1.1 AUTHORITY AND RESPONSIBILITY OF THE PIC (b) The PIC shall be responsible for the operations and safety of the aircraft and for the safety of all crew members, passengers and cargo on board, when the doors are closed. The PIC shall also be responsible for the operation and safety of the aircraft from the moment the aircraft is ready to move for the purpose of taking off until the moment it finally comes to rest at the end of the flight and the engine(s) used as primary propulsion units are shut down (and in the case of helicopters, rotor blades stopped).
- 2.5.4 PCAR Part 8, 8.5.1.18 COMPLETION OF THE TECHNICAL LOG: COMMERCIAL AIR TRANSPORT (a) The PIC shall ensure that all portions of the technical log are completed with known or suspected defects in the aircraft before, during and after flight operations, including: (1) The journey logbook; and (2) The aircraft maintenance records section 8.5.1.19 REPORTING MECHANICAL IRREGULARITIES (a) The PIC shall ensure that all mechanical irregularities occurring during flight time are: (1) For general aviation operations, entered in the aircraft logbook and disposed of in accordance with the MEL or other approved or prescribed procedure. (2) For commercial air transport operations and aerial work operations, entered in the aircraft maintenance records section of the technical log of the aircraft at the appropriate points before, during and at the end of that flight time.
- 2.5.5 PCAR Part 8, 8.5.1.19 REPORTING MECHANICAL IRREGULARITIES (a) The PIC shall ensure that all mechanical irregularities occurring during flight time are: (1) For general aviation operations, entered in the aircraft logbook and disposed of in accordance with the MEL or other approved or prescribed procedure. (2) For commercial air transport operations and aerial work operations, entered in the aircraft

maintenance records section of the technical log of the aircraft at the appropriate points before, during and at the end of that flight time. 9.3.1.5 AIRCRAFT TECHNICAL LOG ENTRIES - JOURNEY RECORDS SECTION (a) Each aircraft journey log book (technical log) shall contain the following items and corresponding Roman numerals for each flight: (See 9.4.1.9 for maintenance section of the aircraft technical log): (1) Aircraft nationality and registration; (2) Date; (3) Names of crewmembers; (4) Duty assignments of crewmembers; (5) Place of departure; (6) Place of arrival; (7) Time of departure; (8) Time of arrival; (9) Hours of flight; (10) Nature of flight (private, aerial work, scheduled, or non-scheduled); (11) Incidents, observations, if any; and (12) Signature of person in charge. (b) Entries in the journey logbook shall be made currently and in ink or indelible pencil. (c) Completed journey log books shall be retained to provide a continuous record of the last six months operations.

- 2.6.1 PCAR Part 8, 8.12.1.4 OPERATIONAL CONTROL DUTIES (a) A flight operations officer/flight dispatcher in conjunction with a method of control and supervision of flight operations in accordance with Subpart 9.1.1.6 (c) shall: (1) Assist the PIC in flight preparation and provide the relevant information required; (2) Assist the PIC in preparing the operational and ATS flight plans; (4) Furnish the PIC while in flight, by appropriate means, with information which may be necessary for the safe conduct of the flight; 8.10.1.7 PERSONS QUALIFIED TO FLIGHT RELEASE (a) No person may act as a flight operations officer in releasing a Scheduled passenger carrying commercial air transport operation unless that person: (1) Holds a flight operations officer license or an ATP license; and (2) Is currently qualified with the Operator for the operation and type of aircraft used. (b) When the Authority requires that a flight operations officer/flight dispatcher, employed in conjunction with an approved method of control and supervision of flight operations, be licensed, that flight operations officer/flight dispatcher shall be licensed in accordance with the provisions of Part 2 of these Regulations. (c) In accepting proof of qualifications other than the option of holding of a flight operations officer/flight dispatcher license, the Authority, in accordance with the approved method of control and supervision of flight operations, shall require that, as a minimum, such persons meet the requirements specified in Part 2 of these Regulations for the flight operations officer/flight dispatcher license. (d) A flight operations officer/flight dispatcher shall not be assigned to duty unless that person has: (1) satisfactorily completed an operator-specific training course that addresses all the specific components of its approved method of control and supervision of flight operations specified in Subpart 9.3.1.23 and IS: 9.3.1.23; (2) made within the preceding 12 months, at least a one-way qualification flight in the flight crew compartment of an airplane over any area for which that individual is authorized to exercise flight supervision. The flight should include landings at as many airports as practicable; (3) demonstrated to the operator a knowledge of: (i) the contents of the operations manual described in Subpart 9.3.1.2; (ii) the radio equipment in the airplanes used; and (iii) the navigation equipment in the airplanes used; (4) demonstrated to the operator a knowledge of the following details concerning operations for which the officer is responsible and areas in which that individual is authorized to exercise flight supervision: (i) the seasonal meteorological conditions and the sources of meteorological information; (ii) the effects of meteorological conditions on radio reception in the airplanes used; (iii) the peculiarities and limitations of each navigation system which is used by the operation; and (iv) the airplane loading instructions; (5) demonstrated to the operator knowledge and skills related to human performance relevant to dispatch duties; and (6) demonstrated to the operator the ability to perform the duties specified in Subpart 8.12.1.4. - (e) A flight operations officer/flight dispatcher assigned to duty shall maintain complete familiarization with all features of the operation which are pertinent to such duties, including knowledge and skills related to human performance. (f) A flight operations officer/flight dispatcher shall

not be assigned to duty after 12 consecutive months of absence from such duty, unless the provisions of paragraph (c) are met.

- 2.7 PCAR Part 8, 8.9.2.14 CARRY-ON BAGGAGE (a) No person may allow the boarding of carry-on baggage in to the passenger cabin unless it is adequately and securely stowed in accordance with the operator's approved Operations Manual procedures.

Chapter 3

- 3.1.1 PCAR Part 8, 8.7.1.1 APPLICABILITY (a) Each civil aircraft shall be operated in accordance with the comprehensive code of performance established by the Authority and in compliance with the acceptable standards, as prescribed in CAR Part 5, Subpart 5.4.1.2 and this Subpart.

Chapter 4

- 4.1.3 PCAR Part 9, 9.3.1.11 COCKPIT CHECK PROCEDURE (e) The operator shall include in the operations manual a minimum equipment list (MEL), approved by the Authority which will enable the PIC to determine whether a flight may be commenced or continued from any intermediate stop should any instrument, equipment or systems become inoperative. Where the State of the Operator is not the State of Registry, the State of the Operator shall ensure that the MEL does not affect the aircraft's compliance with the airworthiness requirements applicable in the State of Registry.
- 4.1.4 PCAR Part 9, 9.3.1.4 AIRCRAFT OPERATING MANUAL (a) Each operator or applicant shall submit proposed aircraft operating manuals for each type and variant of aircraft operated, containing the normal, abnormal and emergency procedures relating to the operation of the aircraft for approval by the Authority. (b) Each Aircraft Operating Manual shall be based upon the aircraft manufacturer's data for the specific aircraft type and variant operated by the operator and shall include specific operating parameters, details of the aircraft systems, and of the checklists to be used applicable to the operations of the operator that are approved by the Authority. The design of the manual shall observe Human Factors principles. (c) The Aircraft Operating Manual of each aircraft operated by the operator shall be issued to the flight crew members, persons assigned operational control functions and operations staff. The manual shall be easily accessible to the flight crew during all flight operations.
- 4.2.1 PCAR Part 7, 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (1) Control the flight path of the aircraft; (2) Carry out any required procedural manoeuvres; (3) Observe the operating limitations of the aircraft in the expected operating conditions; and (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.
- 4.2.2 PCAR Part 7, 7.8.11(a) Part 7,7.8.6(a)(1)(3) Part 7,7.9.1(a)(1)(2)(ii) Part 8,8.9.14 Part 8,8.9.2.17(a) (1)(2)(3)(4)(7)(b). Part 7,7.9.8 7.8.11 FIRST-AID AND EMERGENCY MEDICAL KIT (a) No person may operate and aircraft unless it is equipped with first-aid kits readily accessible to cabin crew located near an exit and, on passenger flights, an approved emergency medical kit for treatment of injuries or medical emergencies that might occur during flight time or in minor accidents. 7.8.6 PORTABLE FIRE EXTINGUISHERS (a) No person may operate an aircraft unless it is equipped with portable fire extinguishers accessible for use in crew, passenger, and cargo compartments as follows: (1) The type and quantity of extinguishing agent shall be suitable for the kinds of fires likely to occur in the compartment where the extinguisher is intended to be used. Note: For passenger compartments, the extinguisher shall be designed to minimize the hazard of toxic gas concentrations. (3) At least one portable fire extinguisher shall be conveniently located on the flight deck for use by the flight

crew. 7.9.1 SEATS. SAFETY BELTS, AND SHOULDER HARNESSSES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (1) A seat with safety belt for each person on board over the age of two (2) years old; and a restraining belt for each berth on board the aircraft (2) A safety harness for each flight crew member seat. (ii) The safety harness for each pilot seat, which includes shoulder straps and a seat belt, should incorporate a device to prevent a suddenly incapacitated pilot from interfering with the flight controls. 8.9.1.4 PASSENGER BRIEFING (a) The PIC shall ensure that crew members and passengers are made familiar, by means of an oral briefing or by other means, with the location and use of the following items: (1) Seat belts or harnesses, as appropriate; (2) Emergency exits; (3) Life jackets, if carriage of life jackets is prescribed; (4) Oxygen dispensing equipment, if provision of oxygen for the passengers is prescribed; and (5) Other emergency equipment provided for individual use, including passenger emergency briefing cards. (b) The PIC shall ensure that all persons on board are aware of the locations and general manner of use of the principal emergency equipment carried for collective use. Note: For commercial air transport operations, the briefing shall contain all subjects approved by the Authority for the specific operations conducted as included in the pertinent Operations Manual. Note: When cabin crew members are required in a commercial air transport operation, the PIC may delegate this responsibility, but shall ascertain that the proper briefing has been conducted prior to take-off. (c) The PIC shall ensure that, during take-off and landing and whenever by reason of turbulence or any emergency occurring during flight, the precaution is considered necessary, all passengers aboard the aircraft shall be secured in their seats by means of the seat belts or harnesses provided. Note: When cabin crew members are required in a commercial air transport operation, the PIC may delegate this responsibility, but shall ascertain that the proper briefing has been conducted. 8.9.2.17 REQUIRED PASSENGER BRIEFINGS (a) No person may commence a take-off unless the passengers are briefed prior to takeoff in accordance with the Operator's Operation Manual procedures on (1) Smoking limitations and prohibitions; (2) Emergency exit location and use; (3) Use of safety belts; (4) Emergency floatation means location and use; (7) If flight is above 10000 FT MSL. the normal and emergency use of oxygen; (b) Immediately before or immediately after turning the seat belt sign off, the PIC or copilot shall ensure that the passengers are briefed to keep their seat belts fastened while seated even. 7.9.8 PROTECTIVE CIRCUIT FUSES [AAC] No person may operate an aircraft in which protective fuses are installed unless there are spare fuses available for use in flight equal to at least 10% of the number of fuses of each rating or three of each rating whichever is the greater.

- 4.2.3 PCAR Part 8, 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (a) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in commercial air transport operations unless it has within it the following current and approved documents: (10) AFM (for airplanes) or RFM (for helicopters). (11) Part(s) of the Operations Manual relevant to operation(s) conducted. to the aircraft. (20) Maps and charts for routes of proposed flight or possibly diverted flights. 8.5.1.8 REQUIRED CREW MEMBER EQUIPMENT (c) Each pilot crew member shall have at his or her station current and suitable charts to cover the route of the proposed flight and any route along which it is reasonable to expect that the flight may be diverted.
- 4.3.1.1.1 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (m) No person shall operate a helicopter of a maximum certificated take-off mass of over 3175 KG, for which the individual certificate of airworthiness is first issued after 1 January 2005, unless it is equipped

- with a Type IVA FDR. Implementing Standard: See IS: 7.7.1 for specific data to be recorded by night recorders.
- 4.3.1.1.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (l) No person shall operate a helicopter of a maximum certificated take-off mass of over 7000 KG, for which the individual certificate of airworthiness is first issued after 1 January 1989, unless it is equipped with a Type IV FDR.
 - 4.3.1.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (b) Flight data recorder media not acceptable for use in aircraft registered in the Republic of the Philippines, are - (1) Engraving metal foil.
 - 4.3.1.3 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall - (3) Be deactivated upon completion of a flight following an accident or incident.
 - 4.3.2.1.1 PCAR Part 7, 7.7.2 COCKPIT VOICE RECORDERS (d) For helicopters not equipped with an FDR, at least main rotor speed shall be recorded on one track of the CVR.
 - 4.3.2.3 PCAR Part 7, 7.7.2 COCKPIT VOICE RECORDERS (c) A CVR shall be capable of retaining the information recorded during at least the last: (2) 2 hours, for airplanes of a maximum certificated take-off mass of over 5700 KG or a helicopter that receives type certification after 1 January 2003.
 - 4.3.3.1.1 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
 - 4.3.3.2 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
 - 4.3.3.3 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
 - 4.3.4.1 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall - (1) Be constructed, located and installed so as to provide maximum practical protection for the recordings in order that the recorded information may be preserved, recovered and transcribed 7.7.4 CONSTRUCTION AND LOCATION OF FDR AND CVR (b) Flight recorders shall be constructed located and installed so as to provide maximum practical protection for the recordings in order that the recorded information may be preserved, recovered and transcribed. Flight recorders shall meet the prescribed crash worthiness and fire protection specifications.
 - 4.3.4.2.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall (3) Be deactivated upon completion of a flight following an accident or incident.
 - 4.3.4.3 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall (4) The FDR system shall be recalibrated at least every five years to determine any discrepancies in the engineering conversion routines for the mandatory parameters, in accordance with the requirements of the aircraft manufacturer to ensure that the parameters are being recorded within the calibration tolerances; and when the parameters of altitude and airspeed are provided by sensors that are dedicated to the FDR system, there

shall be a recalibration performed as recommended by the sensor manufacturer, or at least every two years.

- 4.4.1 PCAR Part 7, 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (1) An airspeed indicator; (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (5) A magnetic compass; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required).

- 4.4.2 PCAR Part 7, 7.2.2(b)(1)(2) Part 7, 7.2.2(a)(3)(4) Part 7, 7.4.1(a)(5)(6)(7) & (b)(1). 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight Instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope).

7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator.

7.4.1 AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (a) All aircraft operated at night shall be equipped with: (5) Illumination for all flight instruments and equipment that are essential for the safe operation of the aircraft; (6) Lights in all passenger compartments; (7) A flashlight for each crew member station (approval not required) (b) No AOC holder may operate an aircraft by night unless, in addition to the equipment specified in paragraph (a) above, it is equipped with; (1) Two landing lights or a single light having two separately energized filaments.

- 4.4.3 PCAR Part 7, 7.2.3(a)(1)(2)(3) Part 7, 7.2.2(a)(3)(4)(5)(6)(7)(b)(1)(2) Part 7, 7.2.1(b) Part 7, 7.4.1(b) (1). 7.2.3 IFR INSTRUMENTS (a) [AAC] All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments. shall be equipped with the flight instruments as listed in 7.2.2 (1) An airspeed indicating system with a means of preventing malfunctioning due to either condensation or icing; (2) A sensitive pressure altimeter with counter drum-pointer or equivalent presentation, calibrated in feet with a sub-scale setting calibrated in hectopascals/millibars, adjustable for any barometric pressure likely to be set during flight (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate. PCAR Part 7 7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) [AAC] Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight Instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope). PCAR Part 7 7.2.1 GENERAL REQUIREMENTS (b) [AAC] When a means is provided for transferring an instrument from its primary operating system to an alternative system, the means shall include a positive positioning control and shall be marked to indicate clearly which system is being used. PCAR Part 7 7.4.1 AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (b) (1) Two landing lights or a single light having two separately energized filaments.

- 4.4.3.1 PCAR Part 7, 7.2.4 STANDBY ATTITUDE INDICATOR (a) No person may operate an airplane with a maximum certified take-off mass exceeding 5700 KG, aircraft having a

- maximum approved passenger seating configuration of more than 9 seats and all helicopters when operating in accordance with IFR, unless it is equipped with a single standby attitude indicator (artificial horizon) that - (3) After a total failure of the normal electrical generating system, is automatically powered for the purpose of operating and illuminating for a minimum of 30 minutes from a source independent of the normal electrical generating system
- 4.5.1 PCAR Part 7, 7.8.18 FLOTATION DEVICE FOR HELICOPTER DITCHING (a) All helicopters flying over water at a distance from land corresponding to more than 10 minutes at normal cruise speed in the case of performance Class 1 or 2 helicopters, or flying over water beyond auto rotational or safe forced landing distance from land in the case of performance Class 3 helicopters, shall be fitted with a permanent or rapidly deployable means of floatation so as to ensure a safe ditching of the helicopter.
- 4.5.2.1 PCAR Part 7, 7.8.16(a)(2) (c)(1) Part 7, 7.8.17(a)(d)(3) Part 7, IS: 7.8.4(b). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided. (c) Helicopters. (1) For offshore operations the life jacket shall be worn unless the occupant is wearing an integrated survival suit. 7.8.17 LIFE RAFT (a) In addition to the equipment prescribed in 7.8.16 and 7.8.18 of this Part, lifesaving rafts in sufficient numbers to carry all persons on board shall be installed in: (1) All airplanes when operating in "Extended over-water operations" as defined in Subpart 8.1.1.2(b). (2) Class 1 and 2 helicopters when they are operated over water at a distance from land corresponding to more than 10 minutes at normal cruise speed. (3) Class 3 helicopters when they are operated over water beyond auto rotational or safe forced landing distance from land. (d) Life rafts shall be equipped with the following life sustaining equipment - (3) A pyrotechnical signaling device. IS: 7.8.4 SURVIVAL EQUIPMENT FLIGHTS OVER LAND (b) Where life rafts are required to be carried in accordance with subpart 7.8.17 (d), they shall be equipped with an attached survival kit containing at least the following: 1) a pyrotechnic signaling device; 2) a radar reflector; 3) a life raft repair kit; 4) a bailing bucket and sponge; 5) a signaling mirror; 6) a whistle; 7) a raft knife; 8) an inflation pump; 9) dye marker; 10) a waterproof flashlight; 11) a two day supply of water, calculated using the overload capacity of the raft, consisting of one pint of water per day for each person or a means of desalting or distilling salt water sufficient to provide an equivalent amount; 12) a fishing kit; 13) a book on sea survival; and 14) a first aid kit containing antiseptic swabs, burn dressing compresses, bandages and anti-motion sickness pills.
- 4.5.2.2 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.
- 4.5.2.2.1 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (c) (1) For offshore operations the life jacket shall be worn unless the occupant is wearing an integrated survival suit.
- 4.5.2.3 PCAR Part 7, 7.8.17 LIFE RAFT (d) Life rafts shall be equipped with the following life sustaining equipment (3) A pyrotechnical signalling device.
- 4.5.2.4 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (1) Shall carry the equipment prescribed in paragraph (2): (iii) When taking off or landing at an aerodrome where the (Authority) has determined the take-off or approach path is so disposed over water that in the event of a mishap there would be the likelihood of a ditching.

- 4.5.2.5 PCAR Part 7, 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.
- 4.5.3.1 PCAR Part 7, 7.8.3 VISUAL SIGNALLING DEVICES (a) No person may operate an aircraft over water or across land areas which have been designated by the. Republic of the Philippines as areas in which search and rescue Would be especially difficult, unless equipped with such signalling devices as may be appropriate to the area overflown, to include (1) Visual signals for use by intercepting and intercepted aircraft; and (2) At least one pyrotechnic signalling device for each life raft required for over water operations.
- 4.6 PCAR Part 7,7.8.3(a) and Part 7,7.8.4(a). 7.8.3 VISUAL SIGNALLING DEVICES (a) No person may operate an aircraft over water or across land areas which have been designated by the. Republic of the Philippines as areas in which search and rescue would be especially difficult, unless equipped with such signaling devices as may be appropriate to the area overflown, to include (1) Visual signals for use by intercepting and intercepted aircraft; and (2) At least one pyrotechnic signaling device for each life raft required for over water operations. 7.8.4 SURVIVAL EQUIPMENT (a) Subject to subsection (b) no person shall operate an aircraft over land.unless there is carried on board survival equipment, sufficient for the survival on the ground of each person on board, given the. geographical area, the season of the year and anticipated seasonal climatic variations, that provides the means for (1) starting a fire; (2) providing shelter; (3) providing or purifying water; and (4) visually signaling distress.
- 4.7.1 PCAR Part 7, IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic E L T and one ELT(S) in a raft or life jacket.
- 4.7.2 PCAR Part 7, IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.
- 4.7.3 PCAR Part 7,7.8.5(a)(1). 7.8.5 EMERGENCY LOCATOR, TRANSMITTER(ELT) (a) Except for 7.8.5 (b) all aircraft on fall flights shall be equipped with a 406 MHZ ELT of a type and quantity as specified in IS:7.8.5, meet the technical standards specified by the Authority and the relevant portions of ICAO Annex 10, Volume 3. (1) The ELT must be registered and programmed with a distinct Unique Identification Number (UIN) which is comprised of a 24 bit code. This code is supplied by the Authority prior to installation and use.
- 4.8.1 PCAR Part 7, IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental

oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. Notes: 1. The supply provided shall take account of the cabin pressure altitude and descent profile for the routes concerned. 2. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 20 minutes at 10000 FT. 3. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 110 minutes at 10000 FT. The oxygen required to meet the Crew Protective Breathing Equipment provisions of this Part may be included in determining the supply required. 4. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 15000 FT (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not operate a non-pressurized helicopter at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

- 4.8.2 PCAR Part 7, IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a)
The supplemental oxygen supply requirements for non-pressurized aircraft are as

follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. Notes: 1. The supply provided shall take account of the cabin pressure altitude and descent profile for the routes concerned. 2. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 20 minutes at 10000 FT. 3. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 110 minutes at 10000 FT. The oxygen required to meet the Crew Protective Breathing Equipment provisions of this Part may be included in determining the supply required. 4. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 15000 FT. (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not operate a non-pressurized helicopter at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in

accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

- 4.8.3 PCAR Part 7, IS: 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (a) The supplemental oxygen supply requirements for non-pressurized aircraft are as follows: (1) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 1. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (2) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 1. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (b) The supplemental oxygen supply requirements for pressurized aircraft are as follows: (1) The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurization failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aircraft will descend in accordance with emergency procedures specified in the Aircraft Flight Manual to a safe altitude for the route to be flown that will allow continued safe flight and landing. (2) Following a cabin pressurization failure, the cabin pressure altitude shall be considered the same as the aircraft altitude, unless it is demonstrated to the Authority that no probable failure of the cabin or pressurization system will result in a cabin pressure altitude equal to the aircraft altitude. Under these circumstances, this lower cabin pressure altitude may be used as a basis for determination of oxygen supply. (3) Flight crew members. (i) Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with Table 2. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply. (ii) The oxygen supply requirements, as specified in Table 2, for aircraft not certificated to fly at altitudes above 25000 FT, may be reduced to the entire flight time between 10000 FT and 14000 FT cabin pressure altitudes for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aircraft is able to descend safely within 4 minutes to a cabin pressure altitude of 14000 FT. Notes: 1. The supply provided shall take account of the cabin pressure altitude and descent profile for the routes concerned. 2. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 20 minutes at 10000 FT 3. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 10000 FT in 10 minutes and followed by 110 minutes at 10000 FT. The oxygen required to meet the Crew Protective Breathing Equipment provisions of this Part may be included in determining the supply required. 4. The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aircraft's maximum certificated operating altitude to 15000 FT (c) The supplemental oxygen supply requirements for non-pressurized helicopters are as follows: (1) An operator shall not

operate a non-pressurized helicopter at pressure altitudes above 10000 FT unless supplemental oxygen equipment, capable of storing and dispensing the oxygen supplies required, is provided. (2) The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures, established for each operation in the Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Operations Manual. (3) A helicopter intended to be operated above 10000 FT pressure altitude shall be provided with equipment capable of storing and dispensing the oxygen supplies required. (4) Oxygen supply requirements: (i) Flight crew members. Each member of the flight crew on flight deck duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of flight deck seats are supplied from the flight crew source of oxygen supply then they shall be considered as flight crew members on flight deck duty for the purpose of oxygen supply. (ii) Cabin crew members, additional crew members and passengers. Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried in addition to the minimum number of cabin crew members required, and additional crew members, shall be considered as passengers for the purpose of oxygen supply.

- 4.9 PCAR Part 7, 7.9.9 ICING PROTECTION EQUIPMENT (a) No person may operate an aircraft in icing conditions unless it is certified and equipped for the prevention or removal of ice on windshields, wings, empennage, propellers, and other parts of the aircraft where ice formation will adversely affect the safety of the aircraft.
- 4.11 PCAR Part 8, 8.2.1.8(a)(9). 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (a) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in commercial air transport operations unless it has within it the following current and approved documents: (9) Noise Certificate, or a document attesting noise certification. When the document, or a suitable statement attesting noise certification as contained in another document approved by the State of Registry, is issued in a language other than English, it shall include an English translation.
- 4.12.1 PCAR Part 7, 7.9.1(a)(3)(4). 7.9.1 SEATS. SAFETY BELTS, AND SHOULDER HARNESES (a) Each aircraft used in passenger carrying operations shall be equipped with the following seats, safety belts, and shoulder harnesses that meet the airworthiness requirements for type certification of that aircraft: (3) A forward or rearward facing (within 15 degrees of the longitudinal axis of the airplane) seat equipped with a safety harness for each cabin crew member station in the passenger compartment. (4) The cabin crew member's seats shall be located near floor level and other emergency exits for emergency evacuation.
- 4.13 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV. (1) The aircraft will be assigned a distinct serialized 24-bit address identity code supplied by the Authority for the transponder, this must be programmed prior to operation of the aircraft. (b) No person may operate an aircraft at altitudes above PL 290 unless it is equipped with a system that is automatically reporting pressure altitudes. (c) No person may operate an aircraft in commercial air transportation unless it is equipped with a pressure-altitude reporting transponder that operates in accordance with the requirements of the Republic of the Philippines.

Chapter 5

- 5.1.1 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) [All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment

- capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHz. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 5.1.2 PCAR Part 7, 7.3.1(b) 7.3.1 RADIO EQUIPMENT (b) [AAC] All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHz Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 5.1.3 Not implemented.
- 5.2.1 PCAR Part 7, 7.2.9(a)(1)(3). 7.2.9 NAVIGATION EQUIPMENT — GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (1) Its operational flight plan, (3) The requirements of air traffic services.
- 5.2.2 PCAR Part 7, 7.2.9 (a)(2) and Part 8, 8.8.1.31(a)(b). 7.2.9 NAVIGATION EQUIPMENT - GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (2) The prescribed Performance-Based Navigation Specification.
- 8.8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (a) No operator shall permit, and no pilot shall conduct operations in airspace designated as Performance-based Navigation (PBN) airspace, unless so approved in the operator's Operations Specifications. (b) For flights in defined portions of airspace or on routes where an PBN type has been prescribed, an aircraft shall be provided with navigation equipment which will enable it to operate in accordance with the prescribed RNP type(s) and the ATS requirements as specified in Subparts 7.2.7, 7.2.8, 7.2.9 and 7.2.10.
- 5.2.3 PCAR Part 8, 8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (c) An operator shall equip the aircraft as prescribed in subparagraph (b) above by incorporating the necessary airworthiness requirements, and submit to the Authority for approval the company manuals and amendments thereof, including the pre-flight and en-route procedures to be followed for such flights, the training and qualifications required of maintenance personnel, flight operations officers/flight dispatchers, and flight crew members; and such other information necessary in the conduct of operations in PBN designated airspace. On successful demonstration of competency of operations in PBN designated airspace by an operator, the Authority shall authorize such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in PBN airspace.
- 5.2.4 PCAR Part 8, 8.8.1.31 OPERATIONS IN PERFORMANCE-BASED NAVIGATION (PBN) AIRSPACE (a) No operator shall permit, and no pilot shall conduct operations in airspace designated as Performance-based Navigation (PBN) airspace, unless so approved in the operator's Operations Specifications. (b) For flights in defined portions of airspace or on routes where a PBN type has been prescribed, an aircraft shall be provided with navigation equipment which will enable it to operate in accordance with the prescribed RNP type(s) and the ATS requirements as specified in Subparts 7.2.7, 7.2.8, 7.2.9 and 7.2.10. (c) An operator shall equip the aircraft as prescribed in subparagraph (b) above by incorporating the necessary airworthiness requirements, and submit to the Authority for approval the company manuals and amendments

thereof, including the pre-flight and en-route procedures to be followed for such flights, the training and qualifications required of maintenance personnel, flight operations officers/flight dispatchers, and flight crew members; and such other information necessary in the conduct of operations in PBN designated airspace. On successful demonstration of competency of operations in PBN designated airspace by an operator, the Authority shall authorize such operations. Note: See ICAO Doc 9613 for information on the approval process for operations in PBN airspace.

- 5.2.5 PCAR Part 7,7.2.1(a)(4). 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.
- 5.2.6 PCAR Part 7, 7.2.3 IFR INSTRUMENTS (c) All aircraft intended to land in IMC or at night shall be provided with radio navigation equipment capable of receiving signals providing guidance to - (1) A point from which a visual landing can be effected; or (2) Each aerodrome at which it is intended to land in IMC; and (3) Any designated alternate aerodromes.
- 5.4 PCAR Part 7,7.1.4(d)(2). 7.1.4 GENERAL INSTRUMENT AND EQUIPMENT REQUIREMENTS (d) An operator shall ensure that a flight does not commence unless the required equipment - (2) Is installed such that the failure of any single unit required for either communication or navigation purposes, or both, will not result in the inability to communicate and/or navigate safely on the route being flown.

Chapter 6

- 6.1.1 PCAR Part 9,9.4.1.2(a) (3)(b)(c). 9.4.1.2 MAINTENANCE RESPONSIBILITY (a) Each operator shall ensure the airworthiness of the aircraft and the serviceability of both operational and emergency equipment by (3) Assuring the accomplishment of all maintenance in accordance with the approved operator's aircraft maintenance program: (b) Each operator shall ensure that the Certificate of Airworthiness for each aircraft operated remains valid in respect to (1) The requirements in paragraph (a); (2) The expiration date of the Certificate: and (3) Any other maintenance condition specified in the Certificate. (c) Each operator shall ensure that the requirements specified in paragraph (a) are performed in accordance with procedures approved by or acceptable to the Authority.
- 6.1.3 PCAR Part 9, 9.4.1.9 RETURN TO SERVICE (RTS) (c) Each operator shall employ a person or group of persons, acceptable to the Authority; to ensure that all maintenance is carried out to an approved standard such that the maintenance requirements of Subpart 9.4.1.2 and requirements of the AOC holder's maintenance control manual are satisfied; and to ensure the functioning of the quality system.
- 6.1.4 PCAR Part 9, 9.4.1.9 RETURN TO SERVICE (RTS) (c) Each operator shall employ a person or group of persons, acceptable to the Authority; to ensure that all maintenance is carried out to an approved standard such that the maintenance requirements of Subpart 9.4.1.2 and requirements of the AOC holder's maintenance control manual are satisfied; and to ensure the functioning of the quality system.
- 6.3.1 PCAR Part 9, 9.4.1.11 OPERATOR'S MAINTENANCE PROGRAM (a) Each operator's aircraft maintenance program and any subsequent amendment shall be submitted to the State of Registry for approval; acceptance by the Authority will be conditioned upon prior approval by the State of Registry, or where appropriate, upon the operator complying with recommendations provided by the State of Registry. The design and application of the operator's maintenance program shall observe Human Factors principles.

- 6.4.2 PCAR Part 9, 9.4.1.7 MAINTENANCE RECORDS (b) Each operator shall ensure that items in subparagraphs (a) (1) to (5) shall be kept for a minimum of 90 days after the unit to which they refer has been permanently withdrawn from service, and the records in subparagraph (a) (6) shall be kept for a minimum of 1 year after the signing of the maintenance release and/or RTS.
- 6.4.3 PCAR Part 9, 9.4.1.7 MAINTENANCE RECORDS (c) Each operator shall ensure that in the event of temporary change of operator, the records specified in paragraph (a) shall be made available to the new operator.
- 6.5.1 PCAR Part 9,9.4.1.4(b)(7) (8). 9.4.1.4 MAINTENANCE CONTROL MANUAL (b) The AOC holder's maintenance control manual shall contain the following information which may be issued in separate parts (7) A description of the procedures for monitoring, assessing and reporting maintenance and operational experience for all aircraft over 5700 KG maximum certificated take-off mass; (8) A description of the procedures for obtaining and assessing continuing airworthiness information and implementing any resulting actions for all aircraft over 5700 KG maximum certificated take-off mass, from the organization responsible for the type design, and shall implement such actions considered necessary by the State of Registry.
- 6.5.2 PCAR Part 9,9.4.1.4(b)(8). 9.4.1.4 MAINTENANCE CONTROL MANUAL (b) The AOC holder's maintenance control manual shall contain the following information which may be issued in separate parts (8) A description of the procedures for obtaining and assessing continuing airworthiness information and implementing any resulting actions for all aircraft over 5700 KG maximum certificated take-off mass, from the organization responsible for the type design, and shall implement such actions considered necessary by the State of Registry.
- 6.6 PCAR Part 9, 9.4.1.10 MODIFICATION AND REPAIRS (a) All modifications and repairs shall comply with airworthiness requirements of this CAR and acceptable to the authority and the State of Registry. Procedures shall be established to ensure that the substantiating data supporting compliance with the airworthiness requirements are retained. However, in the case of a major repair or major modification, the work must have been done in accordance with technical data approved by the Authority.
- 6.8.1 PCAR Part 8,8.3.1.9(a) (2)(i)(ii)(iii)(iv)(b). 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (a) (2) Records containing the following information - (i) The total time-in-service of the airframe, each engine, each propeller, and each rotor. (ii) The current status of all life-limited aeronautical products; (iii) The time since last overhaul of all items installed on the aircraft which are required to be overhauled on a specified time basis; (iv) The current inspection status of the aircraft, including the time since the last inspection required by the inspection program under which the aircraft and its appliances are maintained. (b) The records specified in paragraph (a) of this section shall be retained and transferred with the aircraft at the time the aircraft is sold or leased.
- 6.8.2 PCAR Part 8, 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (d) Except for records maintained by an operator, each registered owner or operator shall retain the following records for a minimum period of 90 days, or until the work is repeated or superseded by other work of equivalent scope and detail, or the unit to which they refer has been permanently withdrawn from service, and the records in paragraph (1) below for a minimum period of one year after the signing of the maintenance release.

Chapter 7

- 7.1.2 PCAR Part 8, 8.4.1.1 COMPOSITION OF THE FLIGHT CREW (b) The flight crew shall include at least one member who holds a valid radio license, issued or rendered valid

by the State of Registry, authorizing operation of the type of radio transmitting equipment to be used.

- 7.3.1 PCAR Part 8, 8.10.1.8.3(a) thru (f). 8.10.1.8.3 FLIGHT CREW MEMBER TRAINING PROGRAMS An operator shall establish and maintain a ground and flight training programme, approved by the Authority, which ensures that all flight crew members are adequately trained to perform their assigned duties. The training program shall: (a) Include ground and flight training facilities and properly qualified instructors as determined by the Authority; (b) Consist of ground and flight training in the type(s) of aircraft on which the flight crew member serves; (c) Include proper flight crew coordination and training in all types of emergency and abnormal situations or procedures caused by power plant, (for helicopters: transmission and rotor), airframe or systems malfunctions, fire or other abnormalities; (d) Include training in knowledge and skills related to visual and instrument flight procedures for the intended area of operation, human performance including threat and error management and in the transport of dangerous goods and, where applicable, procedures specific to environment in which the aircraft is to be operated; (e) Ensure that all flight crew members know the functions for which they are responsible and the relation of these functions to the functions of other crew members, particularly in regard to abnormal or emergency procedures; and (f) Be given on a recurrent basis, as determined by the Authority and shall include an assessment of competence.

- 7.4.1.1 PCAR Part 2, 2.3.1.6(a) and Part 8, 8.4.1.9(a). 2.3.1.6 RECENT EXPERIENCE REQUIREMENTS (a) A pilot shall not operate an aircraft carrying passengers as pilot-in-command or copilot to operate at the flight controls of a type or a variant of a type of aircraft during take-off and landing unless that pilot has operated the flight controls during at least three take-offs and landings within the preceding 90 days on the same type of aircraft or in the flight simulator approved for the purpose.

8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (a) No person may act as PIC or a co-pilot of an aircraft unless, within the preceding 90 days that pilot has: (1) Made three take-offs and landings as the sole manipulator of the flight controls in an aircraft of the same category and class and if a type rating is required, of the same type or in a flight simulation training device approved for the purpose. (2) For a tail-wheel airplane, made the three take-offs and landings in a tail-wheel airplane with each landing to a full stop. (3) For night operations, made the three take-offs and landings required by paragraph (a) (1) above at night.

- 7.4.2.1 PCAR Part 2, 2.3.1.6(a), Part 8, 8.4.1.9(a) and Part 8, 8.4.1.9(e). 2.3.1.6 RECENT EXPERIENCE REQUIREMENTS (a) A pilot shall not operate an aircraft carrying passengers as pilot-in-command or copilot to operate at the flight controls of a type or a variant of a type of aircraft during take-off and landing unless that pilot has operated the flight controls during at least three take-offs and landings within the preceding 90 days on the same type of aircraft or in the flight simulator approved for the purpose.

8.4.1.9 PILOT CURRENCY: TAKE-OFF AND LANDINGS, AND CRUISE RELIEF (a) No person may act as PIC or a co-pilot of an aircraft unless, within the preceding 90 days that pilot has: (1) Made three take-offs and landings as the sole manipulator of the flight controls in an aircraft of the same category and class and if a type rating is required, of the same type or in a flight simulation training device approved for the purpose. (2) For a tail-wheel airplane, made the three take-offs and landings in a tail-wheel airplane with each landing to a full stop. (3) For night operations, made the three take-offs and landings required by paragraph (a) (1) above at night. (e) When a pilot is flying several variants of the same type of aircraft or different types of aircraft with similar characteristics in terms of operating procedures, systems and handling, the

Authority shall decide under which conditions the requirements of paragraph (a) for each variant or each type of aircraft can be combined.

7.4.2.3 PCAR Part 8, 8.10.1.30.2 OPERATIONAL QUALIFICATIONS: HELICOPTER PIC (c) An operator shall not continue to utilize a pilot as a PIC on an operation unless, within the preceding 12 months, the pilot has made at least one representative flight as a pilot member of the flight crew, or as a check pilot, or as an observer on the flight deck. In the event that more than 12 months elapse in which a pilot has not made such a representative flight, prior to again serving as a pilot-in-command on that operation, that pilot must requalify in accordance with paragraph (b) and (c) above.

7.4.2.4 PCAR Part 8, 8.10.1.45 RECORDING OF CREW MEMBER QUALIFICATIONS (a) The operator shall maintain a record, sufficient to satisfy the Authority of the qualification of each crew member and flight operations officer, and of the manner in which such qualification has been achieved, as required by this Part.

7.4.3.1 PCAR Part 8, 8.10.1.20(b) (c) and Part 9.9.3.1.9(b). 8.10.1.20 AIRCRAFT AND INSTRUMENT PROFICIENCY CHECKS (b) An operator shall ensure that piloting technique and the ability to execute emergency procedures is checked in such a way as to demonstrate the pilot's competence on each type or variant of type of aircraft. Where the operation may be conducted under IFR, an operator shall ensure that the pilot's competence to comply with such rules is demonstrated to either a check pilot of the operator or to a representative of the Authority. A pilot may complete both these requirements simultaneously in a specific aircraft type. (c) Such checks shall be performed twice within any period of one (1) year. Any two (2) such checks which are similar and which occur within a period of four (4) consecutive months shall not alone satisfy this requirement.

9.3.1.9 CREW MEMBER CHECKING AND STANDARDIZATION PROGRAM (b) An operator shall check pilots' proficiency on those maneuvers and procedures that are prescribed by the Authority for pilot proficiency checks, which shall include emergency procedures and, where applicable, instrument flight rules. Note: A standardized process is defined to address the operator unique fleet differences and compliance methods. Note: See Part 8 for specific checking requirements.

Chapter 8

8.3 PCAR Part 8, 8.10.1.7(d)(1) thru (6). 8.10.1.7 PERSONS QUALIFIED TO FLIGHT RELEASE (d) A flight operations officer/flight dispatcher shall not be assigned to duty unless that person has: (1) satisfactorily completed an operator-specific training course that addresses all the specific components of its approved method of control and supervision of flight operations specified in Subpart 9.3.1.23 and IS: 9.3.1.23; (2) made within the preceding 12 months, at least a one-way qualification flight in the flight crew compartment of an airplane over any area for which that individual is authorized to exercise flight supervision. The flight should include landings at as many airports as practicable; (3) demonstrated to the operator a knowledge of: (i) the contents of the operations manual described in Subpart 9.3.1.2; (ii) the radio equipment in the airplanes used; and (iii) the navigation equipment in the airplanes used; (4) demonstrated to the operator a knowledge of the following details concerning operations for which the officer is responsible and areas in which that individual is authorized to exercise flight supervision: (i) the seasonal meteorological conditions and the sources of meteorological information; (ii) the effects of meteorological conditions on radio reception in the airplanes used; (iii) the peculiarities and limitations of each navigation system which is used by the operation; and (iv) the airplane loading instructions; (5) demonstrated to the operator knowledge and skills related to human performance relevant to dispatch duties; and (6) demonstrated to the operator the ability to perform the duties specified in Subpart 8.12.1.4.

Chapter 9

- 9.1 PCAR Part 8, 8.2.1.6 CIVIL AIRCRAFT FLIGHT MANUAL, MARKING AND PLACARD REQUIREMENTS (c) Each AFM or RFM shall be updated by implementing changes made mandatory by the State of Registry
- 9.2 PCAR Part 9,9.4.1.4(b) (1) to (13)(d). 9.4.1.4 MAINTENANCE CONTROL MANUAL (b) The AOC holder's maintenance control manual shall contain the following information which may be issued in separate parts (1) A description of the administrative agreements between the AOC holder and the AMO containing a description of the maintenance procedures and the procedures for completing and signing a maintenance release; (2) A description of the procedures to ensure each airplane they operate is in an airworthy condition; (3) A description of the procedures to ensure the operational emergency equipment for each flight is serviceable; (4) The names and duties of the person or persons required to ensure that all maintenance is carried out in accordance with the maintenance control manual; (5) A reference to the maintenance program required in Subpart 9.4.1.11; (6) A description of the methods for completion and retention of the operator's maintenance records required by Subpart 9.4.1.8; (7) A description of the procedures for monitoring, assessing and reporting maintenance and operational experience for all aircraft over 5700 KG maximum certificated take-off mass; (8) A description of the procedures for obtaining and assessing continuing airworthiness information and implementing any resulting actions for all aircraft over 5700 KG maximum certificated take-off mass, from the organization responsible for the type design, and shall implement such actions considered necessary by the State of Registry; (9) A description of the procedures for implementing mandatory continuing airworthiness information as required in Subpart 9.4.1.2(a)(5); (10) A description of establishing and maintaining a system of analysis and continued monitoring of the performance and efficiency of the maintenance program in order to correct any deficiency in that program; (11) A description of aircraft types and models to which the manual applies; (12) A description of procedures for ensuring that unserviceabilities affecting airworthiness are recorded and rectified; and (13) A description of the procedures for advising the State of Registry of significant in-service occurrences. (d) The operator shall ensure that the maintenance control manual is amended as necessary to keep the Information contained therein up to date.
- 9.5 PCAR Part 9,9.3.1.25(c)(1) (2)(3). (c) An operator engaged in international air navigation shall at all times have available for immediate communication to rescue coordination centers, lists containing information on the emergency and survival equipment carried on board any of their airplanes/helicopters including, as applicable: (1) The number, colour and types of life rafts and pyrotechnics; (2) Details of emergency water and medical supplies, water supplies; and (3) The type and frequencies of the emergency portable radio equipment.
- 9.6 PCAR Part 9, 9.2.2.6 COCKPIT VOICE AND FLIGHT DATA RECORDER RECORDS (c) The operator shall ensure, to the extent possible, the preservation of all related flight recorder records and, if necessary, the associated flight recorders (cockpit voice recorder and flight data recorder), and their retention in safe custody pending their disposition as determined in accordance with Annex 13.

Chapter 10

- 10.2 PCAR Part 8, 8.9.2.4 CABIN CREW MEMBERS AT DUTY STATIONS (b) Each cabin crew member assigned to emergency evacuation duties shall occupy a seat provided in accordance with Subpart 7.9.1 during take-off and landing and whenever the RIC so directs, with seat belt or, when provided, safety harness fastened.

- 10.3 PCAR Part 8, 8.10.1.14(c) (1)(2) and Part 8, 8.10.1.34 (c) (1) to (6). 8.10.1.14 INITIAL AIRCRAFT GROUND TRAINING (c) (1) For cabin crew members, initial and re-qualification aircraft ground training shall include the pertinent portions of the operations manuals relating to aircraft-specific configuration, equipment, normal and emergency procedures for the aircraft types within the fleet. Specific course curriculum requirements, for cabin crew members, are contained in IS: 8.10.1.14(c1). (2) For check cabin crew members, initial and re-qualification ground training shall include checking of cabin crew members on the pertinent portions of the operations manuals relating to aircraft-specific configuration, equipment, normal and emergency procedures for the aircraft types within the fleet. Specific course curriculum requirements for check cabin crew member are contained in IS: 8.10.1.14(c2).

8.10.1.34 RECURRENT TRAINING: CABIN CREW (c) The recurrent training program shall ensure that each cabin crew member is: (1) Competent to execute those safety duties and functions which the cabin crew member is assigned to perform in the event of an emergency or in a situation requiring emergency evacuation; (2) Drilled and capable in the use of emergency and life-saving equipment required to be carried, such as life jackets, life rafts, evacuation slides, emergency exits, portable fire extinguishers, oxygen equipment and first-aid kits; (3) When serving on airplanes operated above 3000 M (10000 FT), knowledgeable as regards the effect of lack of oxygen and, in the case of pressurized airplanes, as regards physiological phenomena accompanying a loss of pressurization; (4) Aware of other crew members' assignments and functions in the event of an emergency so far as is necessary for the fulfilment of the cabin crew member's own duties; (5) Aware of the types of dangerous goods which may, and may not, be carried in a passenger cabin and has completed the dangerous goods training programme required by Annex 18; and (6) Knowledgeable about human performance as related to passenger cabin safety duties including flight crew-cabin crew coordination.

Chapter 11

- 11.1 PCAR Part 9, 9.5.1.6 AIRCRAFT SEARCH PROCEDURE CHECKLIST Each operator shall ensure that there is on board a checklist of the procedures to be followed for a bomb in case of a suspected sabotage and for inspecting aircraft for concealed weapons, explosives, or other dangerous devices when a well-founded suspicion exists that the aircraft may be the object of an act of unlawful interference. The checklist shall be supported by guidance on appropriate course of action to be taken should a bomb or suspicious object be found and information on the least-risk bomb location specific to the aircraft.
- 11.3 PCAR Part 9, 9.5.1.5 REPORTING ACTS OF UNLAWFUL INTERFERENCE (a) Following an act of unlawful interference on board an aircraft the PIC or, in his absence, the operator shall submit, without delay, a report of such an act to the designated local authority, and to the Authority.

Section III - International General Aviation

Chapter 1

- 1.1.1 PCAR Part 8, 8.5.1.1(d) and Part 8, 8.5.1.2(a). 8.5.1.1 AUTHORITY AND RESPONSIBILITY OF THE PIC (d) The PIC of an aircraft shall, whether manipulating the controls or not, be responsible for the operation of the aircraft in accordance with the rules of the air, except that the PIC may depart from these rules in emergency circumstances that render such departure absolutely necessary in the interests of safety.

8.5.1.2 COMPLIANCE WITH LOCAL REGULATIONS (a) The PIC shall comply with the relevant laws, regulations and procedures of the States in which the aircraft is operated.

1.1.2 PCAR Part 8, 8.5.1.1 AUTHORITY AND RESPONSIBILITY OF THE PIC (b) The PIC shall be responsible for the operations and safety of the aircraft and for the safety of all crew members, passengers and cargo on board, when the doors are closed. The PIC shall also be responsible for the operation and safety of the aircraft from the moment the aircraft is ready to move for the purpose of taking off until the moment it finally comes to rest at the end of the flight and the engine(s) used as primary propulsion units are shut down (and in the case of helicopters, rotor blades stopped).

1.1.3 PCAR Part 8, 8.5.1.2(b) (1) to (3). 8.5.1.2 COMPLIANCE WITH LOCAL REGULATIONS (b) If an emergency situation which endangers the safety of the aircraft or persons necessitates the taking of action which involves a violation of local regulations or procedures, the PIC shall - (1) Notify the appropriate local authority without delay; (2) Submit a report on any such violation to the appropriate authority of such State, if required by the State in which the incident occurs; and (3) Submit a copy of this report to the State of Registry.

Chapter 2

2.2.1 PCAR Part 8, 8.8.1.7 INSTRUMENT APPROACH OPERATING MINIMA (a) No person may operate to or from an airport/heliport using operating minima lower than those which may be established for that airport/heliport by the State in which it is located, unless that State specifically approves that operation, in accordance with the provisions of Implementing Standard: IS: 8.8.1.7.

2.4 PCAR Part 8, 8.6.2.1(a) (1) to (6). 8.6.2.1 AIRCRAFT AIRWORTHINESS AND SAFETY PRECAUTIONS (a) A flight, or series of flights, shall not be commenced until flight preparation forms have been completed certifying that the PIC is satisfied that: (1) the aircraft is airworthy, duly registered and that appropriate certificates with respect thereto are aboard the aircraft; (2) the instruments and equipment prescribed in Part 7 for the particular type of operation to be undertaken, are appropriate, taking in account the expected flight conditions, installed and sufficient for the flight; (3) any necessary maintenance has been performed and a maintenance release, if applicable, has been issued in respect to the aircraft; (4) the mass of the aircraft and center of gravity location are such that the flight can be conducted safely, taking into account the flight conditions expected; (5) any load carried is properly distributed and safely secured; (6) a check has been completed indicating that the operating limitations, contained in the flight manual or its equivalent, can be complied with for the flight to be undertaken.

2.6.1 PCAR Part 8, 8.6.2.4 WEATHER LIMITATIONS FOR VFR FLIGHTS (a) No person will commence a flight to be conducted in accordance with the VFR unless available current meteorological reports, or a combination of current reports and forecasts, indicate that the meteorological conditions along the route, or that part of the route to be flown under the VFR, will, at the appropriate time, allow VFR operations.

2.6.2.1 PCAR Part 8, 8.6.2.5 IFR DESTINATION AIRPORT/HELIPORT (a) When a destination alternate airport/heliport is required, a flight to be conducted in accordance with IFR shall not be commenced unless the information is available which indicates that will, at the estimated time of arrival, be at or above the airport/heliport operating minima.

2.6.3.3 PCAR Part 8, 8.8.4.16 LANDING DURING INSTRUMENT METEOROLOGICAL CONDITIONS (a) No pilot operating a civil aircraft may land that aircraft when the flight

- p visibility is less than the visibility prescribed in the standard instrument approach procedure being used.
- 2.7.1 PCAR Part 8, 8.6.2.6.1(a) (b). 8.6.2.6.1 IFR DESTINATION ALTERNATE AIRPORT/ HELIPORT For a flight to be conducted in accordance with the IFR, at least one destination alternate airport/heliport shall be selected and specified in the operational and ATS flight plans, unless: (a) the duration of the flight and the meteorological conditions prevailing are such that there is reasonable certainty that, at the estimated time of arrival at the airport/heliport of intended landing, and for a reasonable period before and after such time, the approach and landing may be made under VMC, (for helicopters: the weather conditions in Subpart 8.6.2.6.2 prevails); or (b) the airport/ heliport of intended landing is isolated and there is no suitable destination alternate airport/heliport; in case of helicopter operations, a point of no return (PNR) shall be determined.
- 2.8.1 PCAR Part 8, 8.6.2.13.1 FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (a) A flight shall not be commenced unless, taking into account both the meteorological conditions and any delays that are expected in flight, the aircraft carries sufficient fuel and oil to ensure that it can safely complete the flight, and, applicable special provisions are complied with, as stated in Subpart 8.6.2.14 or 8.6.2.15. In addition, a reserve shall be carried to provide for contingencies.
- 2.8.2 PCAR Part 8, 8.6.2.14(b) (1) to (3). 8.6.2.14 MINIMUM FUEL AND OIL SUPPLY FOR VFR FLIGHTS (b) For helicopters: The fuel and oil carried in order to comply with Subpart 8.6.2.13.1 shall, in the case of VFR operations, be at least the amount sufficient (considering the wind and forecast weather conditions) to allow the helicopter: (1) To fly to the heliport to which the flight is planned; (2) To fly thereafter for a period of 20 minutes at best-range speed; and (3) To have an additional amount of fuel, sufficient to provide for the increased consumption on the occurrence of any of the potential contingencies specified by the operator to the satisfaction of the Authority. For international flights, for at least an additional 10% of the total flight time calculated.
- 2.8.3.3 PCAR Part 8, 8.6.2.15.4 HELICOPTERS: IFR FLIGHTS (c) When no suitable alternate is available, in terms of Subpart 8.6.2.6 (a)(e.g. the destination is isolated), sufficient fuel shall be carried to enable the helicopter to fly to the destination to which the flight is planned and thereafter for a period that will, based on geographic and environmental considerations, enable a safe landing to be made.
- 2.8.4 PCAR Part 8, 8.6.2.13.1(b) (1) to (7). 8.6.2.13.1 FUEL AND OIL PLANNING AND CONTINGENCY FACTORS (b) In computing the fuel and oil required in paragraph (a) at least the following shall be considered: (1) meteorological conditions forecast; (2) expected ATC routings; (3) anticipated traffic delays; (4) for IFR flights, one instalment approach at the destination, including a missed approach; (5) procedures prescribed in the operations manual for loss of pressurization enroute, where applicable; (6) failure of one power-unit en route; and (7) Any other conditions that may delay landing of the aircraft or increase fuel and/or oil consumption.
- 2.9.1 PCAR Part 8.6.2.15.6 IN-FLIGHT FUEL MANAGEMENT (b) The pilot-in-command shall continually ensure that the amount of usable fuel remaining on board is not less than the fuel required to proceed to an aerodrome where a safe landing can be made with the planned final reserve fuel remaining upon landing.
- 2.10.2 PCAR Part 8, 8.6.2.13.2 OXYGEN PLANNING AND CONTINGENCY FACTORS (b) A flight to be operated with a pressurized airplane shall not be commenced unless a sufficient quantity of stored breathing oxygen is carried to supply all the crew members and passengers, as is appropriate to the circumstances of the flight being undertaken, in the event of loss of pressurization, for any period that the atmospheric pressure in

any compartment occupied by them would be less than 700 hPa. In addition, when an airplane is operated at flight altitudes at which the atmospheric pressure is less than 376 hPa, or which, if operated at flight altitudes at which the atmospheric pressure is more than 376 hPa and cannot descend safely within four minutes to a flight altitude at which the atmospheric pressure is equal to 620 hPa, there shall be no less than a 10-minute supply for the occupants of the passenger compartment.

- 2.16.1 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (a) Each required flight crew member shall remain at the assigned duty station during take-off and landing and critical phases of flight.
- 2.16.2 PCAR Part 8, 8.5.1.7 FLIGHT CREW MEMBERS AT DUTY STATIONS (b) Each flight crew member shall remain at his or her station during all phases of flight unless: (1) Absence is necessary for the performance of his or her duties in connection with the operation; (2) Absence is necessary for physiological needs, provided one qualified pilot remains at the controls at all times
- 2.16.3 PCAR Part 8, 8.5.1.6(a). 8.5.1.6 FLIGHT CREW MEMBER USE OF SEAT BELTS AND SAFETY HARNESES (a) Each flight crew member shall have his or her seat belts fastened during take-off and landing and all other times when seated at his or her station.
- 2.17.1 PCAR Part 8, 8.8.4.1 APPLICABILITY (c) Helicopters: One or more instrument approach procedures to serve each final approach and take-off area or heliport utilized for instrument flight operations shall be approved and promulgated by the State in which the heliport is located, or by the State which is responsible for the heliport when located outside the territory of any State.
- 2.17.2 PCAR Part 8, 8.8.4.1 APPLICABILITY (a) All aircraft operated in accordance with instrument flight procedures shall comply with the IFR, and the airport/heliport instrument approach procedures approved by the State in which the operation will take place.
- 2.18 PCAR Part 8, 8.8.1.1 OPERATION OF AIRCRAFT ON THE GROUND (b) For helicopters: No person shall cause a helicopter rotor to be turned under power for the purpose of flight, unless there is a qualified pilot at the controls. The operator shall provide appropriately specific training and procedures to be followed for all personnel, other than qualified pilots, who are likely to carry out the turning of a rotor under power for purposes other than flight.

Chapter 3

- 3.1 PCAR Part 8, 8.7.1.2(a)(2) (3)(4). 8.7.1.2 GENERAL (a) No person may operate an aircraft that: (2) Exceeds the operating limitations contained in the aircraft flight manual, or its equivalent; or (3) Exceeds the terms of its certificate of airworthiness. (4) Exceeds the mass limitations, if applicable, imposed by the terms of its noise certification standards, as contained in the applicable part of ICAO Annex 16, Volume I, unless otherwise approved by the Authority.
- 3.2 PCAR Part 8, 8.2.1.6 CIVIL AIRCRAFT FLIGHT MANUAL, MARKING AND PLACARD REQUIREMENTS (d) Each operator shall display in the aircraft all placards, listings, instrument markings or combination thereof, containing those operating limitations prescribed by the certifying authority for the aircraft's State of Registry for visual presentation.
- 3.3 PCAR Part 8, 8.7.2.3.2 AIRCRAFT PERFORMANCE: HELICOPTERS (d) Where helicopters are operated to or from heliports in a congested hostile environment, the competent authority of the State in which the heliport is situated shall specify the

requirements to enable these operations to be conducted in a manner that gives appropriate consideration for the risk associated with a power-unit failure.

Chapter 4

4.1.1 PCAR Part 7, 7.1.4(a)(b). 7.1.4 GENERAL INSTRUMENT AND EQUIPMENT REQUIREMENTS (a) In addition to the minimum equipment necessary for the issuance of a certificate of airworthiness, the instruments, equipment and flight documents prescribed in Part 7 shall be installed or carried, as appropriate, in aircraft according to the aircraft used and to the circumstances under which the flight is to be conducted. (b) All required instruments and equipment shall be approved and installed in accordance with applicable airworthiness requirements.

4.1.2 PCAR Part 7, 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (1) Control the flight path of the aircraft; (2) Carry out any required procedural manoeuvres; (3) Observe the operating limitations of the aircraft in the expected operating conditions; and (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.

4.1.3.1 PCAR Part 7, 7.8 Part 7, 7.8.11 Part 7, 7.8.6(a) (3)(4) Part 7, 7.9.1(a) (1) Part 8, 8.2.1.8(b) (10)(14) Part 8, 8.8.1.28 Part 8, IS: 8.8.1.28 Part 7, 7.9.8. 7.8 EMERGENCY, RESCUE, AND SURVIVAL EQUIPMENT 7.8.1 EMERGENCY EQUIPMENT: ALL AIRCRAFT (a) Each item of emergency and flotation equipment shall be - (1) Readily accessible to the crew and, with regard to equipment located in the passenger compartment, to passengers without appreciable time for preparatory procedures; (2) Clearly identified and clearly marked to indicate its method of operation; (3) Marked as to date of last inspection; and (4) Marked as to contents when earned in a compartment or container.

7.8.11 FIRST-AID AND EMERGENCY MEDICAL KIT (a) No person may operate and aircraft unless it is equipped with first-aid kits readily accessible to cabin crew located near an exit and, on passenger flights, an approved emergency medical kit for treatment of injuries or medical emergencies that might occur during flight time or in minor accidents.

7.8.6 PORTABLE FIRE EXTINGUISHERS (a) (3) At least one portable fire extinguisher shall be conveniently located on the flight deck for use by the flight crew. (4) At least one portable fire extinguisher shall be conveniently located in the passenger compartment of airplanes having a passenger seating capacity of 30 or less.

7.9.1 SEATS, SAFETY BELTS, AND SHOULDER HARNESSSES (a) (1) A seat with safety belt for each person on board over the age of two (2) years old; and a restraining belt for each berth on board the aircraft.

8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (b) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in general aviation operations unless it has within it the following current and approved documents: (10) AFM (for airplanes) or RFM (for helicopters). (14) Maps and charts for routes of proposed flight or possibly diverted flights.

8.8.1.28 INTERCEPTION (a) When intercepted by a military or government aircraft, each PIC shall comply with the international standards when interpreting and responding to visual signals and communication as specified in IS: 8.8.1.28. (b) No pilot may conduct an international flight unless the procedures and signals relating to interception of aircraft, as specified in IS: 8.8.1.28, are readily available on the flight deck.

IS: 8.8.1.28 INTERCEPTION OF CIVIL AIRCRAFT (a) All concerned in Republic of the Philippines shall observe the following principles regarding the interception of civil aircraft. (1) Interception of civil aircraft will be undertaken only as a last resort. (2) If undertaken, an interception will be limited to determining the identity of the aircraft, unless it is necessary to return the aircraft to its planned track, direct it beyond the boundaries of national airspace, guide it away from a prohibited, restricted or danger area or instruct it to effect a landing at a designated aerodrome. (3) Practice interception of civil aircraft will not be undertaken. (4) Navigational guidance and related information will be given to an intercepted aircraft by (b) radiotelephony, whenever radio contact can be established. (1) In the case where an Intercepted civil aircraft is required to land in the territory over-flown, the aerodrome designated for the landing is to be suitable for the safe landing of the aircraft type concerned. Note: In the unanimous adoption by me 25th Session (Extraordinary) of the ICAO Assembly on 10 May 1984 of Article 83-Bis to the Convention on International Civil Aviation, the Contracting States have recognized that "every State must refrain from resorting to the use of weapons against civil aircraft in flight". (c) All concerned in Republic of the Philippines shall ensure that: (1) A standard method has been established and made available to the public for the maneuvering of aircraft intercepting a civil aircraft that is designed to avoid any hazard for the intercepted aircraft. (2) Provision is made for the use of secondary surveillance radar, where available, to identify civil aircraft in areas where they may be subject to interception. (d) The PIC of an aircraft that is intercepted by another aircraft shall immediately: (1) Follow the instructions given by the intercepting aircraft. Interpreting and responding to visual signals in accordance with the specifications in item (e) below. (2) Notify, if possible, the appropriate air traffic services unit. (3) Attempt to establish radio communication with the intercepting aircraft or with the appropriate intercept control unit. By making a general call on the emergency frequency 121.5 MHZ, giving the identity of the intercepted aircraft and the nature of the flight; and if no contact has been established and if practicable, repeating this call on the emergency frequency 243 MHZ. (4) If equipped with SSR transponder, select Mode A Code 7700, unless otherwise instructed by the appropriate air traffic services unit. (e) if any instructions received by radio from any sources conflict with those given by the intercepting aircraft by visual signals, the PIC of the intercepted aircraft shall request immediate clarification while continuing to comply with the visual instructions given by the intercepting aircraft. (f) If any instructions received by radio from any sources conflict with those given by the intercepting aircraft by radio, the PIC of the intercepted aircraft shall request immediate clarification while continuing to comply with the radio instructions given by the intercepting aircraft. (g) Radio communication during interception. If radio contact is established during interception but communication in a common language is not possible, the PIC of each involved aircraft shall attempt to convey instructions, acknowledgement of instructions and essential information by using the phrases and pronunciations in the table below and transmitting each phrase twice: (h) Radio communication during interception (1) If radio contact is established during interception but communication in a common language is not possible, attempts shall be made to convey instructions, acknowledgement of instructions and essential information by using the phrases and pronunciations in the table below and transmitting each phrase twice:

7.9.8 PROTECTIVE CIRCUIT FUSES No person may operate an aircraft in which protective fuses are installed unless there are spare fuses available for use in flight equal to at least 10% of the number of fuses of each rating or three of each rating whichever is the greater.

- 4.2.2 PCAR Part 7,7.2.2 (a)(3) (4),(b)(1)(2), Part 8.8.15(c) and Part 7,7.4.1(a)(1)(5)(6)(7).
7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall

have separate flight instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope). (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (3) A vertical speed indicator; (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator.

8.8.1.15 USE OF AIRCRAFT LIGHTS (c) No person may park or move an aircraft between the period from sunset to sunrise in, or in a dangerous proximity to, a movement area of an aerodrome, unless the aircraft: (1) Is clearly illuminated; (2) Has lighted navigation lights; or (3) Is in an area that is marked by obstruction lights; or (4) Has lights to indicate when the engine is running.

7.4.1 AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (a) All aircraft operated at night shall be equipped with: (1) A landing light; (5) Illumination for all flight instruments and equipment that are essential for the safe operation of the aircraft; (6) Lights in all passenger compartments; (7) A flashlight for each crew member station (approval not required).

4.2.3 PCAR Part 7, 7.2.3(a)(1)(2)(3) and 7.2.2(a)(2) to (7)(b)(1)(2). 7.2.3 IFR INSTRUMENTS (a) All aircraft when operated in IFR, or at night, or when the aircraft cannot be maintained in a desired attitude without reference to one or more flight instruments, shall be equipped with the flight instruments as listed in 7.2.2 and additionally as follows - (1) An airspeed indicating system with a means of preventing malfunctioning due to either condensation or icing; (2) A sensitive pressure altimeter with counter drum-pointer or equivalent presentation, calibrated in feet with a sub-scale setting calibrated in hectopascals/millibars, adjustable for any barometric pressure likely to be set during flight; (3) A means of indicating whether the supply of power to the gyroscopic instruments is adequate.

7.2.2 MINIMUM FLIGHT AND NAVIGATIONAL INSTRUMENTS (a) No person may operate any aircraft unless it is equipped with the following flight and navigational instruments: (2) A sensitive pressure altimeter, adjustable for any barometric pressure likely to be set during flight; (3) A vertical speed indicator (4) A turn and slip indicator, or a turn coordinator incorporating a slip indicator; (5) A magnetic compass; (6) A means of indicating in the flight crew compartment the outside air temperature; (7) An accurate timepiece indicating the time in hours, minutes and seconds (approval not required). (b) Whenever an aircraft is operating at night or when two pilots are required, each pilot's station shall have separate flight Instruments as listed in (a) and additionally as follows; (1) An attitude indicator (artificial horizon); and (2) A stabilized direction indicator (directional gyroscope).

4.3.1 PCAR Part 7, 7.8.18 FLOTATION DEVICE FOR HELICOPTER DITCHING (a) All helicopters flying over water at a distance from land corresponding to more than 10 minutes at normal cruise speed in the case of performance Class 1 or 2 helicopters, or flying over water beyond auto-rotational or safe forced landing distance from land in the case of performance Class 3 helicopters, shall be fitted with a permanent or rapidly deployable means of floatation so as to ensure a safe ditching of the helicopter.

4.3.2.1 PCAR Part 7, 7.8.16(a)(2) and 7.8.17(a)(1)(2)(3), (c)(d). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be provided for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.

7.8.17 LIFE RAFT (a) In addition to the equipment prescribed in 7.8.16 and 7.8.18 of this Part, lifesaving rafts in sufficient numbers to carry all persons on board shall be installed in: (1) All airplanes when operating in "Extended over-water operations" as

defined in Subpart 8.1.1.2(b). (2) Class 1 and 2 helicopters when they are operated over water at a distance from land corresponding to more than 10 minutes at normal cruise speed. (3) Class 3 helicopters when they are operated over water beyond auto-rotational or safe forced landing distance from land. (c) All lifesaving rafts shall be stowed so as to facilitate their ready use in an emergency. (d) Life rafts shall be equipped with the following life sustaining equipment - (1) A electric survivor locator light; (2) A survival kit equipped in accordance with the requirements of subsection 7.8.4; (3) A pyrotechnical signalling device.

4.3.2.2 PCAR Part 7,7.8.16(a)(1) (iii). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (1) Shall carry the equipment prescribed in paragraph (2): (iii) When taking off or landing at an aerodrome where the (Authority) has determined the take-off or approach path is so disposed over water that in the event of a mishap there would be the likelihood of a ditching.

4.3.2.3 PCAR Part 7,7.8.16(a)(2). 7.8.16 INDIVIDUAL FLOTATION DEVICES (a) Landplanes including helicopters. (2) One life-jacket or equivalent flotation device equipped with a means of electric illumination shall be earned for each person on board, stowed in a position easily accessible from the seat or berth of the person for whose use it is provided.

4.4 PCAR Part 7,7.8.3 and Part 7,7.8.4. 7.8.3 VISUAL SIGNALLING DEVICES (a) No person may operate an aircraft over water or across land areas which have been designated by the Republic of the Philippines as areas in which search and rescue Would be especially difficult, unless equipped with such signaling devices a s may be appropriate to the area overflown, to include - (1) Visual signals for use by intercepting and intercepted aircraft; and (2) At least one pyrotechnic signaling device for each life raft required for overwater operations.

7.8.4 SURVIVAL EQUIPMENT (a) Subject to subsection (b) no person shall operate an aircraft over land unless there is carried on board survival equipment, sufficient for the survival on the ground of each person on board, given the geographical area, the season of the year and anticipated seasonal climatic variations, that provides the means for (1) starting a fire; (2) providing shelter; (3) providing or purifying water; and (4) visually signaling distress. (b) Subsection (a) does not apply in respect of (1) a balloon, a glider, a hang glider, a gyroplane or an ultra-light airplane; (2) an aircraft that is operated-within 25 nautical miles of the aerodrome of departure and that has the capability of radio communication with a surface-based radio station for the duration of the flight. (3) a multi-engined aircraft that is operated south of 66° 30' north latitude or north of 66° 30'south latitude (i) in IFR flight within controlled airspace, or (ii) along designated air routes; (4) an aircraft that is operated by an air operator, where the aircraft is equipped with equipment specified in the air operator's company operations manual, but not with the equipment required by subsection (a); or (5) an aircraft that is operated in a geographical area where and at a time of year when the survival of the persons on board is not jeopardised.

4.5.1 PCAR Part 7, 7.8.12 OXYGEN STORAGE AND DISPENSING APPARATUS (c) No person may operate an aircraft at altitudes above 10,000 feet unless it is equipped with oxygen masks, located so as to be within the immediate reach of flight crew members while at their assigned duty station.

4.6 PCAR Part 8, 8.2.1.8 DOCUMENTS TO BE CARRIED ON AIRCRAFT: ALL OPERATIONS (a) Except as provided in Subpart 8.2.1.6, no person may operate a civil aircraft in commercial air transport operations unless it has within it the following current and approved documents: (9) Noise Certificate, or a document attesting noise certification. When the document, or a suitable statement attesting noise certification

- as contained in another document approved by the State of Registry, is issued in a language other than English, it shall include an English translation.
- 4.7.1.1.1 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (m) No person shall operate a helicopter of a maximum certificated take-off mass of over 3175 KG, for which the individual certificate of airworthiness is first issued after 1 January 2005, unless it is equipped with a Type IVA FDR. Implementing Standard: See IS: 7.7.1 for specific data to be recorded by night recorders.
 - 4.7.1.1.2 PCAR Part 7, 7.7.1 FLIGHT RECORDERS (l) No person shall operate a helicopter of a maximum certificated take-off mass of over 7000 KG, for which the individual certificate of airworthiness is first issued after 1 January 1989, unless it is equipped with a Type IV FDR.
 - 4.7.1.2 PCAR Part 7, 7.7.1(b)(1). 7.7.1 FLIGHT RECORDERS (b) Flight data recorder media not acceptable for use in aircraft registered in the Republic of the Philippines, (1) Engraving metal foil.
 - 4.7.1.3 PCAR Part 7, 7.7.1(a)(6). 7.7.1 FLIGHT RECORDERS (a) [AAC] Flight recorders shall (6) Be capable of retaining the information recorded during the last - (i) Type I and II - 25 hours of operation. (ii) Type IIA - 30 minutes of operation. (iii) Type IV and V - 10 hours of operation.
 - 4.7.2.1.1 PCAR Part 7, 7.7.2 COCKPIT VOICE RECORDERS (d) For helicopters not equipped with an FDR, at least main rotor speed shall be recorded on one track of the CVR.
 - 4.7.2.2 Not implemented.
 - 4.7.2.3 PCAR Part 7, 7.7.2(c)(2). 7.7.2 COCKPIT VOICE RECORDERS (c) A CVR shall be capable of retaining the information recorded during at least the last (2) 2 hours, for airplanes of a maximum certificated take-off mass of over 5700 KG or a helicopter that receives type certification after 1 January 2003.
 - 4.7.3.1.1 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio. (b) Sufficient information to derive the content of the data link communications message and, whenever practical, the time the message was displayed to or generated by the crew shall be recorded. Note: Data link communications include, but are not limited to, automatic dependent surveillance - contract (ADS-C), controller-pilot data link communications (CPDLC), data link-flight information services (D-FIS) and aeronautical operational control messages.
 - 4.7.3.1.2 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio. (b) Sufficient information to derive the content of the data link communications message and, whenever practical, the time the message was displayed to or generated by the crew shall be recorded. Note: Data link communications include, but are not limited to, automatic dependent surveillance - contract (ADS-C), controller-pilot data link communications (CPDLC), data link-flight information services (D-FIS) and aeronautical operational control messages.
 - 4.7.3.2 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on

- a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
- 4.7.3.3 PCAR Part 7, 7.7.3 RECORDING OF DATA LINK COMMUNICATIONS (a) All aircraft which utilize data link communications and are required to carry a CVR shall record on a flight recorder, all data link communications to and from the aircraft, where the aircraft is capable of being modified. The minimum recording duration shall be equal to the duration of the CVR, and shall be correlated to the recorded cockpit audio.
- 4.7.4.2.2 PCAR Part 7,7.7.1(a)(3) and Part 7,7.7.2(b)(2)(3). 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall (3) Be deactivated upon completion of a flight following an accident or incident.
- 7.7.2 COCKPIT VOICE RECORDERS (b) To preserve the CVR recordings - (2) The CVR shall be de-activated upon completion of flight time following an accident or incident; and (3) Following an accident or incident, the CVR shall not be re-activated before their disposition as determined in accordance with ICAO Annex 13.
- 4.7.4.3 PCAR Part 7,7.7.1(a)(4). 7.7.1 FLIGHT RECORDERS (a) Flight recorders shall (4) The FDR system shall be recalibrated at least every five years to determine any discrepancies in the engineering conversion routines for the mandatory parameters, in accordance with the requirements of the aircraft manufacturer to ensure that the parameters are being recorded within the calibration tolerances; and when the parameters of altitude and airspeed are provided by sensors that are dedicated to the FDR system, there shall be a recalibration performed as recommended by the sensor manufacturer, or at least every two years.
- 4.8.1 PCAR Part 7, IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.
- 4.8.2 PCAR Part 7, IS: 7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT.
- 4.8.3 PCAR Part 7, IS:7.8.5 EMERGENCY LOCATOR TRANSMITTER (ELT) The following requirements shall be used to determine the quantity and type of 406 MHZ ELT which must be carried. (a) Except as provided for in (b), all airplanes authorized to carry more than 19 passengers shall be equipped with at least one automatic ELT or two ELTs of any type. (b) All airplanes authorized to carry more than 19 passengers for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least two ELTs, one of which shall be automatic. (c) Except as provided for in (d), all airplanes authorized to carry 19 passengers or less shall be equipped with at least one ELT of any type. (d) All airplanes authorized to carry 19 passengers or less for which the individual certificate of airworthiness is first issued after 1 July 2008 shall be equipped with at least one automatic ELT. (e) Except as provided for in (f) all helicopters operating in performance Class 1, 2 and 3 shall be equipped with at least one automatic ELT or, (f) when engaged in offshore operations, or other overwater operations, with at least one automatic ELT and one ELT(S) in a raft or life jacket.
- 4.9.1 PCAR Part 7, 7.3.3 ALTITUDE REPORTING TRANSPONDER (a) No person may operate an aircraft in airspace that requires a pressure reporting transponder unless that equipment is operative and operates in accordance with the provisions of ICAO Annex 10, Volume IV.
- 4.11 Not implemented.

Chapter 5

- 5.1.1 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 5.1.2 PCAR Part 7, 7.3.1(c)(1). 7.3.1 RADIO EQUIPMENT (c) No person may operate an aircraft in IFR, or in VFR over routes that cannot be navigated by reference to visual landmarks, unless the aircraft is equipped with communication and navigation equipment in accordance with the requirements of air traffic services in the area(s) of operation, but not less than - (1) Two independent radio communication systems necessary under normal operating conditions to communicate with an appropriate ground station from any point on the route including diversions. Note: Each system shall have an independent antenna installation except that, where rigidly supported non-wire antennae or other antenna installations of equivalent reliability are used, only one antenna is required.
- 5.1.3 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (b) All aircraft operated in VFR as a controlled flight or in IFR shall be provided with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on those frequencies prescribed by the Authority, including the aeronautic emergency frequency 121.5 MHZ. Note: This requirement is considered fulfilled if the ability to conduct the communications specified therein is established during radio propagation conditions which are normal for the route.
- 5.1.4 PCAR Part 7, 7.3.1 RADIO EQUIPMENT (c) No person may operate an aircraft in IFR, or in VFR over routes that cannot be navigated by reference to visual landmarks, unless the aircraft is equipped with communication and navigation equipment in accordance with the requirements of air traffic services in the area(s) of operation, but not less than - (1) Two independent radio communication systems necessary under normal operating conditions to communicate with an appropriate ground station from any point on the route including diversions. Note: Each system shall have an independent antenna installation except that, where rigidly supported non-wire antennae or other antenna installations of equivalent reliability are used, only one antenna is required. (2) Secondary Surveillance Radar transponder equipment as required for the route being flown.
- 5.1.6 PCAR Part 7, 7.3.1(i)(1)(2). 7.3.1 RADIO EQUIPMENT (i) For flights in defined portions of airspace or on routes where a Required Communication Performance (RCP) type has been prescribed, an aircraft shall: (1) have installed communication equipment which will enable it to operate in accordance with the prescribed RCP types; and (2) be authorized by the State of the Operator for operations in such airspace.
- 5.2.1 PCAR Part 7, 7.2.9(a)(1)(3). 7.2.9 NAVIGATION EQUIPMENT - GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (1) Its operational flight plan, (3) The requirements of air traffic services.
- 5.2.2 PCAR Part 7, 7.2.9(a) (2) and (b). 7.2.9 NAVIGATION EQUIPMENT - GENERAL (a) No person may operate an aircraft unless it is equipped with navigation equipment that will enable it to proceed in accordance with - (2) The prescribed Performance-Based Navigation Specification, and (b) No person may operate an aircraft in Performance-

Based Navigation specification airspace unless authorized by CAAP for such operations.

- 5.2.6 PCAR Part 7,7.2.1(a)(4). 7.2.1 GENERAL REQUIREMENTS (a) All aircraft shall be equipped with flight and navigational instruments which will enable the flight crew to - (4) In the event of the failure of one item of equipment at any stage of the flight operate the aircraft with the remaining navigation equipment in accordance with its operational flight plan and the requirements of ATS.

- 5.2.7 PCAR Part 7.2.8 NAVIGATION EQUIPMENT FOR OPERATIONS IN RVSM AIRSPACE (g) On flights in which it is intended to land in instrument meteorological conditions, an airplane shall be provided with radio equipment capable of receiving signals providing guidance to a point from which a visual landing can be effected. This equipment shall be capable of providing such guidance for each aerodrome at which it is intended to land in instrument meteorological conditions and for any designated alternate aerodromes.

Chapter 6

- 6.1.1 PCAR Part 5,5.5.1.2(a) and Part 8,8.3.1.6. 5.5.1.2 RESPONSIBILITY (a) The owner of an aircraft or, in the case of a leased aircraft, the lessee, shall be responsible for maintaining the aircraft in an airworthy condition by ensuring that - (1) All maintenance, overhaul, modifications and repairs which affect airworthiness are performed as prescribed by the State of Registry; (2) Authorized maintenance personnel make appropriate entries in the aircraft maintenance records certifying that the aircraft is airworthy; (3) A maintenance release is completed to certify that the maintenance work performed has been completed satisfactorily and in accordance with the prescribed methods; and (4) In the event there are open discrepancies, the maintenance release includes a list of the uncorrected maintenance items and these items are made a part of the aircraft permanent record.

8.3.1.6 CONTINUOUS AIRWORTHINESS MAINTENANCE INSPECTION (a) The registered owner or operator of each large airplane certificated with a maximum take-off mass of over 5700 KG, turbine-powered multi-engine airplane, and turbine-powered rotorcraft shall select, identify in the aircraft maintenance records, and use one of the following continuous airworthiness maintenance inspection programs for the inspection of the aircraft: (1) A current inspection program recommended by the manufacturer; (2) A continuous airworthiness maintenance program for that make and model of aircraft currently approved by the Authority for use by an AOC holder, or (3) Any other inspection program established by the registered owner or operator of that aircraft and approved by the Authority. (b) Each owner/operator shall include in the selected inspection program the name and address of the person responsible for the scheduling of the inspections required by the program and provide a copy of the program to the person performing inspection on the aircraft. Note: The aircraft manufacturer's inspection program and any other inspection program approved by the Authority will specify who can perform aircraft maintenance, inspections and return of the aircraft to service.

- 6.2.1 PCAR Part 8,8.3.1.9(a)(1)(2) (i) to (v). 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (a) Except for records maintained by an operator, each registered owner or operator shall retain the following records for a minimum period of 90 days, or until the work is repeated or superseded by other work of equivalent scope and detail, or the unit to which they refer has been permanently withdrawn from service, and the records in paragraph (1) below for a minimum period of one year after the signing of the maintenance release: (1) Records of the maintenance, preventive maintenance, minor modifications, and records of the 100-hour; annual, and other required or approved inspections, as appropriate, for each aircraft (including the

airframe) and each engine, propeller, rotor, and appliance of an aircraft to include - (i) A description (or reference to data acceptable to the Authority) of the work performed, (ii) The date of completion of the work performed; and (iii) The signature and certificate number of the person signing the Maintenance Release; and (iv) As necessary the identification of the person at the AMO approving the aircraft for return to service. (2) Records containing the following information - (i) The total time-in-service of the airframe, each engine, each propeller, and each rotor. (ii) The current status of all life-limited aeronautical products; (iii) The time since last overhaul of all items installed on the aircraft which are required to be overhauled on a specified time basis; (iv) The current inspection status of the aircraft, including the time since the last inspection required by the inspection program under which the aircraft and its appliances are maintained. (v) The current status of applicable Airworthiness Directives including, for each the method of compliance, the Airworthiness Directive number, and revision date. If the Airworthiness Directive involves recurring action, the time and date when the next action is required.

6.2.2 PCAR Part 8, 8.3.1.9 MAINTENANCE AND INSPECTION RECORDS RETENTION (a) Except for records maintained by an operator, each registered owner or operator shall retain the following records for a minimum period of 90 days, or until the work is repeated or superseded by other work of equivalent scope and detail, or the unit to which they refer has been permanently withdrawn from service, and the records in paragraph (1) below for a minimum period of one year after the signing of the maintenance release.

6.3 PCAR Part 8, 8.3.1.1 APPLICABILITY (f) The owner of an airplane over 5700 KG maximum certificated take-off mass, or in the case where it is leased, the lessee, shall, as prescribed by the State of Registry, ensure that the information resulting from maintenance and operational experience with respect to continuing airworthiness, is transmitted as required by Subpart 8.3.16.

6.4 PCAR Part 9, 9.4.1.10 MODIFICATION AND REPAIRS (a) All modifications and repairs shall comply with airworthiness requirements of this CAR and acceptable to the authority and the State of Registry. Procedures shall be established to ensure that the substantiating data supporting compliance with the airworthiness requirements are retained. However, in the case of a major repair or major modification, the work must have been done in accordance with technical data approved by the Authority.

Chapter 7

7.1 PCAR Part 8, 8.4.1.2 FLIGHT CREW QUALIFICATIONS (a) The PIC shall ensure that the licenses of each flight crew member have been issued or rendered valid by the State of Registry, contain the proper ratings, and that all flight crew members have maintained recent experience and competence.

7.2 PCAR Part 8, 8.4.1.1 COMPOSITION OF THE FLIGHT CREW (a) The number and composition of the flight crew shall not be less than that specified in the flight manual (for helicopters: operations manual). The flight crew members shall include flight crew members in addition to the minimum numbers specified in the flight manual or other documents associated with the certificate of airworthiness, when necessitated by considerations related to the type of aircraft used, the type of operation involved and the duration of flight between points where flight crew members are changed.

7. Annex 7 - Aircraft Nationality and Registration Marks, 6th Edition, July 2012, Amendment 7.

Chapter 3

3.1 PCAR Part 4, 4.3.1.3 NATIONALITY AND REGISTRATION MARKS TO BE USED (a) Every aircraft duly registered by the Authority shall display the registration mark

assigned to such aircraft. The nationality and registration marks appearing on the aircraft shall consist of a group of characters.

- 3.3 PCAR Part 4, 4.3.1.3 NATIONALITY AND REGISTRATION MARKS TO BE USED (c)
The nationality mark for Philippine registered aircraft is RP.
- 3.5 PCAR Part 4, 4.3.1.3 NATIONALITY AND REGISTRATION MARKS TO B E USED (d)
The registration mark shall be a combination of letters and numbers assigned by the Authority.
- 3.6 PCAR Part 4, 4.3.1.2 NATIONALITY AND REGISTRATION MARKS TO B E USED (b)
Unless otherwise authorized by the Authority, no person may place on any aircraft a design, mark, or symbol that modifies or confuses the nationality and registration marks. The marks shall not be confused with the International Five Letter Code of Signals or Distress Codes.
- 4.1 PCAR Part 4, 4.3.1.2 GENERAL (c) Permanent marking of aircraft nationality and registration shall be painted on the aircraft or affixed by other means insuring a similar degree of permanence, have no ornamentation: (2) Be legible.
- 4.3.2 PCAR Part 4, 4.3.1.7 (b)(1) (2). 4.3.1.7 LOCATION OF MARKS ON FIXED-WING AIRCRAFT (b) The marks required by paragraph (a) of this section shall be displayed as follows: (1) If displayed on the vertical tail surfaces, horizontally on both surfaces, horizontally on both surfaces of a single vertical tail or on the outer surfaces of a multi-vertical tail. (2) If displayed on the fuselage surfaces, horizontally on both sides of the fuselage between the trailing edge of the wing and the leading edge of the horizontal stabilizer.
- 4.3.3 PCAR Part 4, 4.3.1.7(b)(3). 4.3.1.7 LOCATION OF MARKS ON FIXED-WING AIRCRAFT (b) (3) If engine pods or other appurtenances are located in the area described in paragraph (b) (2) and are an integral part of the aircraft, the operator may place the marks on those pods or appurtenances.

Chapter 5

- 5 PCAR Part 4, 4.3.1.5 SIZE OF MARKS (b) Height. The character marks shall be of equal height and on.
- 5.1.2 PCAR Part 4, 4.3.1.5 SIZE OF MARKS (b) (3) Unmanned free balloons, shall be determined by the Authority taking into consideration the size of the surface to which the identification plate is affixed.
- 5.1.3 PCAR Part 4, 4.3.1.6 DEVIATIONS FOR SIZE AND LOCATION OF MARKS (a) If either one of the surfaces authorized for displaying required marks is large enough for display of marks meeting the size requirements of this section and the other is not, the operator shall place full-size marks on the larger surface.(b) If neither surface is large enough for full-size marks, the Authority may approve marks as large as practicable for display on the larger of the two surfaces.
- 5.2.1 PCAR Part 4, 4.3.1.5 (b)(1)(i). 4.3.1.5 SIZE OF MARKS (b) Height. The character marks shall be of equal height and on - (1) Heavier-than-air aircraft, shall be at least: (i) 50 centimeters high if on the wings.
- 5.2.2 PCAR Part 4, 4.3.1.5 (b)(1)(ii). 4.3.1.5 SIZE OF MARKS (b) Height. The character marks shall be of equal height and on - (1) Heavier-than-air aircraft, shall be at least (ii) 30 centimeters high if on the fuselage (or equivalent structure) and vertical tail surfaces.
- 5.2.3 PCAR Part 4, 4.3.1.6(a)(b). 4.3.1.6 DEVIATIONS FOR SIZE AND LOCATION OF MARKS (a) If either one of the surfaces authorized for displaying required marks is large enough for display of marks meeting the size requirements of this section and the

other is not, the operator shall place full-size marks on the larger surface. (b) If neither surface is large enough for full-size marks, the Authority may approve marks as large as practicable for display on the larger of the two surfaces.

Chapter 6

- 6.1 PCAR Part 4, 4.3.1.3(e) and Part 4.3.1.4. 4.3.1.3 NATIONALITY AND REGISTRATION MARKS TO BE USED (e) The nationality and registration marks borne by aircraft shall consist of the following: (1) The Roman capital letters "RP" immediately preceding identification number for aircraft used solely for governmental purposes, and or belonging to the Philippine government, provinces, municipalities, or political subdivisions (except those belonging to the Philippine Army). (2) The Roman capital letters "RP-C" will classify aircraft, except gliders, as complying fully with the airworthiness requirements as the case may be for the purpose of carrying persons or property, for or without hire. (3) Gliders complying fully with airworthiness requirements shall display the Roman capital letters "RP-G" immediately preceding the license number or numbers. (4) The Roman capital letters "RP-R" will classify an aircraft as complying in some limited respect with the airworthiness requirements as the case may be, provided each deficiency is compensated for by means of suitable operation limitations. The certificate of an aircraft in this classification will specify the use or uses, for such aircraft is deemed airworthy by such use or uses, shall exclude carrying of passengers, and shall be for industrial purposes only. (5) The Roman capital letters "RP-X" will classify an aircraft as one which has applied for an experimental certificate. (6) Non-type certificated aircraft will only be registered as "RP-S" (refer to Subpart 11.10.2(b)). (7) The airworthiness symbol will be followed by a number or numbers issued for the purposes of further identification.

4.3.1.4 DISPLAY OF MARKS: GENERAL (a) Each owner shall display on that aircraft marks consisting of the Roman capital letter denoting nationality of the Republic of the Philippines followed by the registration number of the aircraft in Arabic numerals. Each suffix letter used in the marks displayed must also be a Roman capital letter. (b) If, because of the aircraft configuration, it is not possible to mark the aircraft in accordance with this Part, the owner may apply to the Authority for a different procedure.

- 6.2 PCAR Part 4, 4.3.1.5 SIZE OF MARKS (c) Width. Characters must be two-thirds as wide as they are high, except the number "1", which must be one-sixth as wide as it is high, and the letters "M" and "W" which may be as wide as they are high.

- 6.3 PCAR Part 4, 4.3.1.2 (c)(1) and Part 4, 4.3.1.5 (d). 4.3.1.2 GENERAL (c) Permanent marking of aircraft nationality and registration shall be painted on the aircraft or affixed by other means insuring a similar degree of permanence, have no ornamentation: (1) Contrast in color with the background.

4.3.1.5 SIZE OF MARKS (d) Thickness. Characters shall be formed by solid lines one-sixth as thick as the character is high.

- 6.4 PCAR Part 4, 4.3.1.5 SIZE OF MARKS (e) Spacing. The space between each character may not be less than one-fourth of the character width.

Chapter 7

- 7 PCAR Part 4, 4.3.1.1 GENERAL (a) No person may operate a civil aircraft that is eligible for registration under the laws of the Republic of the Philippines unless it has been registered by its owner/operator under the provisions of the laws of the Republic of the Philippines and the Authority has issued a Certificate of Aircraft Registration for that aircraft which shall be carried aboard that aircraft for all operations. (b) The certificate of aircraft registration shall be in English.

Chapter 8

- 8.1 CAR 4.3.1.2 (d) Difference on the aircraft certificate of registration template: Item No. 4 of the certificate may contain name of the owner or lessee. The lowest part of the certificate contains remarks as to the type of aircraft operation (e.g. commercial, general aviation, aerial works) - Conveyances will be recorded as attachment to the certificate.
- 8.2 PCAR Part 4,4.2.1.1(a)(b). 4.2.1.1 GENERAL (a) No person may operate a civil aircraft that is eligible for registration under the laws of the Republic of the Philippines unless it has been registered by its owner/operator under the provisions of the laws of the Republic of the Philippines and the Authority has issued a Certificate of Aircraft Registration for that aircraft which shall be carried aboard that aircraft for all operations. (b) The certificate of aircraft registration shall be in English.

Chapter 9

- 9.1 PCAR Part 4,4.3.1.11(a)(1) (2)(3). 4.3.1.11 IDENTIFICATION PLATE REQUIRED (a) The operator shall affix to each aircraft registered under the laws of the Republic of the Philippines an identification plate (1) Containing the marks of nationality and registration; (2) Made of fireproof metal or other fireproof material of suitable physical properties; (3) Secured to the aircraft in a prominent position, near the main entrance, or, in the case of a free balloon, affixed conspicuously to the exterior of the payload.

8. Annex 8 - Airworthiness of Aircraft, 13th Edition, July 2022, Amendment 109.

Chapter 3

- 3.2.3 PCAR Part 5 Paragraph 5.4.1.6 (a)(1). The validity period of the Certificate of Airworthiness is extended until 31 March 2022 pursuant to the Civil Aviation Authority of the Philippines Memorandum Circular No. 04-2022 dated 17 January 2022.

9. Annex 9 - Facilitation, 16th Edition, July 2022 - Nil.

10. Annex 10 - Aeronautical Telecommunications

Volume I - Radio Navigation Aids, 8th Edition, July 2023, Amendment 92.

Chapter 2

- 2.1.4.2 The CAAP approves the GNSS-based operations and shall ensure that GNSS data relevant to those operations are recorded.

Note 1: These recorded data are primarily intended for use in accident and incident investigations. They may also support periodic confirmation that accuracy, integrity, continuity and availability are maintained within the limits required for the operations approved.

Note 2: Guidance material on the recording of GNSS parameters is contained in Attachment D, 11.

- 2.1.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.1.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.1.3.5.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 3.1.3.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.3.12.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.5.1.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.5.4.2 Only CAT I ILS in use.
- 3.1.5.5.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.5.5.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.5.8.1 Only CAT I ILS in use.
- 3.1.6.2 Odd tenths are used.
- 3.1.7.6.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.3.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.3.6.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.4.8.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.4.8.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.5.3.1.2.3 Only DME/N used.
- 3.5.3.6.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.5.3.7.2 Only DME/N used.
- 3.5.4.1.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.5.4.1.5.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.5.4.5.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.3.5.4.4.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.11.4.9.4.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Volume II - Communication Procedures including those with PANS status, 7th Edition, July 2016, Amendment 92

Chapter 2

- 2.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.3.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 3.3.6.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.4.10.1.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 4.4.10.1.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 5

- 5.2.1.6.2.2.2 The PANS was adopted to mandatory requirements in the Philippine regulations.

Chapter 8

- 8.1.1.1.2 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.1.1.2 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.1.1.3 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.1.1.4.1 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.1.1.4.2 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.1.1.4.3 The PANS was adopted to mandatory requirements in the Philippine regulations.

- 8.2.11.3 The PANS was adopted to mandatory requirements in the Philippine regulations.

Volume III - Communication Systems (Part I - Digital Data Communication Systems), 2nd Edition, July 2007, Amendment 91

Chapter 3

- 3.4.11 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 3.4.12 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 4.5.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

- 4.5.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 4.5.2.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

- 4.6.3.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

4.6.4.1.2.1.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

4.6.4.1.2.3.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

4.6.4.1.2.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

4.6.4.1.2.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

4.6.4.1.2.5.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

4.6.4.1.2.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

4.6.4.1.2.6.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

4.6.4.1.2.7.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

4.6.4.1.3.1.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

Chapter 5

5.2.2.3.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.2.2.3.2.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.2.5.1.1.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.2.5.1.4.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

5.2.10.3 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.

Chapter 6

6.4.3.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.9.2.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.9.5.1.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

6.9.5.1.5.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 7

- 7.5.3.2 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.
- 7.5.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.5.4.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 8

- 8.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.7.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.8 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.9 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.10 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.2.11 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.3.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.3.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.3.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.3.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.3.2.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 8.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.1.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.3.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.3.6.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 8.6.4.2.7.2.3.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.
- 8.6.4.3.2.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 9

- 9.1.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 11

- 11.3.1.14.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 11.3.2.4.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 12

- 12.2.1.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Volume III - Communication Systems (Part II - Voice Communication Systems), 2nd Edition, July 2007, Amendment 91

Chapter 2

- 2.2.1.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.2.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.1.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.2.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.3.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.2.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.4.1.2.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 4th quarter of 2022.
- 2.4.1.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.1.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.1.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.1.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.1.8 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Volume IV - Surveillance and Collision Avoidance Systems, 5th Edition, July 2014, Amendment 91

Chapter 2

- 2.1.2.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.1.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.1.3.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.1.4.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.1.4.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.1.5.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.2.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.2.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.1.1.7.14.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.1.5.2.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 1st quarter of 2023.
- 3.1.2.5.2.1.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.5.2.1.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.5.2.1.5.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.7.6.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.7.9.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.10.1.1.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.10.3.10.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.10.3.10.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.10.5.2.2.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.1.2.11.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.3.2.1.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.5.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.8.4.2.3.3.1 Structure of MU for a resolution message is 4.3.8.4.2.3.3.1 different. Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 1st quarter of 2023.
- 4.4.2.6.4.4.1 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 1st quarter of 2023.

4.4.2.6.4.5 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 1st quarter of 2023.

4.4.2.6.4.7 No national regulations that will address the corresponding standard (SARP) has been promulgated yet; Amendment of the national regulations that will include this standard (SARP) is planned for incorporation by the 1st quarter of 2023.

Volume V - Aeronautical Radio Frequency Spectrum Utilization, 3rd Edition, July 2013, Amendment 89

Chapter 4

4.3.3.1 CAR-ANS Part 13 has additional provisions on b) "... or 5.4.3.3.2 Groups 6 to 10. These DME channels will be permitted to be used on the basis of a regional agreement when they have become applicable in accordance with the conditions specified at CAR-ANS Part 13, 13.3.3.2."

11. Annex 11 - Air Traffic Services, 15th Edition, July 2018, Amendment 52

Chapter 2

2.7.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.3.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.3.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.5.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.11.5.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.12.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.12.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.12.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.13.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.14.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.14.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

2.16.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 2.16.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.19.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.19.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.19.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.24.1.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.33.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.3.5.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.4.2.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.7.5.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.9.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.10.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 4

- 4.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.2.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 4.3.2.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.2.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.3.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.4.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.4.8 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.3.6.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 4.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 5

- 5.2.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 6

- 6.1.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.1.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.1.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.1.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.2.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.2.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.2.3.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.3.1.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 6.2.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.4.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 7

- 7.1.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.1.3.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.1.4.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.1.4.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 7.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.

12. Annex 12 - Search and Rescue, 8th Edition, July 2004 Amendment 18

Chapter 3

- 3.3.2 Even though this standard is implemented in practice, it has not been incorporated into national regulations.
- 3.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

13. Annex 13 - Aircraft Accident and Incident Investigation, 12th Edition, July 2020 Amendment 18

Chapter 4

- 4.5 PCAR PART 13.060 OCCURRENCE OUTSIDE THE REPUBLIC OF THE PHILIPPINES (a) (1) 13.060 OCCURRENCE OUTSIDE THE REPUBLIC OF THE PHILIPPINES (a) Upon receipt of a notification of an accident or a serious incident which occurs outside Republic of the Philippines involving a Republic of the Philippines aircraft or an aircraft operated by a Republic of the Philippines operator, the Chairman shall do the following: (1) acknowledge the receipt of the notification; (2) where the State of Occurrence, the State of Registry or the State of the Operator which is investigating the accident or serious incident is a Contracting State, provide the State with the following information with the least possible delay: (i) any relevant information he has regarding the aircraft and flight crew involved in the accident or serious incident; and (ii) if Republic of the Philippines is the State of the Operator, details of any dangerous goods on board the aircraft; (3) inform the State referred to in 13.060 (a)(2) - (i) whether the Authority intends to appoint or has appointed an accredited representative; and (ii) if such accredited representative will be or has been appointed and will be traveling to the State in which the investigation is being carried out, the contact details and the expected date of arrival of the accredited representative in such State.

Chapter 5

- 5.1.1 PCAR 13.095 (c) The Chairman may, with the approval of the Authority, delegate the task of carrying out an investigation into an accident or a serious incident to another Contracting State by mutual arrangement and consent.
- 5.28 PCAR 13.145 (c) In the interest of accident prevention, the Aircraft Accident Investigation and Inquiry Board (AAIIB) shall make the Final Report publicly available as soon as possible, and, if possible, within twelve months. Making a Final Report publicly available can be achieved by posting the Final Report on the internet and does not necessarily require a hard-copy publication of the Final Report.

Chapter 6

- 6.3.1 PCAR 13.140 A (a) The investigator-in-charge of the State conducting the investigation shall, as soon as possible after the completion of an investigation into an accident, do the following: (1) send a copy of the draft Final Report to the Contracting State that instituted the investigation and all the Contracting States that participated in the investigation, inviting their significant and substantiated comments on the draft Final Report.
- 6.3.2 PCAR 13.140 A (a) The investigator-in-charge of the State conducting the investigation shall, as soon as possible after the completion of an investigation into an accident, do the following: (1) send a copy of the draft Final Report to the Contracting State that instituted the investigation and all the Contracting States that participated in the investigation, inviting their significant and substantiated comments on the draft Final Report.

Chapter 7

- 7.6 PCAR 13.125 B (b) An accredited representative shall be entitled to participate in all aspects of an investigation under the control of the investigator-in-charge and shall be entitled, in particular, to: (1) visit the scene of the accident; (2) examine the wreckage; (3) obtain witness information and suggest areas for questioning witnesses; (4) have full access to all relevant evidence as soon as possible; (5) receive copies of all pertinent documents.

14. Annex 14 - Aerodromes

Volume I - Aerodromes Design and Operations, 9th Edition, July 2022, Amendment 17

Chapter 2

- 2.6.7 For development.

Chapter 3

- 3.4.4 30 M runway width may have a 150 M runway strip width 45 M or more runway width may have a 300 M runway strip width.
- 3.4.5 Partially only, 30 M width runway may have a 90 M runway strip width. RESAs are planned to be implemented for all aerodromes as required by the CAR to be certified by 2021.
- 3.5.1 Implementation of RESA requirement has not been completed for all aerodromes.

Chapter 4

- 4.2.13 Take-off climb surface if the runway is meant for take-off.

Chapter 5

- 5.1.4.2 A signal area must be 9 M in diameter.

Volume II - Heliports, 5th Edition, July 2020, Amendment 9

Chapter 3

- 3.1.20 An object situated in a helicopter clearway which will endanger helicopters in the air shall be regarded as an obstacle and shall be removed.
- 3.2.5 For helicopters with a MTOM of 3175 KG or less, the TLOF shall be of sufficient size to contain an area within which can be accommodated a circle of diameter of not less than 1.0 D of the largest helicopter the helideck is intended to serve.
- 3.2.7 The FATO shall be located so as to avoid, as far as is practicable, the influence of environmental effects, including turbulence, over the FATO, which can have an adverse impact on helicopter operations.
- 3.2.12 For any TLOF 1.0 D or greater and any TLOF designed for use by helicopters having a D-value of greater than 16.0 M, objects installed in the obstacle-free sector whose function requires them to be located on the edge of the TLOF shall be as low as possible and in any case not exceed a height of 15 CM.
- 3.3.11 The FATO shall be located so as to avoid, as far as is practicable, the influence of environmental effects, including turbulence, over the FATO, which can have an adverse impact on helicopter operations.

Chapter 4

- 4.2.6 Existing objects above any of the surfaces in 15.12.1.1 to 15.12.1.2 shall be removed except when the object is shielded by an existing immovable object or after an aeronautical study approved by CAAP determines that the object will not adversely affect the safety or significantly affect the regularity of operations of helicopters.
- 4.2.11 Note: An elevated heliport shall have at least two approach and take-off climb surfaces to avoid downwind conditions, minimize crosswind conditions and permit for a balked landing. See Helicopter Manual (Doc 9261) for guidance.

Chapter 5

- 5.1.1.3 Where TLOF and/or FATO may be subject to a disturbed airflow, then additional wind direction indicators located close to the area shall be provided to indicate the surface wind on the area.
- 5.1.1.6 The color of the wind direction indicator shall be so selected as to make it clearly visible and understandable from a height of at least 200 M (650 FT) above the heliport, having regard to background. Where, practicable, a single color, white or orange, shall be used. Where a combination of two colors is required to give adequate conspicuity against changing backgrounds, they must be orange and white, red and white, or black and white, and should be arranged in five alternate bands the first and last band being the darker color.
- 5.2.2.8 On a helideck or a shipboard heliport where the D-value is 16.0 M or larger, the size of the heliport identification "H" marking shall have a height of 4 M with an overall width not exceeding 3 M and a stroke width not exceeding 0.75 M. Where the D-value is less than 16.0 M, the size of the heliport identification H marking shall have a height of 3 M with an overall width not exceeding 2.25 M and a stroke width not exceeding 0.5 M.
- 5.2.3.2 A maximum allowable mass marking (a)... (b) shall be displayed at a surface level heliport.
- 5.2.3.3 A maximum allowable mass marking shall: (a)... (b)... (c) be located within the TLOF or FATO and so arranged as to be readable from the preferred final approach direction.

- 5.2.3.6 A maximum allowable mass marking shall: (a)... (b)... (c)... (d)... (e) be expressed to the nearest 100 KG. The marking shall be presented to one decimal place and rounded to the nearest 100 KG followed by the letter "t". When the maximum allowable mass is expressed to 100 KG, the decimal place shall be preceded with a decimal point marked with a 30 CM square.
- 5.2.3.7 A maximum allowable mass marking shall: (a)... (b)... (c)... (d)... (e) be expressed to the nearest 100 KG. The marking shall be presented to one decimal place and rounded to the nearest 100 KG followed by the letter "t". When the maximum allowable mass is expressed to 100 KG, the decimal place shall be preceded with a decimal point marked with a 30 CM square.
- 5.2.3.8 All FATOs except runway-type FATOs (a) The numbers and letter of the marking shall have a color contrasting with the background and be in the form and proportion shown in MOS Figure 15-23, for a D-value of more than 30 M. For a D-value between 15 M to 30 M the height of the numbers and the letter of the marking shall be a minimum of 90 CM, and for a D-value of less than 15 M the height of the numbers and the letter of the marking shall be a minimum of 60 CM, each with a proportional reduction in width and thickness.
- 5.2.3.9 Runway-type FATOs: (a) The numbers and the letter of the marking shall have a color contrasting with the background and shall be in the form and proportion shown in MOS Figure 15-23.
- 5.2.5.8 Runway-type FATOs: (a)... (b) FATO perimeter markers shall: (i)... (ii)... (iii) should be a single color, orange or red, or two contrasting colors, orange and white or alternatively red and white shall be used except where such colors would merge with the background.
- 5.2.6.1 A FATO designation marking shall be provided at a heliport where it is necessary to designate the FATO to the pilot.
- 5.2.10.2 Where a limited obstacle sector (LOS) exists on a helideck the marking should be located on the obstacle side of the "heliport identification marking". For a non-purpose built heliport located on a ship's side the marking shall be located on the inboard side of the heliport identification marking in the area between the TLOF perimeter marking and the boundary of the LOS.
- 5.2.10.5 Runway-type FATOs: (a) The characters of the marking shall be not less than 3 M in height.
- 5.2.10.6 All FATOs except runway-type FATOs (a) The characters of the marking shall not be less than 1.5 M in height at surface level heliports and not less than 1.2 M on elevated heliports, helidecks and shipboard heliports. The color of the marking should contrast with the background and preferably be white.
- 5.2.11.6 The color of the chevron 5.2.11.6 shall be black.
- 5.2.12.1 A surface marking shall be provided to assist the pilot to identify the location of the helideck or shipboard heliport during an approach by day.
- 5.2.12.2 A surface marking shall be applied to the dynamic load bearing area bounded by the TLOF perimeter marking.
- 5.2.12.3 The helideck or shipboard heliport surface bounded by the TLOF perimeter marking shall be of dark green using a high friction coating.

Note: Where the application of a surface coating may have a degrading effect on friction qualities the surface might not be painted. In such cases the best operating practice to enhance the conspicuity of markings is to outline deck markings with a contrasting color.

- 5.3.7.4 The light distribution of final approach and take-off area lights shall be as shown in MOS Figure 15-30, Illustration 4.
- 5.3.7.5 The lights shall not exceed a height of 25 CM and should be inset when a light extending above the surface would endanger helicopter operations. Where a FATO is not meant for lift-off or touchdown, the lights should not exceed a height of 25 CM above ground level.
- 5.3.8.1 Where an aiming point marking is provided at a heliport intended for use at night, aiming point lights shall be provided.
- 5.3.8.4 The light distribution of aiming point lights shall be as shown in MOS Figure 15-30, Illustration 4.
- 5.3.9.4 TLOF ASPSL and/or LPs to identify the TDPM and/or floodlighting shall be provided at a surface-level heliport intended for use at night when enhanced surface texture cues are required.
- 5.3.9.11 When LPs are used on an elevated heliport or helideck to enhance surface texture cues, the panels shall not be placed adjacent to the perimeter lights. They shall be placed around a Touchdown marking or coincident with heliport identification marking. 15.14.8.8.
- 5.3.9.15 The chromaticity and luminance of colors of LPs shall conform to MOS 9.2.6.
- 5.3.9.19 When located within the safety area of a surface level or elevated heliport, the TLOF floodlights shall not exceed a height of 25 CM.
- 5.3.9.22 The light distribution of the perimeter lights shall be as shown in MOS Figure 15-30, Illustration 5.
- 5.3.9.23 The light distribution of the LPs shall be as shown in MOS Figure 15-30, Illustration 6.
- 5.3.9.25 The average horizontal illuminance of the floodlighting shall be at least 19 lux, with a uniformity ratio (average to minimum) of not more than 8:1 measured on the surface of the TLOF.
- 5.3.9.26 Lighting used to identify the TDPC should comprise a segmented circle of omnidirectional ASPSL strips showing yellow. The segments shall consists of ASPSL strips, and the total length of the ASPSL strips being not less than 50% of the circumference of the circle.
- 5.3.9.27 If utilized, the heliport identification marking lighting shall be omnidirectional showing green.
- 5.3.10.1 Floodlighting shall be provided on a helicopter stand intended to be used at night.
- 5.3.11.4 The average horizontal illuminance shall be at least 10 lux, measured on the surface of the winching area.

DOC 9981 - PANS-Aerodromes, 3rd Edition 2020, Amendment 4.

15. Annex 15 - Aeronautical Information Services, 16th Edition, July 2018, Amendment 42.

Chapter 1

- 1.2.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 1.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations and the reference table used was the units of measurement table contained in the AIP Philippines (GEN 2.1).

Chapter 2

- 2.3.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.5 Exchange of aeronautical data and aeronautical information has been established by connecting the NOF and AIS Operations (AIS-OPS) of CAA of the Philippines (CAAP) Central Office with Manila Aeronautical Fixed Service (AFS) Station in order to facilitate the international exchange of aeronautical data and aeronautical information.
- 2.3.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.3.9 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 2.5.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 3

- 3.6.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 3.6.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 5

- 5.2.1.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.2.5.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.2.1 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.2.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.3.5 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.3.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.3.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.3.9 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.4.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.4.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.4.8 The recommendation was adopted to mandatory requirements in the Philippine regulations.

- 5.3.3.4.9 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.3.4.11 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.4.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.3.5.2 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.4.1.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 5.4.2.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.

Chapter 6

- 6.2.6 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.2.7 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.3.3.3 The recommendation was adopted to mandatory requirements in the Philippine regulations.
- 6.3.3.4 The recommendation was adopted to mandatory requirements in the Philippine regulations.

16. Annex 16 - Environmental Protection - Nil.

17. Annex 17 - Aviation Security, 12th Edition, July 2022, Amendment 18 - Nil.

18. Annex 18 - The Safe Transport of Dangerous Goods by Air, 4th Edition, July 2011, Amendment 12.

Chapter 2

- 2.1.1 PCAR Part 18.001 SUBPART A: GENERAL 18.001 APPLICABILITY (a) This Part prescribes the requirements of the Republic of the Philippines that apply to the carriage of dangerous goods by air in international and domestic operations of aircraft as specified in – (1) The International Civil Aviation Organization Document, Technical Instructions for the Safe Transport of Dangerous Goods by Air, and all applicable amendments; and (2) As amplified by, the Dangerous Goods Regulations of the International Air Transport Association. (b) This Part is applicable to all Dangerous Goods (DG) and No-DG carry operators of aircraft in - (1) Aerial work; (2) Commercial air transport; and (3) General aviation.

19. Annex 19 - Safety Management, 2nd Edition, July 2016 - Nil.

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